

8		7		6		5		4		3		2		1									
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.												REV	ECN	DESCRIPTION OF REVISION	CK APPD								
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.															DATE								
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.															2012-07-03								
D8 MLB ULTIMATE																							
LAST_MODIFIED= Tue Jul 3 15:14:05 2012																							
Page				Contents				Sync				Page				Contents				Sync			
1				Table of Contents				MASTER				63				AUDIO: Speaker ID				D8_MLB			
2				System Block Diagram				K70_MLB				64				PM Regulator Enables				D8_MLB			
3				Power Block Diagram				D8_MLB				65				PM Power Good				D8_MLB			
4				BOM Configuration				D8_MLB-ULTIMATE				66				VReg CPU Core/AXG Cntl				D8_MLB			
5				DEBUG LEDS				D8_MLB				67				VReg CPU Core Phases				D8_MLB			
6				Power Connectors/Aliases				D8_MLB				68				VReg CPU AXG Phases				D8_MLB			
7				Holes/PD parts				D8_MLB				69				VReg CPU 1.05V S0				D8_MLB			
8				Unused Signal Aliases				D8_MLB				70				VReg CPU VccSA S0				D8_MLB			
9				Signal Aliases				D8_MLB				71				VReg 3.3V S5/SV S4				D8_MLB			
10				CPU DMI/PEG/FDI/RSVD				D8_MLB				72				VReg VDDQ and 1.8V S0				D8_MLB			
11				CPU CLOCK/MISC/JTAG				D8_MLB				73				VREG 3.42V G3HOT				D8_MLB			
12				CPU DDR3 INTERFACES				D8_MLB				74				FET-Controlled S0 and S4				D8_MLB			
13				CPU POWER				D8_MLB				75				Internal DP MUXing				D8_MLB			
14				CPU GROUNDS				D8_MLB				76				TBT DDC Crossbar				D8_MLB			
15				STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU				D8_MLB				77				Thunderbolt Connector A				D8_MLB			
16				CPU NON-GFX DECOUPLING				D8_MLB				78				Internal DP Support				D8_MLB			
17				GFX DECOUPLING & PCH PWR ALIAS				D8_MLB				79				Thunderbolt Connector B				D8_MLB			
18				PCH SATA/PCIE/CLK/LPC/SPI				D8_MLB				80				Backlight Controller MCU				D8_MLB			
19				PCH DMI/FDI/GRAPHICS				D8_MLB				81				Backlight LED Driver				D8_MLB			
20				PCH PCI/USB				D8_MLB				82				Backlight Controller				D8_MLB			
21				PCH MISC				D8_MLB				83				KEPLER PCI-E				D8_YAN			
22				PCH POWER				D8_MLB				84				KEPLER FRAME BUFFER A/B				D8_YAN			
23				PCH GROUNDS				D8_MLB				85				KEPLER FRAME BUFFER C/D				D8_YAN			
24				PCH DECOUPLING				D8_MLB				86				GDDR5 Frame Buffer A				D8_YAN			
25				CPU and PCH XDP				D8_MLB				87				GDDR5 Frame Buffer B				D8_YAN			
26				CHIPSET SUPPORT				D8_MLB				88				GDDR5 FRAME BUFFER C				D8_YAN			
27				USB 2.0 HUB (BT/SMC)				D8_MLB				89				GDDR5 FRAME BUFFER D				D8_YAN			
28				CPU Memory S3 Support				D8_MLB				90				KEPLER EDP/DP/GPIO				D8_YAN			
29				DDR3 SO-DIMM Connector A Slot0				D8_MLB				91				KEPLER GPIO/STRAPPING				D8_YAN			
30				DDR3 SO-DIMM Connector A Slot1				D8_MLB				92				KEPLER MISC				D8_YAN			
31				DDR3 SO-DIMM CONNECTOR B SLOT0				D8_MLB				93				KEPLER CORE POWER				D8_YAN			
32				DDR3 SO-DIMM CONNECTOR B SLOT1				D8_MLB				94				KEPLER FBVDD/Q POWER				D8_YAN			
33				DDR3 ALIASES AND BITSWAPS				D8_MLB				95				KEPLER PEX PWR/GNDS				D8_YAN			
34				DDR3/FRAMEBUF VREF MARGINING				D8_MLB				96				VReg GPU Core Phases				D8_MLB			
35				AIRPORT/BT				D8_MLB				97				VReg GPU Core Phases				D8_MLB			
36				Thunderbolt Host (1 of 2)				D8_MLB				98				VREG GPU CORE PHASE 4				D8_MLB			
37				Thunderbolt Host (2 of 2)				D8_MLB				99				GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS				D8_MLB			
38				Thunderbolt Power Support				D8_MLB				100				D8 RULE DEFINITIONS				D8_MLB			
39				ETHERNET PHY (CAESAR IV+)				D8_MLB				101				DDR3 Constraints				D8_MLB			
40				Ethernet Support & Connector				D8_MLB				102				CPU PCIe Constraints				D8_MLB			
41				SD READER CONNECTOR				D8_MLB				103				CPU MISC/DMI/FDI/XDP Constraints				D8_MLB			
42				Camera Controller				D8_MLB				104				SATA/FDI/XDP Constraints				D8_MLB			
43				Camera Controller Support				D8_MLB				105				PCH and BR Constraints				D8_MLB			
44				SATA Connectors				D8_MLB				106				USB/Camera Constraints				D8_MLB			
45				EXTERNAL USB PORTS A & B				D8_MLB				107				SMBus/Sensor Constraints				D8_MLB			
46				EXTERNAL USB PORTS C & D				D8_MLB				108				VReg Constraints				D8_MLB			
47				SMC				D8_MLB				109				CPU VReg Constraints				D8_MLB			
48				SMC Support				D8_MLB				110				Platform VReg Constraints				D8_MLB			
49				SPI and Debug Connector				D8_MLB				111				TBT/DP Constraints				D8_MLB			
50				SMBus Connections				D8_MLB				112				GDDR5/GPU Constraints				D8_MLB			
51				I and V Sense 1				D8_MLB				113				GDDR5 FB C/D CONSTRAINTS				D8_MLB			
52				HDD/SSD Temp Sense				D8_MLB				114				BLC Constraints				D8_MLB			
53				Temperature Sensors				D8_MLB				115				GPU VREG CONSTRAINTS				D8_MLB			
54				System Fan				D8_MLB				116				ETHERNET/SD CONSTRAINTS				D8_MLB			
55				I and V Sense 2				D8_MLB				117				AUTO-CONSTRAINTS 1				D8_MARK			
56				AUDIO: CODEC/REGULATORS				D8_MLB				118				AUTO-CONSTRAINTS 2				D8_MARK			
57				AUDIO: HEADPHONE AMP				D8_MLB				119				AUTO-CONSTRAINTS 3				D8_MARK			
58				AUDIO: LEFT SPKR AMP				D8_MLB				120				AUTO-CONSTRAINTS 4				D8_MARK			
59				AUDIO: RIGHT SPKR AMP				D8_MLB				121				AUTO-CONSTRAINTS 5				D8_MARK			
60				AUDIO: Jack, Mikey, CHS Switch				D8_MLB				122				AUTO-CONSTRAINTS 6				D8_MARK			
61				Audio: Spkr/Mic Conn.				D8_MLB				123				AUTO-CONSTRAINTS 7				D8_MARK			
62				AUDIO: Detects/Grounding				D8_MLB															
DRAWING																							
TITLE=K72																							
ABBREV=DRAWING																							
LAST_MODIFIED= Tue Jul 3 15:14:05 2012																							
8		7		6		5		4		3		2		1									

SCH, D8, MLB ULTIMATE

Apple Inc.

051-9505

D

REVISION

6.1.0

BRANCH

PAGE

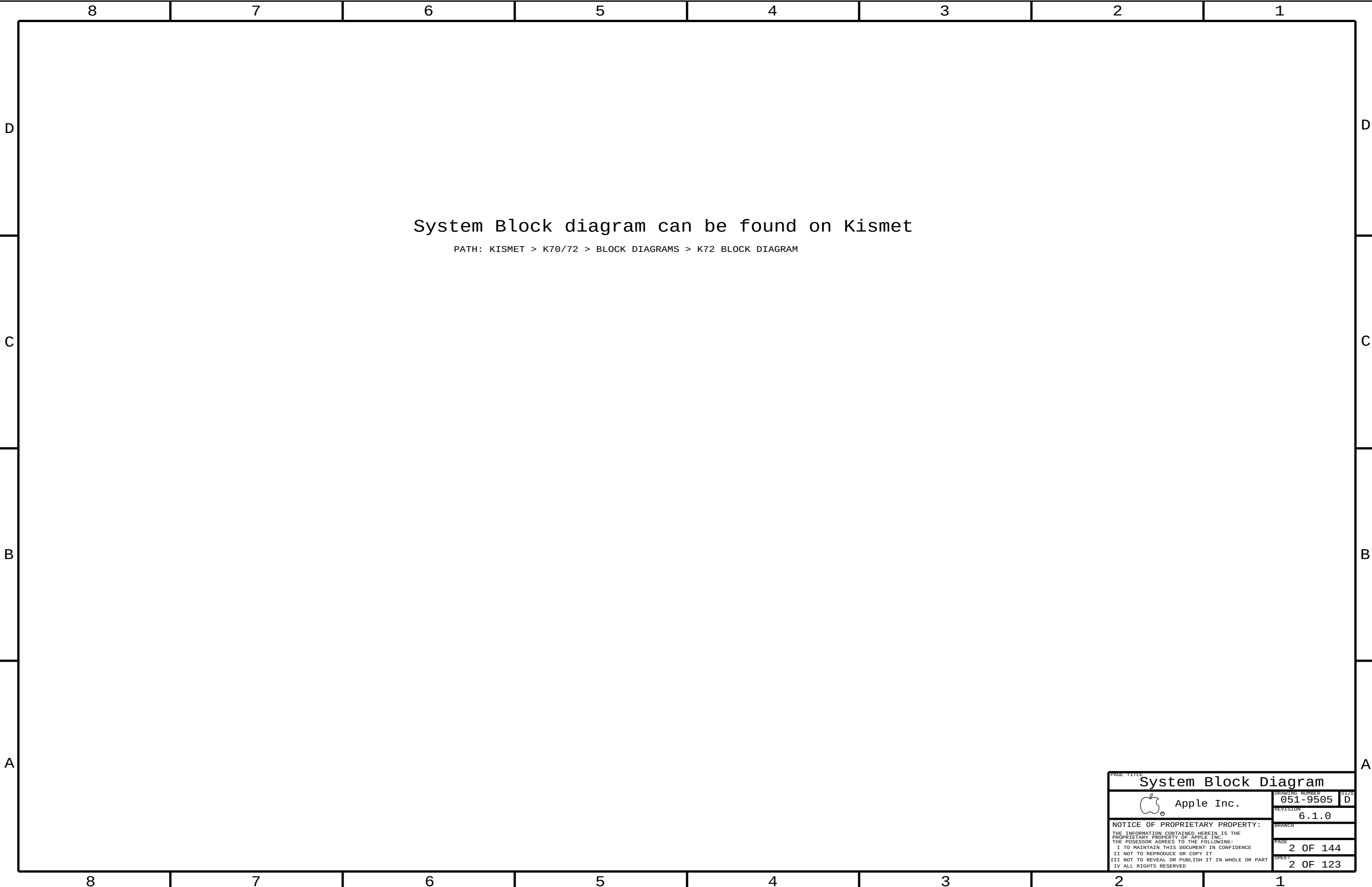
1 OF 144

SHEET

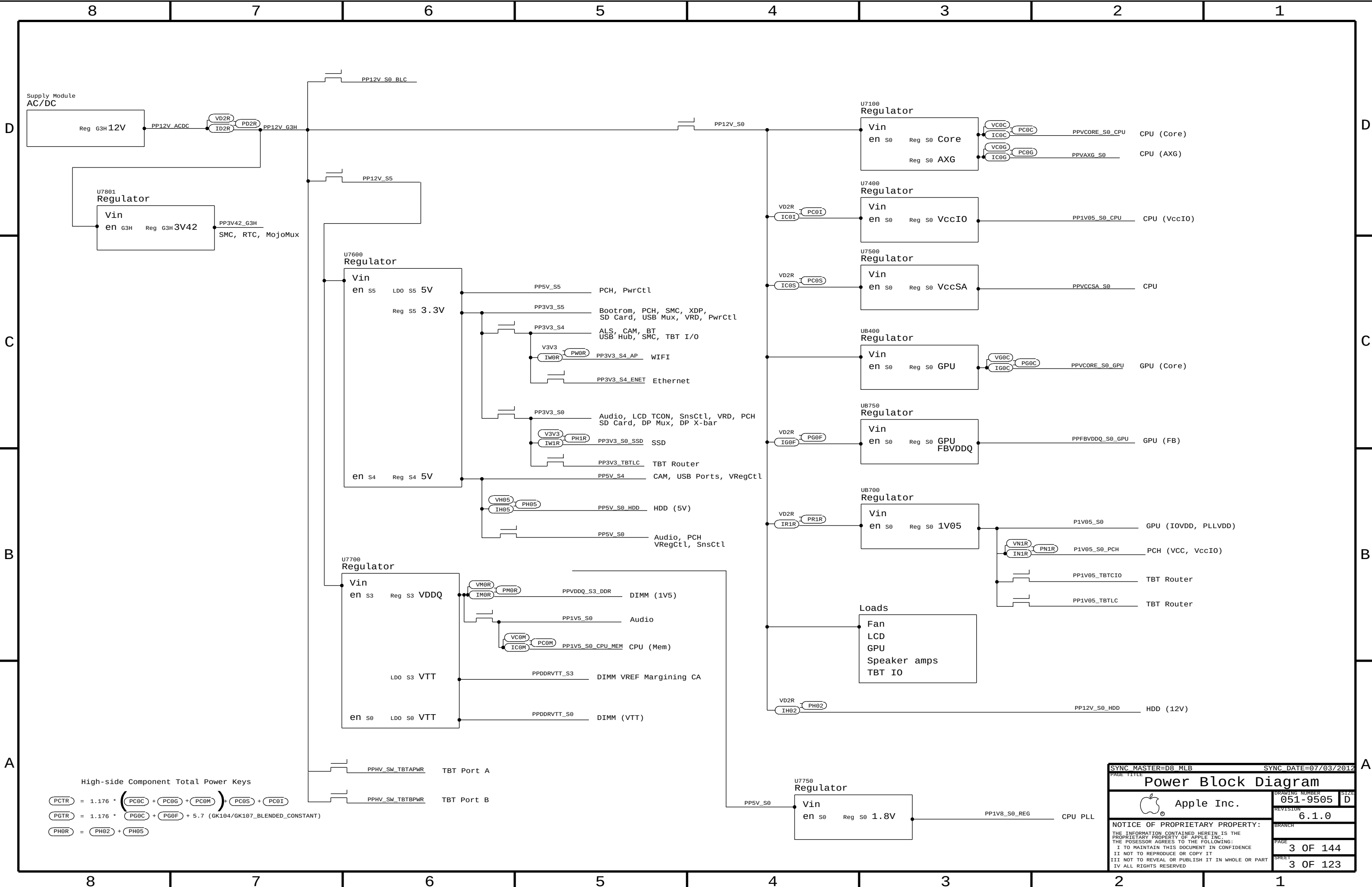
1 OF 123

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IN ALL RIGHTS RESERVED



PAGE TITLE		System Block Diagram	
 Apple Inc.	DRAWING NUMBER		051-9505
	REVISION		6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		2 OF 144	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		2 OF 123	
IV ALL RIGHTS RESERVED			



High-side Component Total Power Keys

$$\text{PCTR} = 1.176 * \left(\text{PC0C} + \text{PC0G} + \text{PC0M} \right) + \text{PC0S} + \text{PC0I}$$
$$\text{PGTR} = 1.176 * \left(\text{PG0C} + \text{PG0F} \right) + 5.7 \text{ (GK104/GK107_BLENDED_CONSTANT)}$$
$$\text{PH0R} = \text{PH02} + \text{PH05}$$

SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

Power Block Diagram

Apple Inc.

DRAWING NUMBER
051-9505

REVISION
6.1.0

BRANCH

PAGE
3 OF 144

SHEET
3 OF 123

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-3662	PCBA, MLB, ULTIMATE, 3. 4G, GTX, SAM, 2GB, D8	D8_COMMON, D8, CPU:4C_3P4GHZ, GPU:104GTX, FB:2G_SAMSUNG, EEEE:F0V5
639-3950	PCBA, MLB, ULTIMATE, 3. 2G, GTX, SAM, 2GB, D8	D8_COMMON, D8, CPU:4C_3P2GHZ, GPU:104GTX, FB:2G_SAMSUNG, EEEE:F49R
639-3560	PCBA, MLB, ULTIMATE, 3. 4G, GT, SAM, 1GB, D8	D8_COMMON, D8, CPU:4C_3P4GHZ, GPU:104GT, FB:1G_SAMSUNG, EEEE:DYN3
639-3949	PCBA, MLB, ULTIMATE, 3. 2G, GT, SAM, 1GB, D8	D8_COMMON, D8, CPU:4C_3P2GHZ, GPU:104GT, FB:1G_SAMSUNG, EEEE:F49P
639-4087	PCBA, MLB, ULTIMATE, 3. 4G, GTX, HYN, 2GB, D8	D8_COMMON, D8, CPU:4C_3P4GHZ, GPU:104GTX, FB:2G_HYNIX, EEEE:F64W
639-4091	PCBA, MLB, ULTIMATE, 3. 2G, GTX, HYN, 2GB, D8	D8_COMMON, D8, CPU:4C_3P2GHZ, GPU:104GTX, FB:2G_HYNIX, EEEE:F652
639-4086	PCBA, MLB, ULTIMATE, 3. 4G, GT, HYN, 1GB, D8	D8_COMMON, D8, CPU:4C_3P4GHZ, GPU:104GT, FB:1G_HYNIX, EEEE:F64V
639-4090	PCBA, MLB, ULTIMATE, 3. 2G, GT, HYN, 1GB, D8	D8_COMMON, D8, CPU:4C_3P2GHZ, GPU:104GT, FB:1G_HYNIX, EEEE:F651
085-4435	PCBA, MLB, DEV, D8. ULTIMATE	DEVELOPMENT, D8_DEVEL

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_DYW3	CRITICAL	EEEE: DYW3
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F0V5	CRITICAL	EEEE: F0V5
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F49P	CRITICAL	EEEE: F49P
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F49R	CRITICAL	EEEE: F49R
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F4MW	CRITICAL	EEEE: F4MW
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F4TY	CRITICAL	EEEE: F4TY
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F64W	CRITICAL	EEEE: F64W
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F652	CRITICAL	EEEE: F652
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F64V	CRITICAL	EEEE: F64V
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F651	CRITICAL	EEEE: F651

BOM Groups

BOM GROUP	BOM OPTIONS
D8_COMMON	COMMON,ALTERNATE,D8_COMMON1,D8_PROGPARTS
D8_COMMON1	XDP,RSMRST:GATE,SPEAKERID,VREF:CPU,TBTHV:P12V,PRE_BOOST:Y
D8_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,BLCMCU:PROG
D8_DEVEL	XDP_CONN,LPCPLUS,VREFMRGN:EXT,BKLT_PWM,DEVEL_AUDIO,TEMPSNSDEV

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33754356	1	1VB, SR0TS, PRQ, N1, 3, 2, 77W, 4+1, 1, 1, 6H, LG	CPU	CRITICAL	CPU: 4C_3P2GHZ
33754247	1	1VB, SR0PK, PRQ, E1, 3, 4, 77W, 4+2, 1, 15, 1M, LG	CPU	CRITICAL	CPU: 4C_3P4GHZ

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4277	1	IC, PANTHER POINT, C1, 5L3C7, PRQ, 8082277	U1800	CRITICAL	
338S1113	1	IC, TBT, CR-4C, B1, PRQ, 288 FCBGA, 12X12MM	U3600	CRITICAL	
343S0616	1	IC, BCM57766A1, ENET&SD, 8X8	U3900	CRITICAL	
337S4333	1	IC, GPU, NV GK104 7-4-PS-A2	UA000	CRITICAL	GPU:104GT
337S4333	1	IC, GPU, NV GK104 7-4-PS-A2	UA000	CRITICAL	GPU:104GT2
337S4332	1	IC, GPU, NV GK104 8-4-PS-A2,	UA000	CRITICAL	GPU:104GTX

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3640	1	IC,EEPROM,CR,V11.1 (B1),D8	U3690	CRITICAL	TBTR0M:PROG
33S0865	1	IC,EEPROM,SERIAL,8KB,MLP8	U3690	CRITICAL	TBTR0M:BLANK
341S3641	1	IC,PROGRMD,EFI ROM,PIB,D7/D8	U5110	CRITICAL	BOOTROM:PROG
33S0867	1	IC,64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK
341S3394	1	IC,PROGRMD,SMC,A3,V2.2A32,D8	U4900	CRITICAL	SMC:PROG
33S1098	1	IC,SMC,LX4F51AH5BBCIGA3	U4900	CRITICAL	SMC:BLANK
341S3646	1	IC,CAMERA FLASH,V7226,D7/D8	U4202	CRITICAL	CAMROM:PROG
33S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK
341S3645	1	IC,ENET 1MBIT, SPI,ROM, V.1.3 D8	U3990	CRITICAL	CIVROM:PROG
33S0862	1	IC,SERIAL FLASH,2MBIT, 2.7V, REF F	U3990	CRITICAL	CIVROM:BLANK
341S3627	1	IC,BLC,MCU, PRPGRAMMED, V0283, D8	U9700	CRITICAL	BLCMCU:PROG
33S3978	1	IC,BLC MCU LPC2132FBD64/01, LQFP64	U9700	CRITICAL	BLCMCU:BLANK

ALTERNATE: 335S0812

ALTERNATE: 335S0854

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0073	1	SOCKET.MOLEX, LGA1155,CPU- LF	U1000	CRITICAL	

CPU SOCKET ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
511S0071	511S0073		ALL	TYCO SOCKET
511S0072	511S0073		ALL	FOXCONN SOCKET

D8 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9505	1	SCH, MLB, D8, ULTIMATE	SCH1	CRITICAL	D8
820-3299	1	PCBF, MLB, D8, ULTIMATE	PCB1	CRITICAL	D8

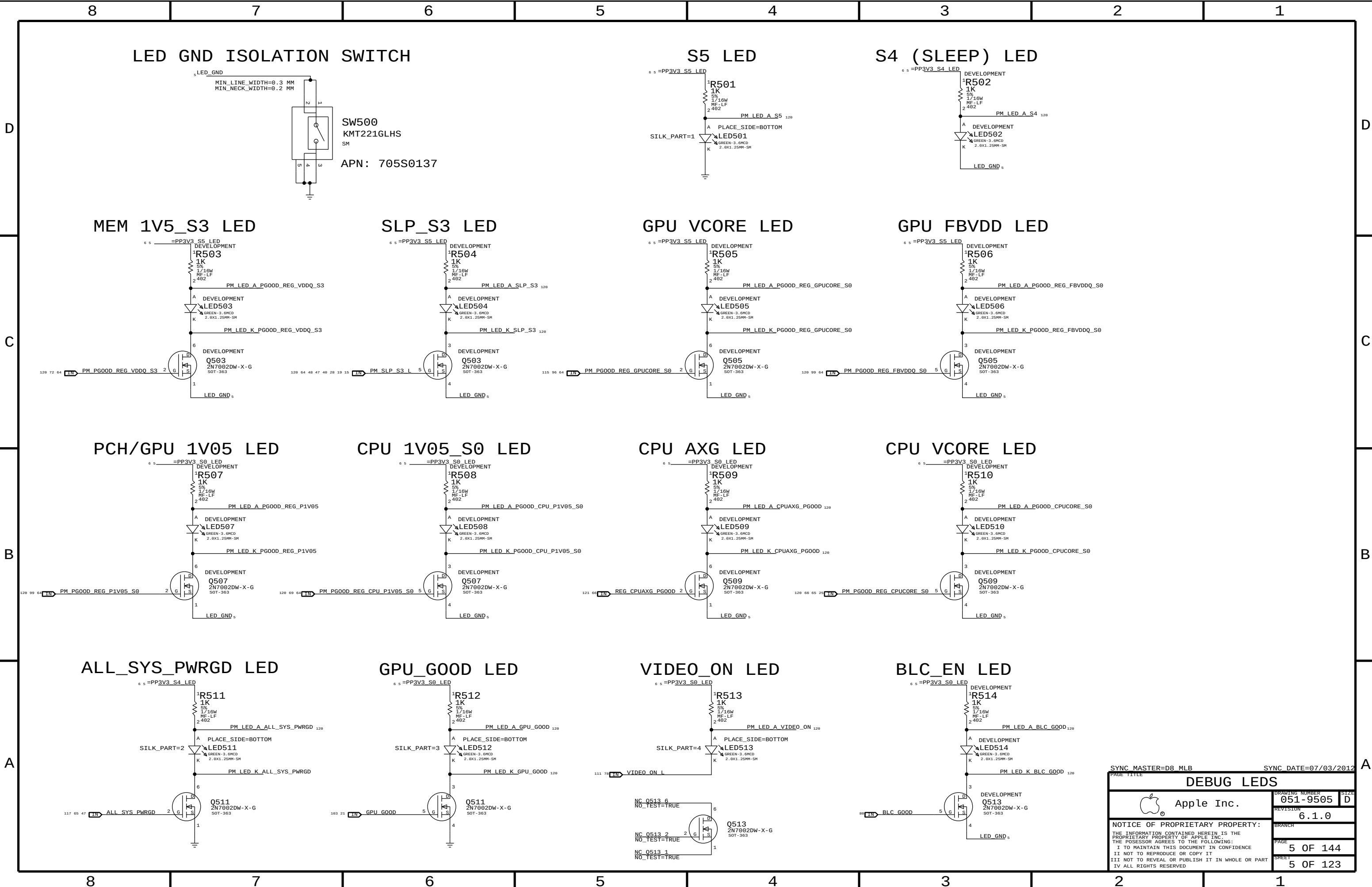
D8 ALTERNATES

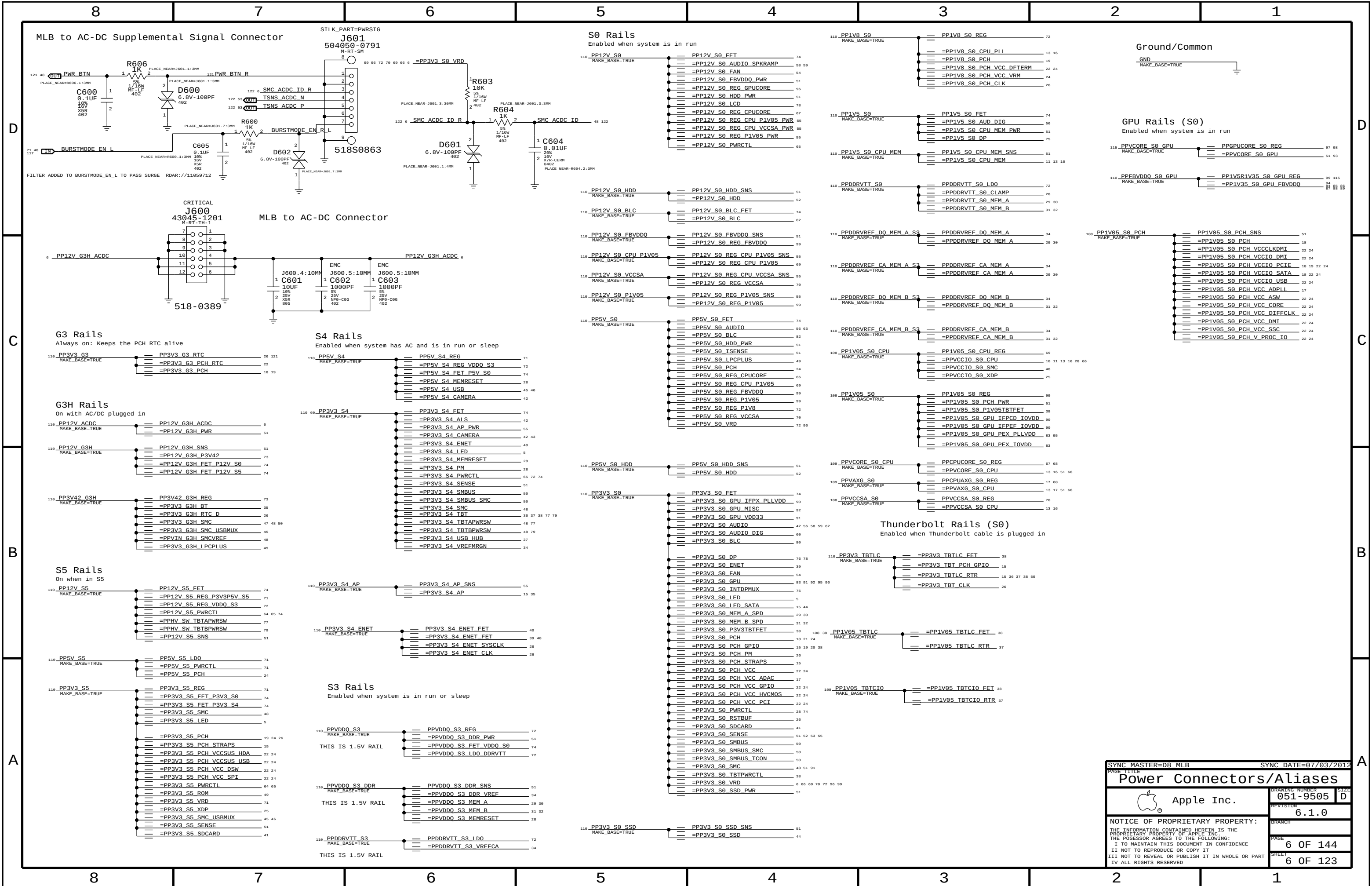
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB diodes
157S0084	157S0058		ALL	Enet Magnetics
341S3644	341S3645		U3990	CIVROM
376S0975	376S1081		ALL	P/NCH DUAL FET
152S1668	152S1591		ALL	33UH BLC INDUCTOR
128S0365	128S0368		ALL	150UF CAPS BLK

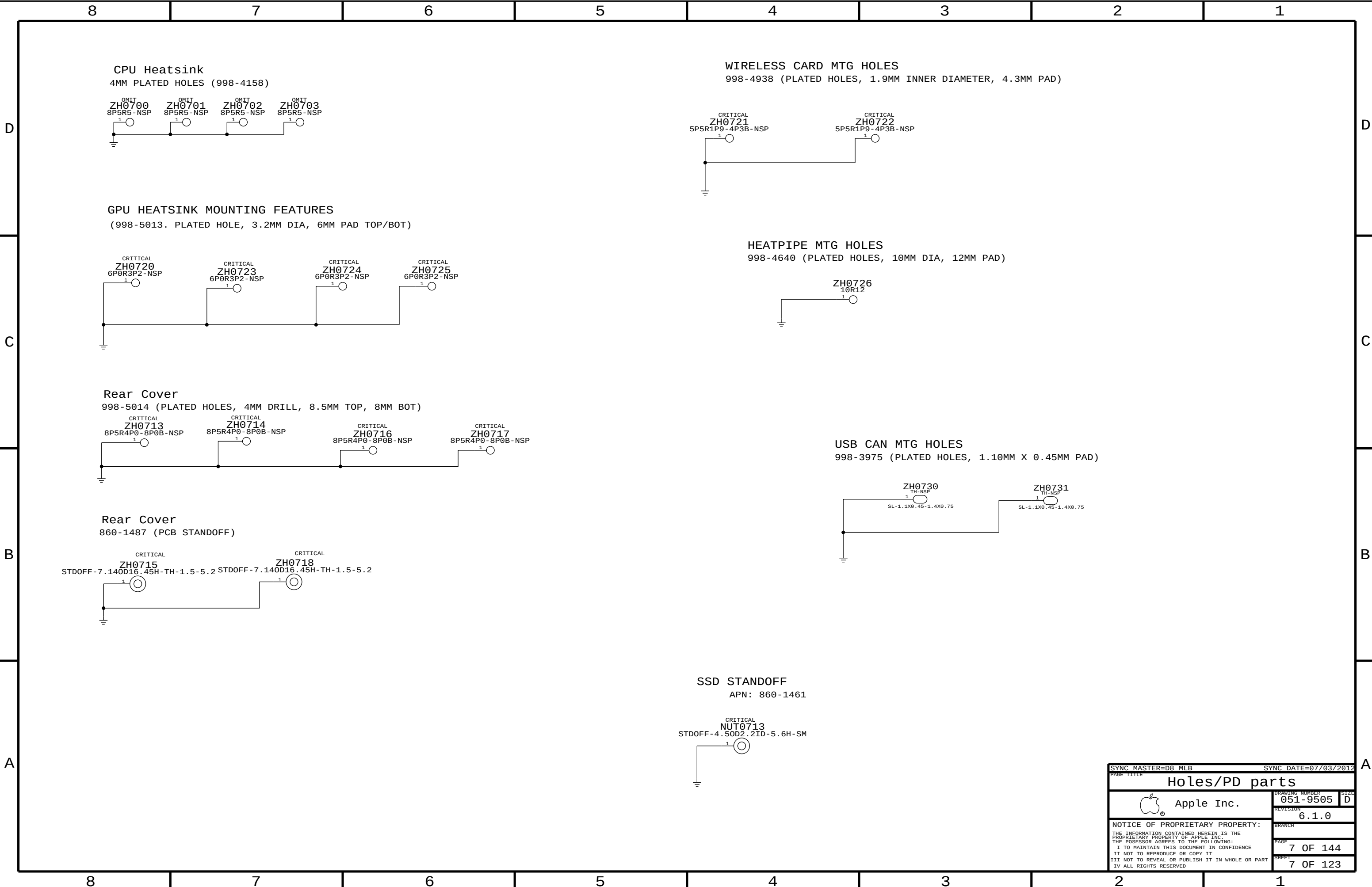
VRAM Module Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33350619	4	IC, SGRAM, GDDR5, 32MX32, 1.5GHZ, G-DIE, HF	UA300, UA350, UA400, UA450	CRITICAL	FB:1G_SAMSUNG
33350619	4	IC, SGRAM, GDDR5, 32MX32, 1.5GHZ, G-DIE, HF	UA500, UA550, UA600, UA650	CRITICAL	FB:1G_SAMSUNG
33350626	4	IC, GDDR5, 32MX32, 1.5GHZ, VEGA 44N, B-DIE	UA300, UA350, UA400, UA450	CRITICAL	FB:1G_HYNIX
33350626	4	IC, GDDR5, 32MX32, 1.5GHZ, VEGA 44N, B-DIE	UA500, UA550, UA600, UA650	CRITICAL	FB:1G_HYNIX
33350631	4	IC, SGRAM, GDDR5, 64MX32, D-DIE	UA300, UA350, UA400, UA450	CRITICAL	FB:2G_SAMSUNG
33350631	4	IC, SGRAM, GDDR5, 64MX32, D-DIE	UA500, UA550, UA600, UA650	CRITICAL	FB:2G_SAMSUNG
33350630	4	IC, GDDR5, 64MX32, A-DIE	UA300, UA350, UA400, UA450	CRITICAL	FB:2G_HYNIX
33350630	4	IC, GDDR5, 64MX32, A-DIE	UA500, UA550, UA600, UA650	CRITICAL	FB:2G_HYNIX

SYNC MASTER=D8 MLB ULTIMATE		SYNC DATE=06/15/2012	
PAGE TITLE			
 Apple Inc.		DRAWING NUMBER	
		051-9505	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		6.1.0	
		BRANCH	
		PAGE	
		4 OF 144	
		SHEET	
		4 OF 123	







SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
Holes/PD parts			
 Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	7 OF 144
		SHEET	7 OF 123

8	7	6	5	4	3	2	1		
CPU Reserved		PCH PCIE		PCH Unused Display		PCH SATA		PCH Test Points	
10 TP_CPU_RSVD<16..1> == NC_CPU_RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE1_D2RN == NC_PCIE1_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_RED == NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_R2D_CN == NC_SATA_C_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP1 == NC_PCH_TP1 MAKE_BASE=TRUE NO_TEST=TRUE	
10 TP_CPU_RSVD<46..19> == NC_CPU_RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE1_D2RP == NC_PCIE1_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_GREEN == NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_R2D_CP == NC_SATA_C_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP2 == NC_PCH_TP2 MAKE_BASE=TRUE NO_TEST=TRUE	
10 CPU_CFG<15..12> == TP_CPU_CFG<15..12> MAKE_BASE=TRUE		10 TP_PCIE1_R2D_CN == NC_PCIE1_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_BLUE == NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_D2RN == NC_SATA_C_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP3 == NC_PCH_TP3 MAKE_BASE=TRUE NO_TEST=TRUE	
CPU Memory		10 TP_PCIE1_R2D_CP == NC_PCIE1_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_HSYNC == NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_D2RP == NC_SATA_C_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP4 == NC_PCH_TP4 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQ_CB<7..0> == NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_D2RN == NC_PCIE2_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_VSYNC == NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_R2D_CN == NC_SATA_D_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP5 == NC_PCH_TP5 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQS_N<8> == NC_MEM_A_DQS_N<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_D2RP == NC_PCIE2_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_DDC_CLK == NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_R2D_CP == NC_SATA_D_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP6 == NC_PCH_TP6 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQS_P<8> == NC_MEM_A_DQSPX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_R2D_CN == NC_PCIE2_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_DDC_DATA == NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_D2RN == NC_SATA_D_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP7 == NC_PCH_TP7 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQ_CB<7..0> == NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_R2D_CP == NC_PCIE2_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_MLN<3..0> == NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_D2RP == NC_SATA_D_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP8 == NC_PCH_TP8 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQS_N<8> == NC_MEM_B_DQS_N<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI_MIDBUS_CLK100M_N == NC_DMI_MIDBUS_CLK100M_N MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_MLP<3..0> == NC_DP_IG_B_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_R2D_CN == NC_SATA_E_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP9 == NC_PCH_TP9 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQS_P<8> == NC_MEM_B_DQSPX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI_MIDBUS_CLK100M_P == NC_DMI_MIDBUS_CLK100M_P MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_AUX_N == NC_DP_IG_B_AUX_N MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_R2D_CP == NC_SATA_E_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP10 == NC_PCH_TP10 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE0N == NC_PCIE_CLK100M_PE0N MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_AUX_P == NC_DP_IG_B_AUX_P MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_D2RN == NC_SATA_E_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP11 == NC_PCH_TP11 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE0P == NC_PCIE_CLK100M_PE0P MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_HPD == NC_DP_IG_B_HPD MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_D2RP == NC_SATA_E_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP12 == NC_PCH_TP12 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE4N == NC_PCIE_CLK100M_PE4N MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_DDC_CLK == NC_DP_IG_B_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP13 == NC_PCH_TP13 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE4P == NC_PCIE_CLK100M_PE4P MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_DDC_DATA == NC_DP_IG_B_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP14 == NC_PCH_TP14 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE5N == NC_PCIE_CLK100M_PESN MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_MLN<3..0> == NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP15 == NC_PCH_TP15 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE5P == NC_PCIE_CLK100M_PESP MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_MLP<3..0> == NC_DP_IG_C_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP16 == NC_PCH_TP16 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE6N == NC_PCIE_CLK100M_PEGN MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_AUX_N == NC_DP_IG_C_AUXN MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP17 == NC_PCH_TP17 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE6P == NC_PCIE_CLK100M_PEPX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_AUX_P == NC_DP_IG_C_AUXP MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP18 == NC_PCH_TP18 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE7N == NC_PCIE_CLK100M_PETN MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_HPD == NC_DP_IG_C_HPD MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP19 == NC_PCH_TP19 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE7P == NC_PCIE_CLK100M_PETP MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_CTRL_CLK == NC_DP_IG_C_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP20 == NC_PCH_TP20 MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_C_CTRL_DATA == NC_DP_IG_C_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE					
		10 TP_PE_TX_N<3..0> == NC_PE_TNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_MLN<3..0> == NC_DP_IG_D_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		PCH Reserved		PCH PCI	
		10 TP_PE_TX_P<3..0> == NC_PE_TPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_MLP<3..0> == NC_DP_IG_D_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_0 == NC_PCH_RESERVE_0 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_AD<31..0> == NC_PCI_AD<31..0> MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PE_RX_N<3..0> == NC_PE_RNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_AUXN == NC_DP_IG_D_AUXN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_1 == NC_PCH_RESERVE_1 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_C_BE_L<3..0> == NC_PCI_C_BE_L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PE_RX_P<3..0> == NC_PE_RPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_AUXP == NC_DP_IG_D_AUXP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_2 == NC_PCH_RESERVE_2 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_PAR == NC_PCI_PAR MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_D_HPD == NC_DP_IG_D_HPD MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_3 == NC_PCH_RESERVE_3 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_RESET_L == NC_PCI_RESET_L MAKE_BASE=TRUE NO_TEST=TRUE	
		PCH USB		10 DP_IG_D_CTRL_CLK == NC_DP_IG_D_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_4 == NC_PCH_RESERVE_4 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCH_PCI_GNT0_L == NC_PCH_PCI_GNT0_L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_4_N == NC_USB_PCH_4NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_CTRL_DATA == NC_DP_IG_D_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_5 == NC_PCH_RESERVE_5 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_INIT3V3_L == NC_PCH_INIT3V3_L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_4_P == NC_USB_PCH_4PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_6 == NC_PCH_RESERVE_6 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_LPC_DREQ0_L == NC_LPC_DREQ0_L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_5_N == NC_USB_PCH_5NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_TVCLKINN == NC_SDVO_TVCLKINN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_7 == NC_PCH_RESERVE_7 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_5_P == NC_USB_PCH_5PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_TVCLKINP == NC_SDVO_TVCLKINP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_8 == NC_PCH_RESERVE_8 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_6_N == NC_USB_PCH_6NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_STALLN == NC_SDVO_STALLN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_9 == NC_PCH_RESERVE_9 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_6_P == NC_USB_PCH_6PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_STALLP == NC_SDVO_STALLP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_10 == NC_PCH_RESERVE_10 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_11_N == NC_USB_PCH_11NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_INTN == NC_SDVO_INTN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_11 == NC_PCH_RESERVE_11 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_11_P == NC_USB_PCH_11PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_INTP == NC_SDVO_INTP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_12 == NC_PCH_RESERVE_12 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_12_N == NC_USB_PCH_12NX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_13 == NC_PCH_RESERVE_13 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_12_P == NC_USB_PCH_12PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_14 == NC_PCH_RESERVE_14 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_13_N == NC_USB_PCH_13NX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_15 == NC_PCH_RESERVE_15 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_13_P == NC_USB_PCH_13PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_16 == NC_PCH_RESERVE_16 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_17 == NC_PCH_RESERVE_17 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_18 == NC_PCH_RESERVE_18 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_19 == NC_PCH_RESERVE_19 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_20 == NC_PCH_RESERVE_20 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_21 == NC_PCH_RESERVE_21 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_22 == NC_PCH_RESERVE_22 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_23 == NC_PCH_RESERVE_23 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_24 == NC_PCH_RESERVE_24 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_25 == NC_PCH_RESERVE_25 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_26 == NC_PCH_RESERVE_26 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_27 == NC_PCH_RESERVE_27 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_28 == NC_PCH_RESERVE_28 MAKE_BASE=TRUE NO_TEST=TRUE			

SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

PAGE TITLE

Unused Signal Aliases

 Apple Inc.

DRAWING NUMBER
051-9505

SIZE
D

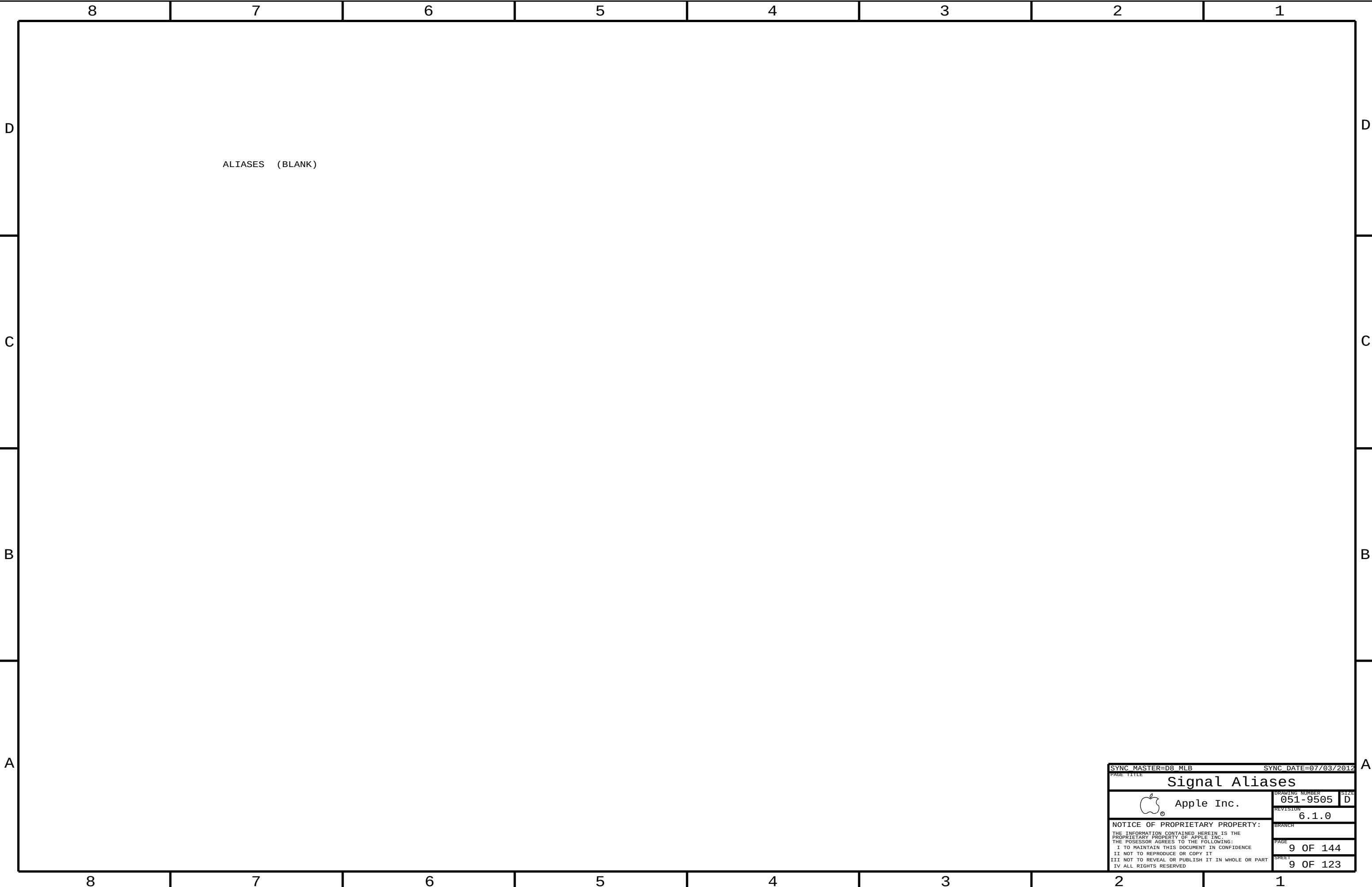
REVISION
6.1.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE
8 OF 144

SHEET
8 OF 123



SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

PAGE TITLE

Signal Aliases

 Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9505

REVISION

6.1.0

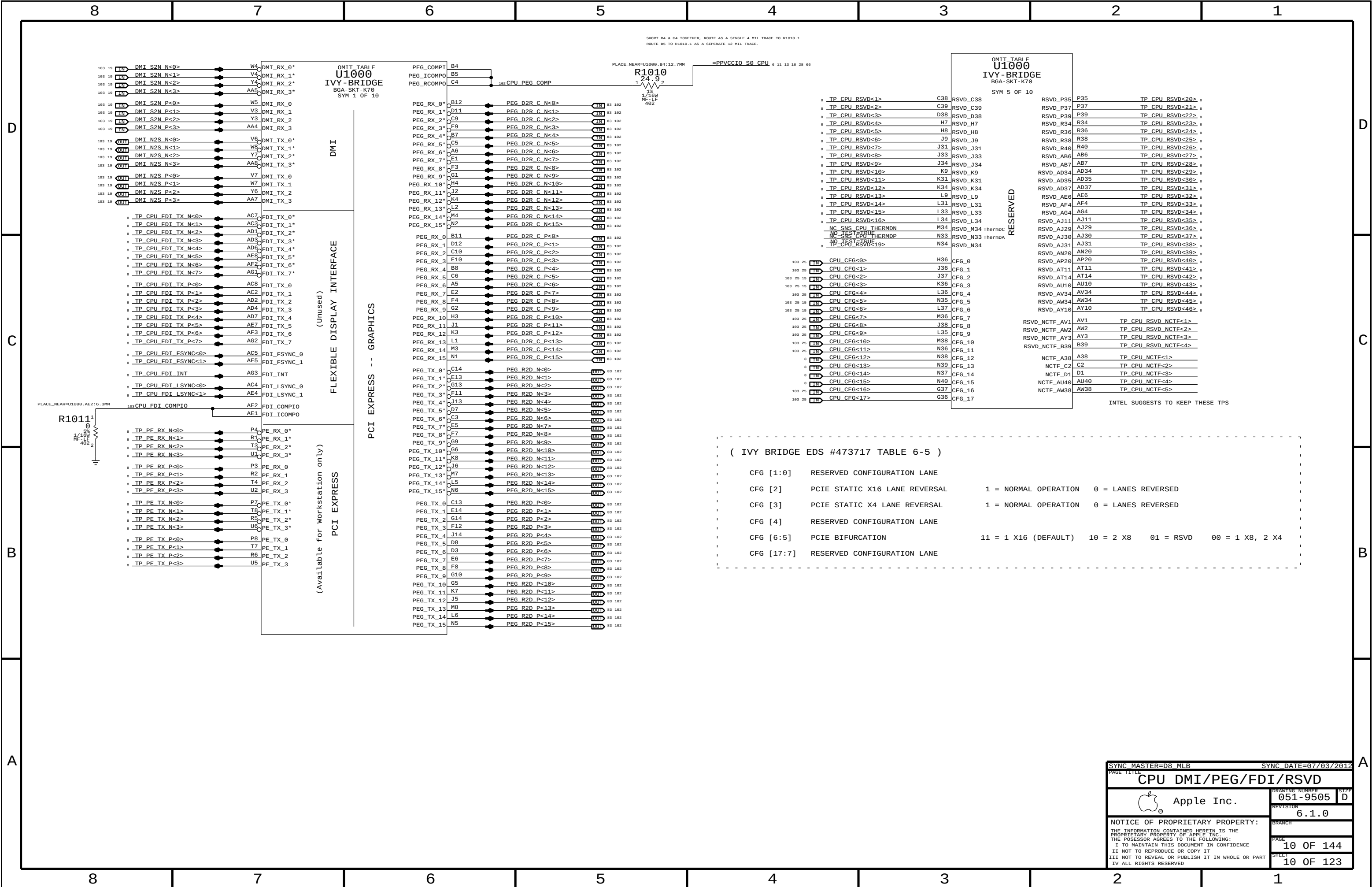
BRANCH

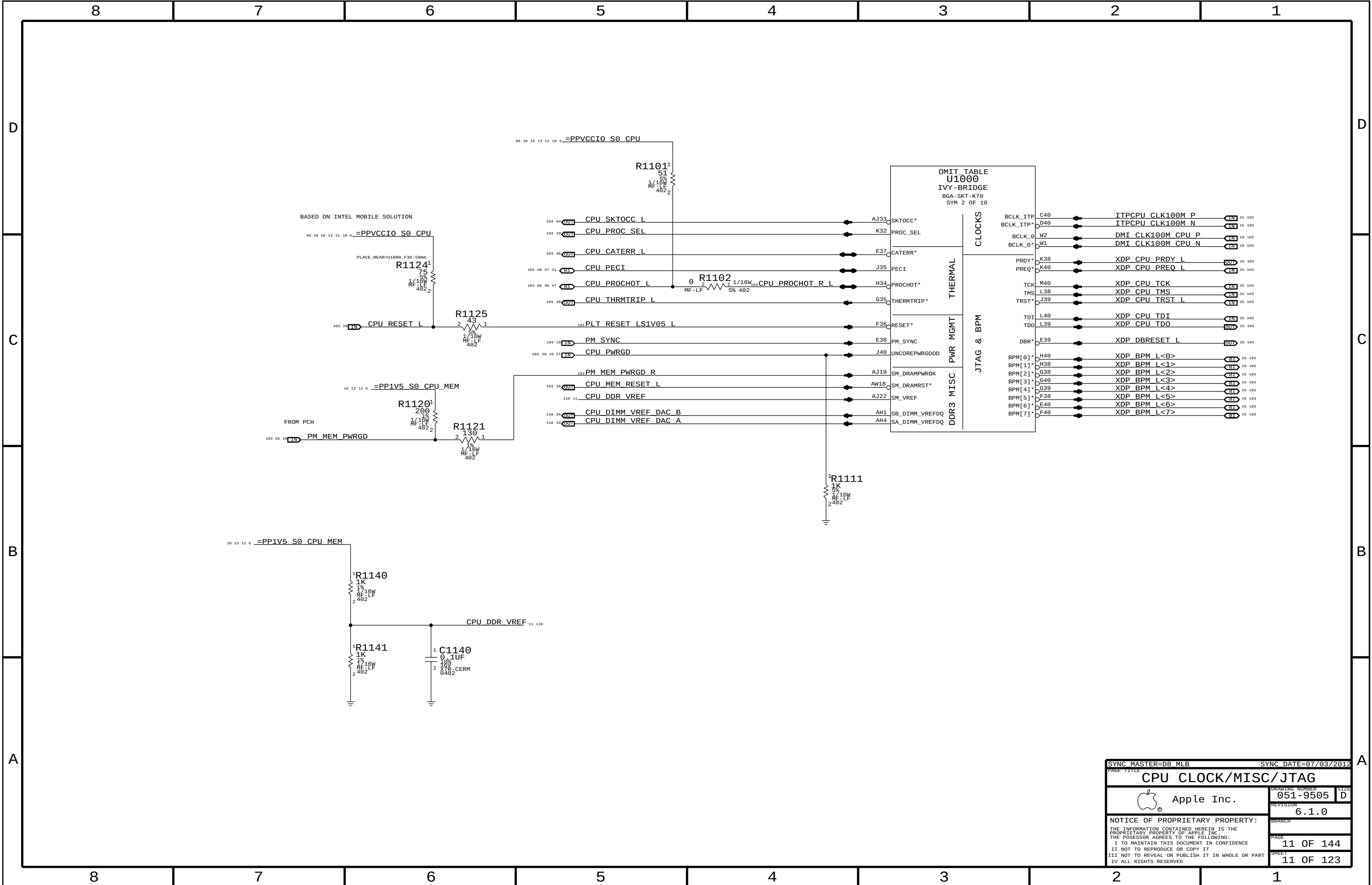
PAGE

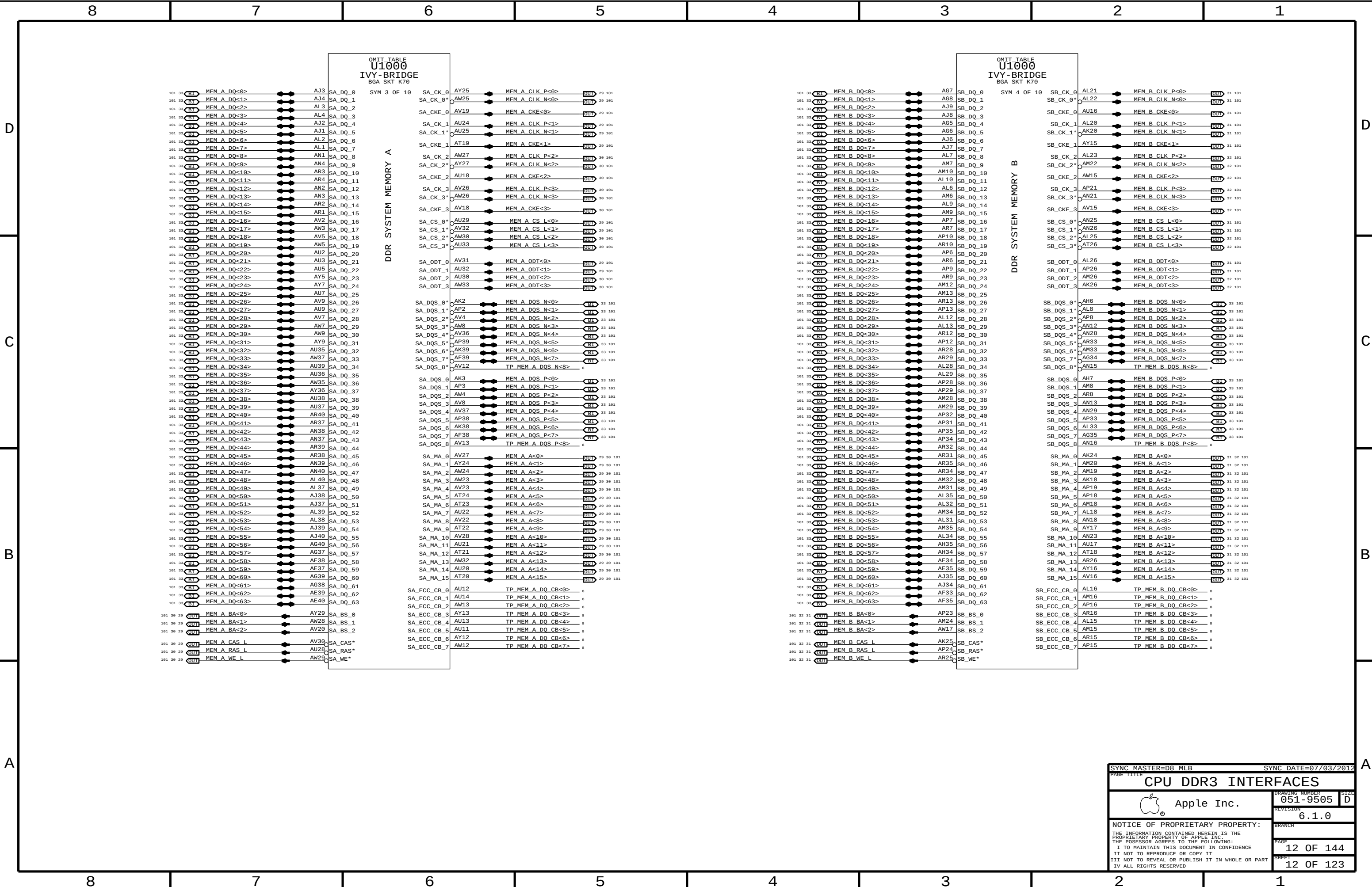
9 OF 144

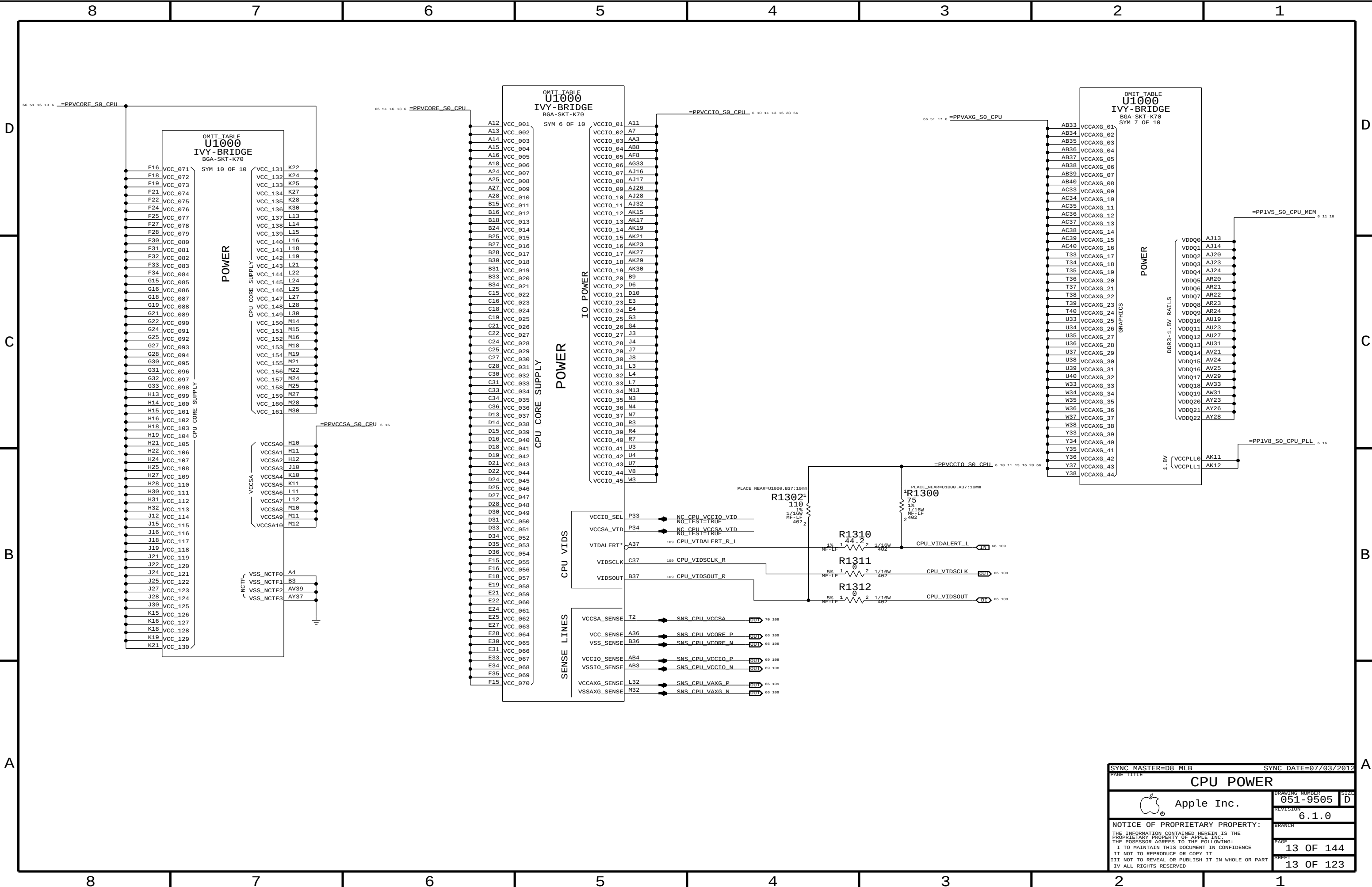
SHEET

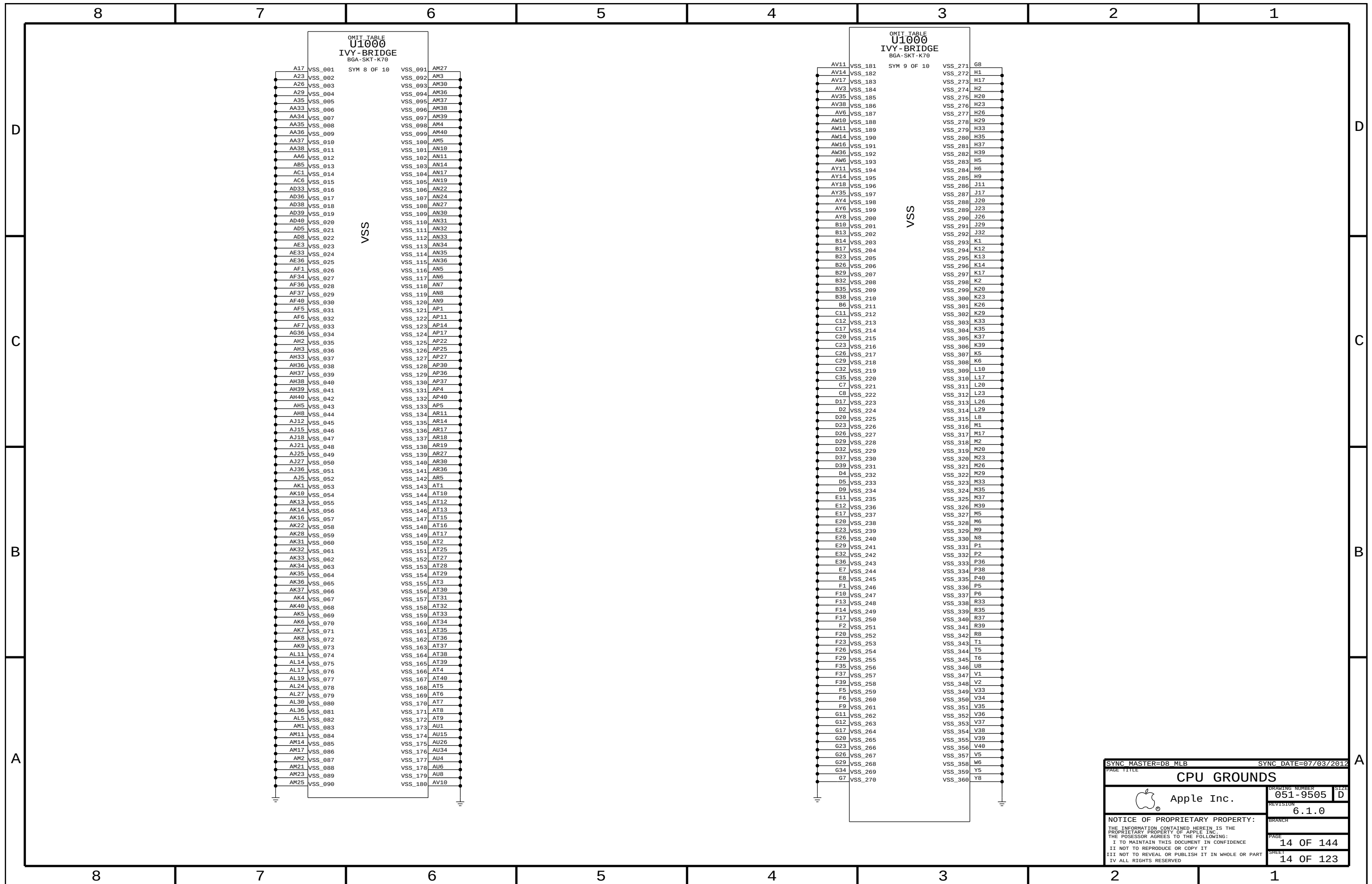
9 OF 123

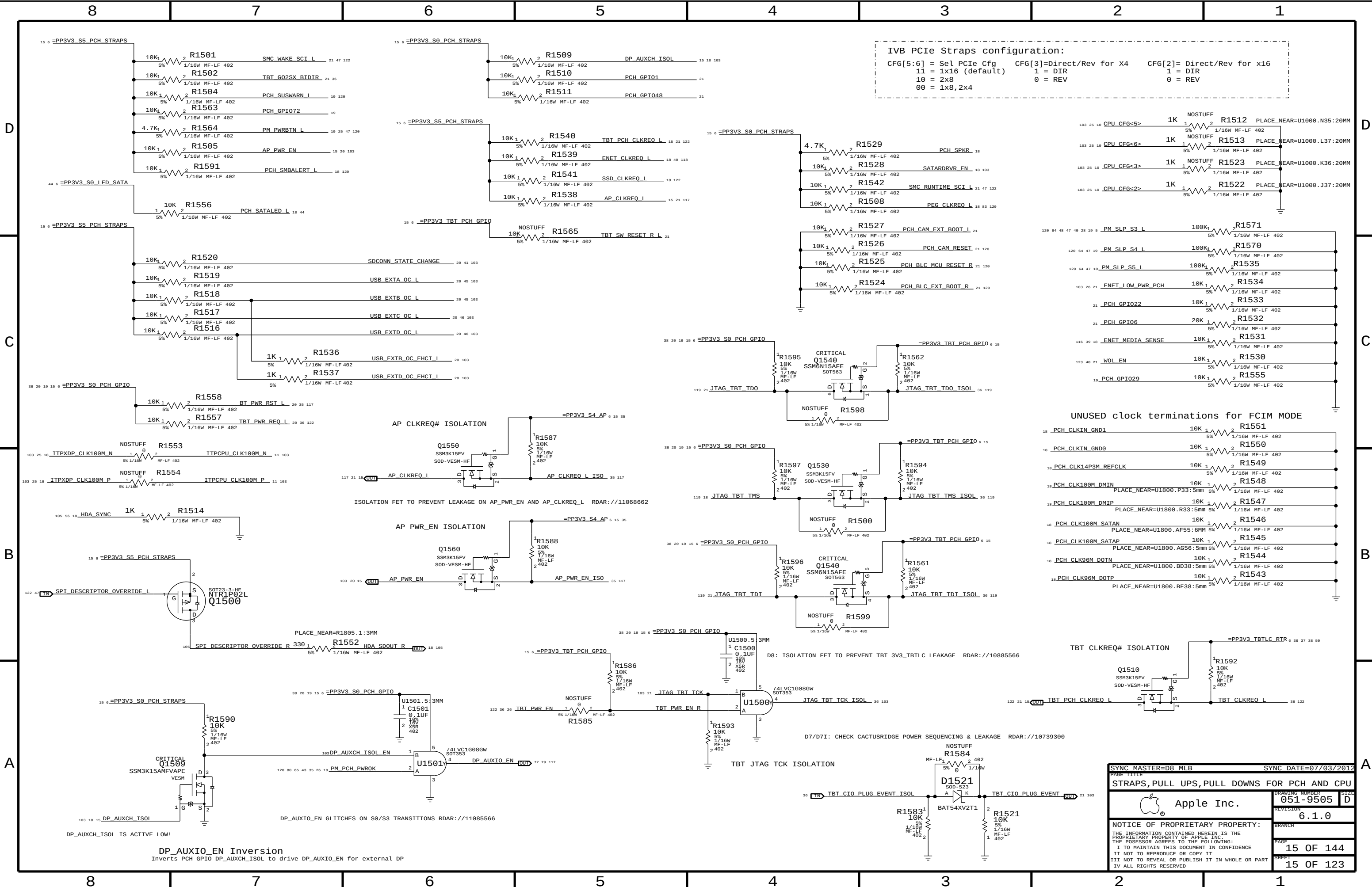








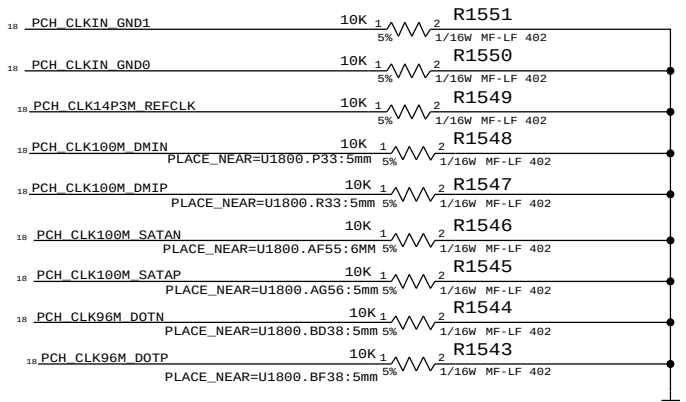


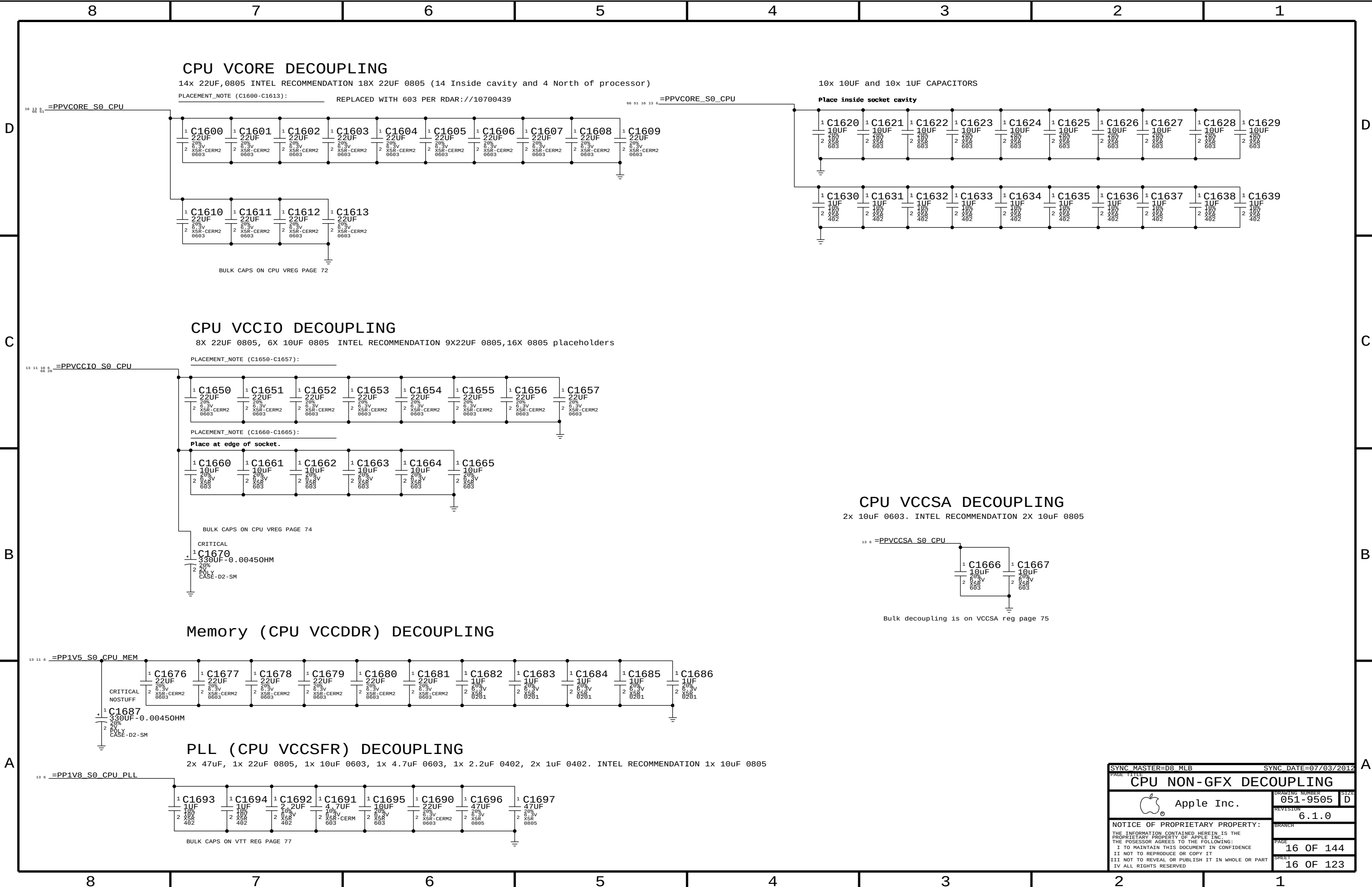


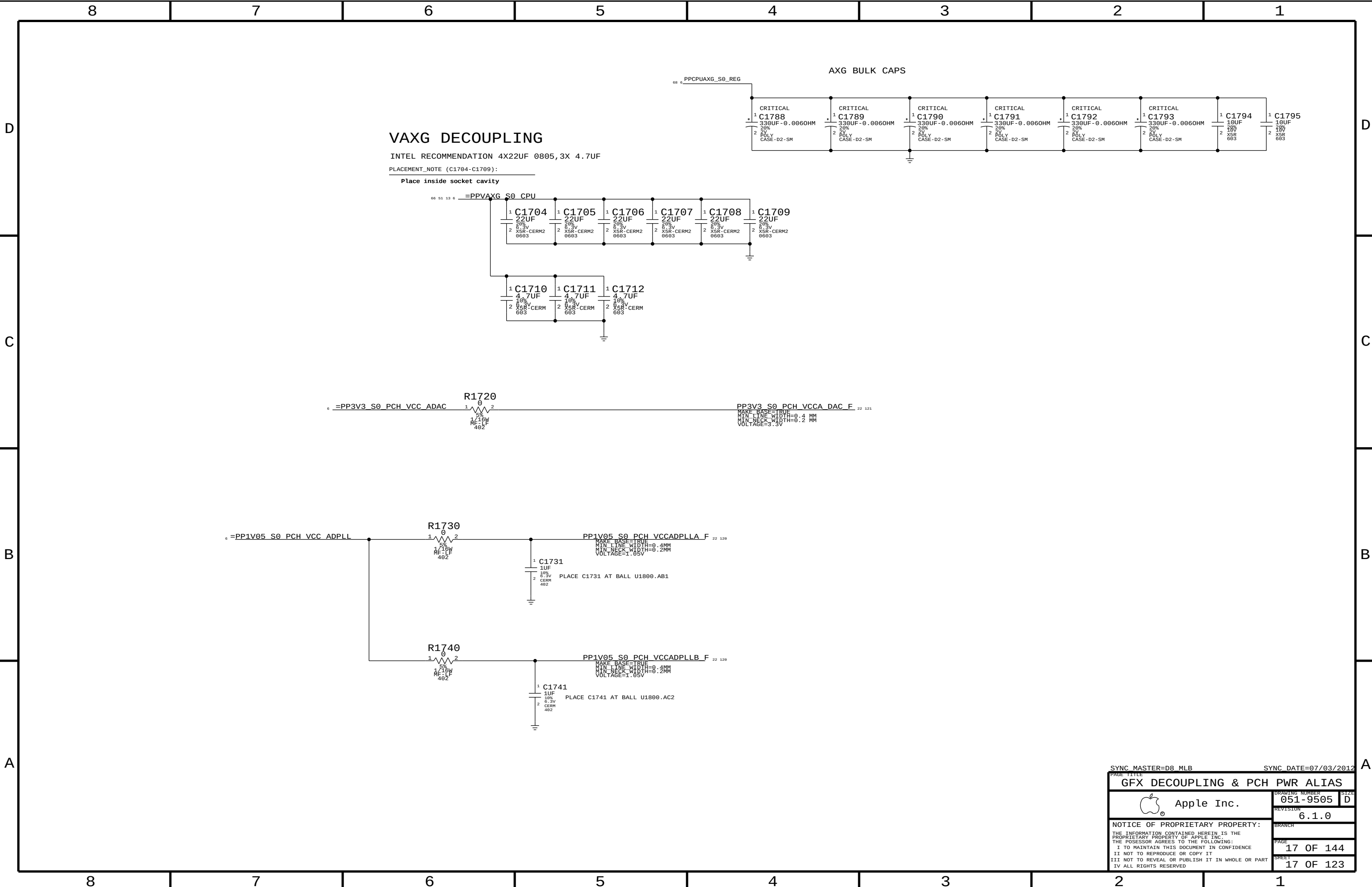
IVB PCIe Straps configuration:

CFG[5:6] = Sel PCIe Cfg	CFG[3]=Direct/Rev for X4	CFG[2]= Direct/Rev for x16
11 = 1x16 (default)	1 = DIR	1 = DIR
10 = 2x8	0 = REV	0 = REV
00 = 1x8,2x4		

UNUSED clock terminations for FCIM MODE



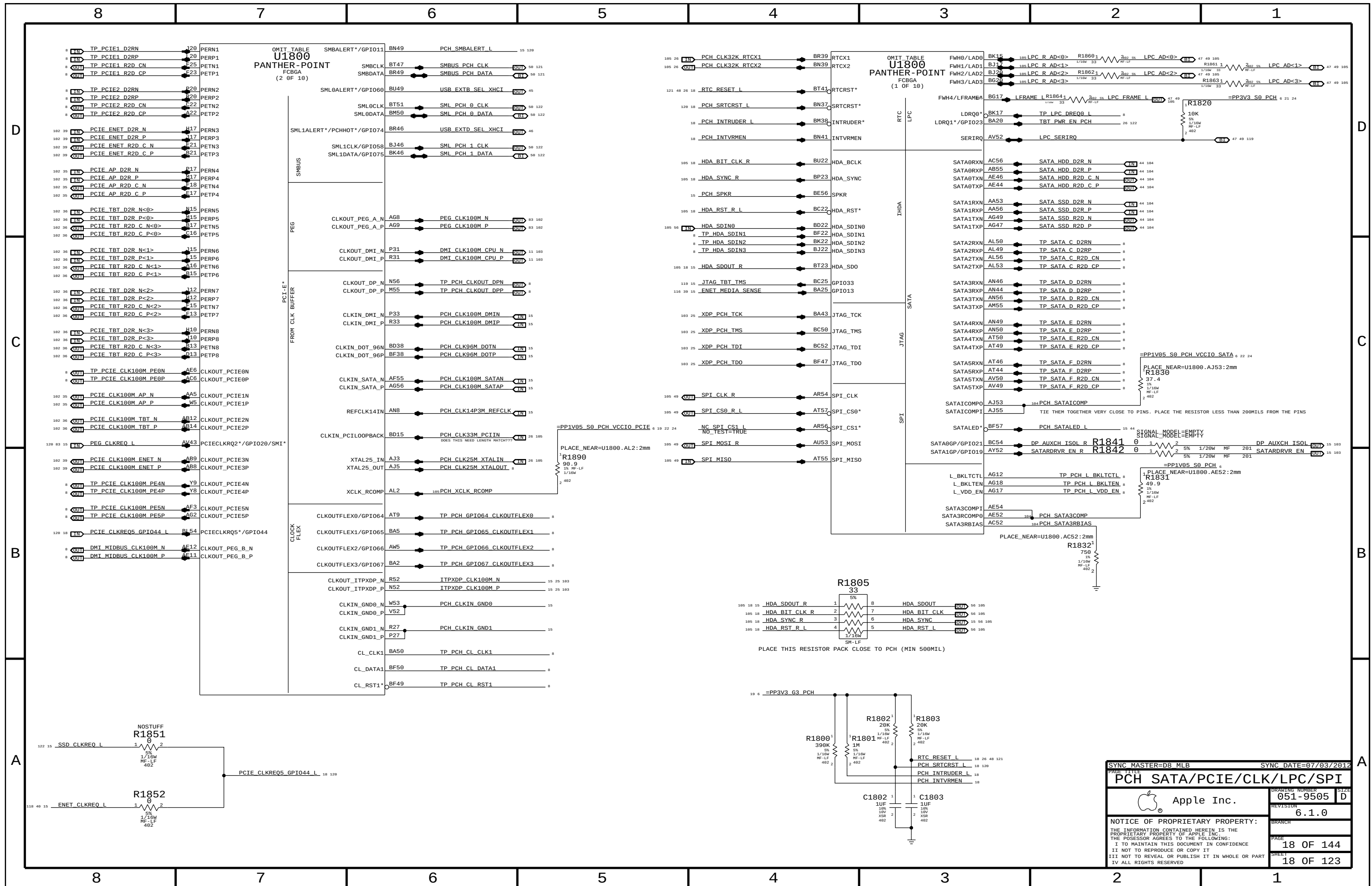


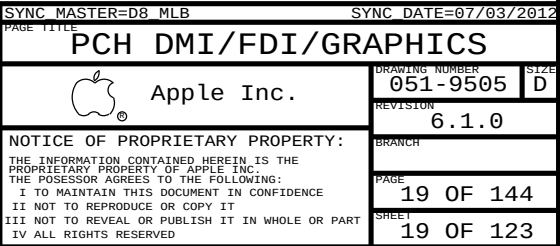


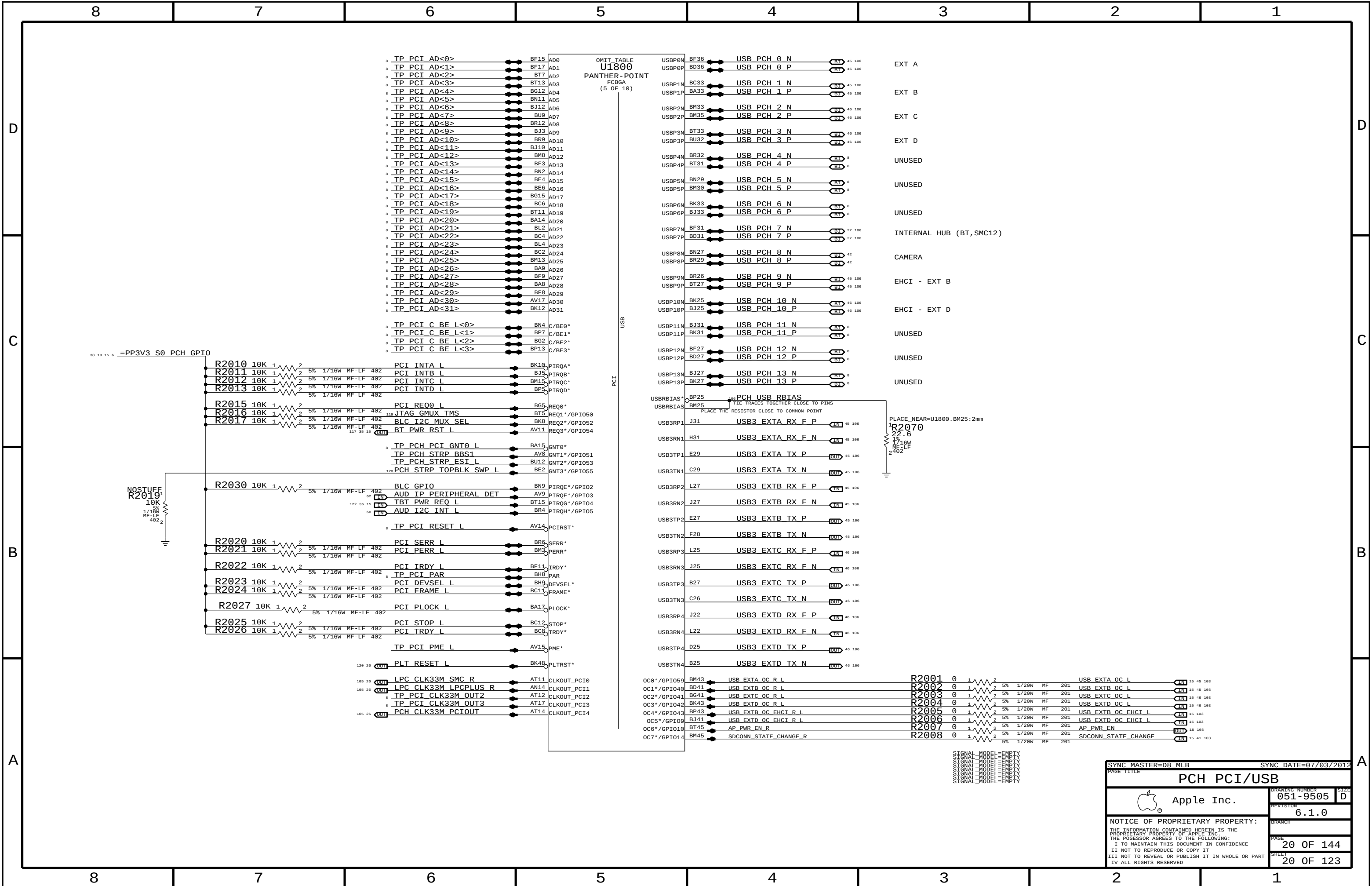
SYNC MASTER=D8_MLB

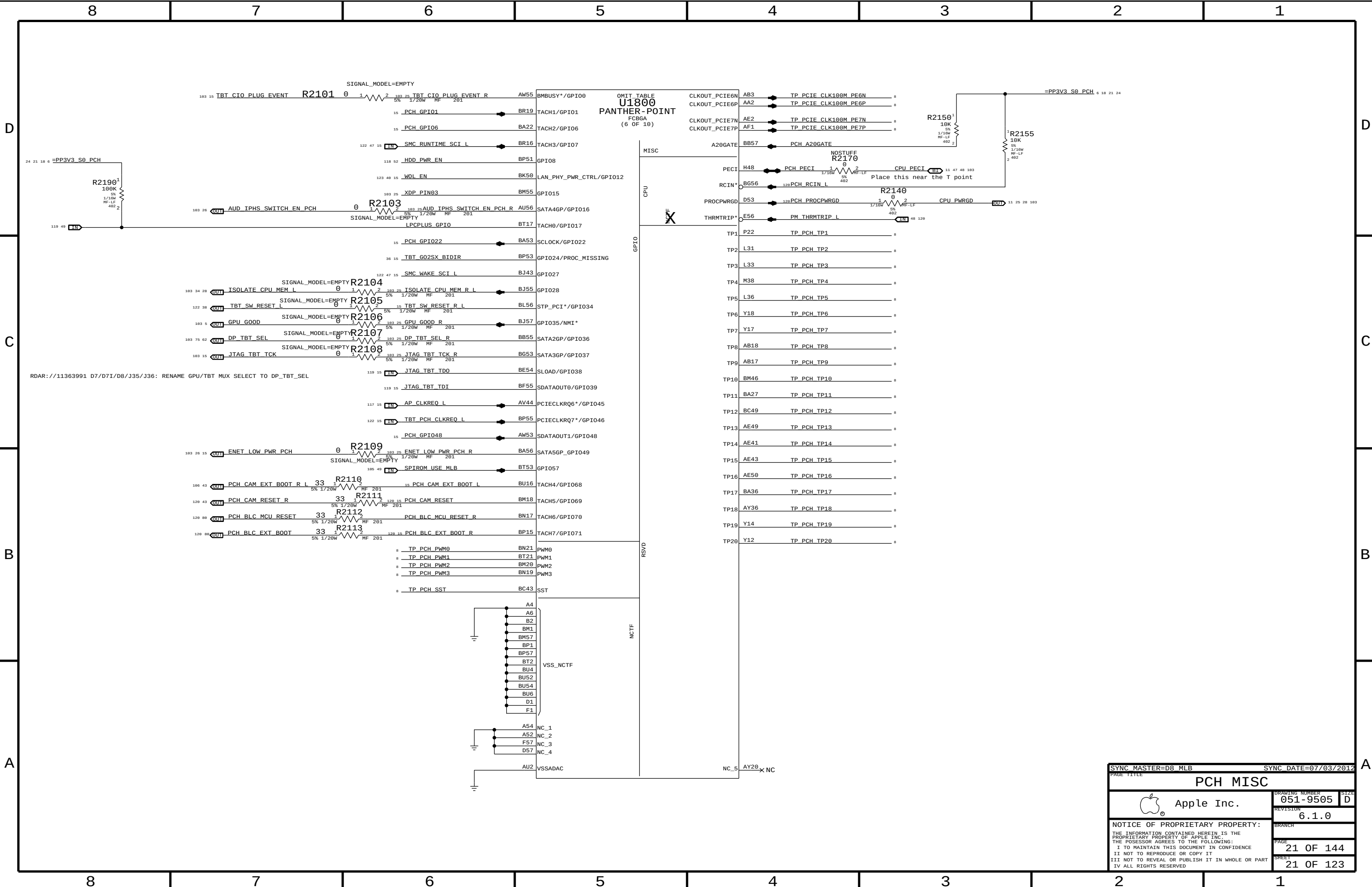
SYNC DATE=07/03/2012

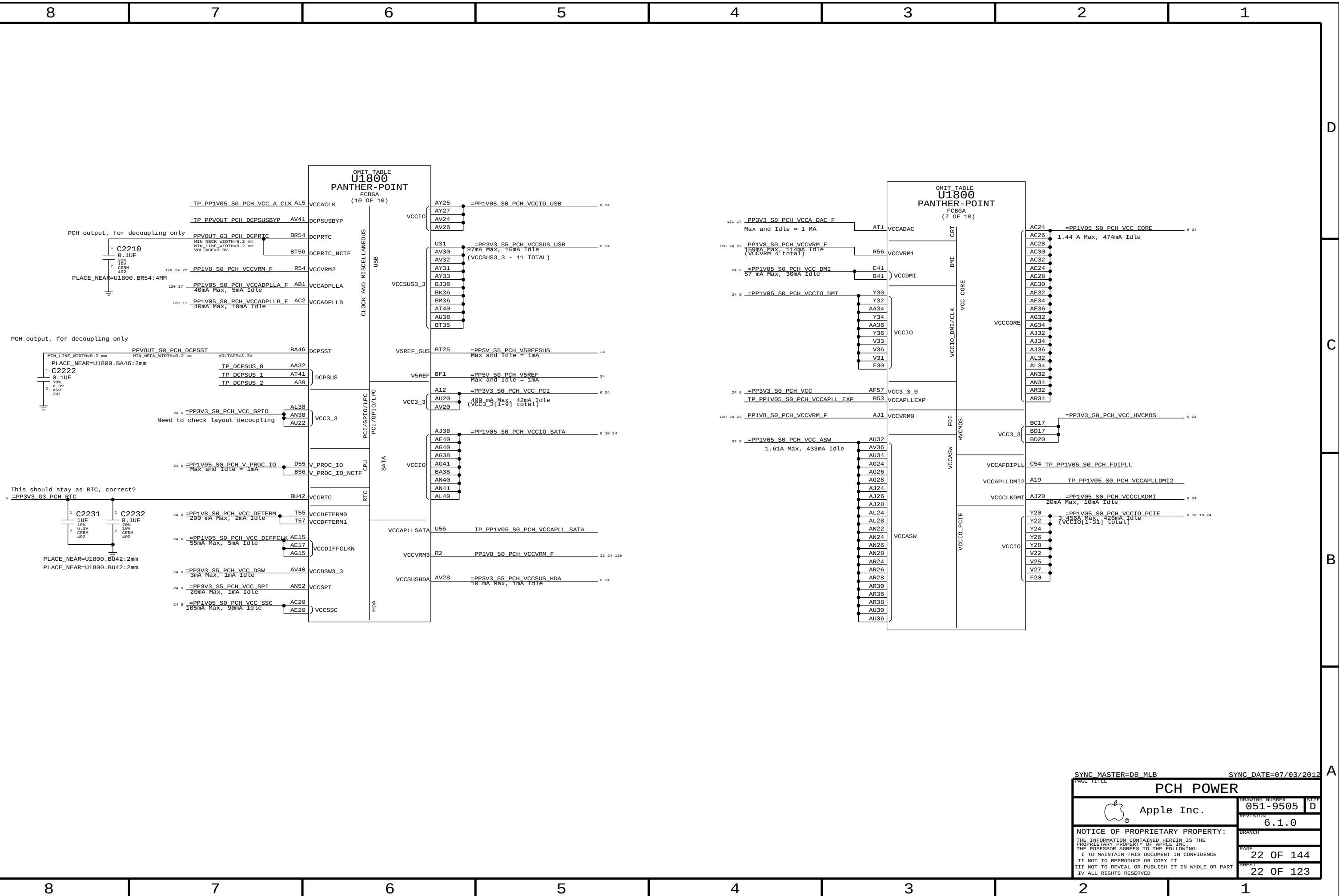
PAGE TITLE		
GFX DECOUPLING & PCH PWR ALIAS		
 Apple Inc.	DRAWING NUMBER	051-9505
	REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	
	PAGE	17 OF 144
	SHEET	17 OF 123

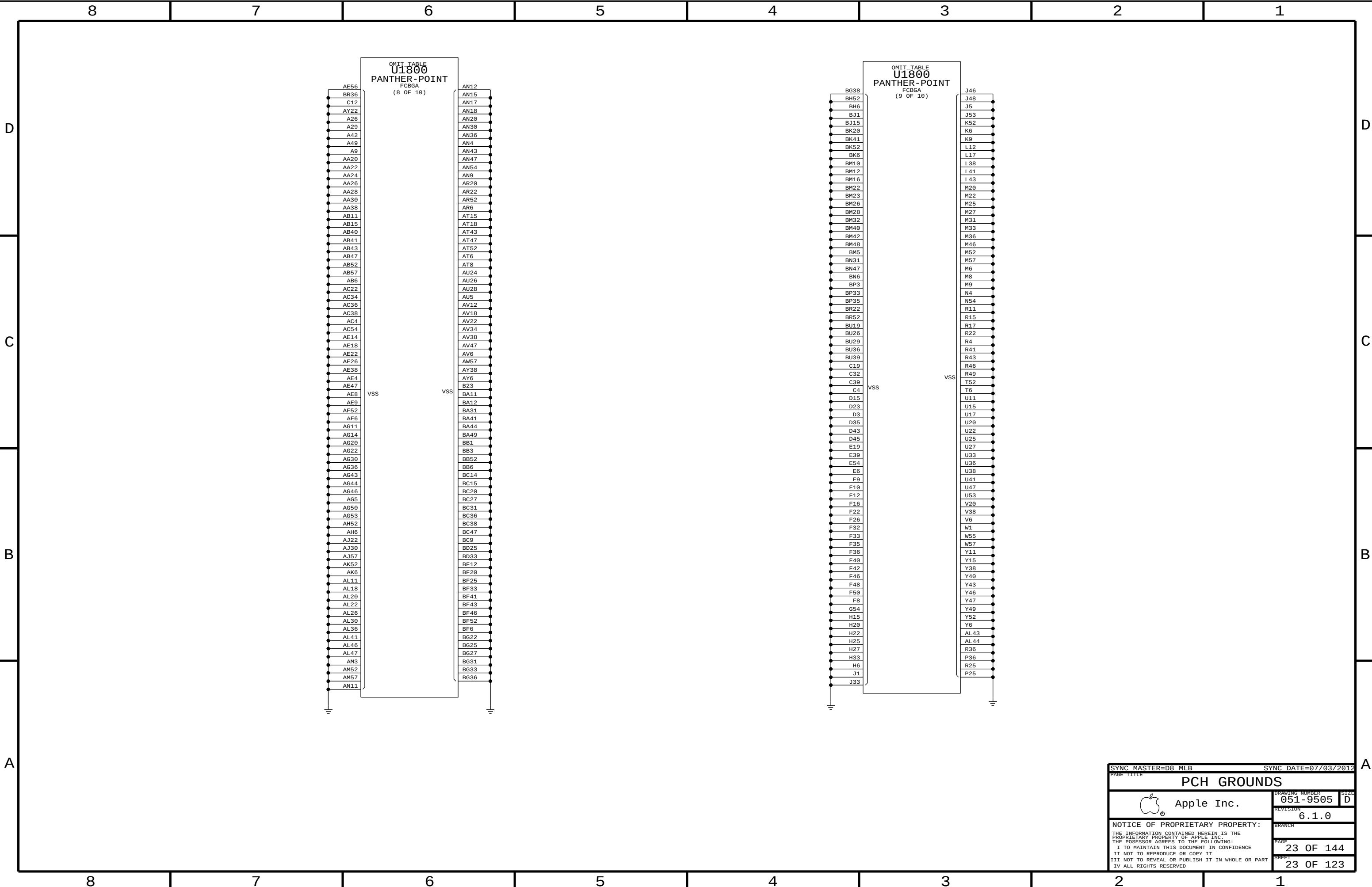








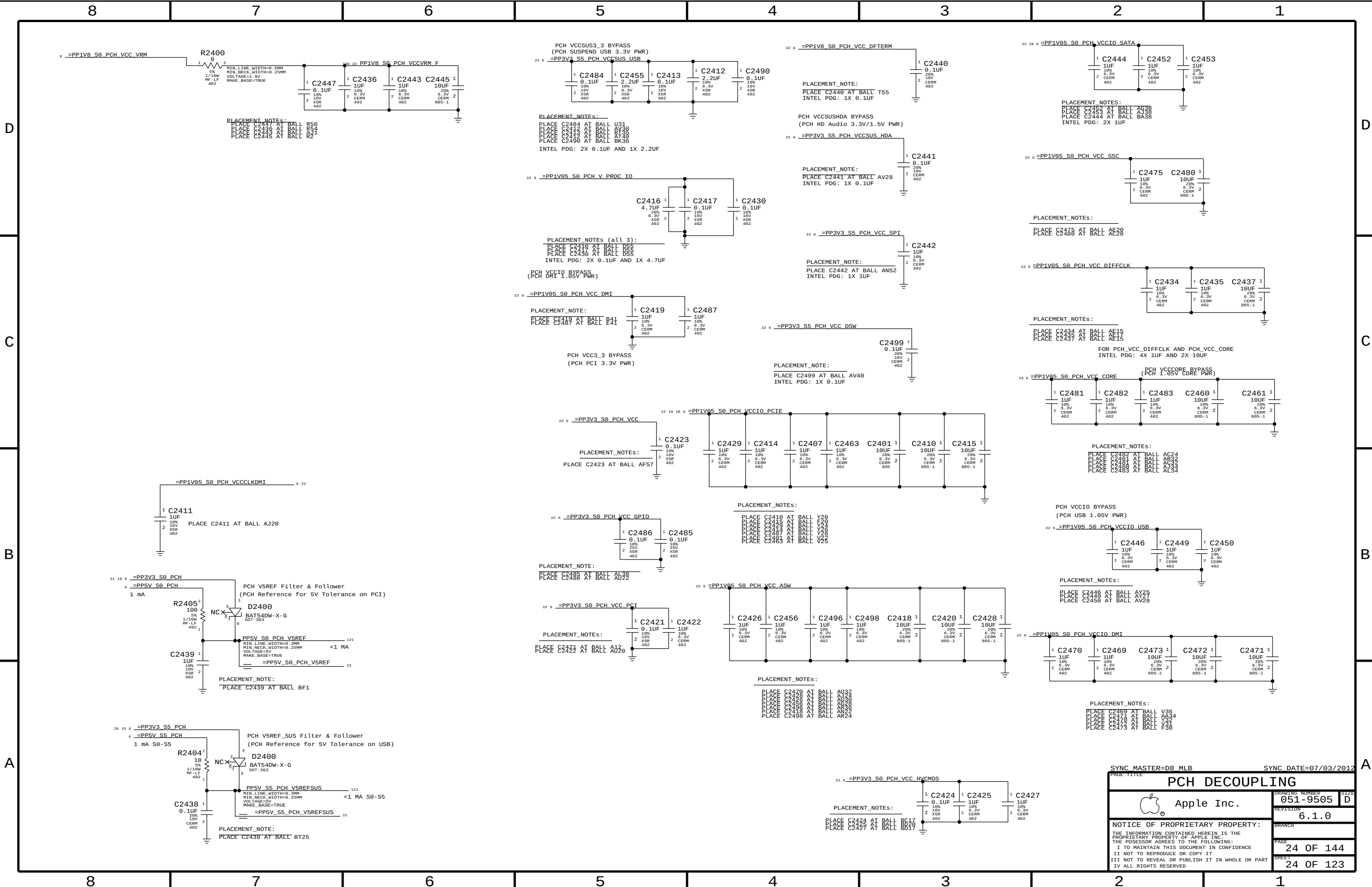




SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

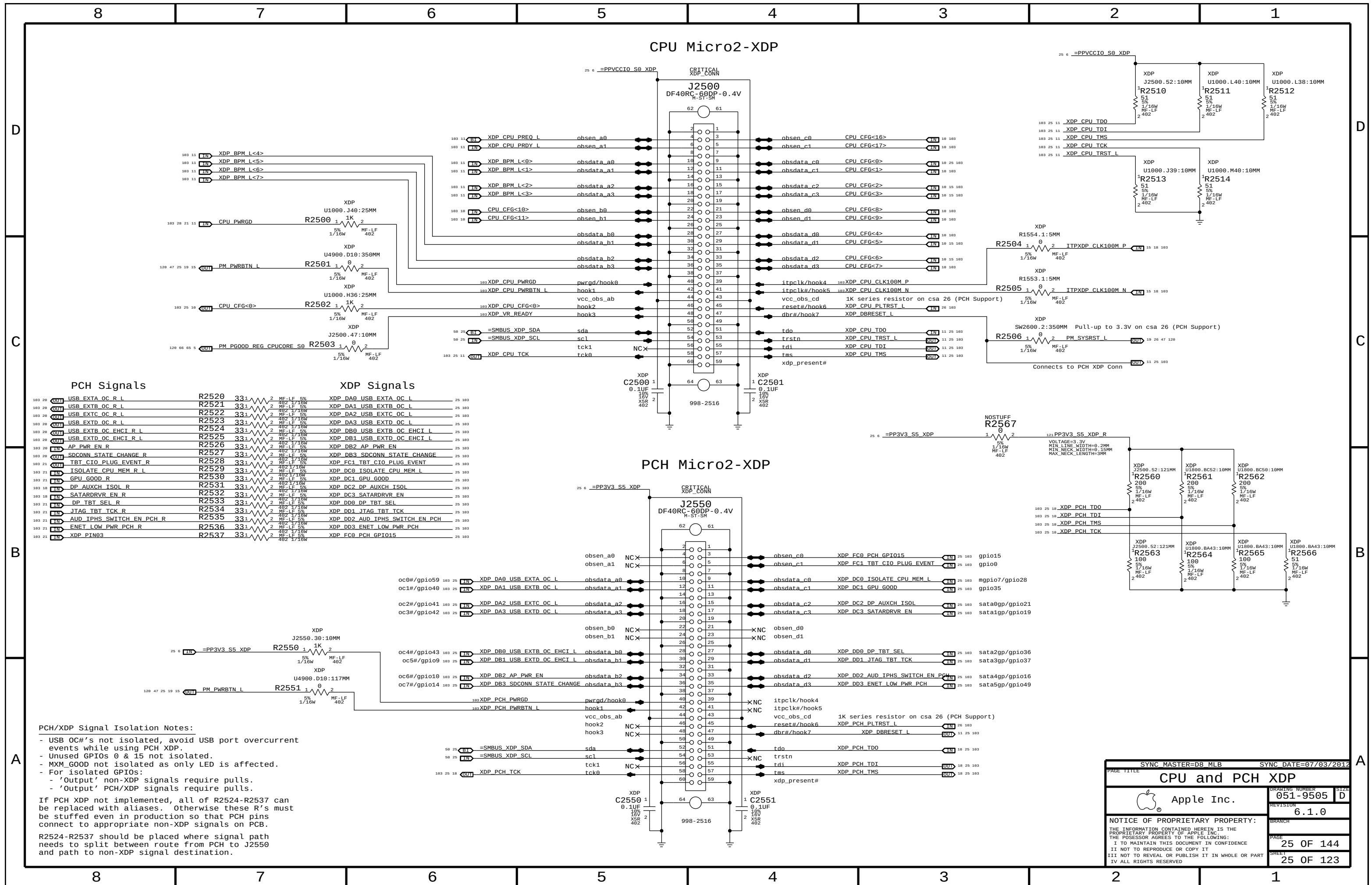
PAGE TITLE		PCH GROUNDS	
 Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	23 OF 144
		SHEET	23 OF 123



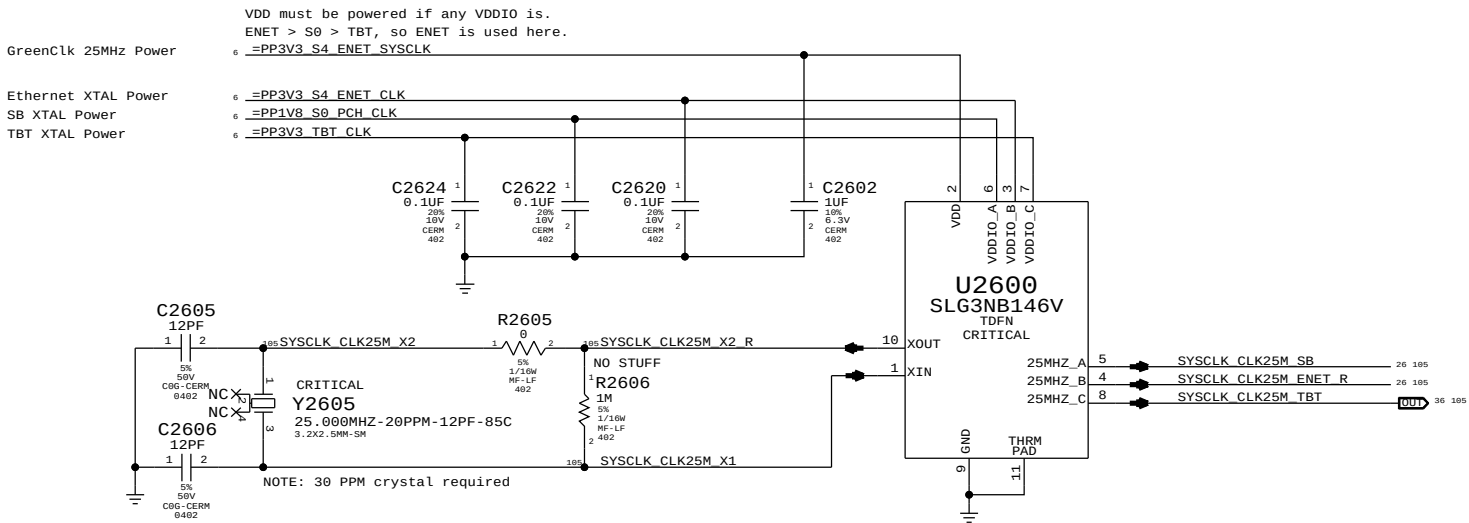
SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

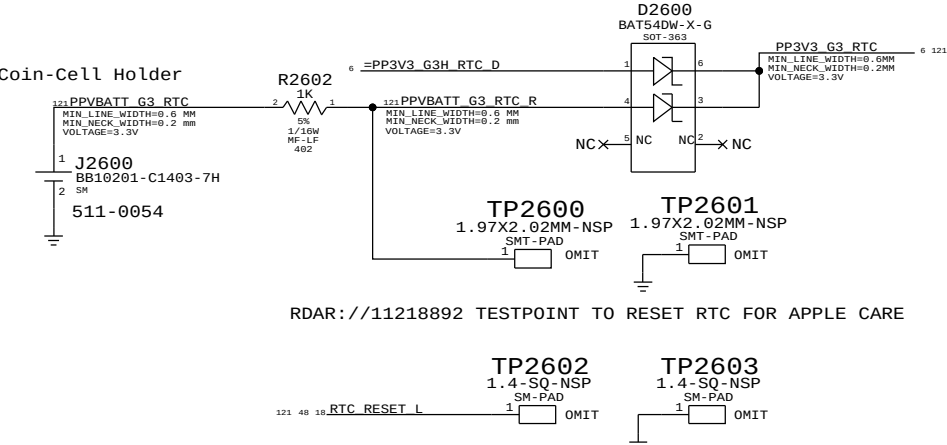
PAGE TITLE		PCH DECOUPLING	
 Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	24 OF 144
		SHEET	24 OF 123



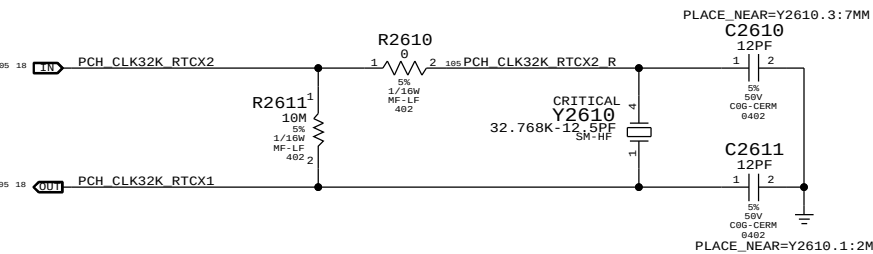
System 25MHz Clock Generator



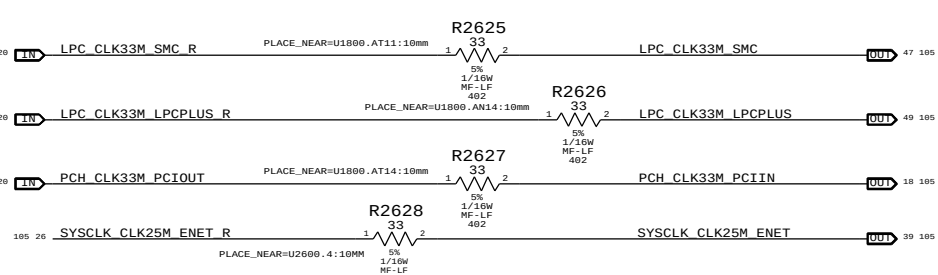
RTC Power Sources



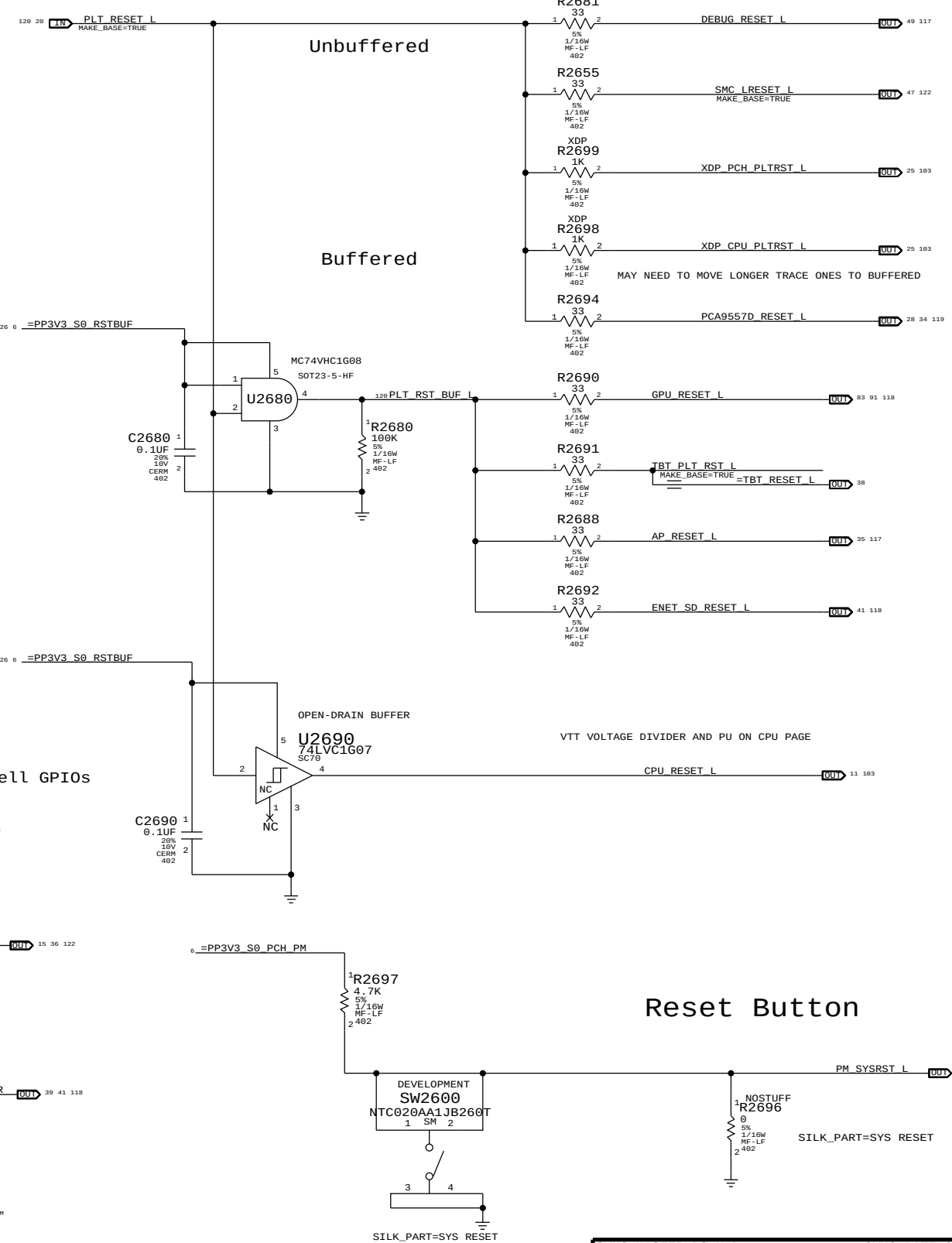
PCH RTC Crystal



Clock series termination

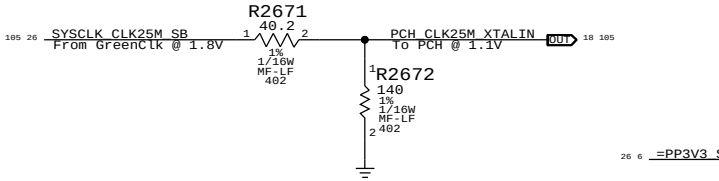


Platform Reset Connections

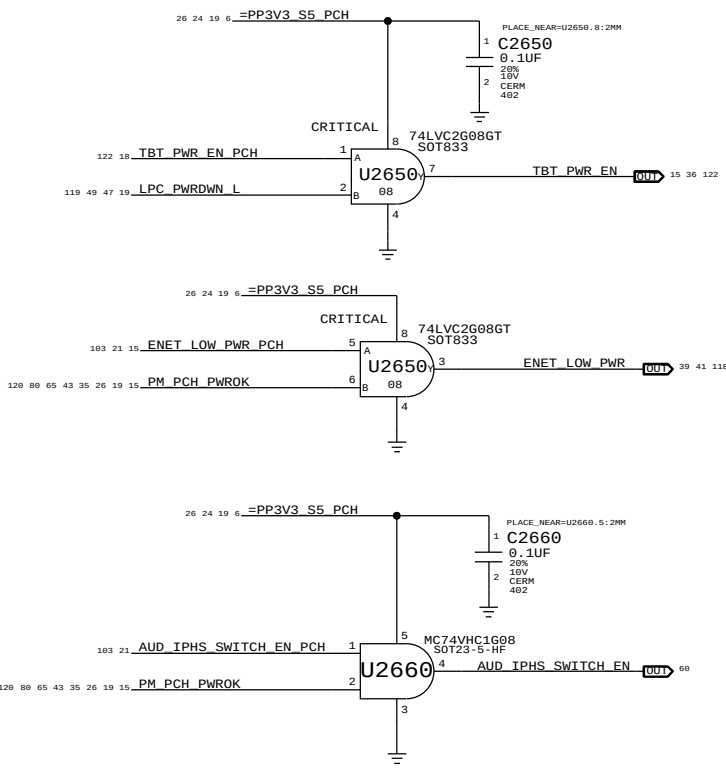


Reset Button

PCH 25MHZ CLOCK



GPIO Isolation to prevent glitches on critical core well GPIOs



SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE		CHIPSET SUPPORT	
Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	26 OF 144
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	26 OF 123
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

MEM_RESET_L Generator

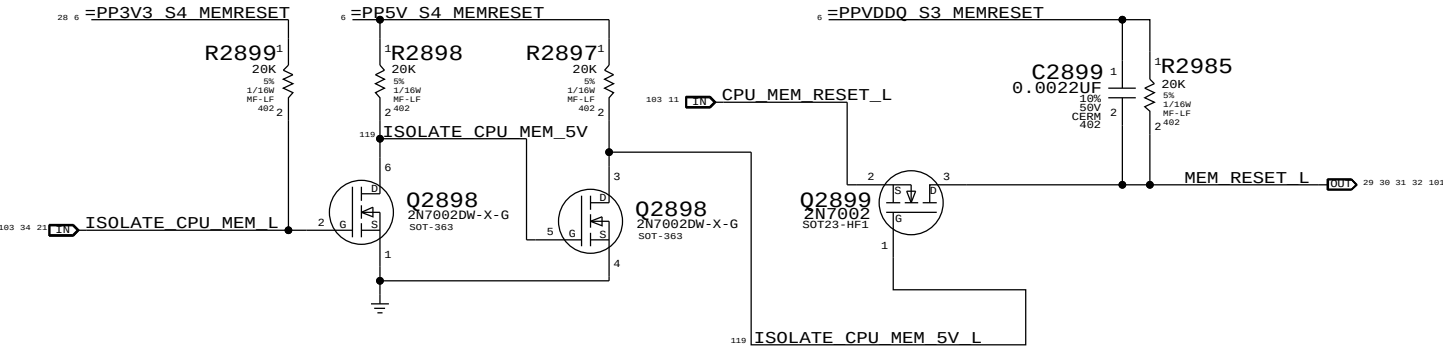
The circuits below handle MEMVTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behaviour of signals.

WHEN HIGH: MEM_RESET_L NOT ISOLATED.

WHEN LOW: MEM_RESET_L IS ISOLATED.

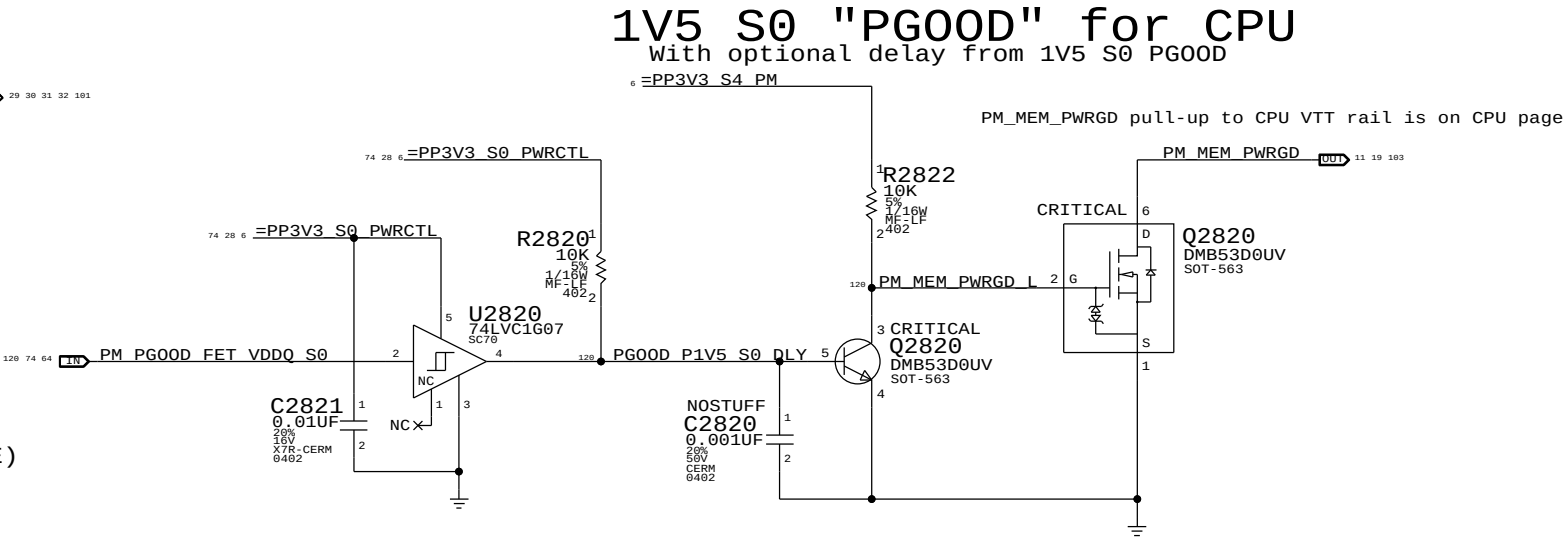
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L (Block CPU from driving MEM_RESET_L in S3)



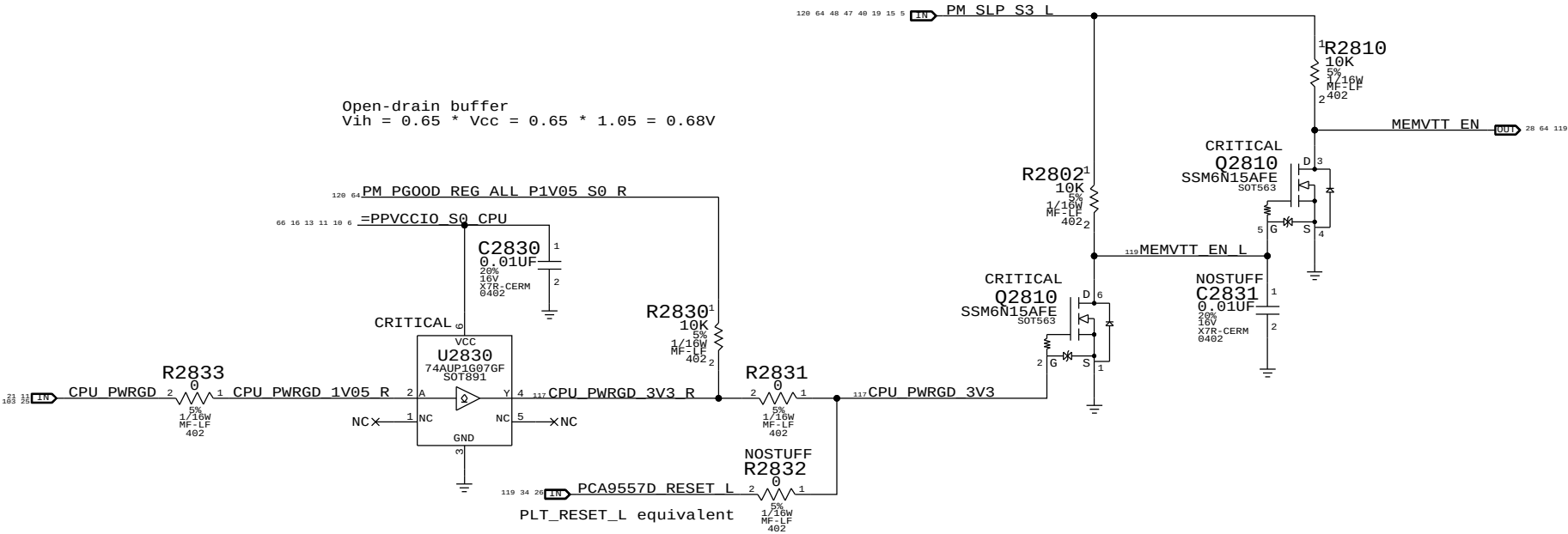
MEMVTT_EN Generator

rdar://11117167 Enables MEMVTT when PCH drives CPU PWRGD.

CPU does not drive MEM_CKE until VCCORE activated but CPU 1V5 (VDDQ) leaks into it. Clamping MEMVTT will keep the MEM_CKE low until CPU actively controls it. MEMVTT Clamp actively holds MEMVTT rail low until MEMVTT is enabled. MEMVTT_EN = CPU_PWRGD * PM_SLP_S3_L (VTT is enabled when PCH tells CPU to enable VCCORE)

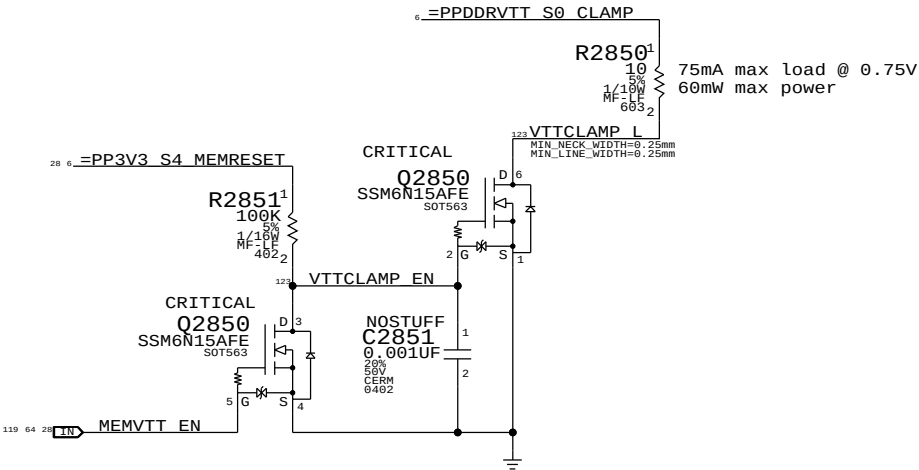


Open-drain buffer
Vih = 0.65 * Vcc = 0.65 * 1.05 = 0.68V



MEMVTT Clamp

Ensures CKE signals are held low in S3 and in S0 before CPU PWRGD

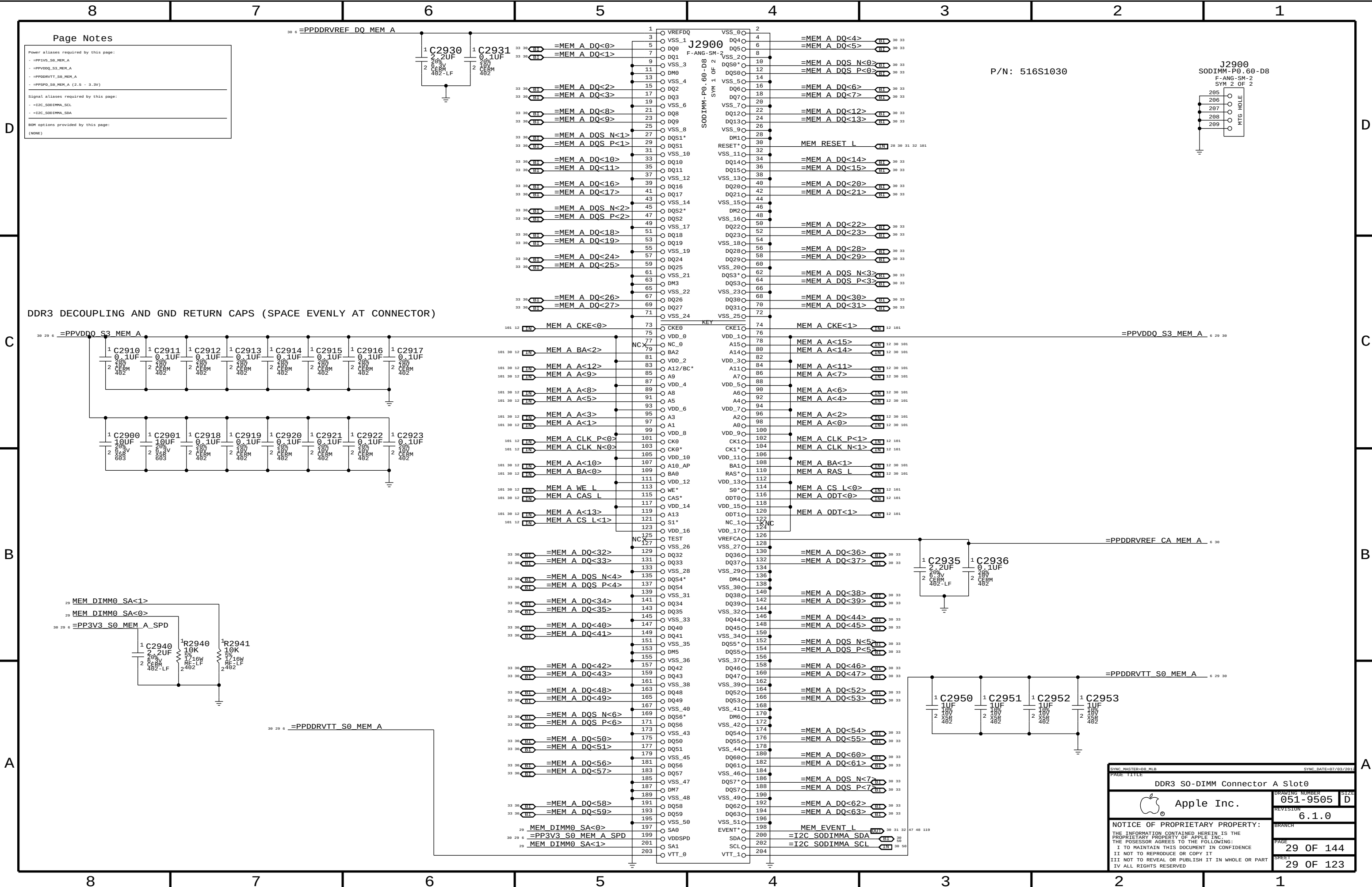


Step	ISOLATE_CPU_MEM_L	PM_SLP_S3_L	CPU_PWRGD	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	CPU_MEM_RESET_L	1	1
1	1	1	1	1	1	1
2	0	1	1	1	1	1
3	0	0	0	X	1	0
4	0	0	0	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: On a S5->S0 transition, ISOLATE_CPU_MEM_L will default low. Rails will power-up as if from S3, but MEM_RESET_L now needs to be asserted in S0. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC_MASTER=D8_MLB		SYNC_DATE=07/03/2012	
PAGE TITLE		CPU Memory S3 Support	
DRAWING NUMBER		051-9505	SIZE D
REVISION		6.1.0	BRANCH
PAGE		28 OF 144	SHEET
SHEET		28 OF 123	IV ALL RIGHTS RESERVED



Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

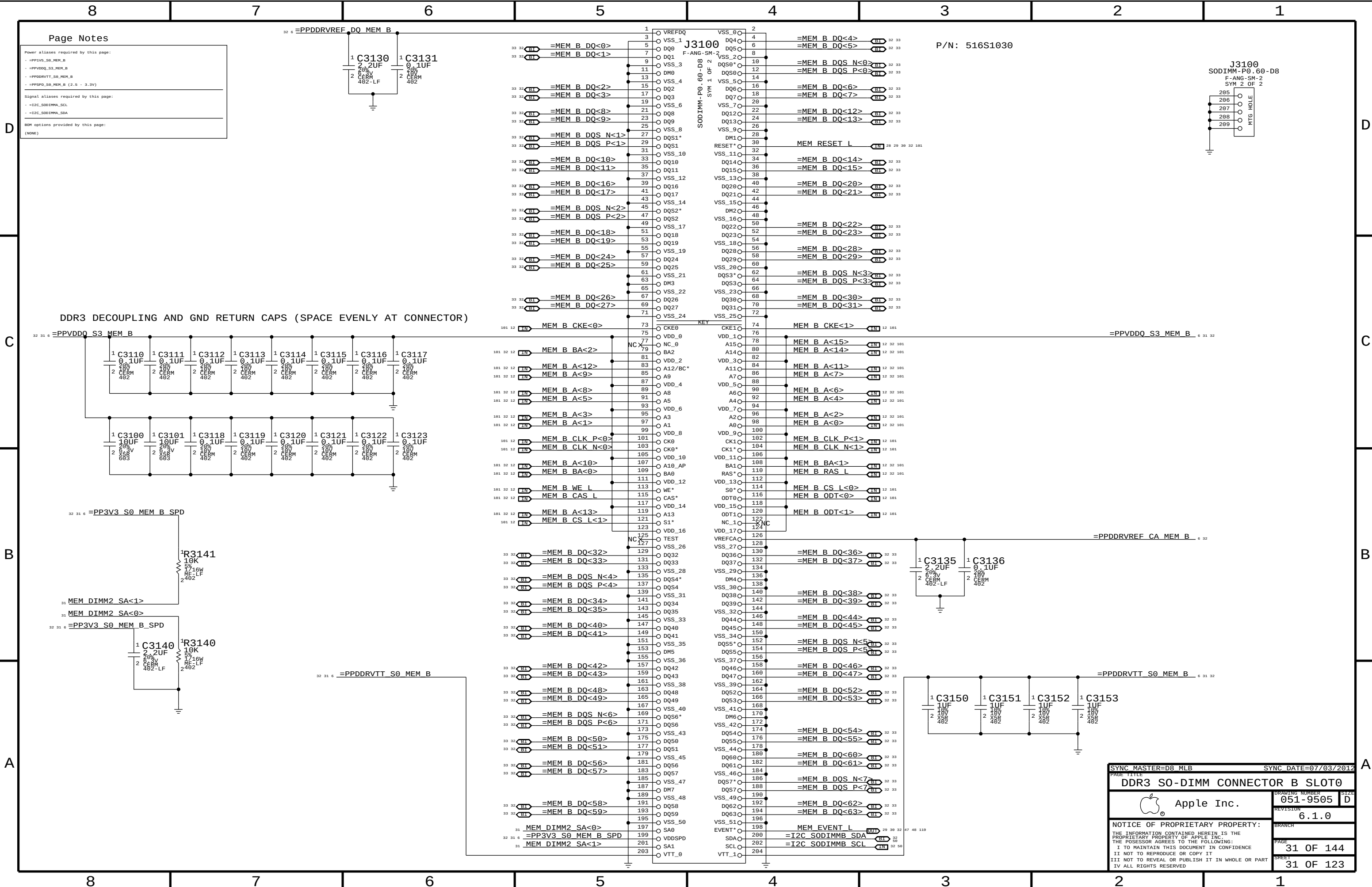
- =I2C_S0DIMMA_SCL
- =I2C_S0DIMMA_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

SYNC MASTER=DS_MLB		SYNC DATE=07/03/2015	
PAGE TITLE			
DDR3 S0-DIMM Connector A Slot0			
 Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
		BRANCH	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
		PAGE	29 OF 144
		SHEET	29 OF 123

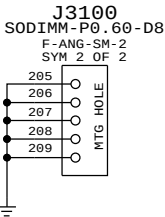


Page Notes

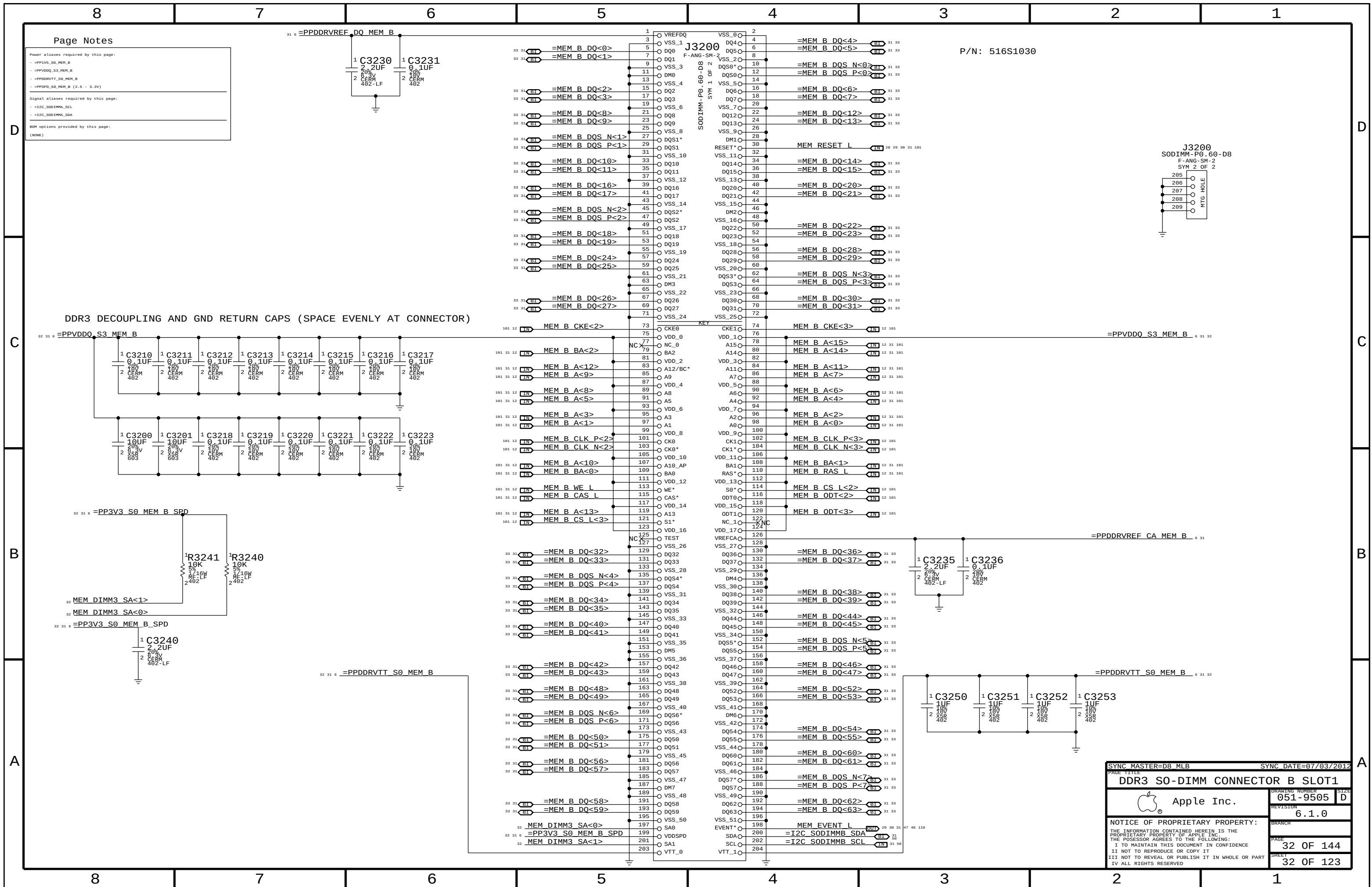
Power aliases required by this page:
- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)
Signal aliases required by this page:
- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA
BOM options provided by this page:
(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

P/N: 516S1030



SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE		DRAWING NUMBER	
DDR3 SO-DIMM CONNECTOR B SLOT0		051-9505	
Apple Inc.		REVISION	
		6.1.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		31 OF 144	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		31 OF 123	
IV ALL RIGHTS RESERVED			



8		7		6		5		4		3		2		1	
D	101 12	MEM A DQS N<0>	==	=MEM A DQS N<0>	29 30	101 12	MEM B DQS N<0>	==	=MEM B DQS N<0>	31 32	D				
	101 12	MEM A DQS P<0>	==	=MEM A DQS P<0>	29 30	101 12	MEM B DQS P<0>	==	=MEM B DQS P<0>	31 32					
	101 12	MEM A DQ<7>	==	=MEM A DQ<7>	29 30	101 12	MEM B DQ<7>	==	=MEM B DQ<3>	31 32					
	101 12	MEM A DQ<6>	==	=MEM A DQ<6>	29 30	101 12	MEM B DQ<6>	==	=MEM B DQ<6>	31 32					
	101 12	MEM A DQ<5>	==	=MEM A DQ<4>	29 30	101 12	MEM B DQ<5>	==	=MEM B DQ<5>	31 32					
	101 12	MEM A DQ<4>	==	=MEM A DQ<1>	29 30	101 12	MEM B DQ<4>	==	=MEM B DQ<4>	31 32					
	101 12	MEM A DQ<3>	==	=MEM A DQ<3>	29 30	101 12	MEM B DQ<3>	==	=MEM B DQ<7>	31 32					
	101 12	MEM A DQ<2>	==	=MEM A DQ<2>	29 30	101 12	MEM B DQ<2>	==	=MEM B DQ<2>	31 32					
	101 12	MEM A DQ<1>	==	=MEM A DQ<5>	29 30	101 12	MEM B DQ<1>	==	=MEM B DQ<1>	31 32					
	101 12	MEM A DQ<0>	==	=MEM A DQ<0>	29 30	101 12	MEM B DQ<0>	==	=MEM B DQ<0>	31 32					
C	101 12	MEM A DQS N<1>	==	=MEM A DQS N<1>	29 30	101 12	MEM B DQS N<1>	==	=MEM B DQS N<1>	31 32	C				
	101 12	MEM A DQS P<1>	==	=MEM A DQS P<1>	29 30	101 12	MEM B DQS P<1>	==	=MEM B DQS P<1>	31 32					
	101 12	MEM A DQ<15>	==	=MEM A DQ<15>	29 30	101 12	MEM B DQ<15>	==	=MEM B DQ<11>	31 32					
	101 12	MEM A DQ<14>	==	=MEM A DQ<14>	29 30	101 12	MEM B DQ<14>	==	=MEM B DQ<10>	31 32					
	101 12	MEM A DQ<13>	==	=MEM A DQ<13>	29 30	101 12	MEM B DQ<13>	==	=MEM B DQ<13>	31 32					
	101 12	MEM A DQ<12>	==	=MEM A DQ<12>	29 30	101 12	MEM B DQ<12>	==	=MEM B DQ<12>	31 32					
	101 12	MEM A DQ<11>	==	=MEM A DQ<11>	29 30	101 12	MEM B DQ<11>	==	=MEM B DQ<15>	31 32					
	101 12	MEM A DQ<10>	==	=MEM A DQ<10>	29 30	101 12	MEM B DQ<10>	==	=MEM B DQ<14>	31 32					
	101 12	MEM A DQ<9>	==	=MEM A DQ<9>	29 30	101 12	MEM B DQ<9>	==	=MEM B DQ<8>	31 32					
	101 12	MEM A DQ<8>	==	=MEM A DQ<8>	29 30	101 12	MEM B DQ<8>	==	=MEM B DQ<9>	31 32					
B	101 12	MEM A DQS N<2>	==	=MEM A DQS N<2>	29 30	101 12	MEM B DQS N<2>	==	=MEM B DQS N<2>	31 32	B				
	101 12	MEM A DQS P<2>	==	=MEM A DQS P<2>	29 30	101 12	MEM B DQS P<2>	==	=MEM B DQS P<2>	31 32					
	101 12	MEM A DQ<23>	==	=MEM A DQ<23>	29 30	101 12	MEM B DQ<23>	==	=MEM B DQ<22>	31 32					
	101 12	MEM A DQ<22>	==	=MEM A DQ<22>	29 30	101 12	MEM B DQ<22>	==	=MEM B DQ<18>	31 32					
	101 12	MEM A DQ<21>	==	=MEM A DQ<21>	29 30	101 12	MEM B DQ<21>	==	=MEM B DQ<21>	31 32					
	101 12	MEM A DQ<20>	==	=MEM A DQ<20>	29 30	101 12	MEM B DQ<20>	==	=MEM B DQ<20>	31 32					
	101 12	MEM A DQ<19>	==	=MEM A DQ<18>	29 30	101 12	MEM B DQ<19>	==	=MEM B DQ<23>	31 32					
	101 12	MEM A DQ<18>	==	=MEM A DQ<19>	29 30	101 12	MEM B DQ<18>	==	=MEM B DQ<19>	31 32					
	101 12	MEM A DQ<17>	==	=MEM A DQ<17>	29 30	101 12	MEM B DQ<17>	==	=MEM B DQ<17>	31 32					
	101 12	MEM A DQ<16>	==	=MEM A DQ<16>	29 30	101 12	MEM B DQ<16>	==	=MEM B DQ<16>	31 32					
A	101 12	MEM A DQS N<3>	==	=MEM A DQS N<3>	29 30	101 12	MEM B DQS N<3>	==	=MEM B DQS N<3>	31 32	A				
	101 12	MEM A DQS P<3>	==	=MEM A DQS P<3>	29 30	101 12	MEM B DQS P<3>	==	=MEM B DQS P<3>	31 32					
	101 12	MEM A DQ<31>	==	=MEM A DQ<31>	29 30	101 12	MEM B DQ<31>	==	=MEM B DQ<31>	31 32					
	101 12	MEM A DQ<30>	==	=MEM A DQ<30>	29 30	101 12	MEM B DQ<30>	==	=MEM B DQ<30>	31 32					
	101 12	MEM A DQ<29>	==	=MEM A DQ<29>	29 30	101 12	MEM B DQ<29>	==	=MEM B DQ<29>	31 32					
	101 12	MEM A DQ<28>	==	=MEM A DQ<28>	29 30	101 12	MEM B DQ<28>	==	=MEM B DQ<28>	31 32					
	101 12	MEM A DQ<27>	==	=MEM A DQ<27>	29 30	101 12	MEM B DQ<27>	==	=MEM B DQ<27>	31 32					
	101 12	MEM A DQ<26>	==	=MEM A DQ<26>	29 30	101 12	MEM B DQ<26>	==	=MEM B DQ<26>	31 32					
	101 12	MEM A DQ<25>	==	=MEM A DQ<25>	29 30	101 12	MEM B DQ<25>	==	=MEM B DQ<25>	31 32					
	101 12	MEM A DQ<24>	==	=MEM A DQ<24>	29 30	101 12	MEM B DQ<24>	==	=MEM B DQ<24>	31 32					

SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

DDR3 ALIASES AND BITSWAPS

 Apple Inc.

DRAWING NUMBER
051-9505

SIZE
D

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

REVISION
6.1.0

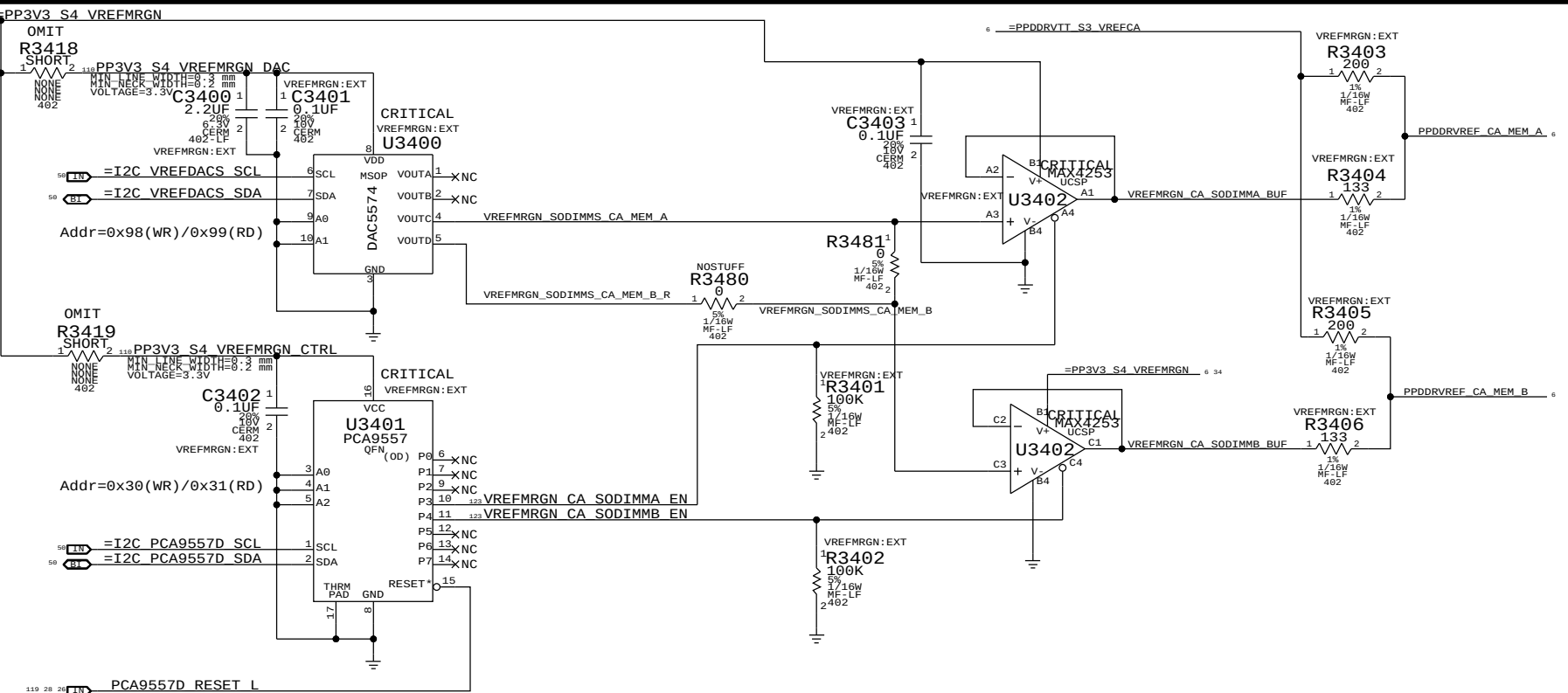
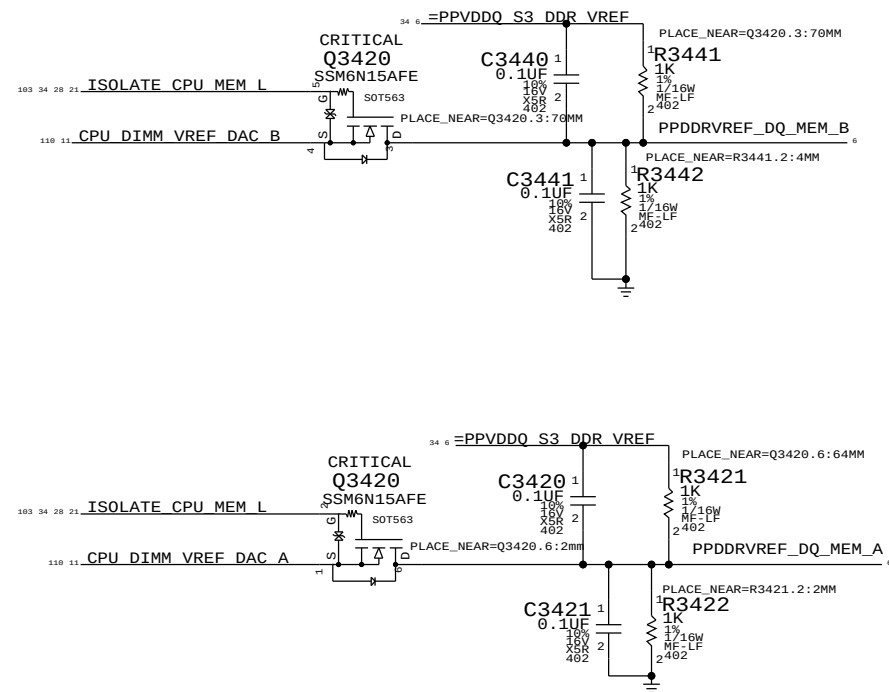
BRANCH

PAGE
33 OF 144

SHEET
33 OF 123

Driven by CPU

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step



RST* on 'platform reset' so that system watchdog will disable margining.

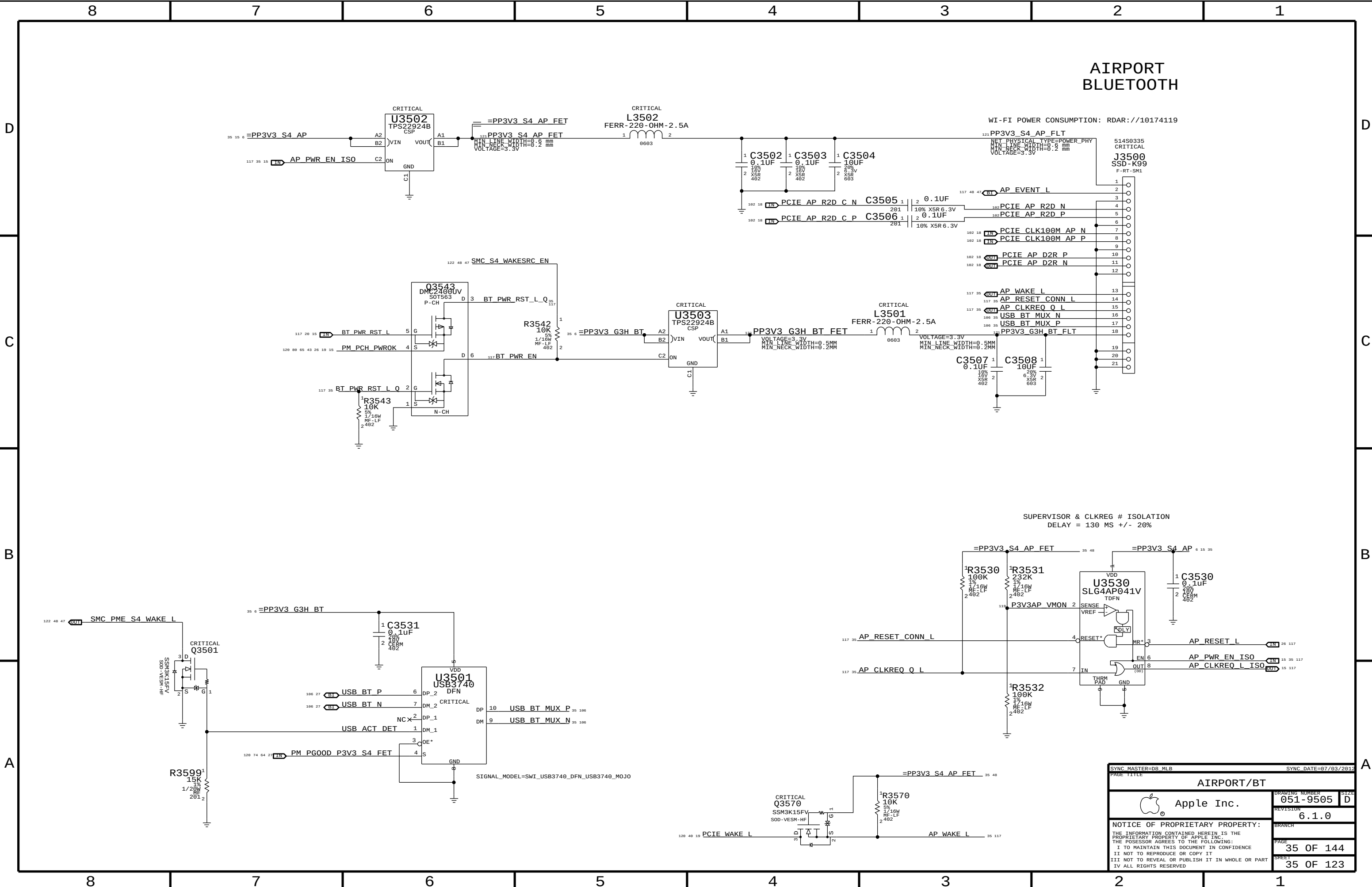
NOTE: MEMVREG and FRAMEBUF share a DAC output, cannot enable both at the same time!

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FLM, 0, 5%, 402, SM, LF	R3403, R3405		VREFMRGN:N

	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	C	C	D	D
PCA9557D Pin:	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)		1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)		1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)		0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)		+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output		8.59mV / step @ output	1.51mV / step @ output

SYMC MASTER=D8 MLB		SYMC DATE=07/03/2012	
PAGE TITLE		DRAWING NUMBER	
DDR3/FRAMEBUF VREF MARGINING		051-9505	
 Apple Inc.		SIZE	
		D	
		REVISION	
		6.1.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED		PAGE	
		34 OF 144	
		SHEET	
		34 OF 123	

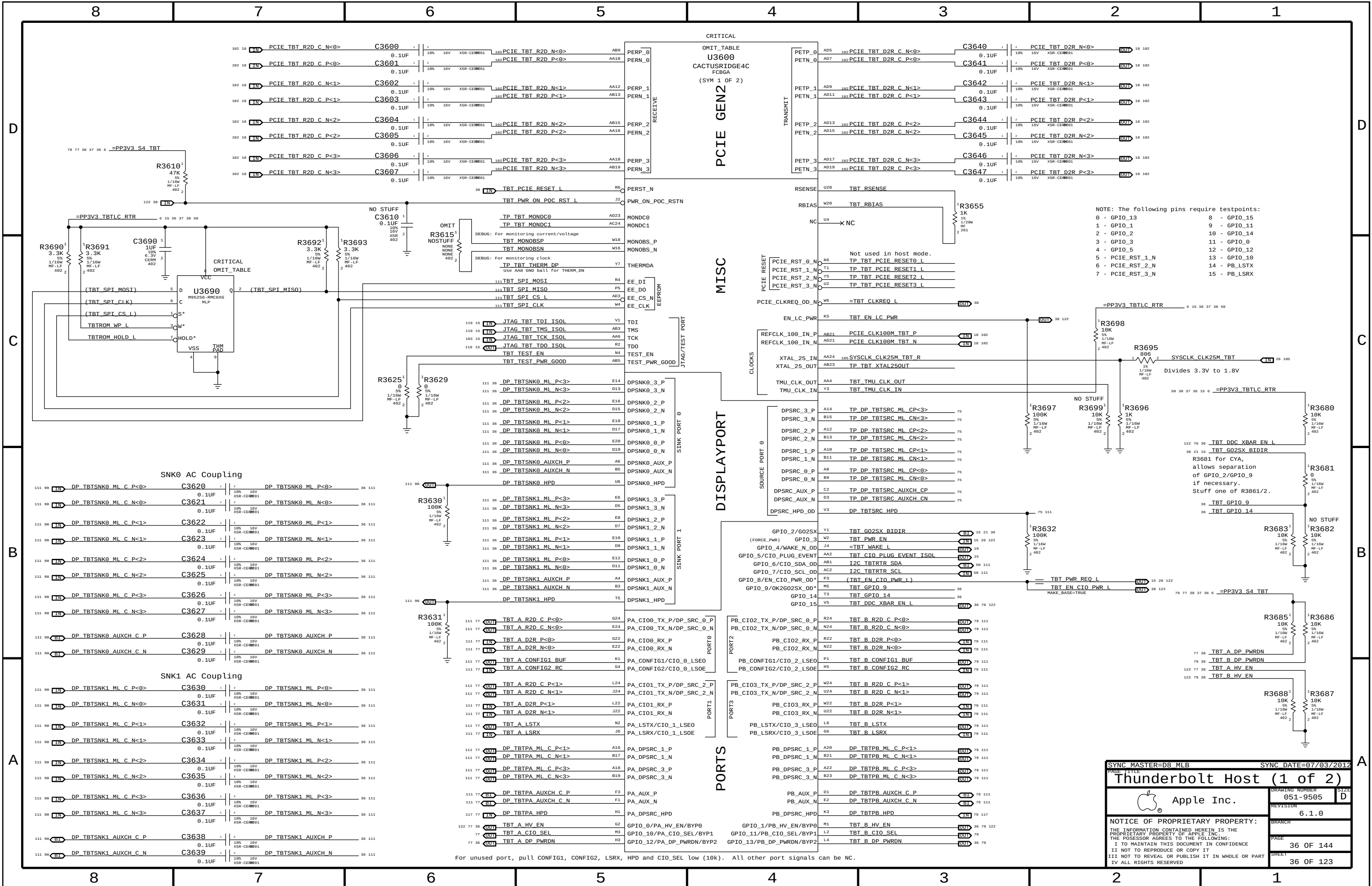


AIRPORT
BLUETOOTH

WI-FI POWER CONSUMPTION: RDAR://10174119

SUPERVISOR & CLKREG # ISOLATION
DELAY = 130 MS +/- 20%

SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
AIRPORT/BT		DRAWING NUMBER	SIZE
 Apple Inc.		051-9505	D
		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		6.1.0	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		35 OF 144	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		35 OF 123	



Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFTET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTTPWRCTL
- =PP1V05_TBT_P1V05TBTFTET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

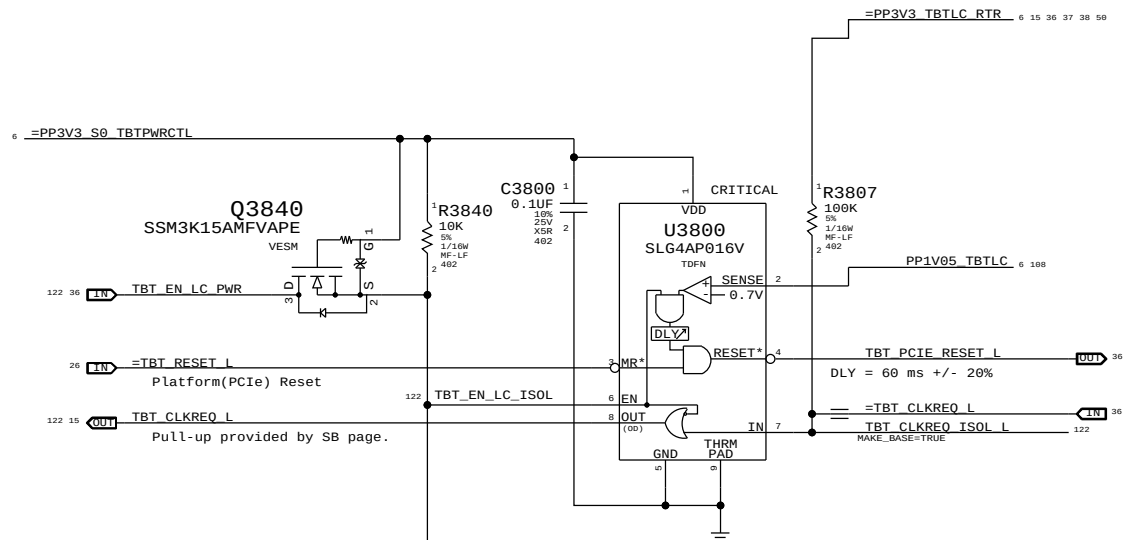
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

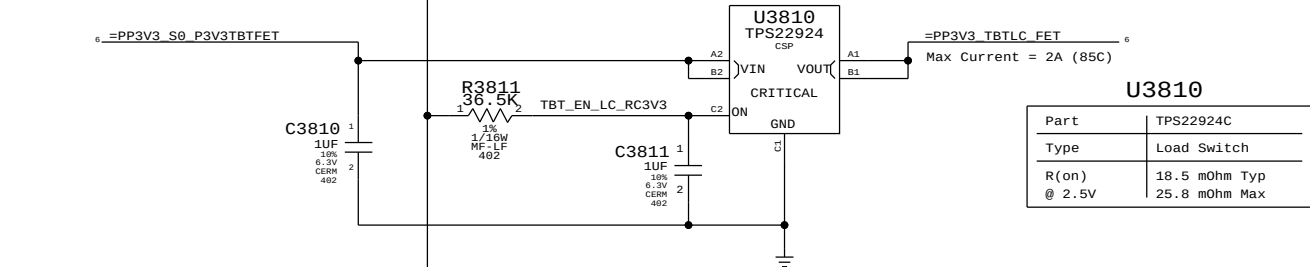
BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

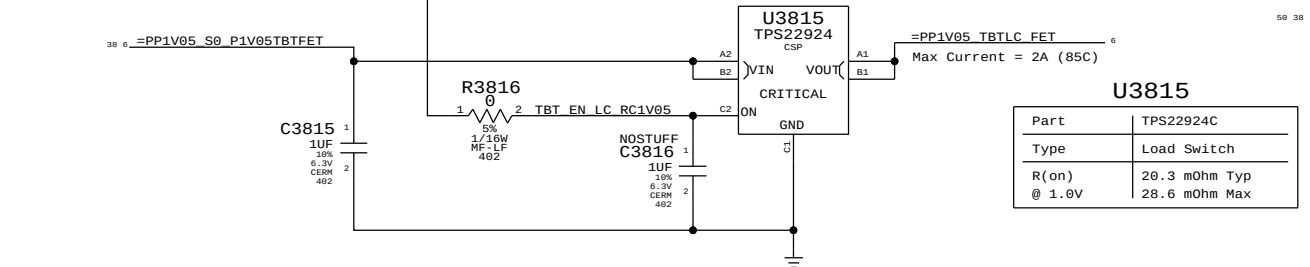
Supervisor & CLKREQ# Isolation



3.3V TBT "LC" Switch

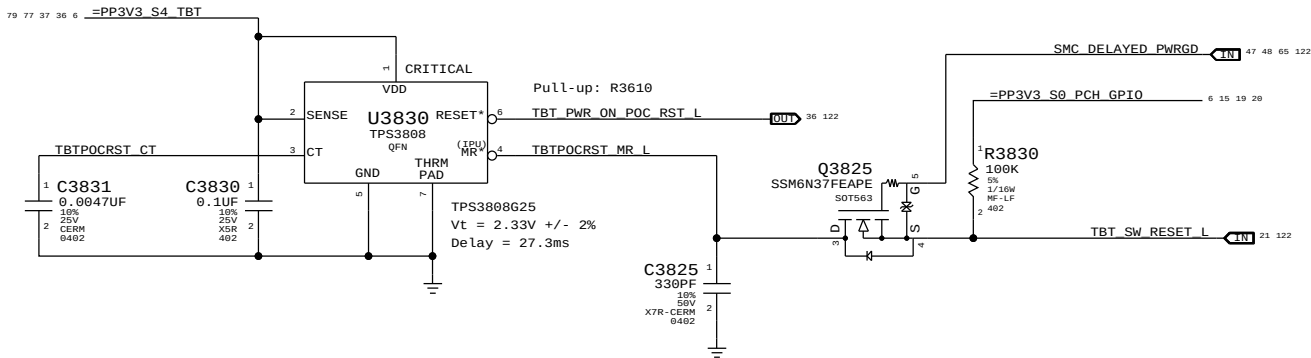


1.05V TBT "LC" Switch

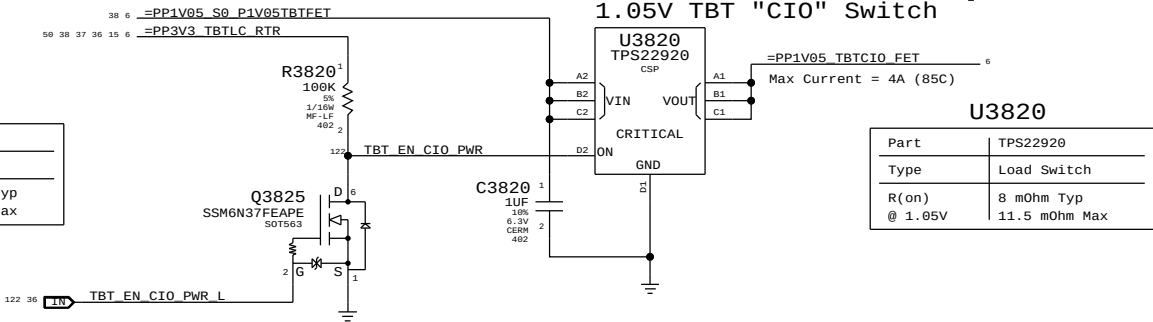


TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.



1.05V TBT "CIO" Switch



Thunderbolt Power Support

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

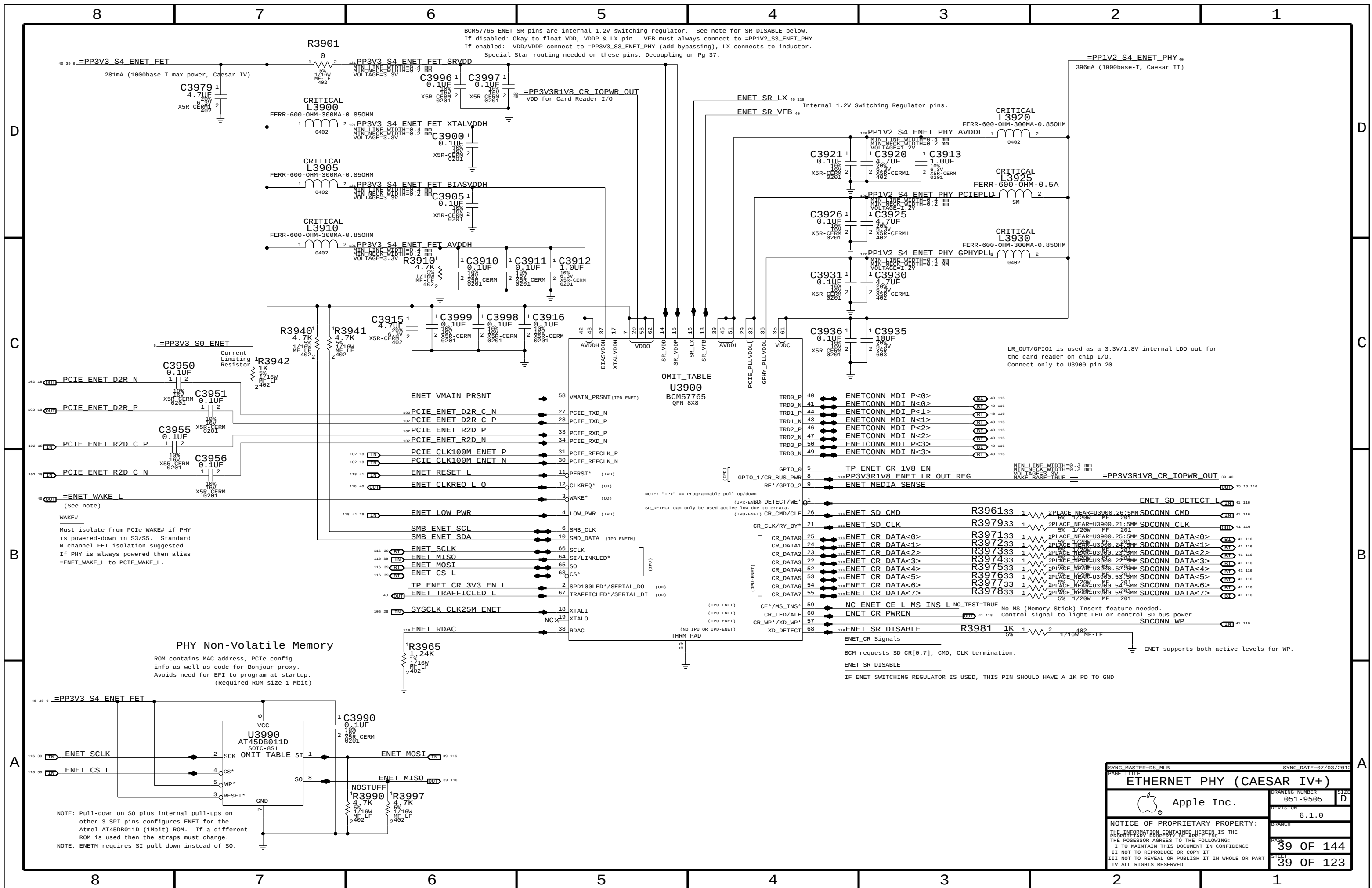
IV ALL RIGHTS RESERVED

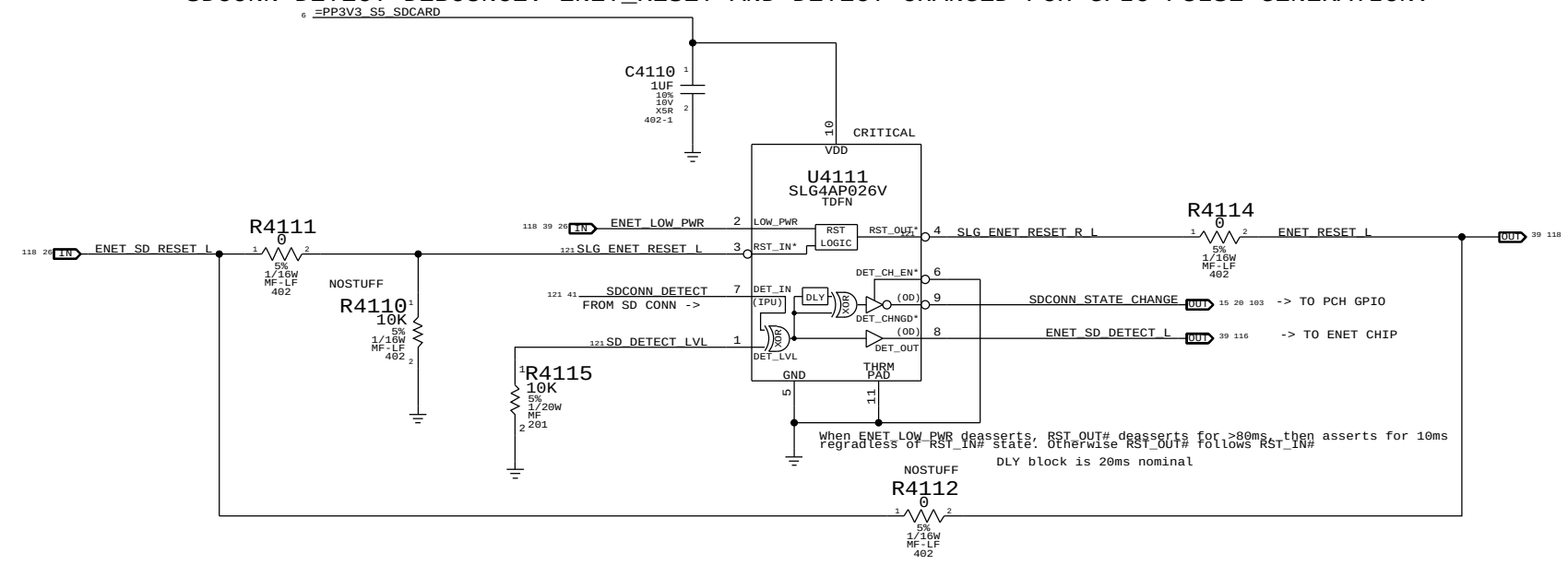
DRAWING NUMBER 051-9505

REVISION 6.1.0

PAGE 38 OF 144

SHEET 38 OF 123



[illegible]

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
SD READER CONNECTOR			
 Apple Inc.		DRAWING NUMBER 051-9505	SIZE D
		REVISION 6.1.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 41 OF 144	
		SHEET 41 OF 123	

D



D

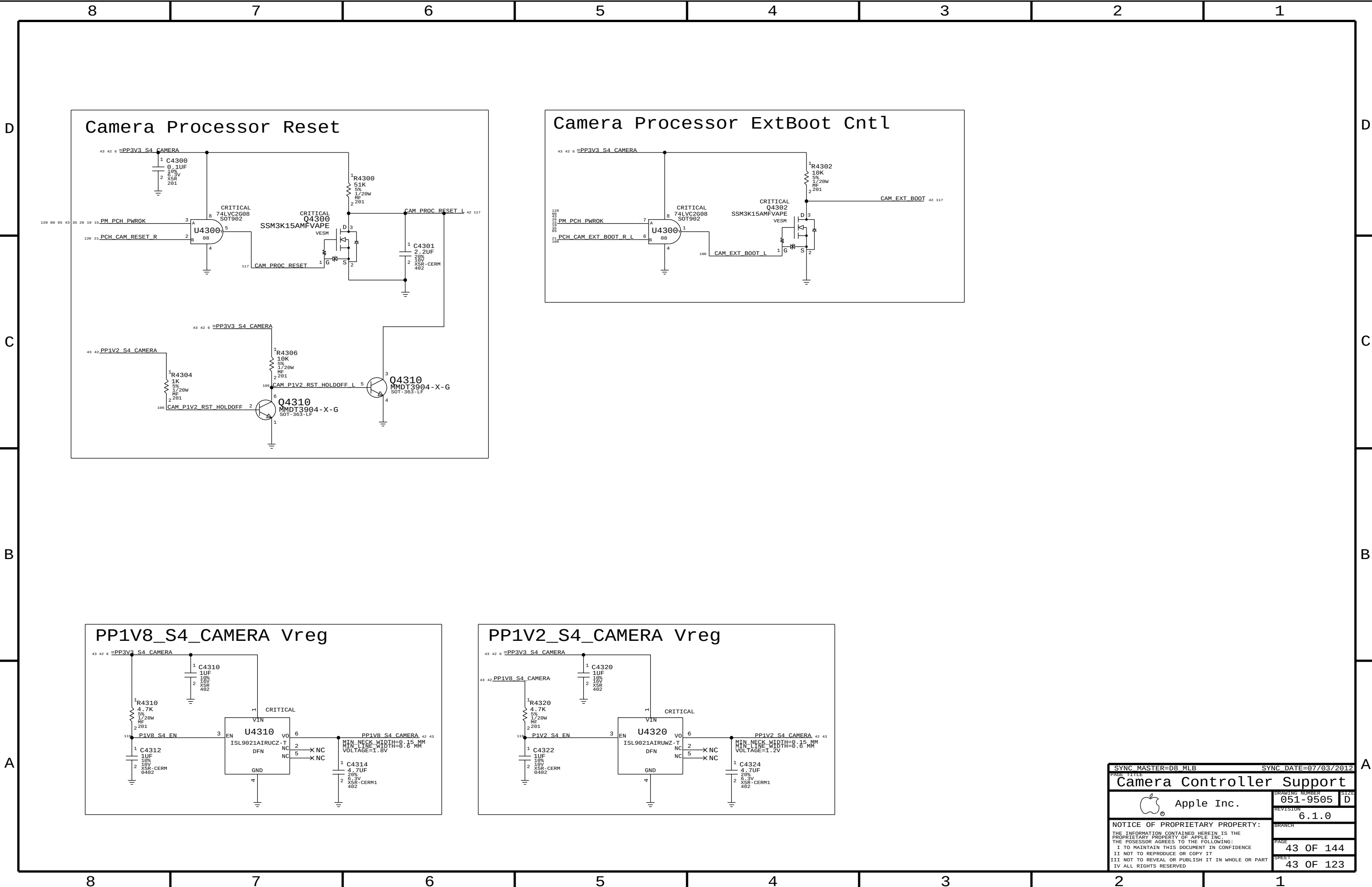


B

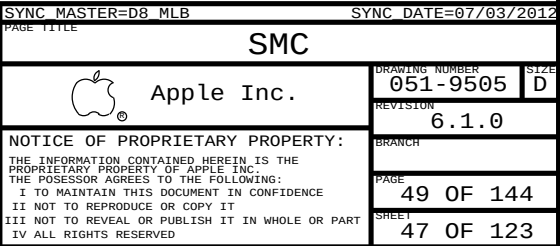


A

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
Camera Controller			
	DRAWING NUMBER		SIZE
	051-9505		D
Apple Inc.		REVISION	
		6.1.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE			
PROPRIETARY PROPERTY OF APPLE INC.			
THE POSSESSOR AGREES TO THE FOLLOWING:			
I. I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II. NOT TO REPRODUCE OR COPY IT			
III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV. ALL RIGHTS RESERVED			



SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE		Camera Controller Support	
Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	43 OF 144
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	43 OF 123
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



D

C

B

A

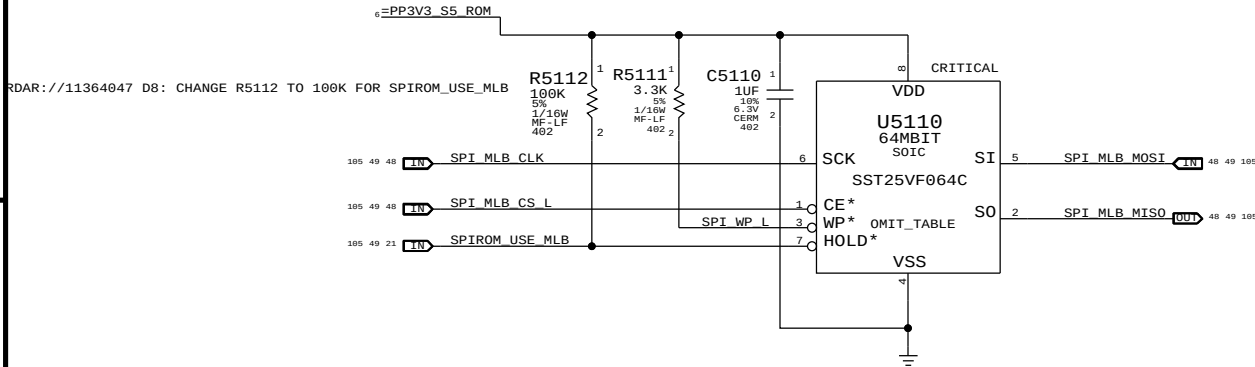
D

C

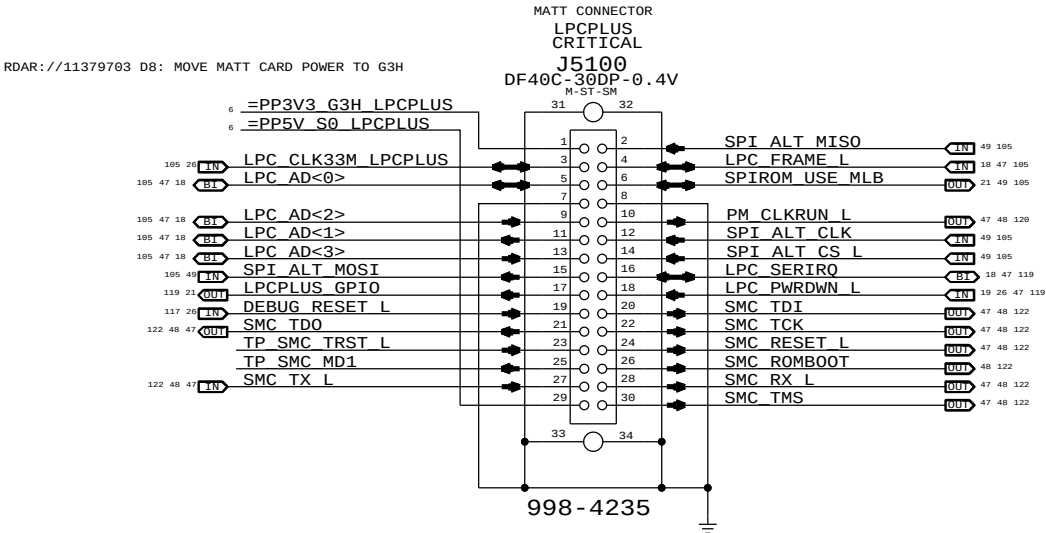
B

A

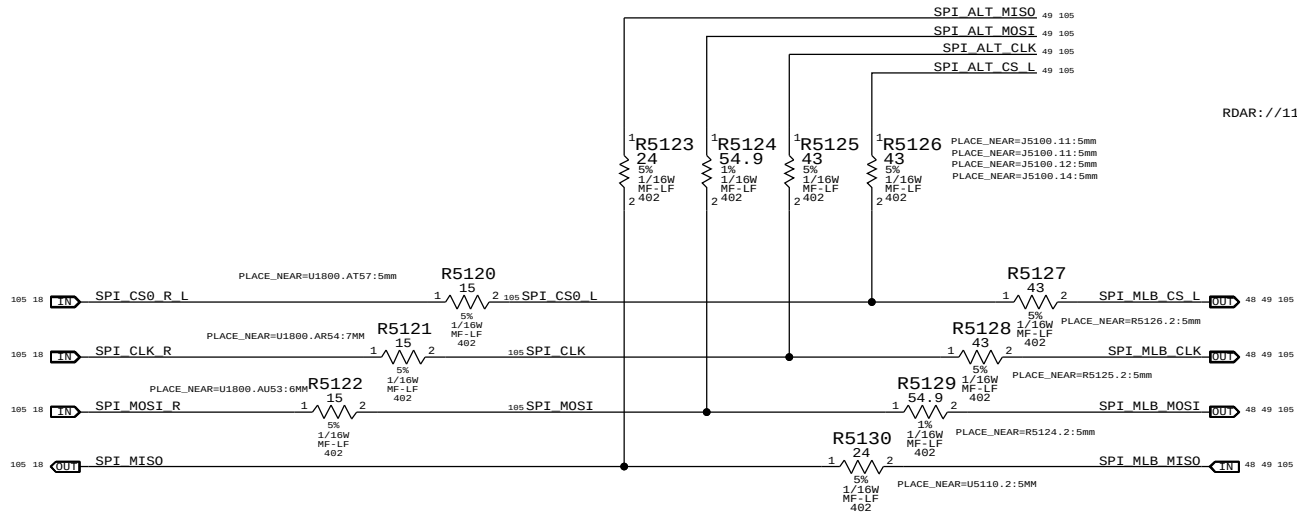
SPI BootROM



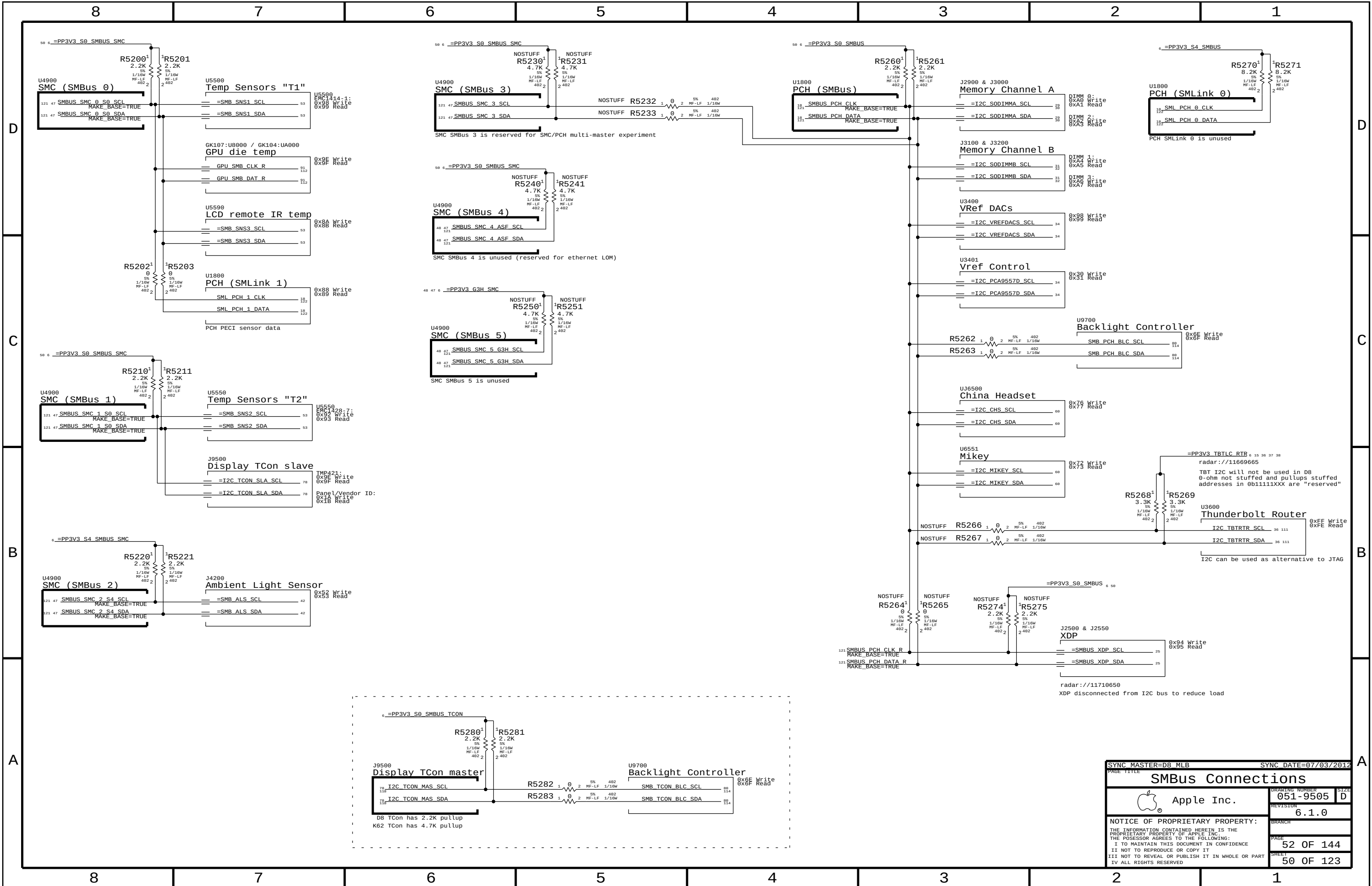
LPC+SPI Connector

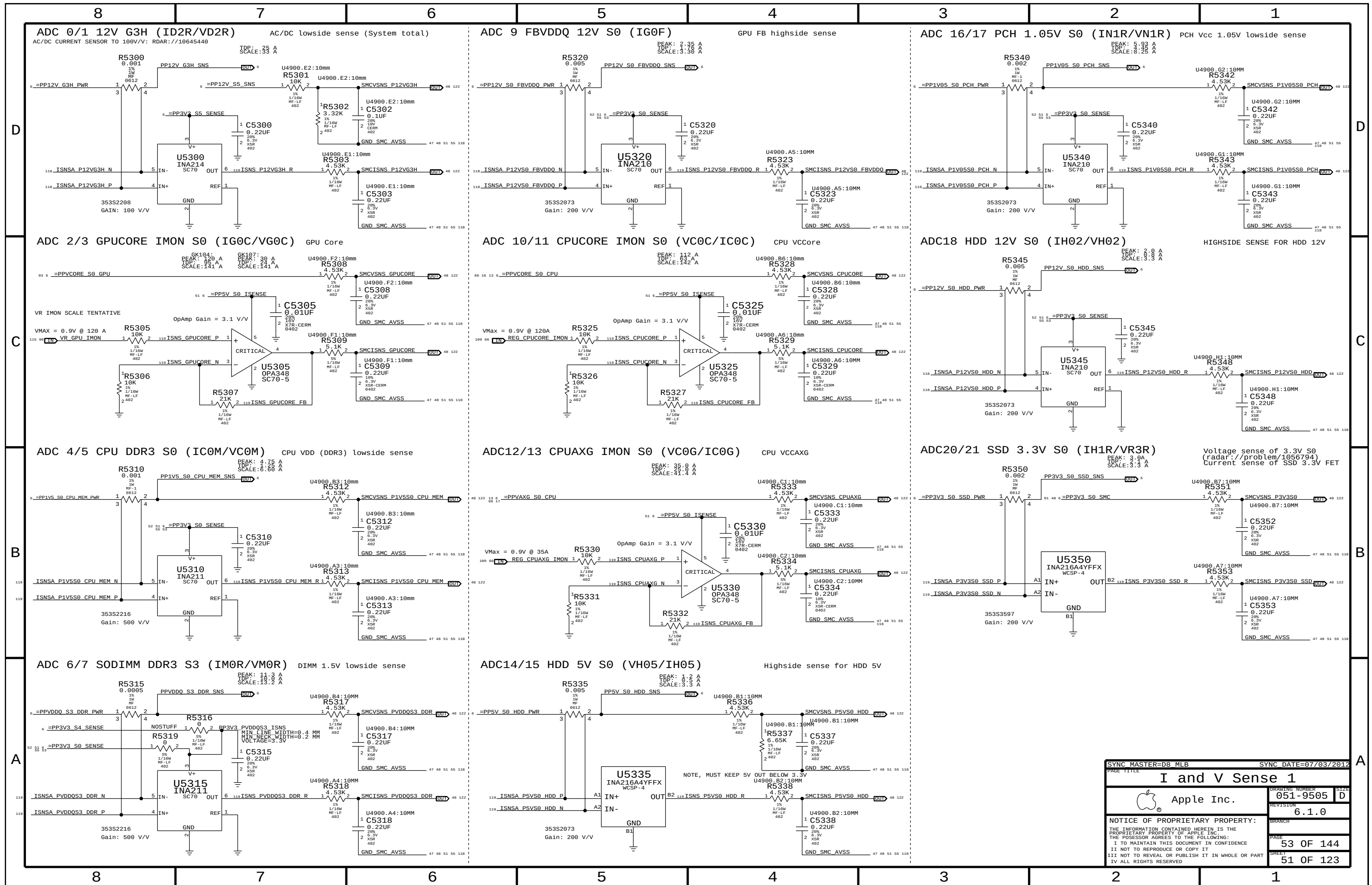


SPI Series Termination



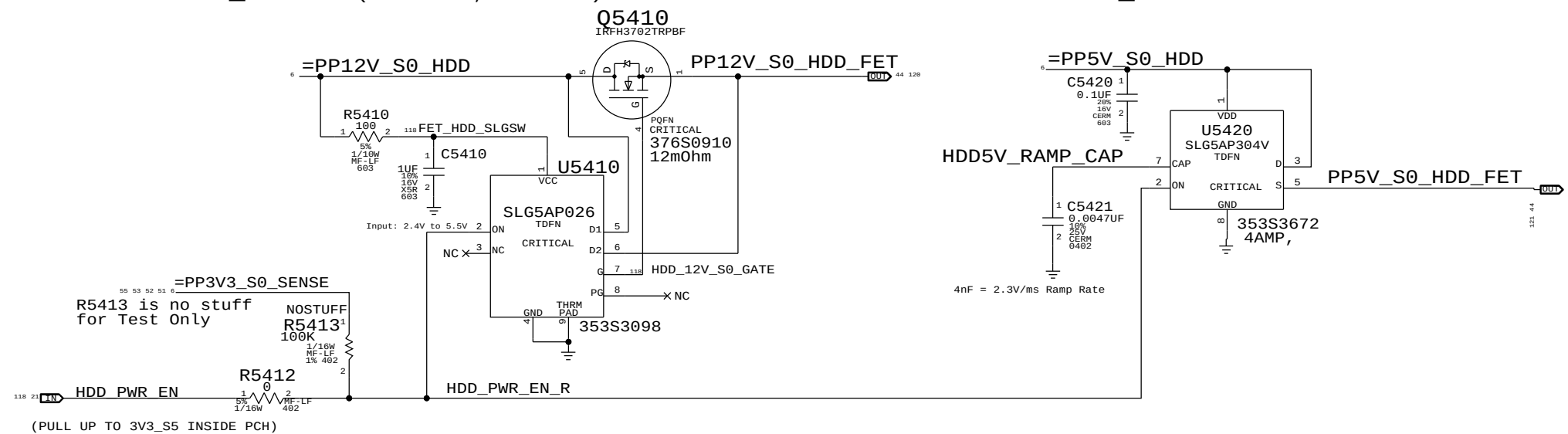
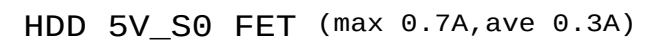
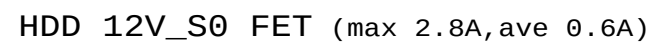
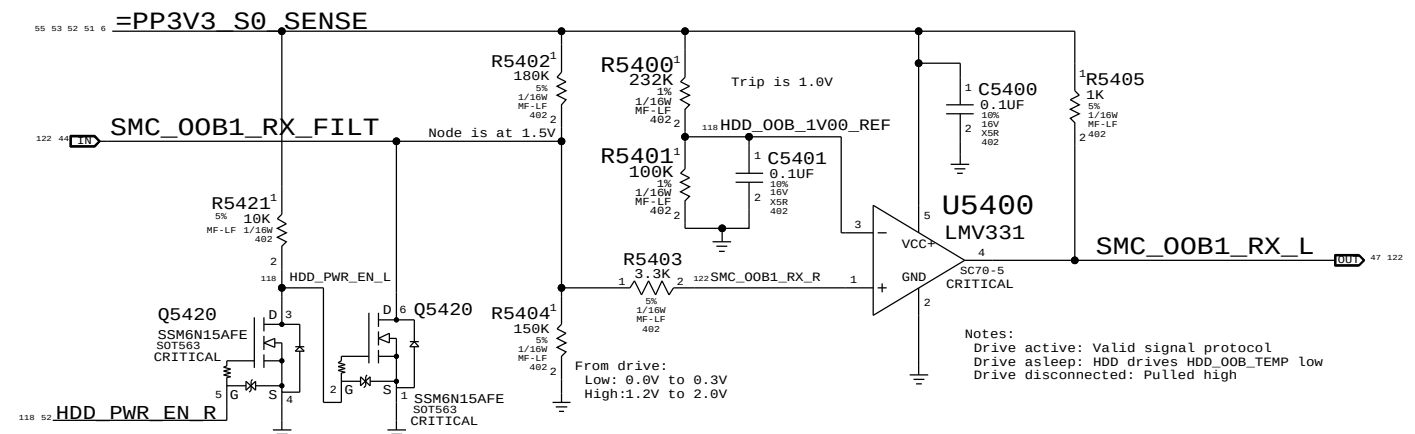
SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE		SPI and Debug Connector	
Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	51 OF 144
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	49 OF 123
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			







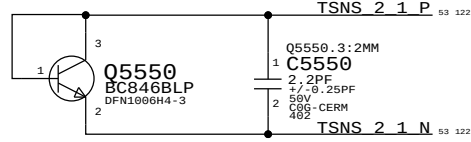
```
Temperature read from SATA power connector pin 11
```



REMOVE R5413 AND SHORT R5412 AFTER HDD_PWR_EN WORKS

SYNCH-MASTER=D8-MLB- - - - -		SYNCH-DATE=07/03/2012		A	
PAGE TITLE					
HDD/SSD Temp Sense					
 Apple Inc.		DRAWING NUMBER		SIZE	
		051-9505		D	
		REVISION			
		6.1.0			
NOTICE OF PROPRIETARY PROPERTY:				BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE III NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED					
				PAGE	
				54 OF 144	
				SHEET	
				52 OF 123	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode



PLACEMENT_NOTE=Place Q5550 near GPU and GDDR5

PLACEMENT_NOTE=Place Q5555 near bottom of board

Q5560
BC846BLP
DFN100GH4-3

05560 : 2MM
2.2PF
7/0.25PF
50V
C0G-CERM
462

TSNs 2 3 P

TSNs 2 3 N

PLACEMENT_NOTE=Place Q5560 near BLC VR

TSNS 2 4 P 5 122

SIGNAL_MODEL=EMPTY

1 3

Q5570 BC846BLP DFN1066H4-3

2

2.2PF

50V C6G-CERM 482

TSNS 2 4 N 5 122

1 2

C5570 Q5570.3:2MM

2.2PF

50V C6G-CERM 482

PLACEMENT_NOTE=Place Q5570 near SO-DIMM connectors (top left)

TSNS 2 5 P N3 122

SIGNAL_MODEL=EMPTY

1 C5575 Q5575.3:2MM

2.2PF

50V

462

TSNS 2 5 N3 122

Q5575 BC846BLP DFN1006H4-3

PLACEMENT_NOTE=Place Q5575 near S0-DIMM connectors (top right)

TSNS 2 6 P 53 122

SIGNAL_MODEL=EMPTY

1 C5580 Q5580.3:2MM

2 2.2PF

3 1/20.25PF

4 50V

5 C96-CERM

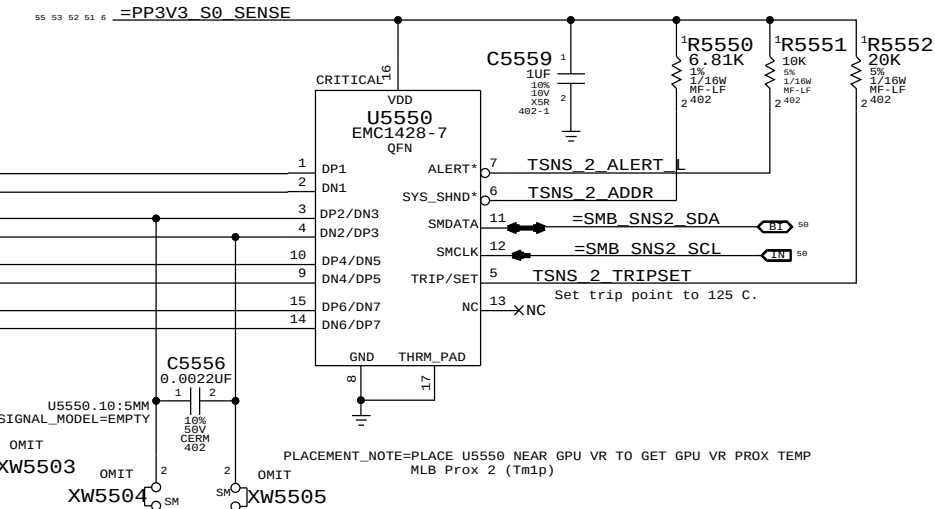
6 462

TSNS 2 6 N 53 122

Q5580 BC846BLP DFM1906H4-3

PLACEMENT_NOTE=Place Q5580 near S0-DIMM connectors (bottom left)

PLACEMENT_NOTE=Place Q5585 near S0-DIMM connectors (bottom right)



EMC1428-7: 6.8K PULL UP: I2C ADDRESS: WRITE: 0x92, READ: 0x93

CRITICAL

J5500
53780-8602
M-RT-3W

3

1 1.22 TSNS SKIN P
2 1.22 TSNS SKIN N

4

J5500 .1:15MM
L5500
FERR-220-OHM

1 0482 2

J5500 .2:15MM
L5500
FERR-220-OHM

1 2

TSNS 1 1 P 53 122

SIGNAL_MODEL=EMPTY
L5500 .2:2MM

1 C5501
0.0022UF
50V
CERAM
482

2 TSNS 1 1 N 53 122

518S0698
SILK_PART=SkinTemp

TSNS 1 2 P

SIGNAL_MODEL=EMPTY
05585-3, 2RM

1 C5505
2 2.2PF
1 7.0-25PF
56V
C66-CERM
482

TSNS 1 2 N

PLACEMENT_NOTE=PLACE Q5505 NEAR CPU

TSNS ACDC P

TSNS ACDC N

J601, 3.2:2MM
L5510
FERR-220-OHM

1 2

0402

TSNS 1 3 P 53 122

L5510, 2:2MM
C510
0.0022UF

1 2

50V
18V
CERN
402

J602, 1.4:20MM
L5511
FERR-220-OHM

1 2

0402

TSNS 1 3 N 53 122

[illegible]

Make sure these caps are OK with U5500 Vendor!

Note: Internal sensor of the EMC 1414 will be used as MLB sensor.
MLB Prox 1 (Tm0p)

[illegible]

NOTE - Follow TI layout guide(SBOU108.pdf) for this part!!!

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
Temperature Sensors			
 Apple Inc.		DRAWING NUMBER 051-9505	SIZE D
		REVISION 6.1.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		55 OF 144	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		53 OF 123	
IV ALL RIGHTS RESERVED			

D

C

B

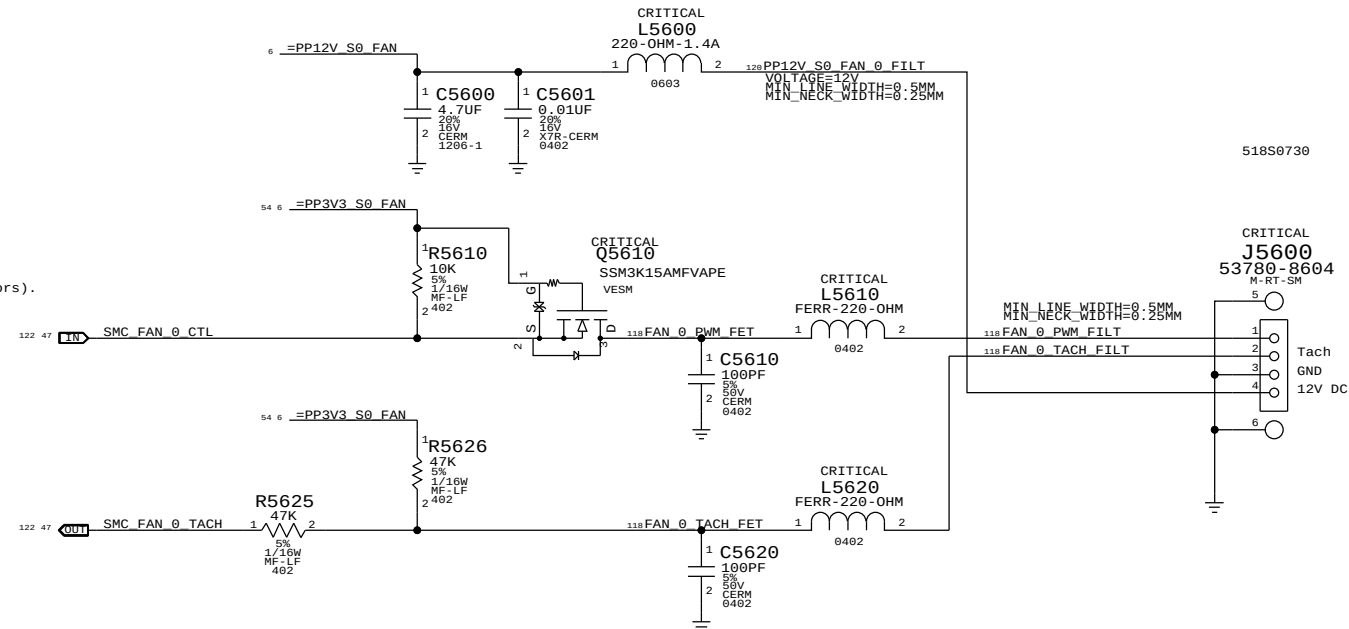
A

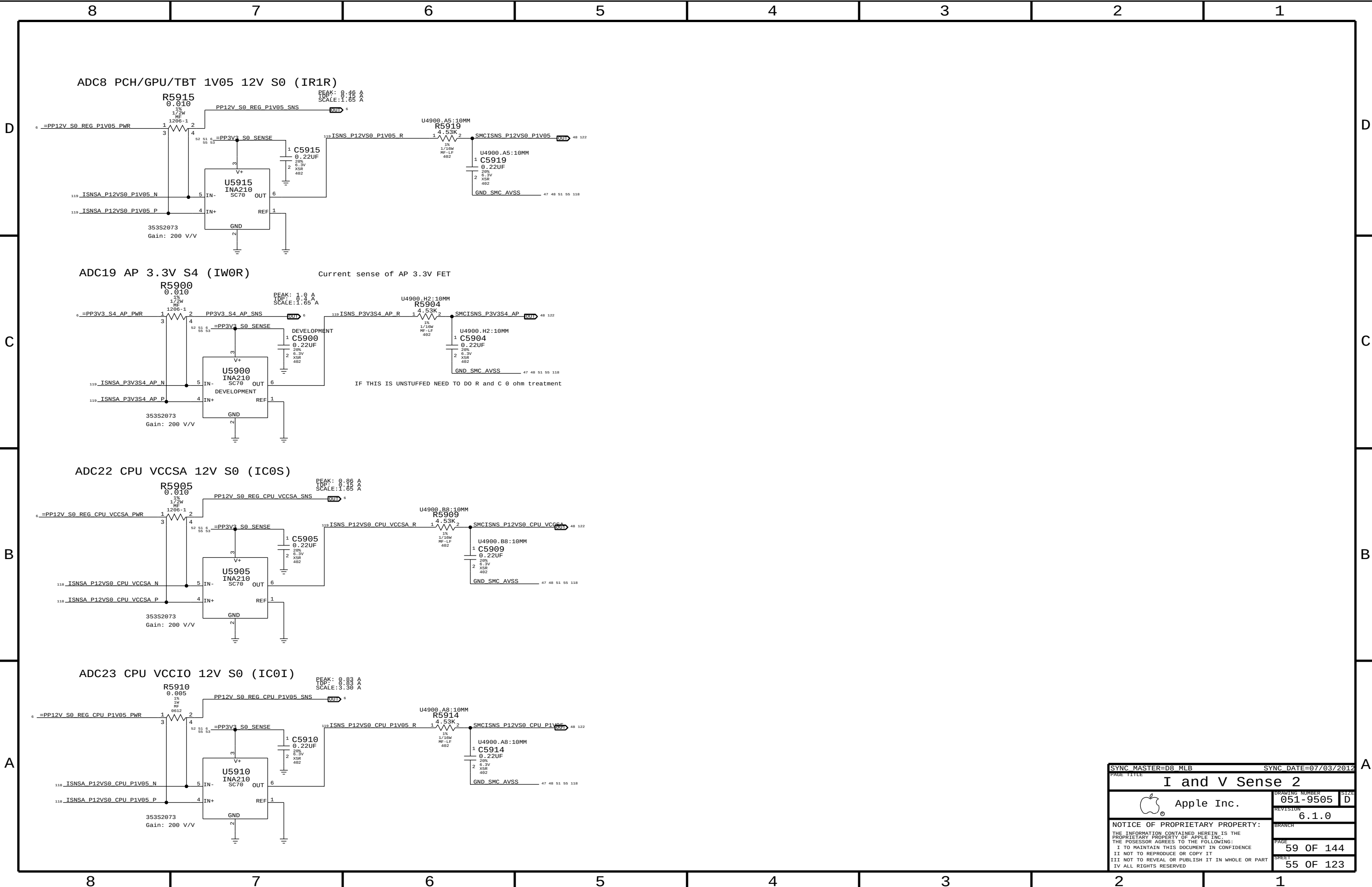
```

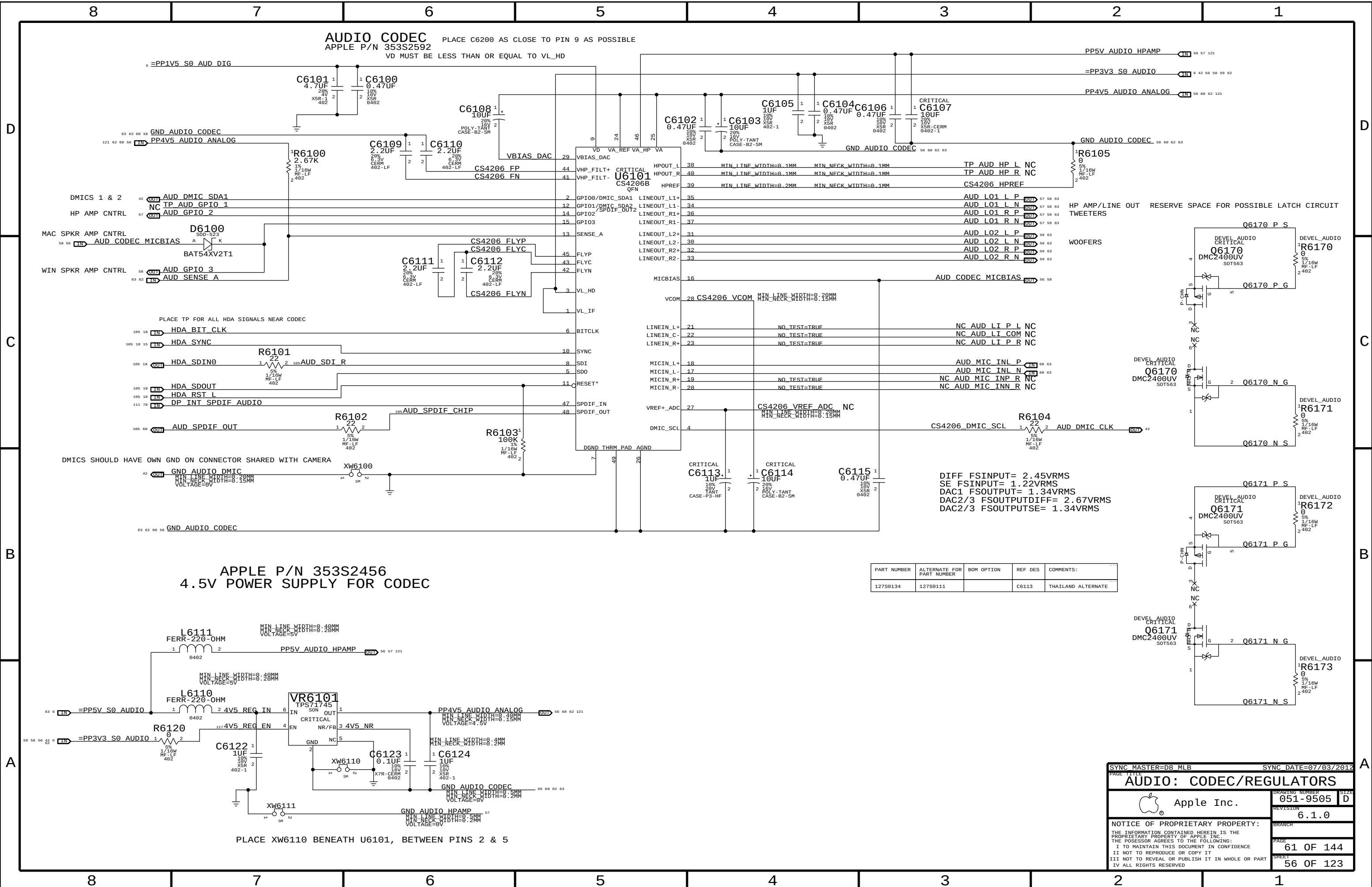
47 [IN] SMC_FAN_1_CTL == NC_SMC_FAN_1_CTL
                        MAKE_BASE=TRUE NO_TEST=TRUE

47 [OUT] SMC_FAN_1_TACH == NC_SMC_FAN_1_TACH
                         MAKE_BASE=TRUE NO_TEST=TRUE

```







PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0134	127S0111		C6113	THAILAND ALTERNATE

SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

AUDIO: CODEC/REGULATORS

Apple Inc.

051-9505

6.1.0

61 OF 144

56 OF 123

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

LEFT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPFF, TWEETERS = ~847 HZ (4700 PF)
FC_HPFF, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

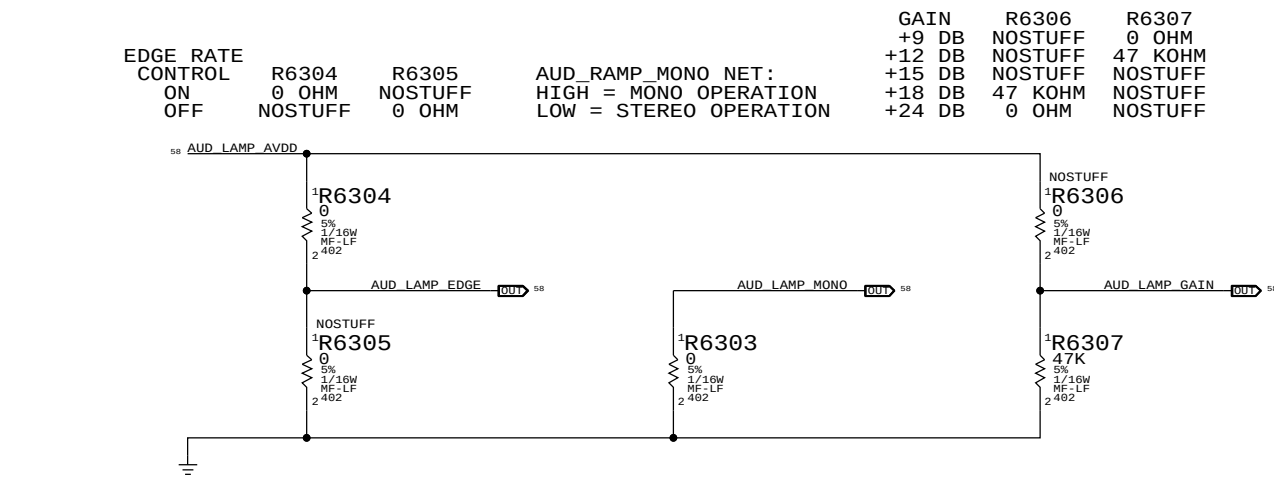
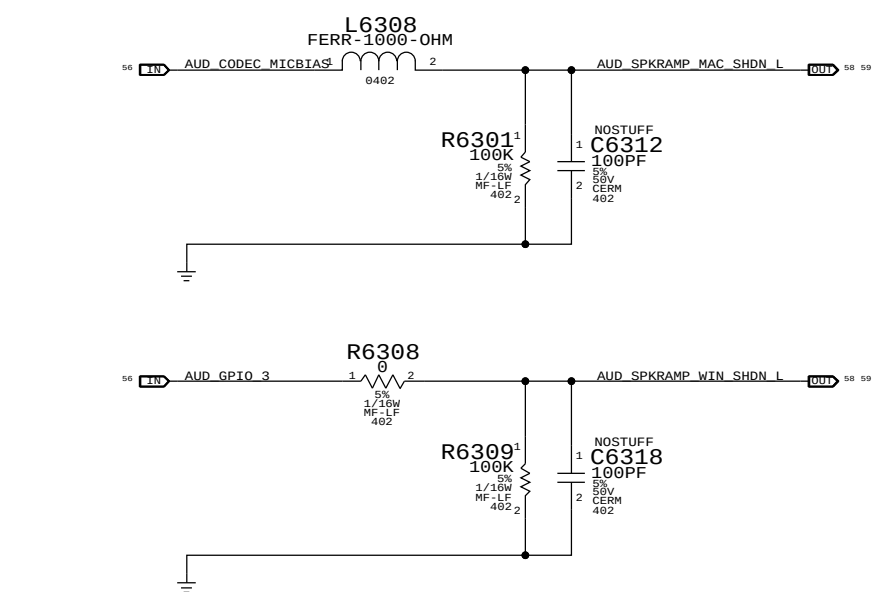
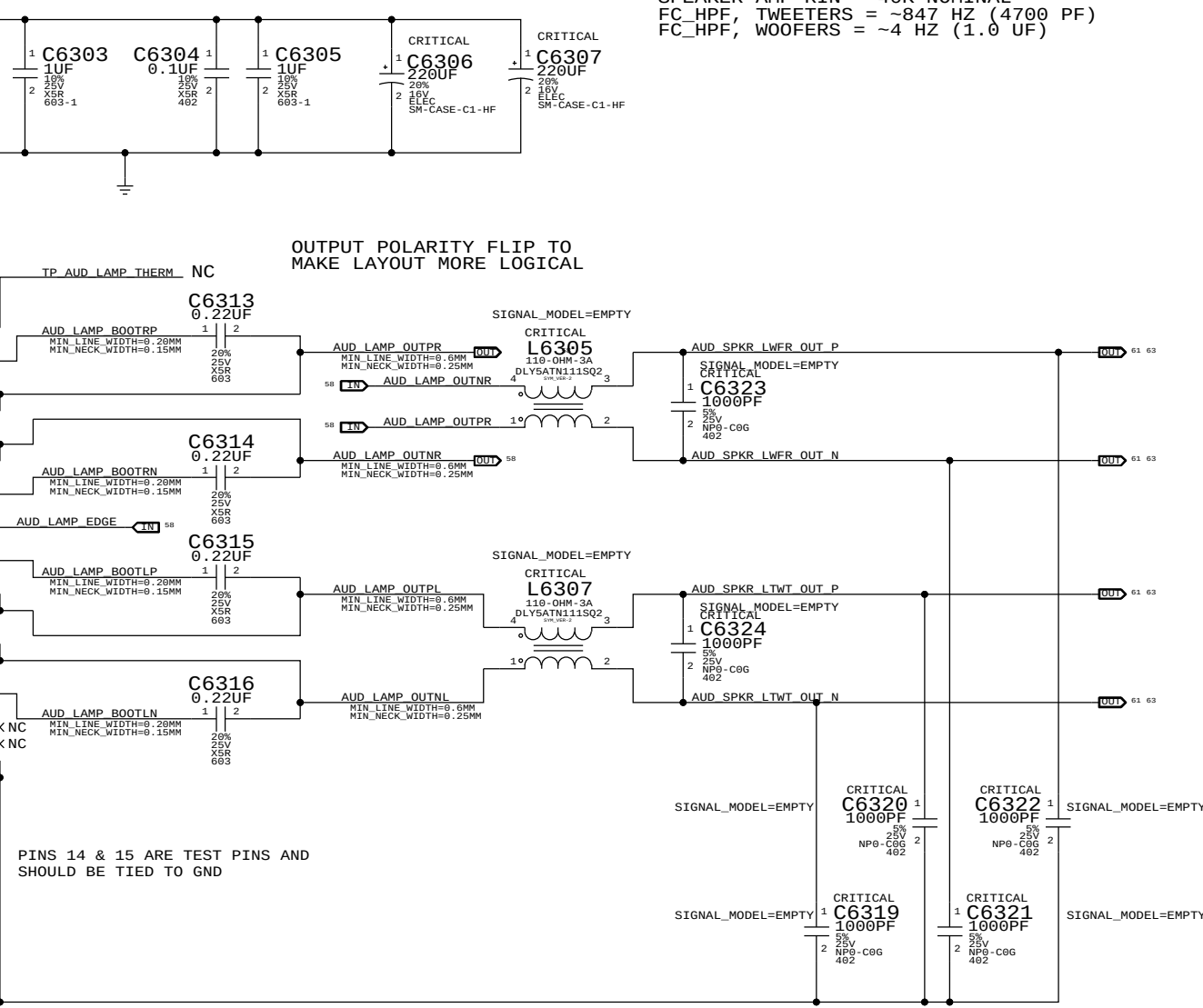
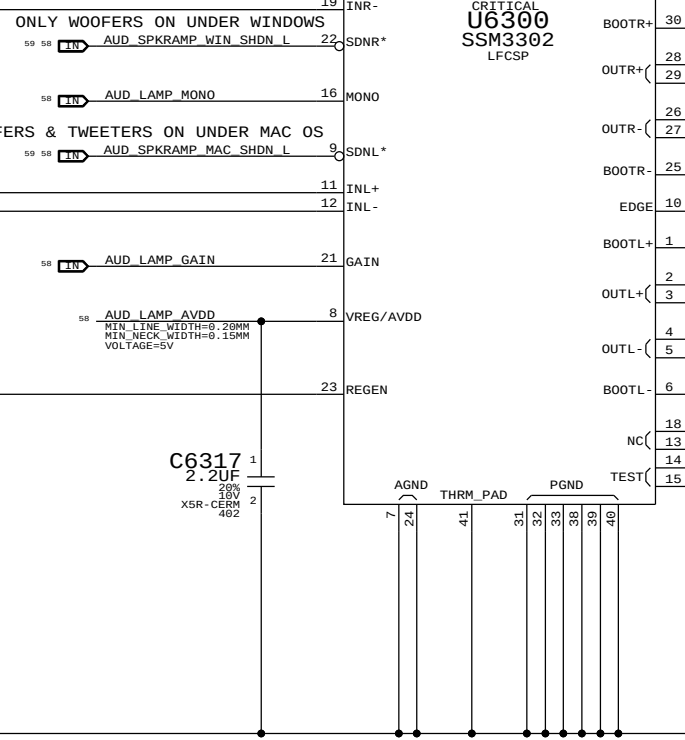
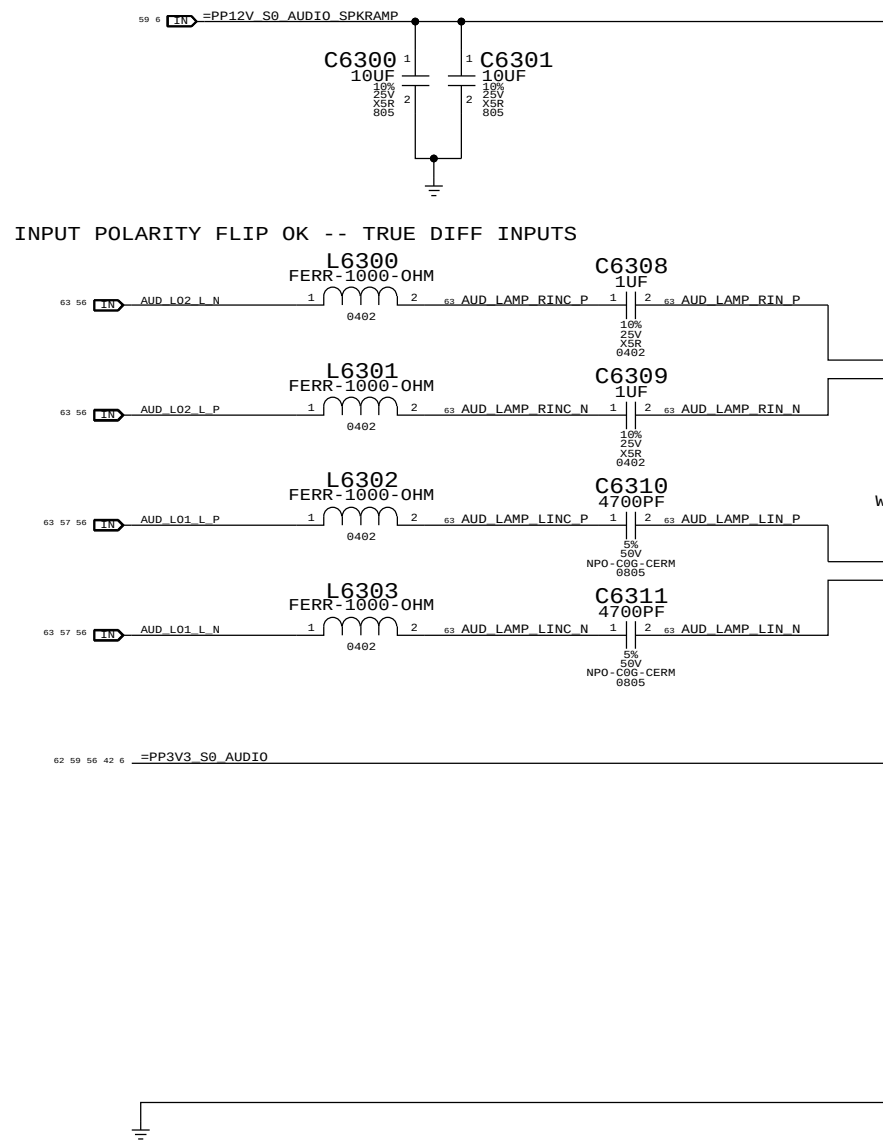
C

B

B

A

A

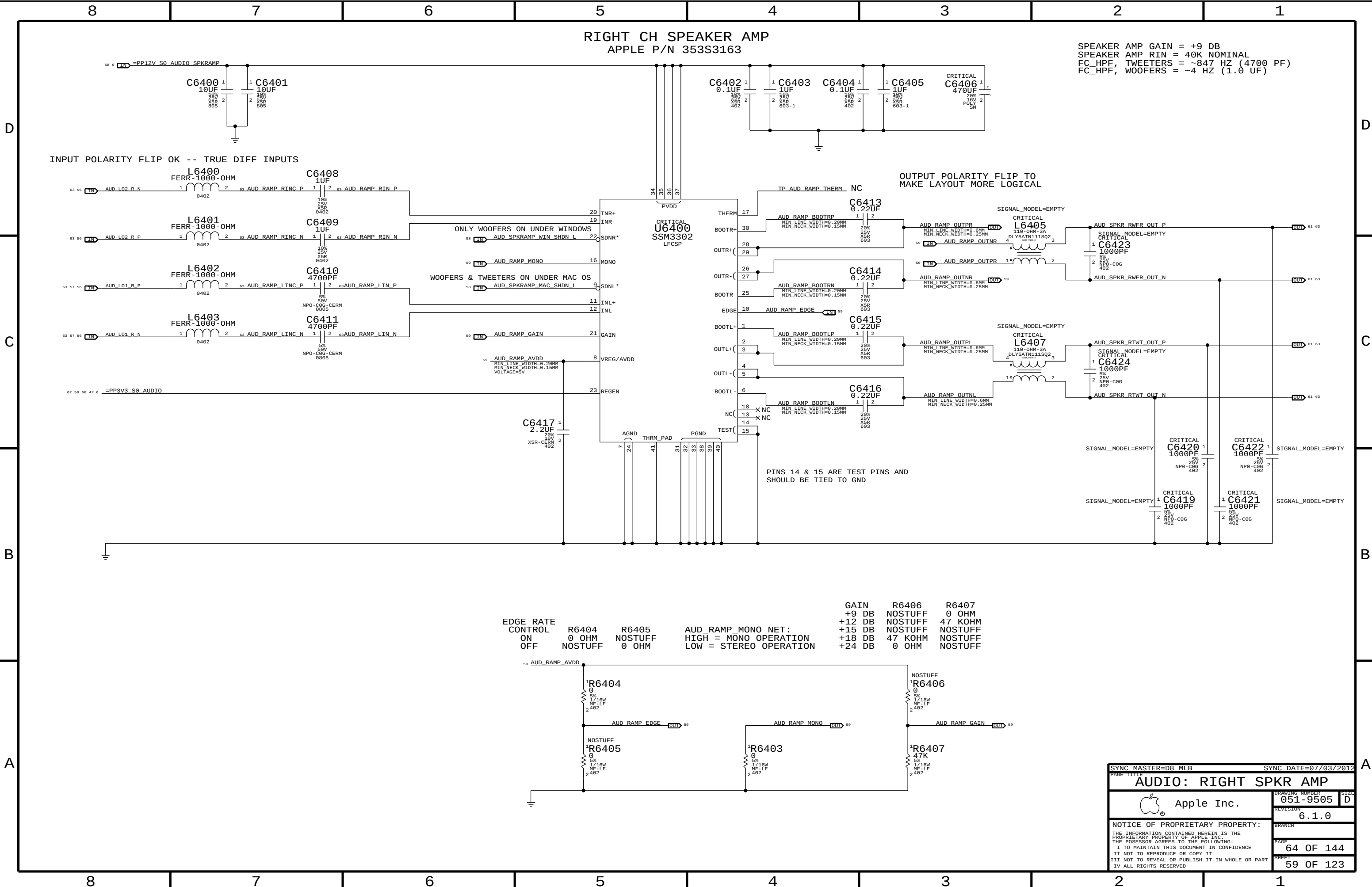


GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6306
NOSTUFF
NOSTUFF
NOSTUFF
NOSTUFF

R6307
0 OHM
47 KOHM
NOSTUFF
NOSTUFF

SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE		AUDIO: LEFT SPKR AMP	
Apple Inc.		DRAWING NUMBER	051-9505
NOTICE OF PROPRIETARY PROPERTY:		REVISION	6.1.0
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	63 OF 144
II NOT TO REPRODUCE OR COPY IT		SHEET	58 OF 123
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72

D

D

C

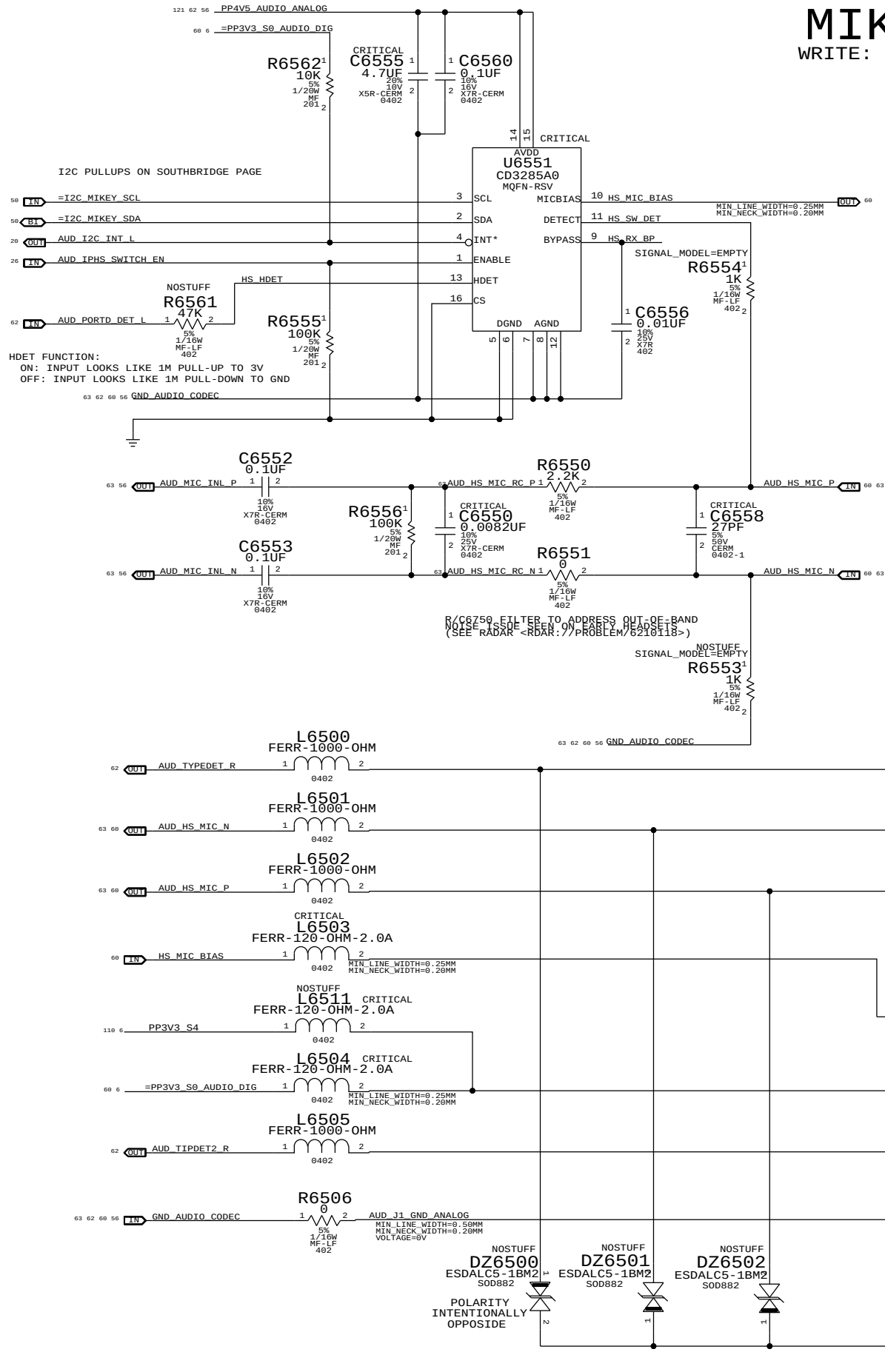
C

B

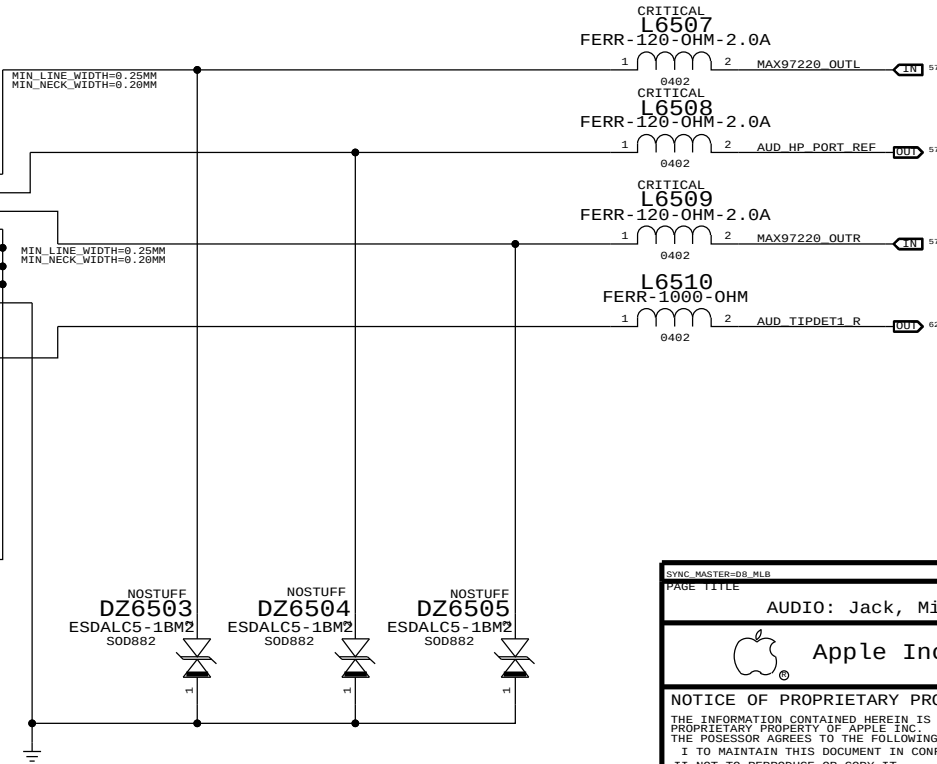
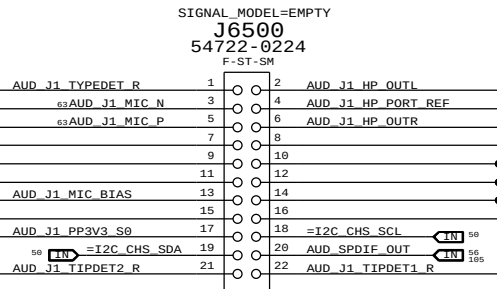
B

A

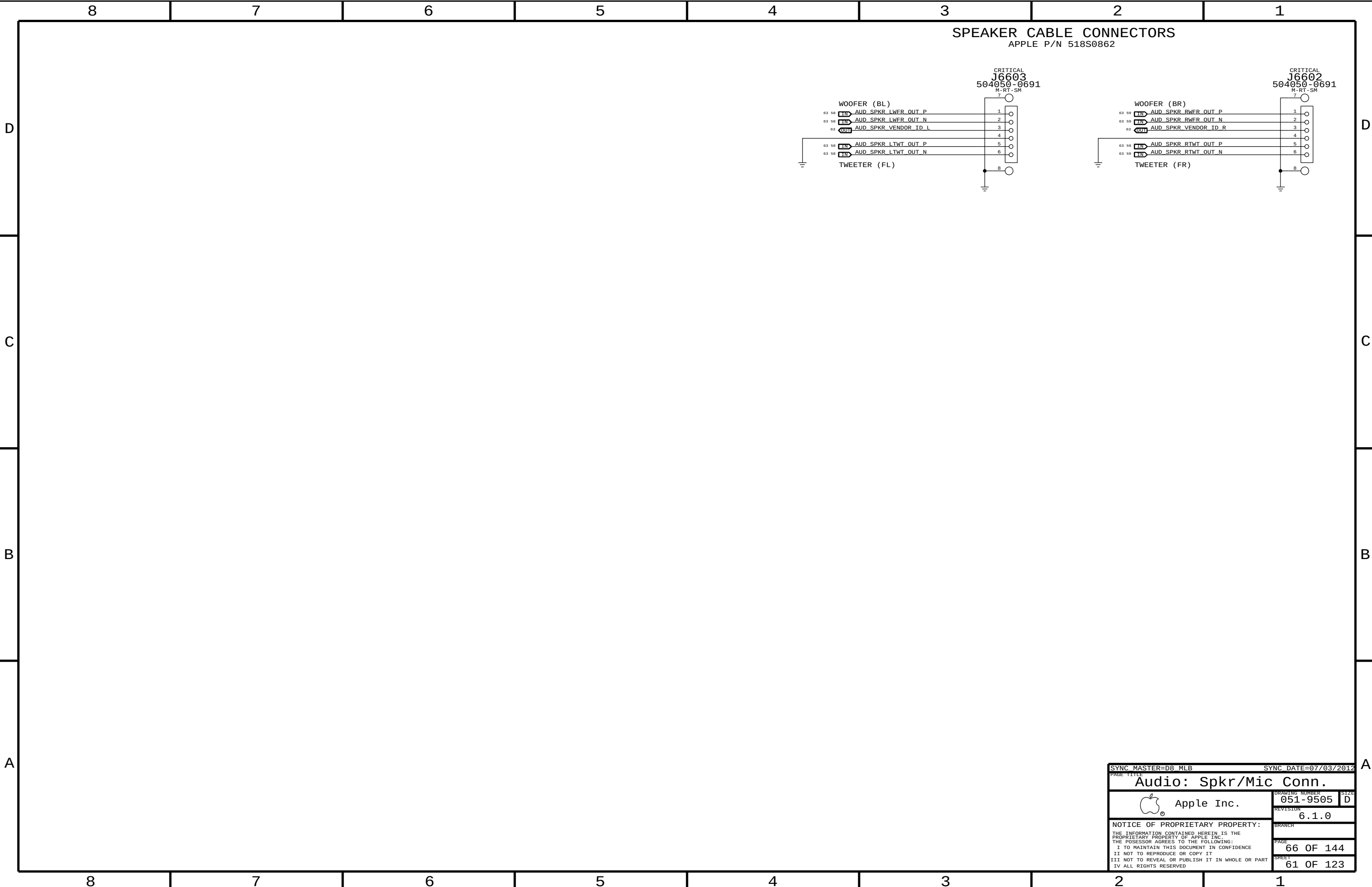
A

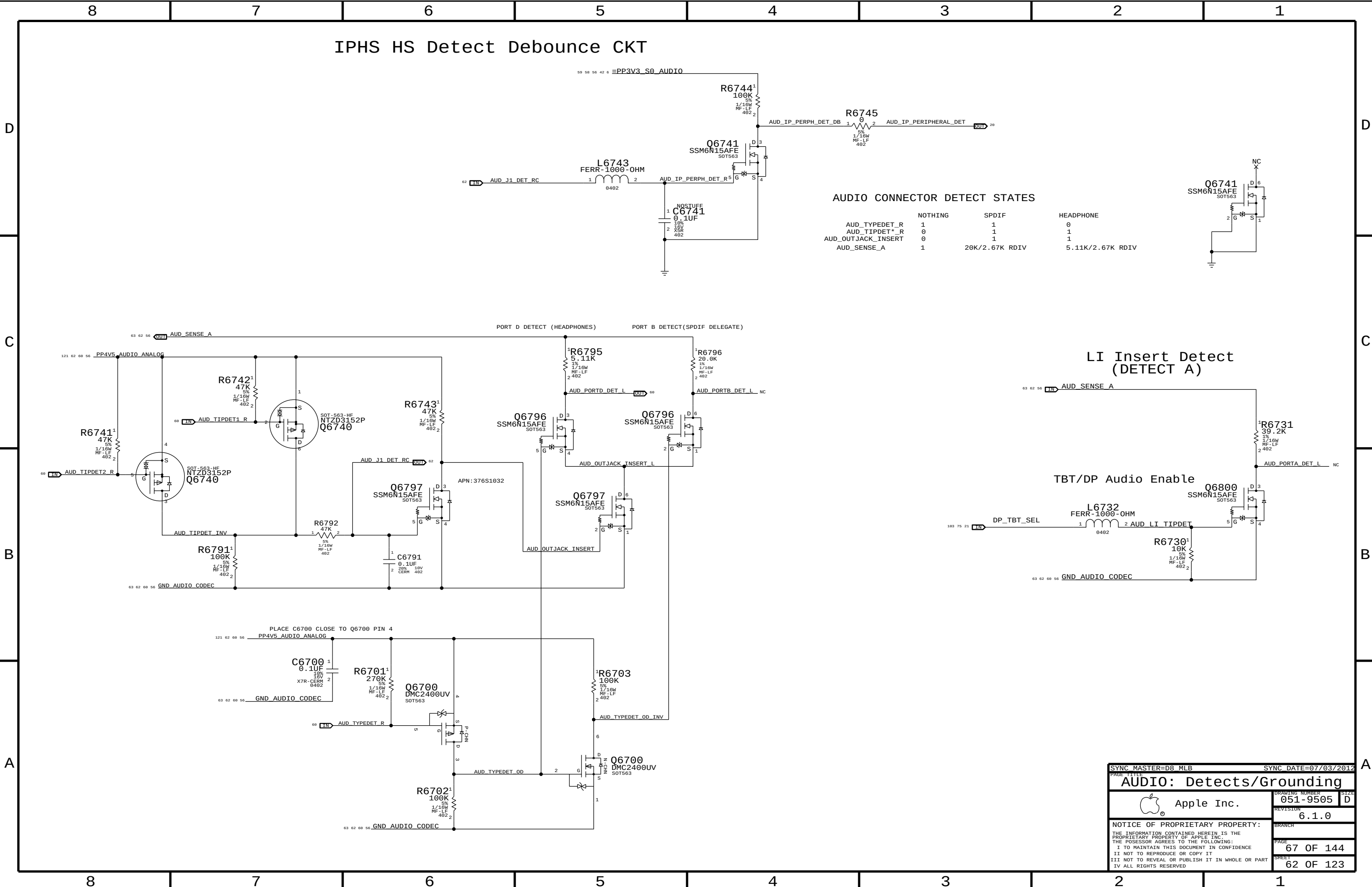
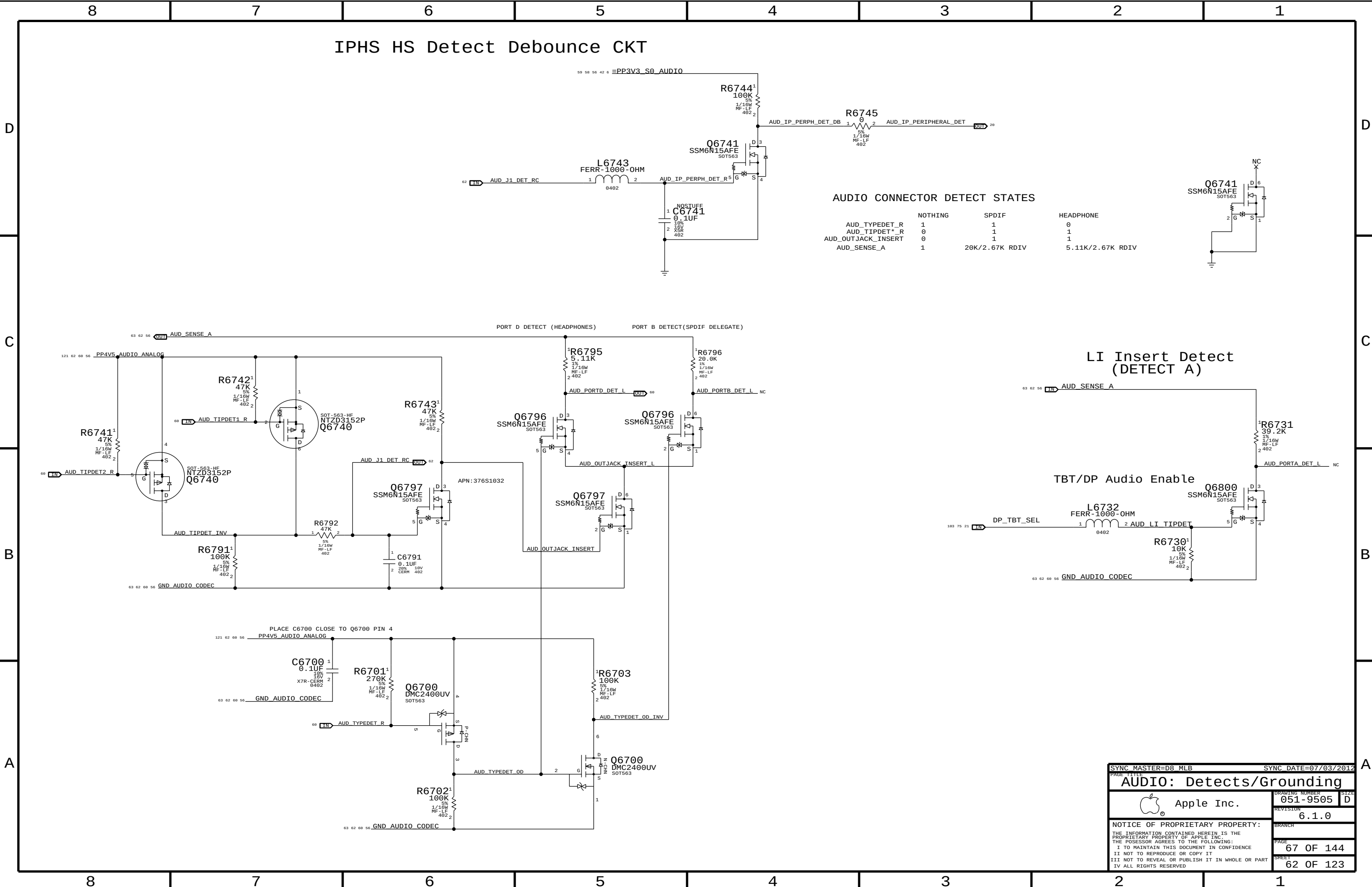


APPLE P/N 518S0687



PAGE TITLE		PAGE TITLE	
AUDIO: Jack, Mikey, CHS Switch		DRAWING NUMBER	
Apple Inc.		051-9505	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		6.1.0	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		BRANCH	
II NOT TO REPRODUCE OR COPY IT		PAGE	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		65 OF 144	
IV ALL RIGHTS RESERVED		SHEET	
		60 OF 123	



[illegible]

IPHS HS Detect Debounce CKT

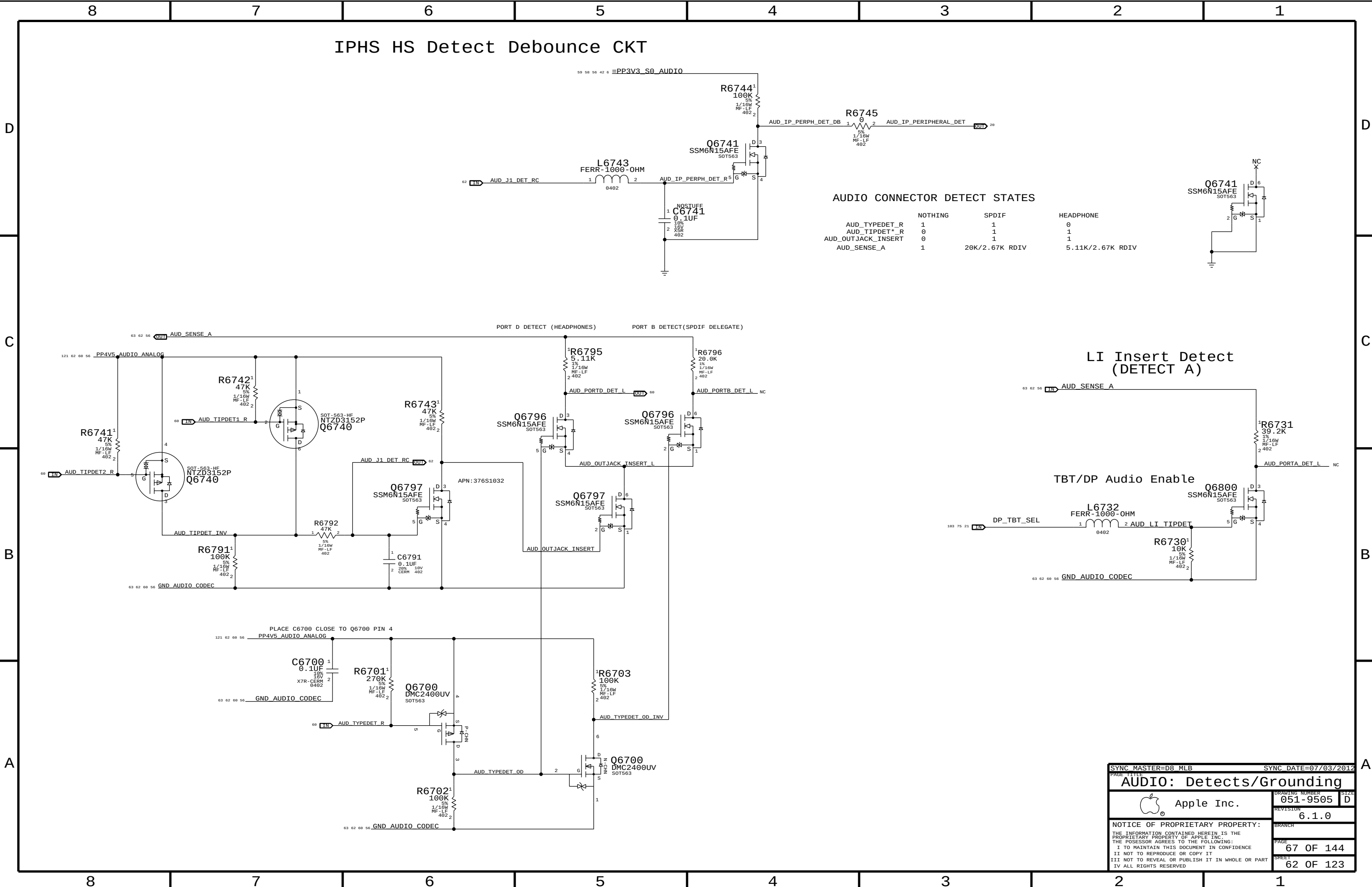
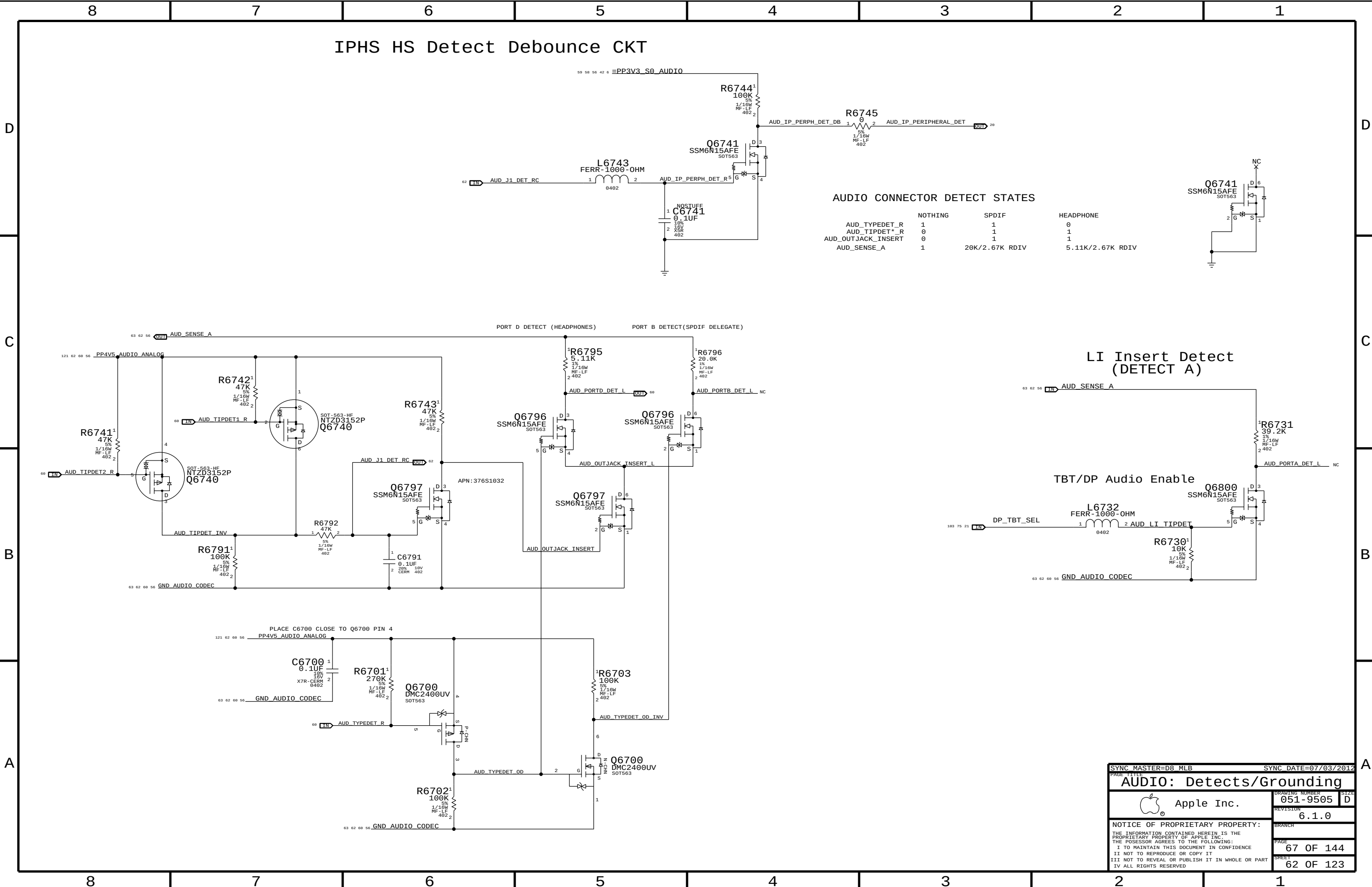
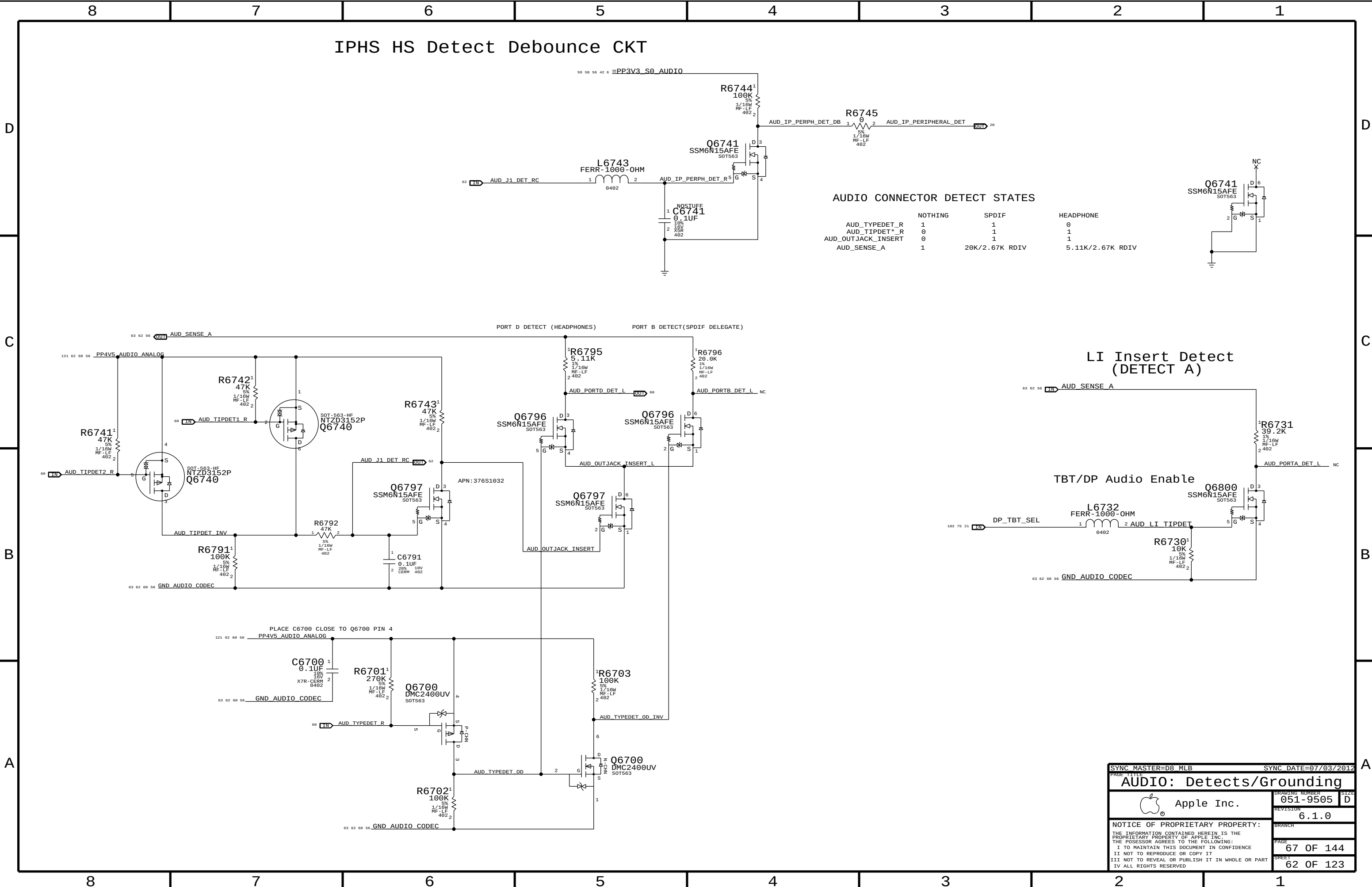
AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

LI Insert Detect (DETECT A)

TBT/DP Audio Enable

NOTICE OF PROPRIETARY PROPERTY:
 THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
 I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
 II NOT TO REPRODUCE OR COPY IT
 III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
 IV ALL RIGHTS RESERVED

[illegible]

IPHS HS Detect Debounce CKT

AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPERDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

LI Insert Detect (DETECT A)

TBT/DP Audio Enable

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

SYNC MASTER=D8 MLB SYNC DATE=07/03/2012

PAGE TITLE: **AUDIO: Detects/Grounding**

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH:

PAGE: 67 OF 144

SHEET: 62 OF 123

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUD10D1FF	*	y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKR0UTD1FF	*	y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

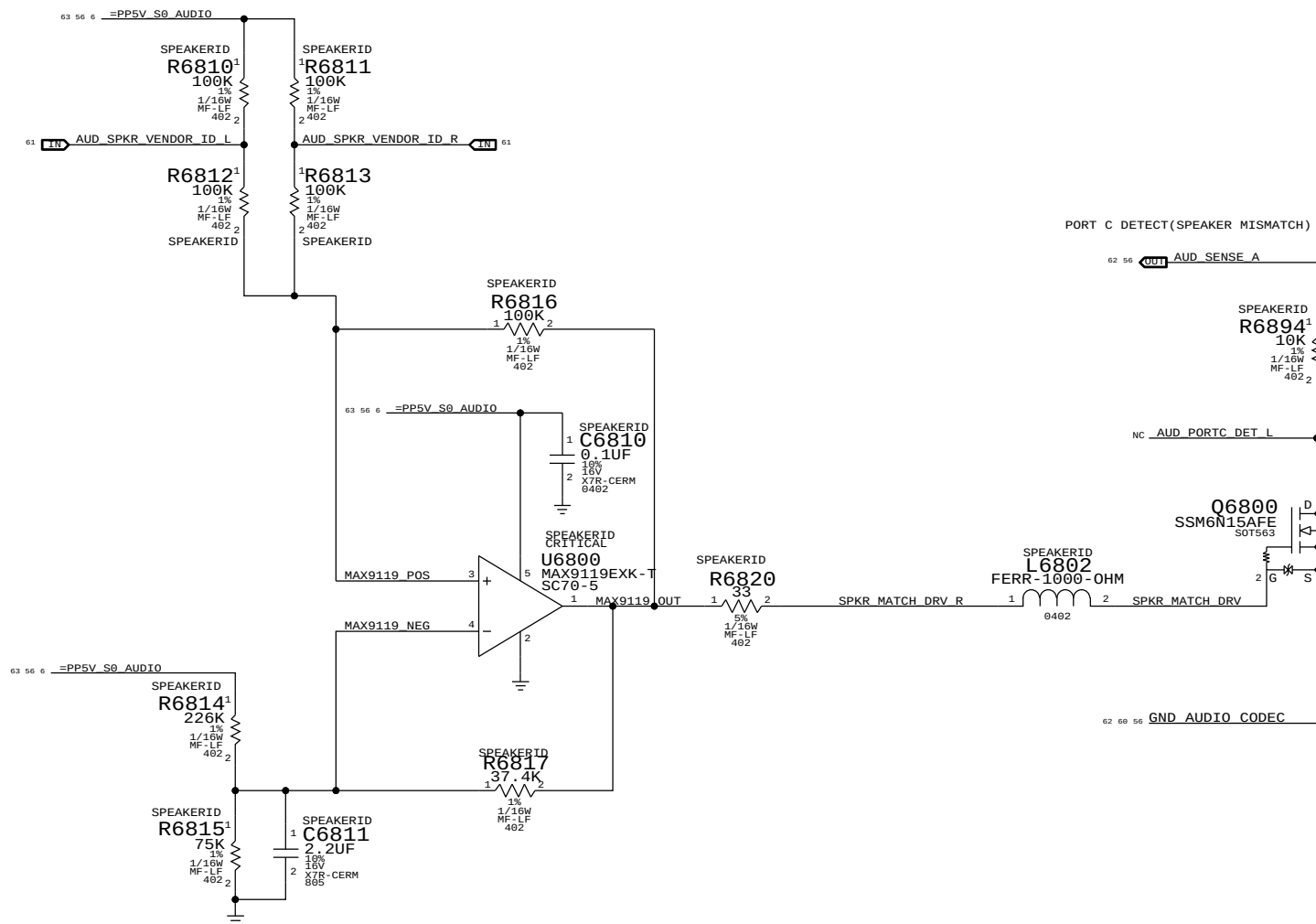
CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPDIF IN	0x07 (7)	0x0F (15)	N/A	0x09 (DET A)
INTERNAL MIC ARRAY	0x08 (8)	0x0E (14, LEFT) & RIGHT	N/A	N/A
EXTERNAL MIC	0x06 (6)	0x0D (13, V22, B, LEFT)	PANTHER POINT GPIO 16	PANTHER POINT GPIO 3 (PCVR INT), PANTHER POINT DET

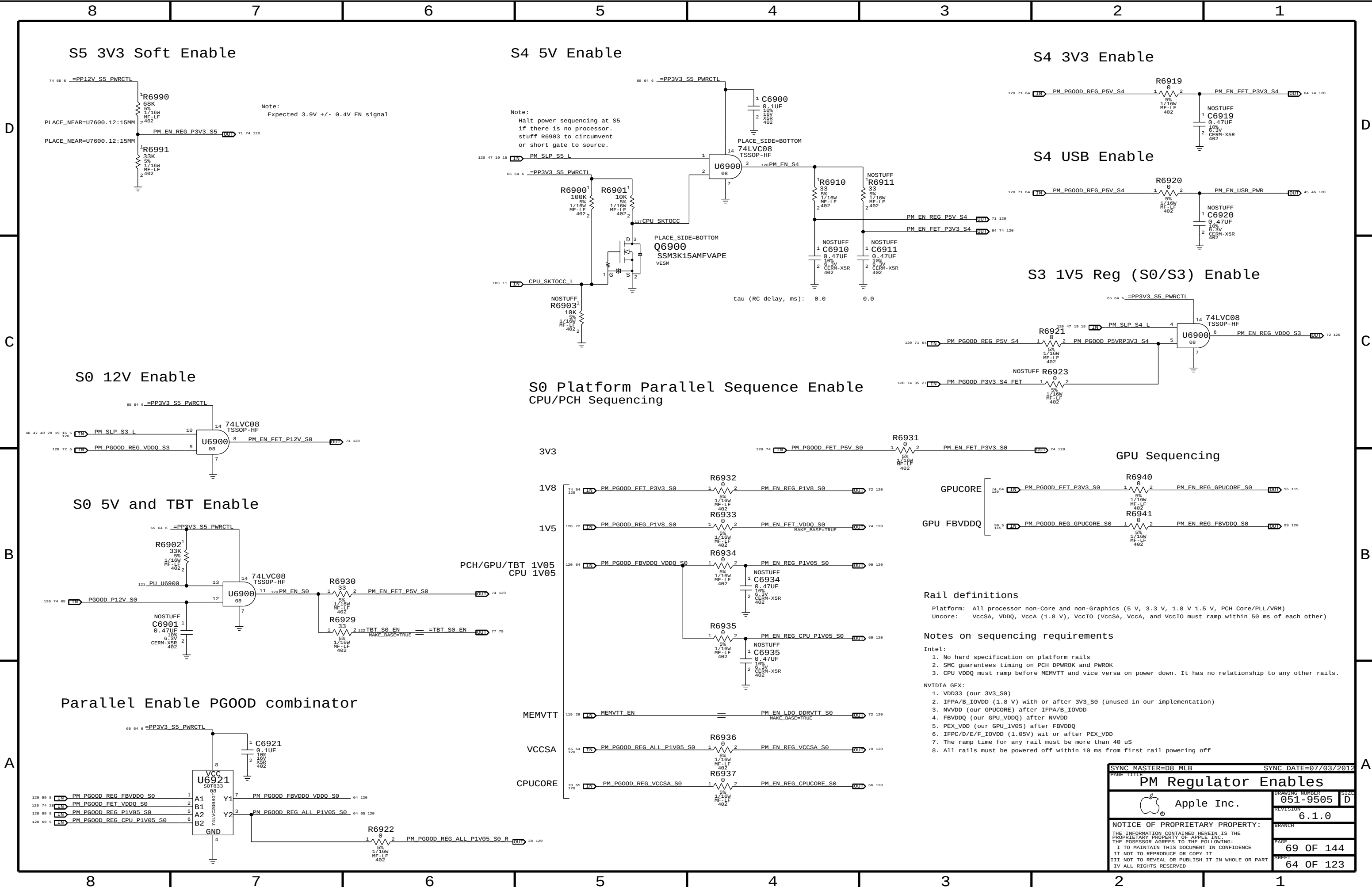
OTHER DETECT

FUNCTION	CONVERTER	PIN	COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A		N/A	0X0C (DET C)

CIRCUIT THEORY OF OPERATION AVAILABLE IN <RDAR://PROBLEM/9776522>



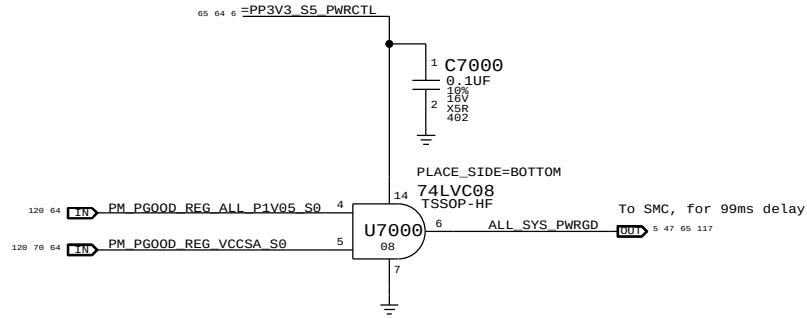
ELECTRICAL_CONSTRAINT_SET		PHYSICAL	SPACING	NET_TYPE	
1600	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L01_L_P	56 57 58
1601	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L01_L_N	56 57 58
1602	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L01_L_C_P	57
1603	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L01_L_C_N	57
1604	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L01_R_P	56 57 59
1605	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L01_R_N	56 57 59
1606	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L01_R_C_P	57
1607	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L01_R_C_N	57
1608	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L02_L_P	56 58
1609	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L02_L_N	56 58
1610	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L02_R_P	56 59
1611	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_L02_R_N	56 59
1612	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_RAMP_LINC_P	59
1613	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_RAMP_LINC_N	59
1614	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_RAMP_RINC_P	59
1615	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_RAMP_RINC_N	59
1616	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_RAMP_LIN_P	59
1617	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_RAMP_LIN_N	59
1618	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_RAMP_RIN_P	59
1619	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_RAMP_RIN_N	59
1620	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_LINC_P	58
1621	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_LINC_N	58
1622	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RINC_P	58
1623	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RINC_N	58
1624	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_LIN_P	58
1625	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_LIN_N	58
1626	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_P	58
1627	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1628	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1629	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1630	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1631	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1632	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1633	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1634	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1635	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1636	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1637	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1638	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1639	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1640	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1641	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1642	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1643	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1644	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1645	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1646	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1647	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1648	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1649	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1650	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1651	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1652	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1653	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1654	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1655	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1656	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1657	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1658	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1659	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1660	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1661	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1662	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1663	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1664	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1665	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1666	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1667	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1668	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1669	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1670	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1671	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1672	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1673	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1674	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1675	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1676	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1677	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1678	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1679	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1680	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1681	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1682	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1683	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1684	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1685	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1686	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1687	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1688	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1689	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1690	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1691	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1692	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1693	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1694	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1695	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1696	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1697	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1698	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1699	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1700	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1701	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1702	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1703	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1704	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1705	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1706	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1707	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1708	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1709	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1710	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1711	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1712	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1713	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1714	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1715	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1716	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1717	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1718	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1719	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1720	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1721	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1722	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1723	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1724	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1725	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1726	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1727	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1728	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1729	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1730	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1731	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1732	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1733	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1734	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1735	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1736	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1737	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1738	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1739	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1740	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1741	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1742	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1743	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1744	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1745	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1746	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1747	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1748	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1749	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1750	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1751	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1752	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1753	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1754	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1755	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1756	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1757	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1758	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1759	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1760	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1761	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1762	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1763	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1764	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1765	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1766	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1767	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1768	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1769	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1770	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1771	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1772	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1773	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1774	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1775	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1776	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1777	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1778	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1779	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1780	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1781	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1782	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1783	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1784	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1785	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1786	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1787	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1788	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1789	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1790	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1791	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1792	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1793	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1794	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1795	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1796	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1797	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1798	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1799	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1800	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1801	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1802	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1803	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58
1804	AUDIO_DIFEP428	AUDIOTIEF	AUD10	AUD_LAMP_RIN_N	58



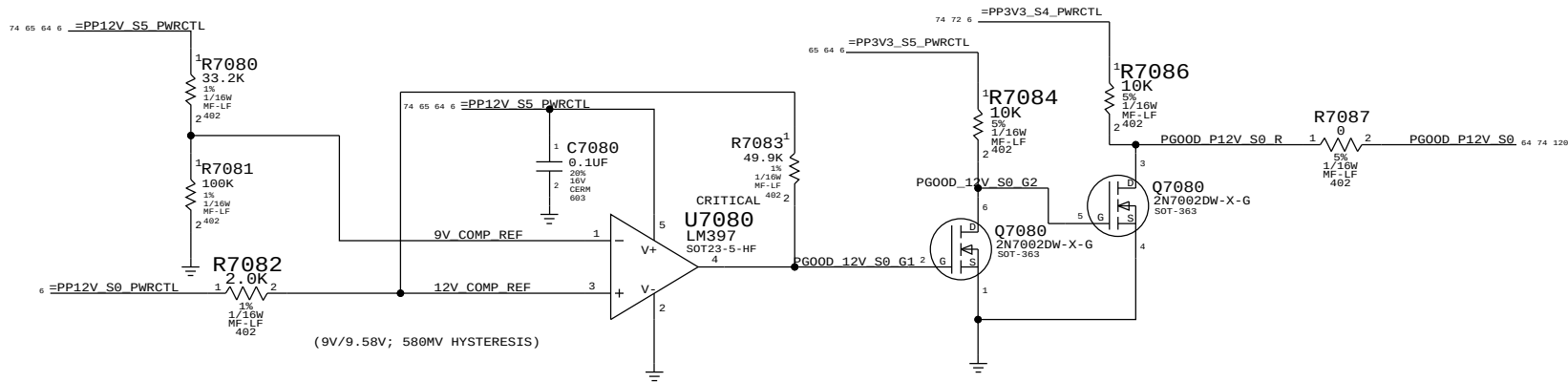
SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE		PM Regulator Enables	
 Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	69 OF 144
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	64 OF 123
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

Platform Power Good derive SMC ALL_SYS_PWRGD

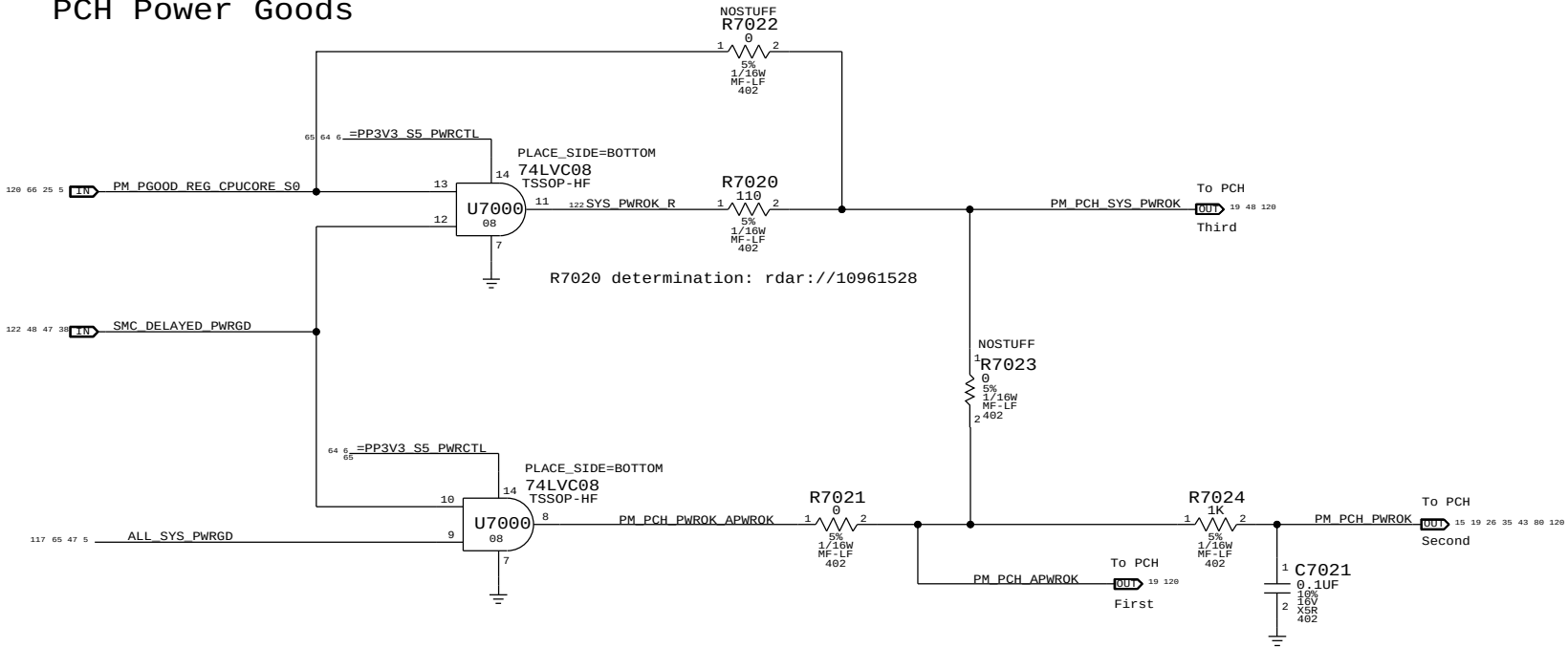
The end of the power sequence for S0 rails except CPU CORE.



PGOOD comparators for PP12V_S0



PCH Power Goods



radar://11043352 Need AND Gate to deassert PM_PCH_PWROK to PCH when unexpected power loss happens

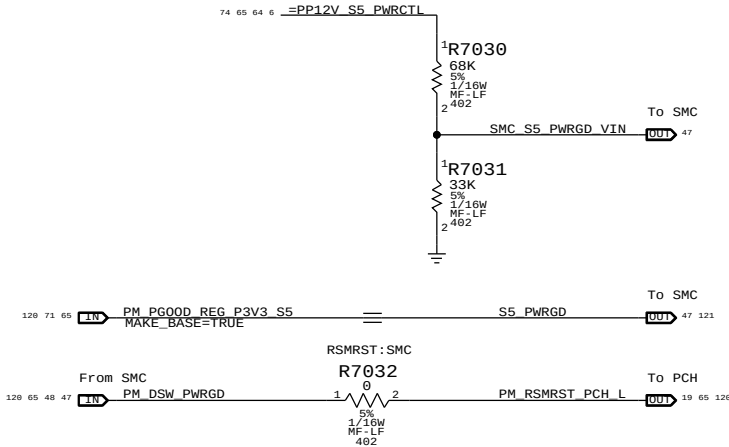
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

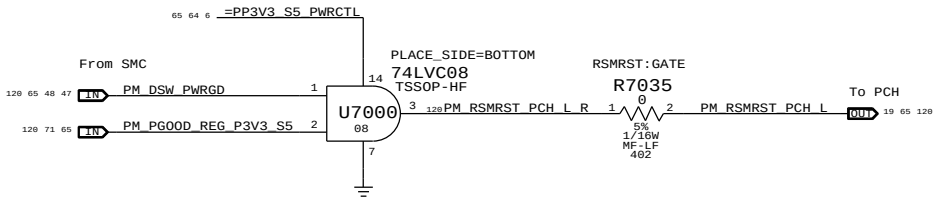
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

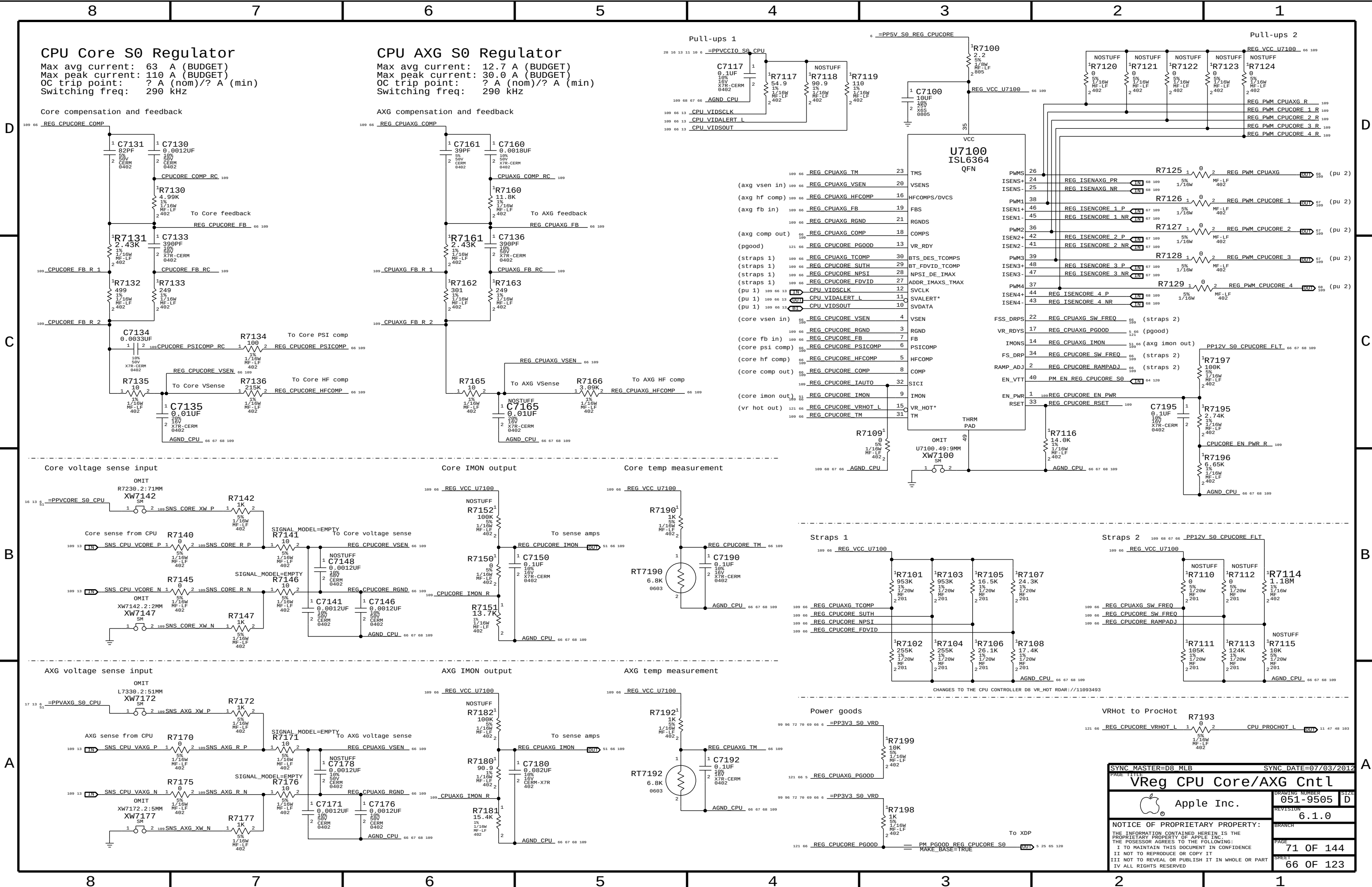
Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.
SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



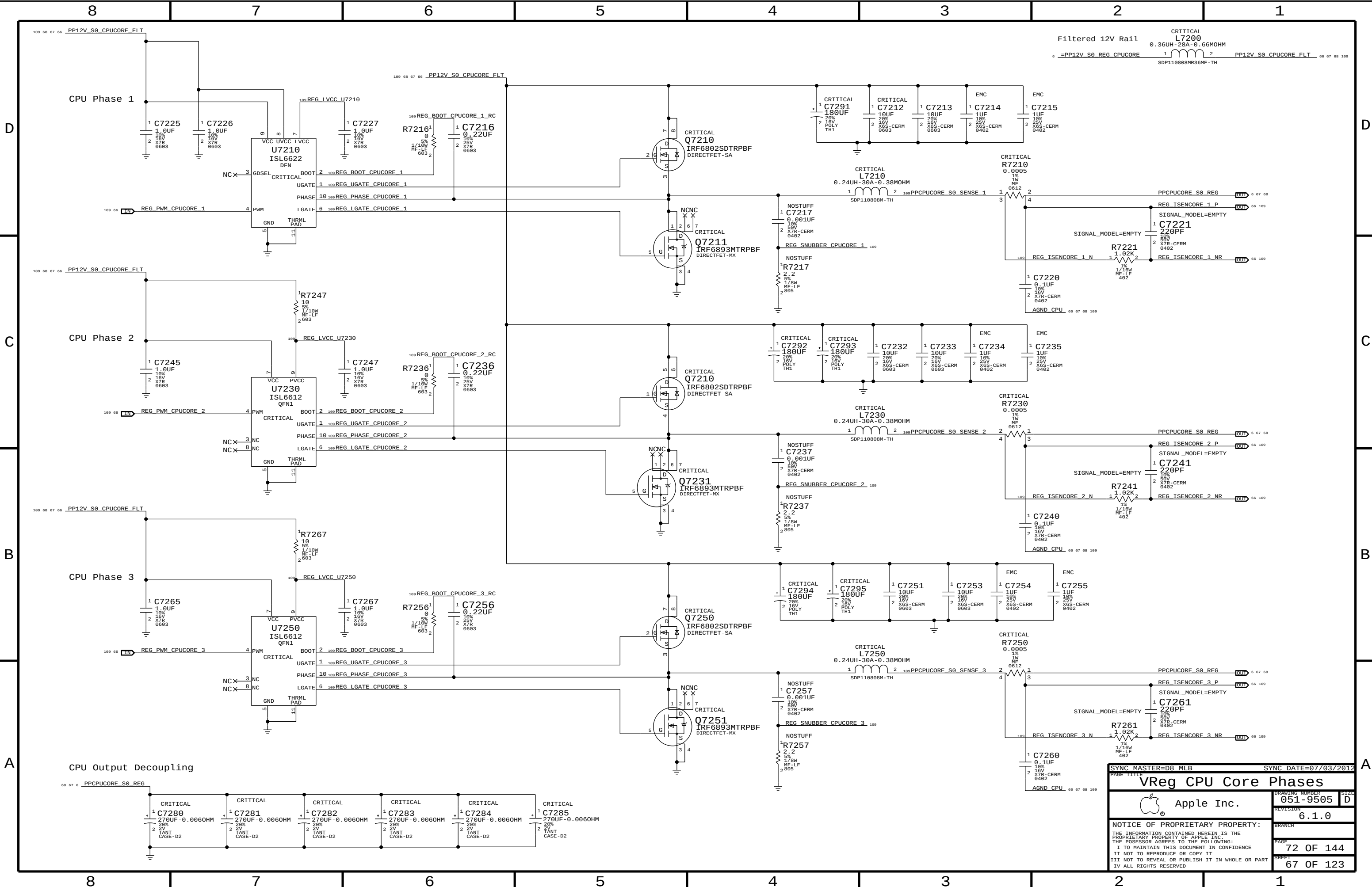
SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE		PM Power Good	
Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	70 OF 144
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	65 OF 123
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



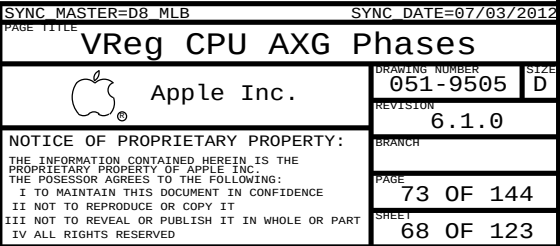
CPU Core S0 Regulator
Max avg current: 63 A (BUDGET)
Max peak current: 110 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz

CPU AXG S0 Regulator
Max avg current: 12.7 A (BUDGET)
Max peak current: 30.0 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE		VReg CPU Core/AXG Cntl	
Apple Inc.		DRAWING NUMBER	051-9505
NOTICE OF PROPRIETARY PROPERTY:		REVISION	6.1.0
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	71 OF 144
II NOT TO REPRODUCE OR COPY IT		SHEET	66 OF 123
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

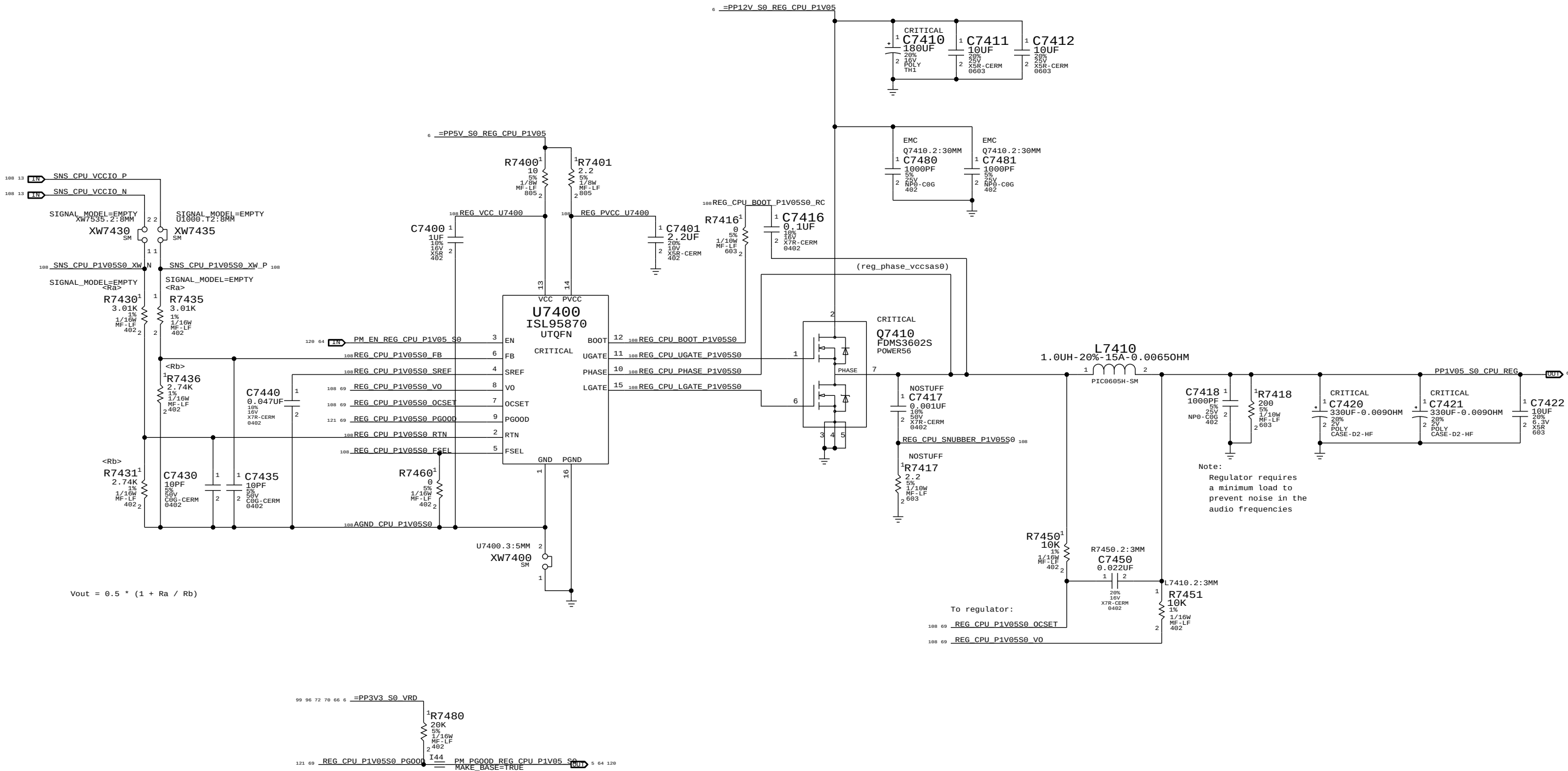


SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
VReg CPU Core Phases			
 Apple Inc.		DRAWING NUMBER	051-9505
		SIZE	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	6.1.0
		BRANCH	
		PAGE	72 OF 144
		SHEET	67 OF 123



CPU VccIO (1.05V) S0 Regulator

Max avg current: 8.10 A (BUDGET)
Max peak current: 8.50 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
VReg CPU 1.05V S0			
 Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	74 OF 144
		SHEET	69 OF 123

D

C



A

-

D

C



A

D

D

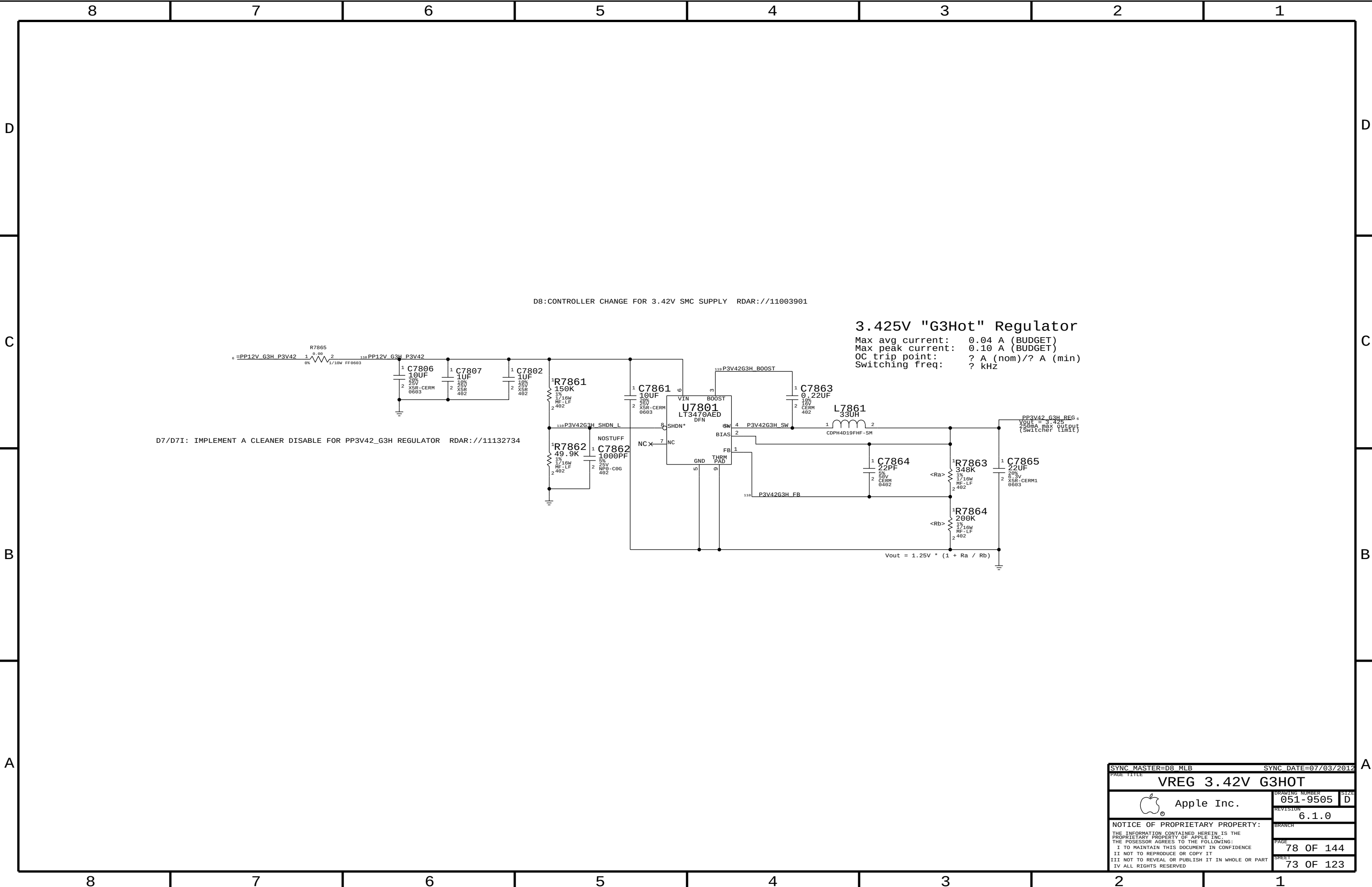
C

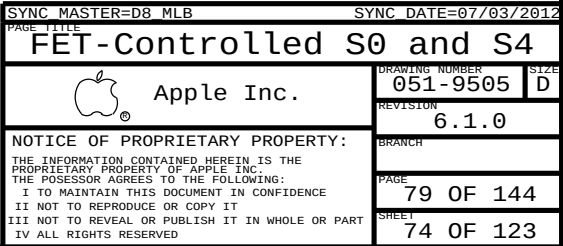


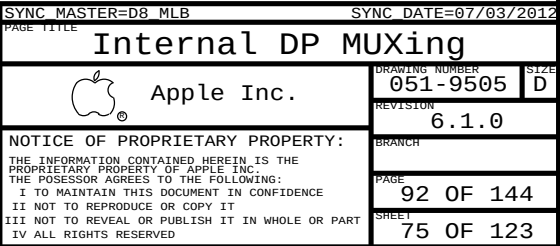
A

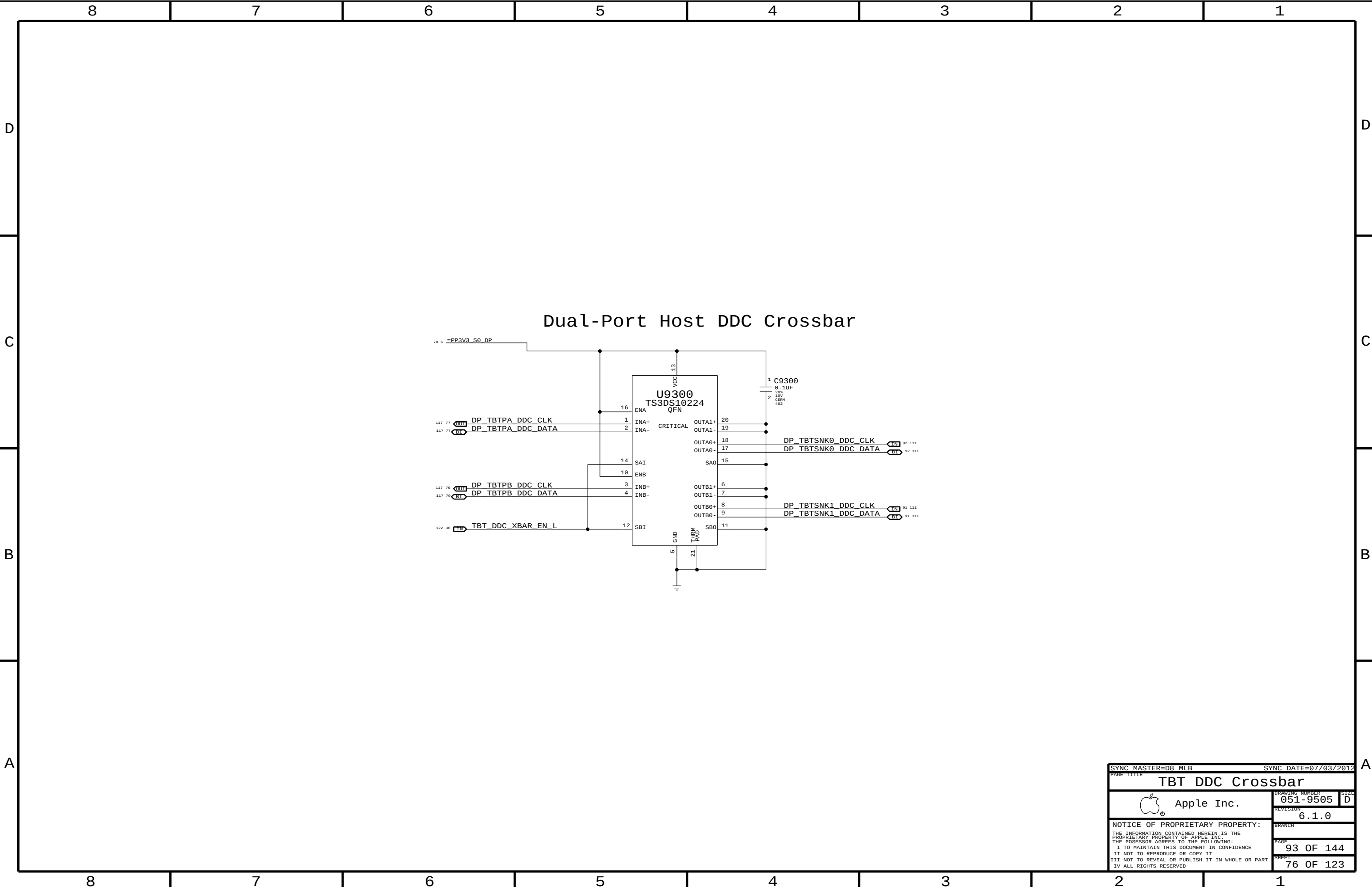
D

A





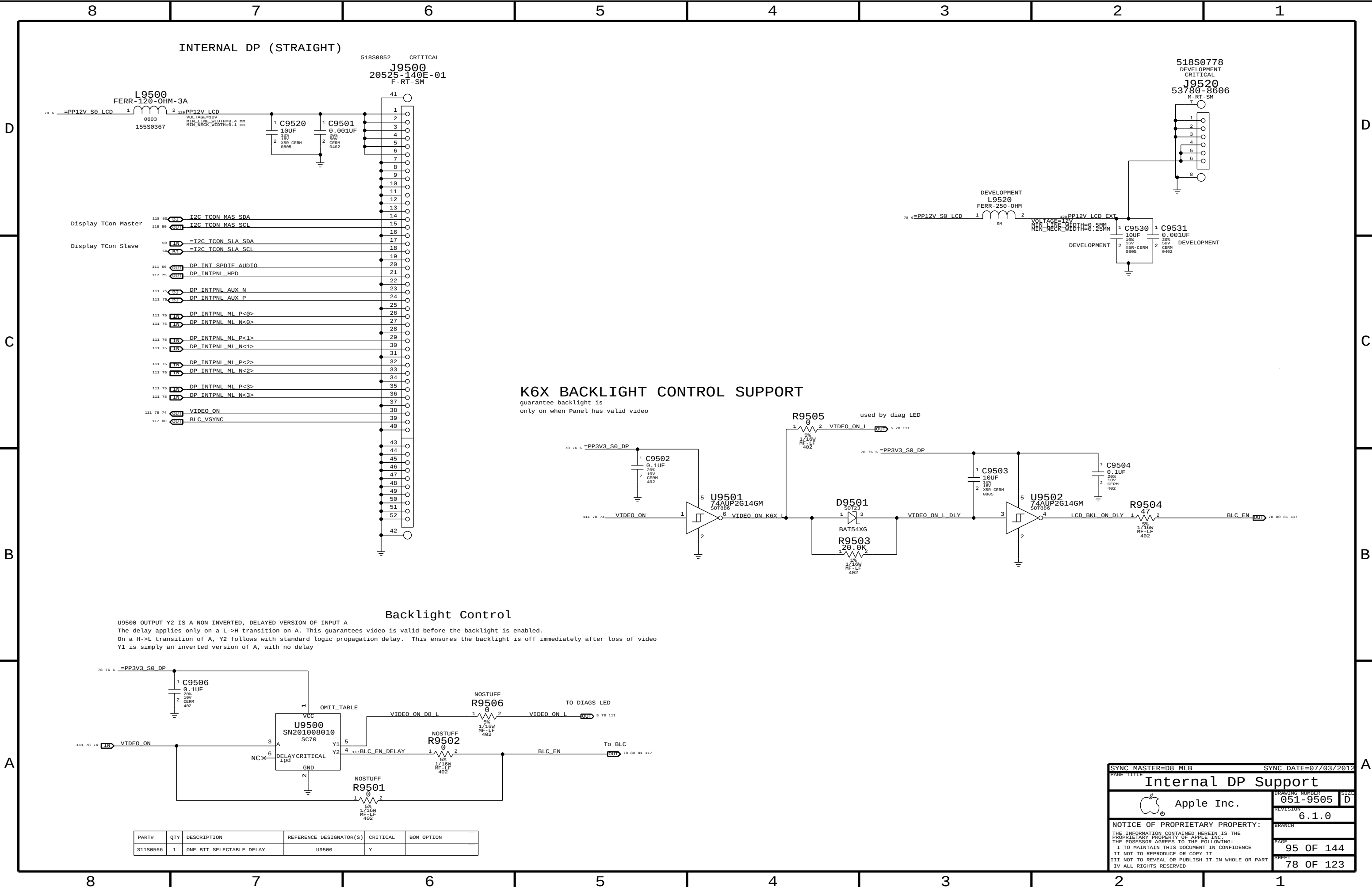




8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

CB

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



INTERNAL DP (STRAIGHT)

518S0852 CRITICAL
J9500
20525-140E-01
F-RT-SM

518S0778 DEVELOPMENT
CRITICAL
J9520
53780-8606
M-RT-SM

K6X BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video

Backlight Control

U9500 OUTPUT Y2 IS A NON-INVERTED, DELAYED VERSION OF INPUT A
The delay applies only on a L->H transition on A. This guarantees video is valid before the backlight is enabled.
On a H->L transition of A, Y2 follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video
Y1 is simply an inverted version of A, with no delay

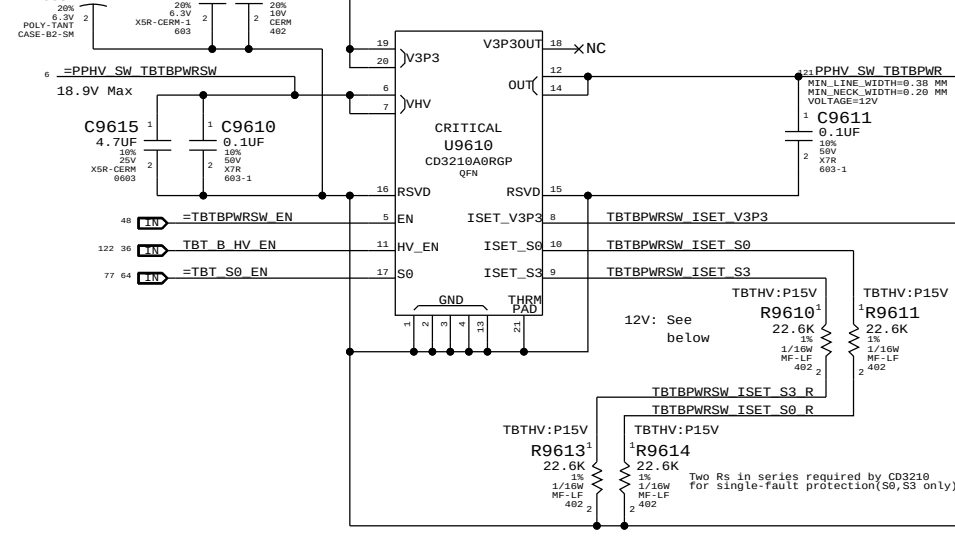
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
311S0566	1	ONE BIT SELECTABLE DELAY	U9500	Y	

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
Internal DP Support			
 Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	95 OF 144
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	78 OF 123
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.

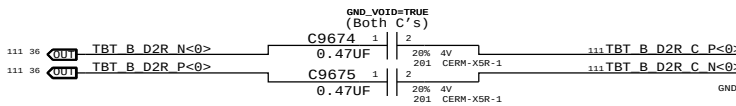
	Nominal	Min	Max
IV3P3	1100mA	1030mA	1200mA
IHV50	890mA	830mA	930mA (assumes 15V, 12W minimum)
IHV53	890mA	830mA	930mA (assumes 3S, 9-12.6V, 7.5-11.7V)



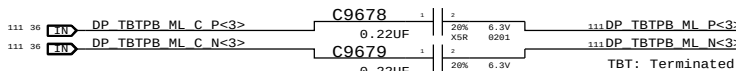
For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9610,R9613		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9611,R9614		TBTHV:P12V

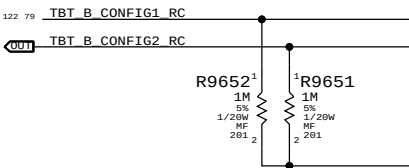
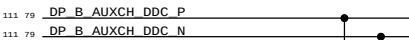
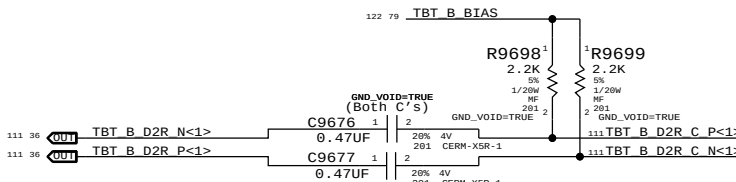
	Nominal	Min	Max
IHV50/S3	1120mA	1090mA	1170mA (12W minimum)



NOTE: Polarity Swapped for Layout!



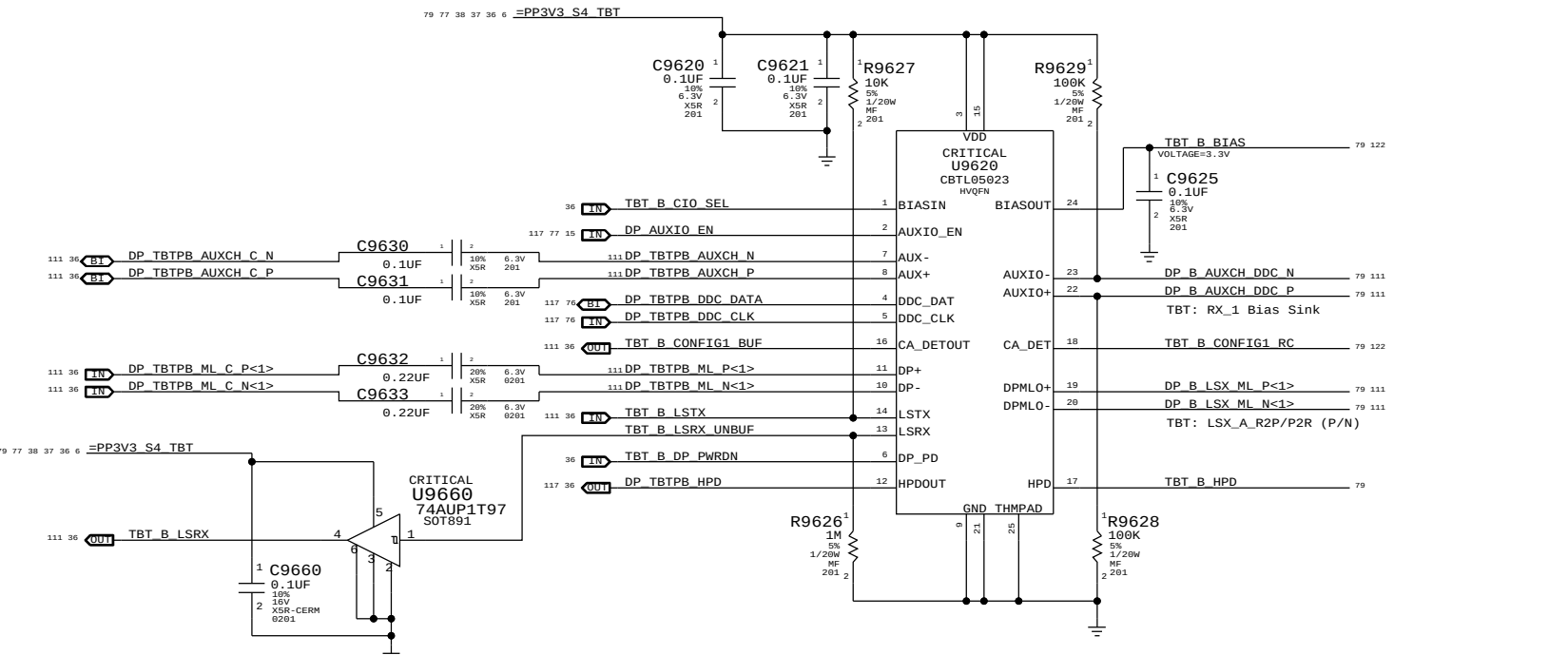
NOTE: Polarity Swapped for Layout!



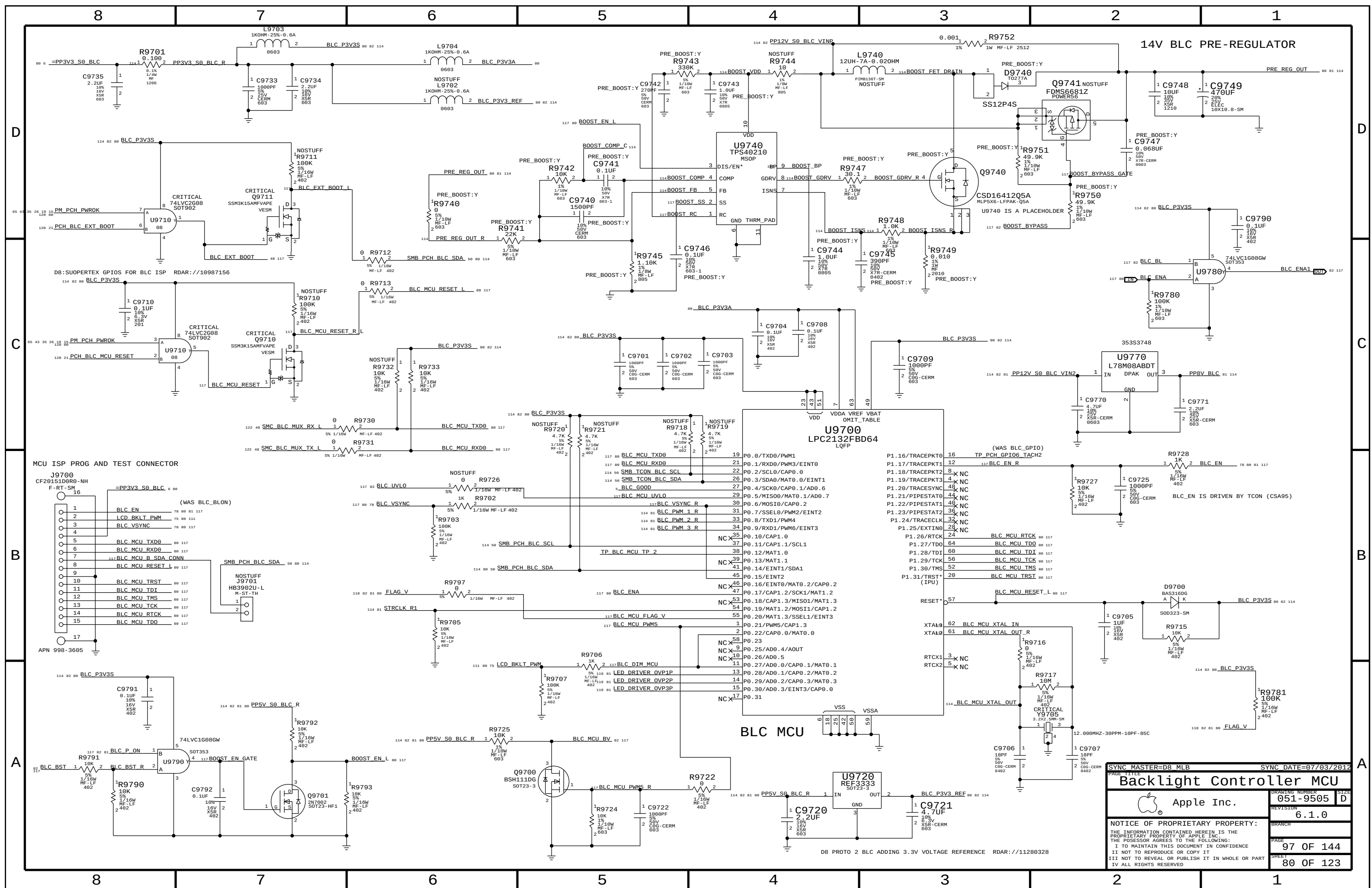
DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

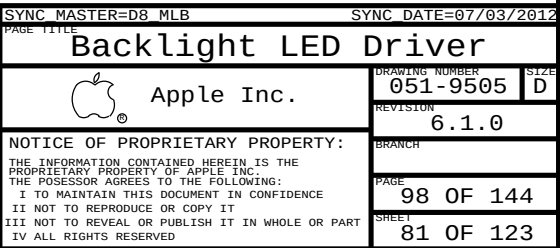
Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

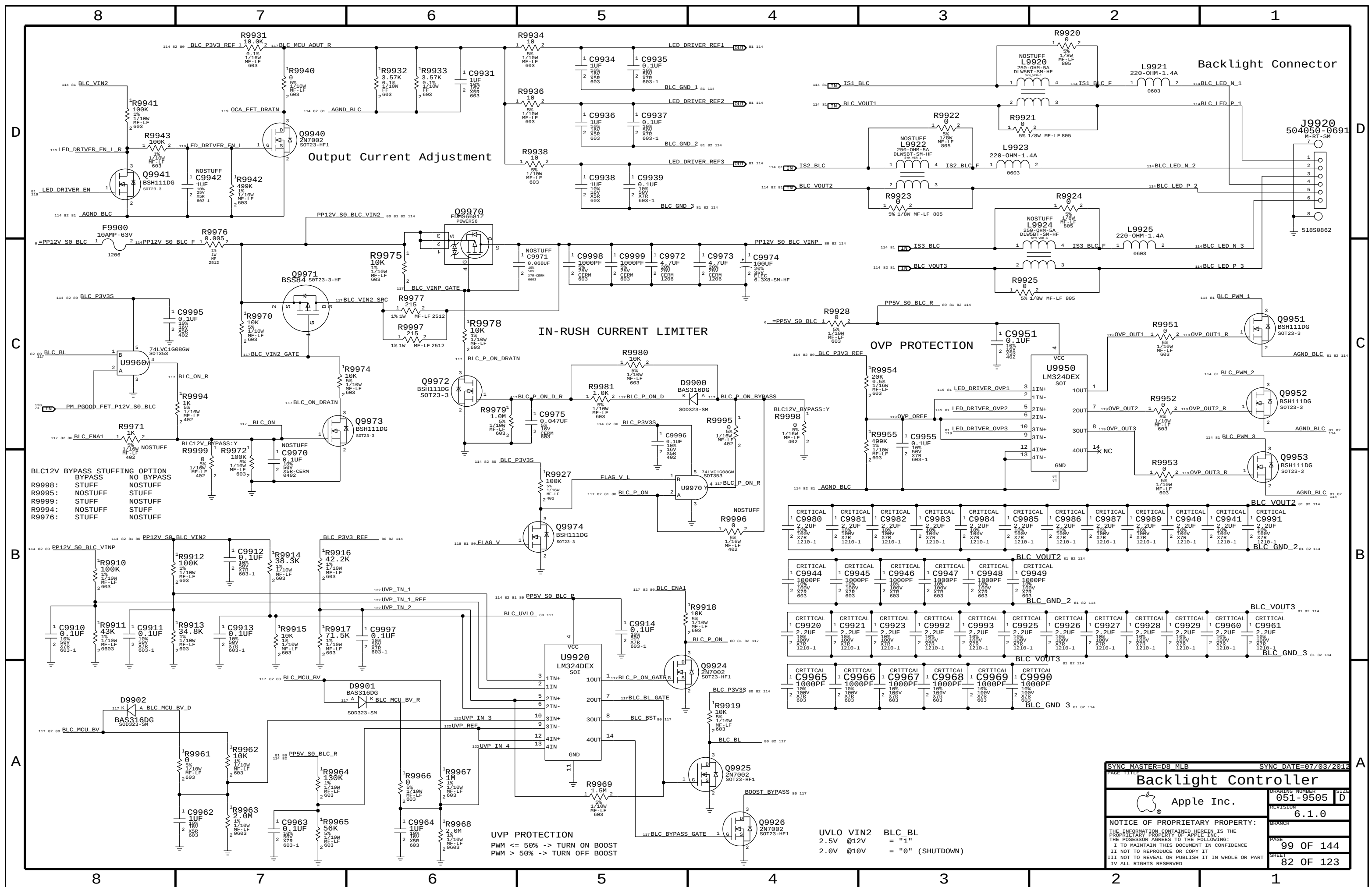
Thunderbolt Connector B

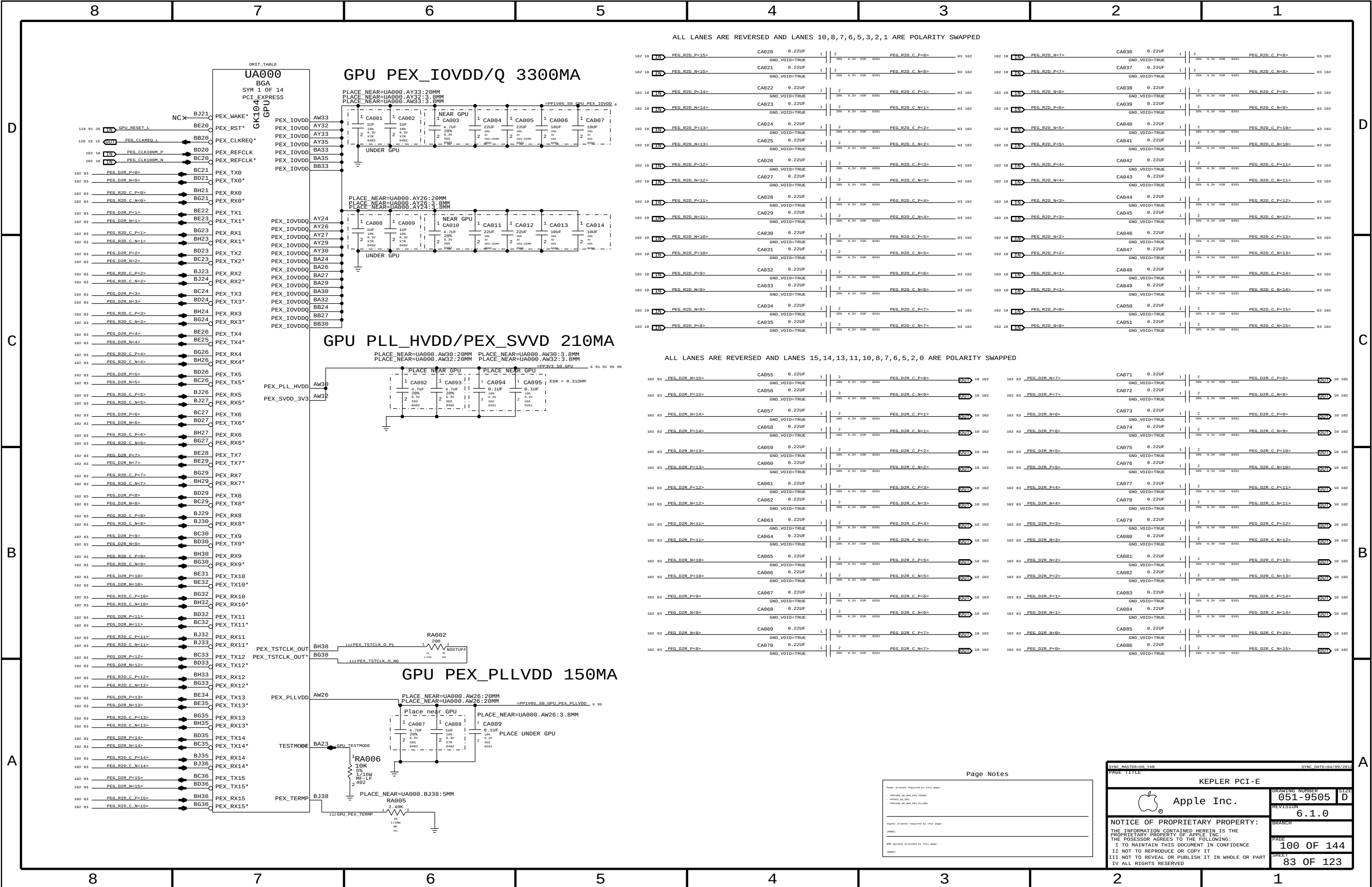


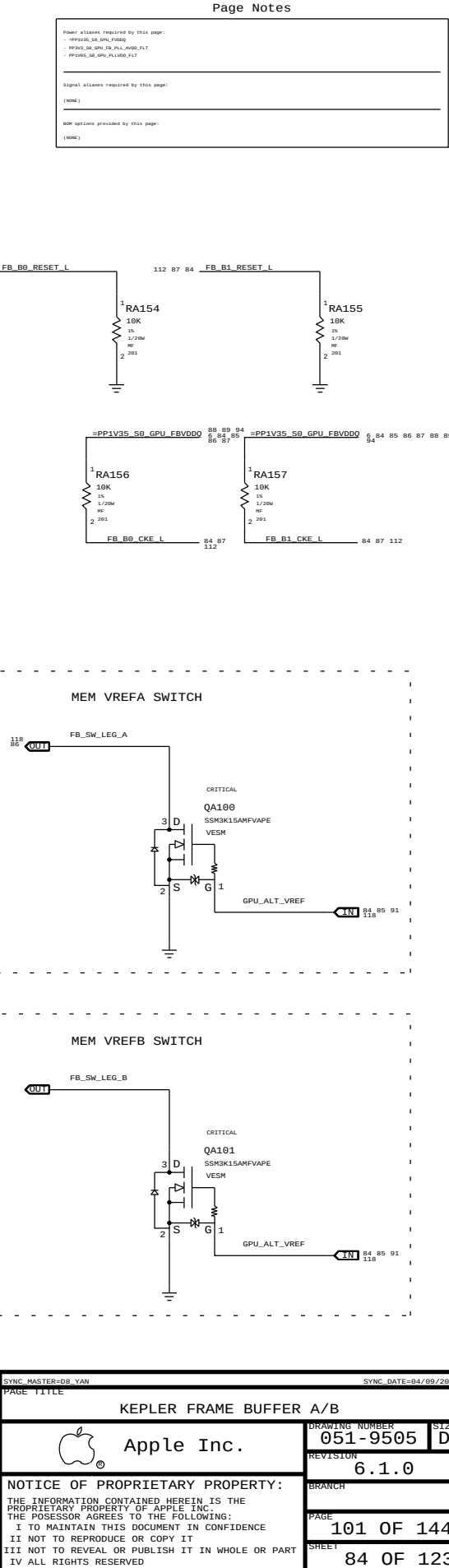
SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE		Thunderbolt Connector B	
Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	96 OF 144
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	79 OF 123
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

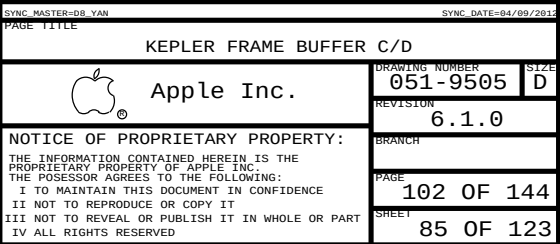


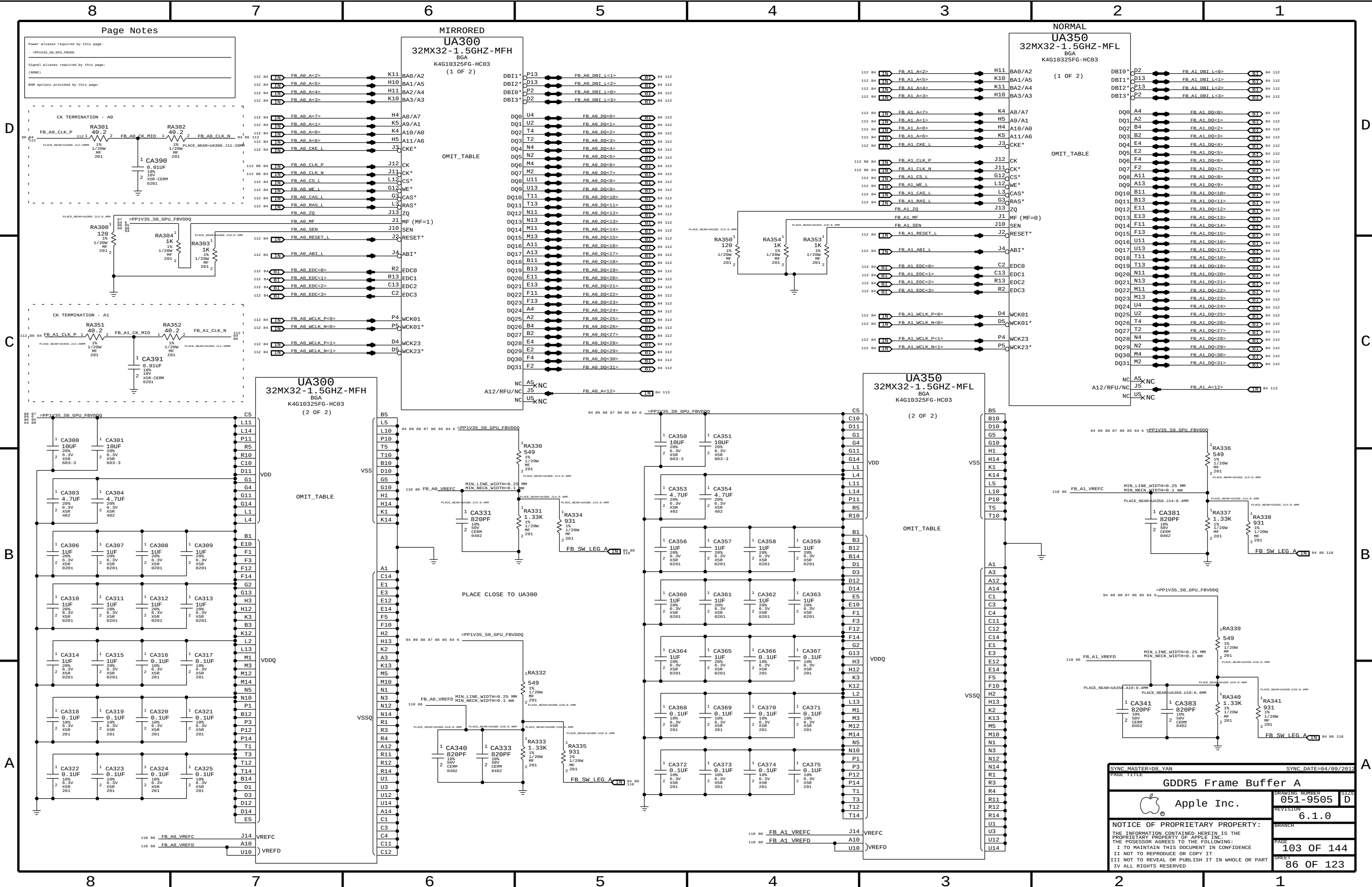








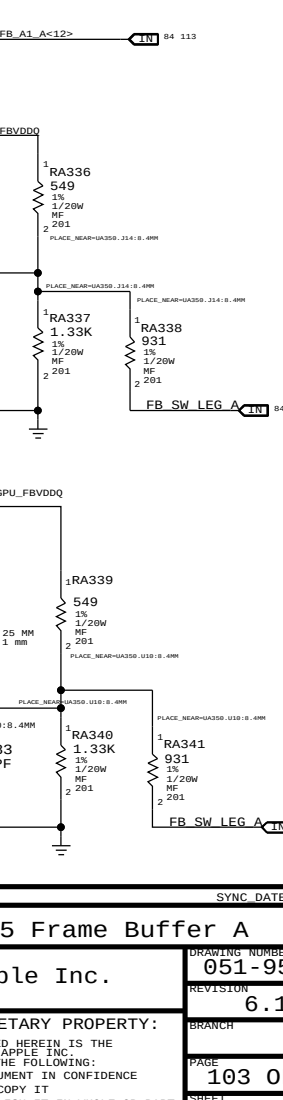
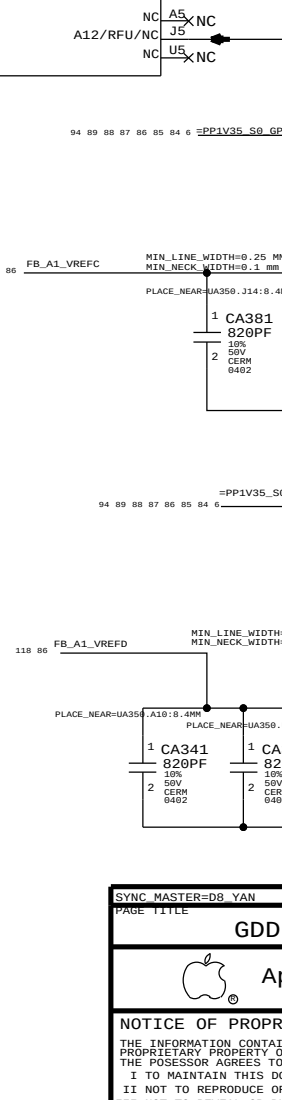
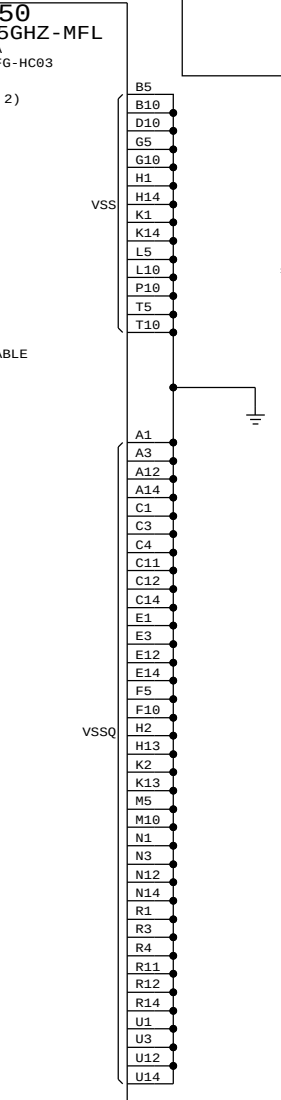
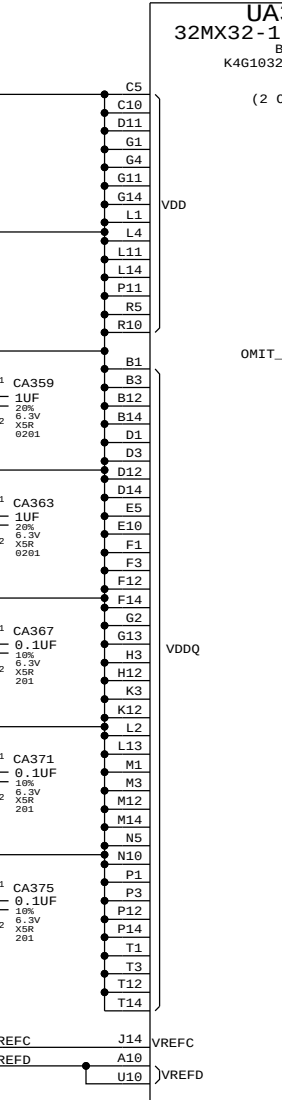
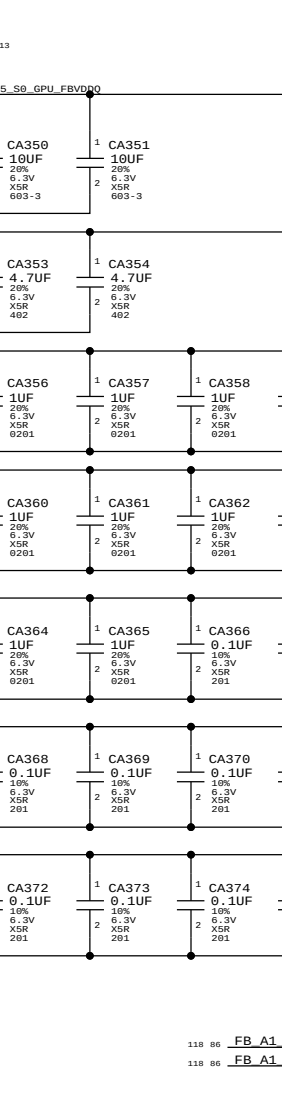
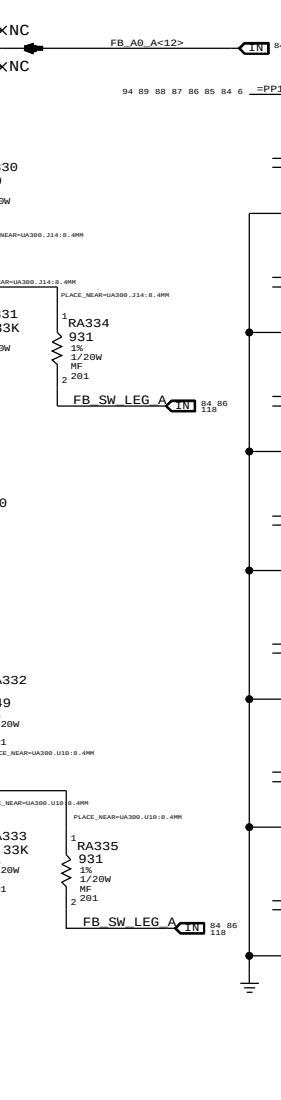
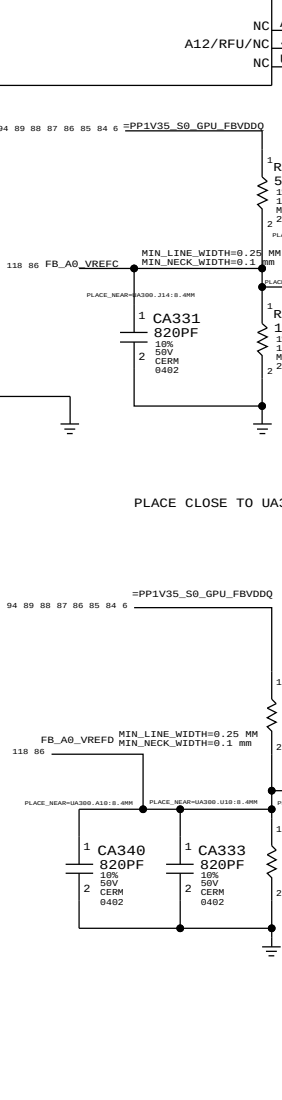
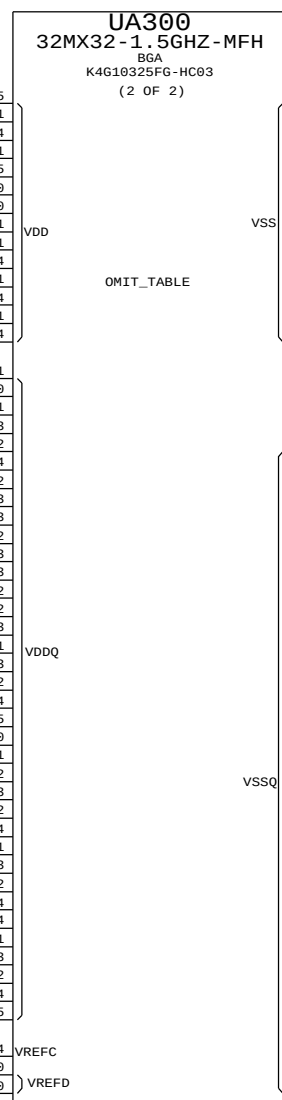
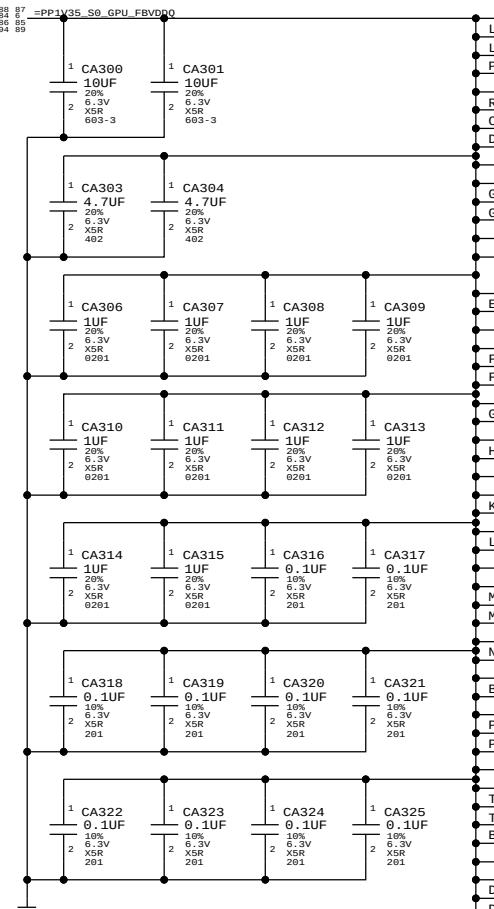
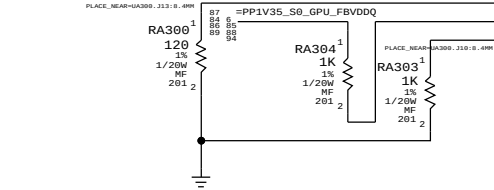
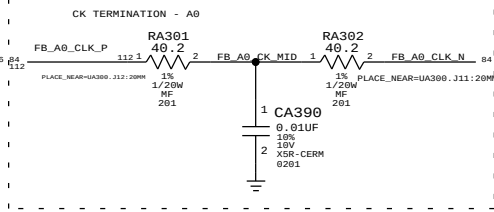




Power aliases required by this page:
- =PP1V35_S0_GPU_FBVDDQ

Signal aliases required by this page:
(NONE)

BOM options provided by this page:



Apple Inc. logo and text.

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

SYNC MASTER=D8_YAN SYNC DATE=04/09/2012

PAGE TITLE

GDDR5 Frame Buffer A

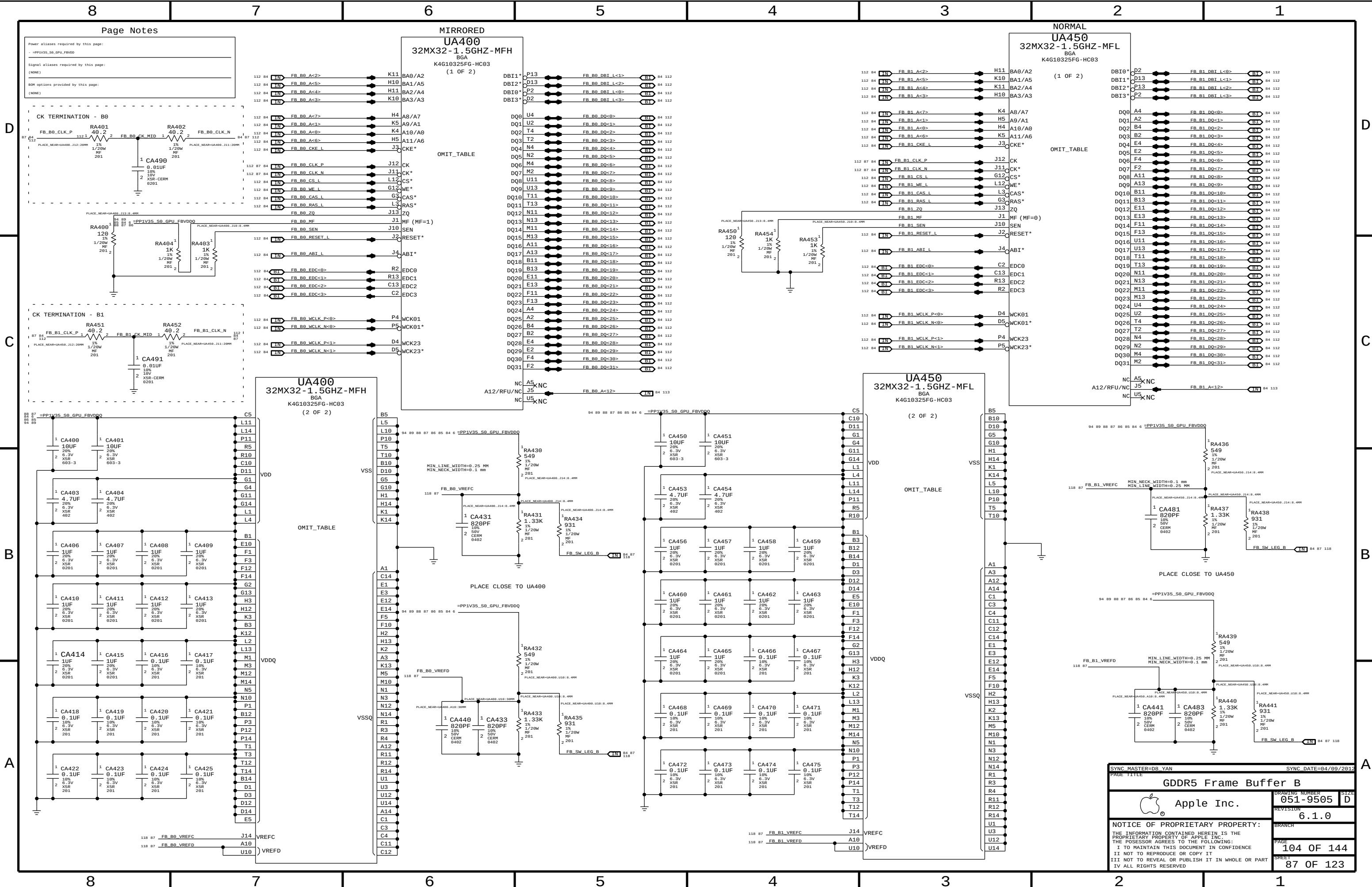
DRAWING NUMBER 051-9505

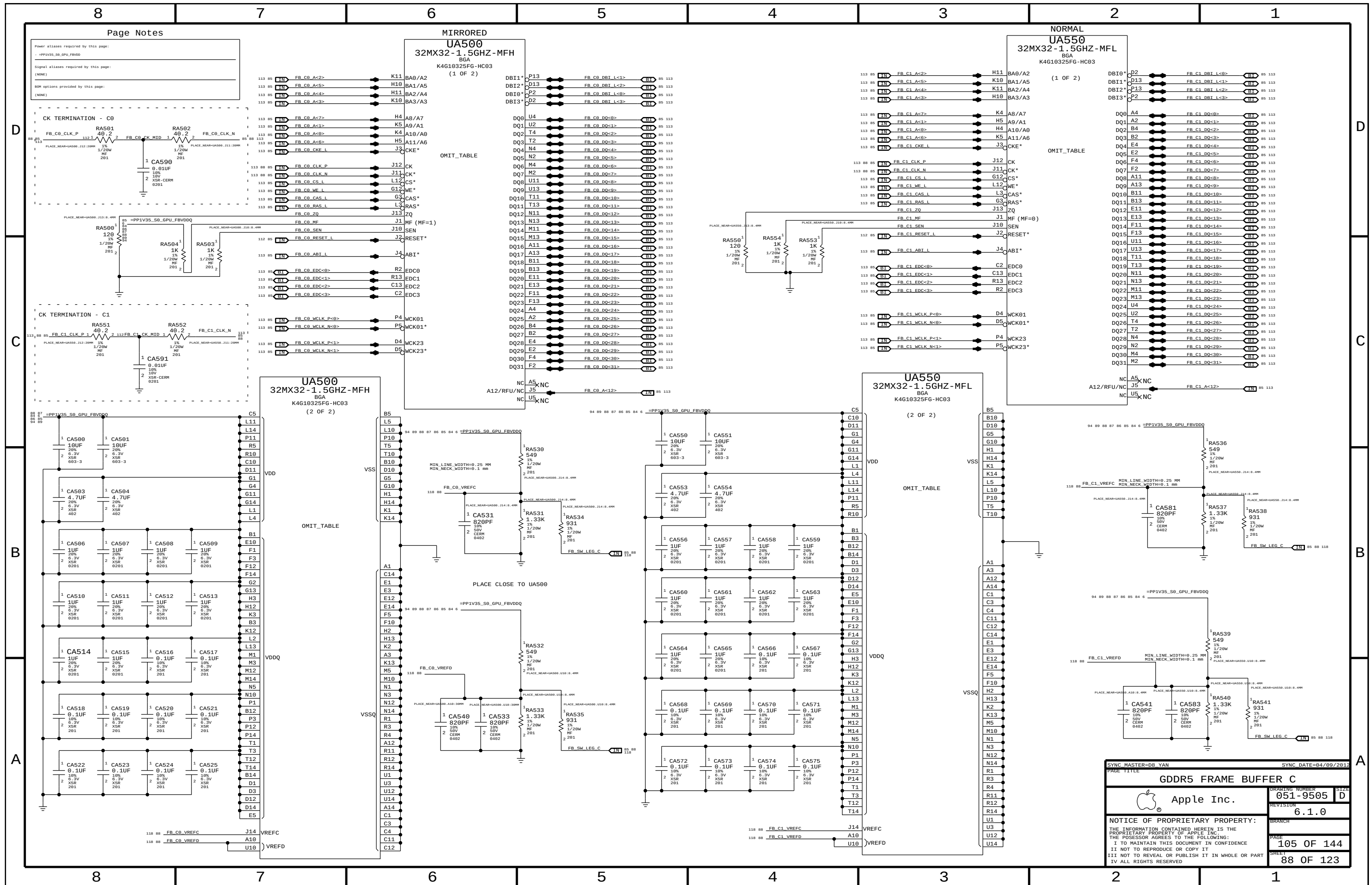
REVISION 6.1.0

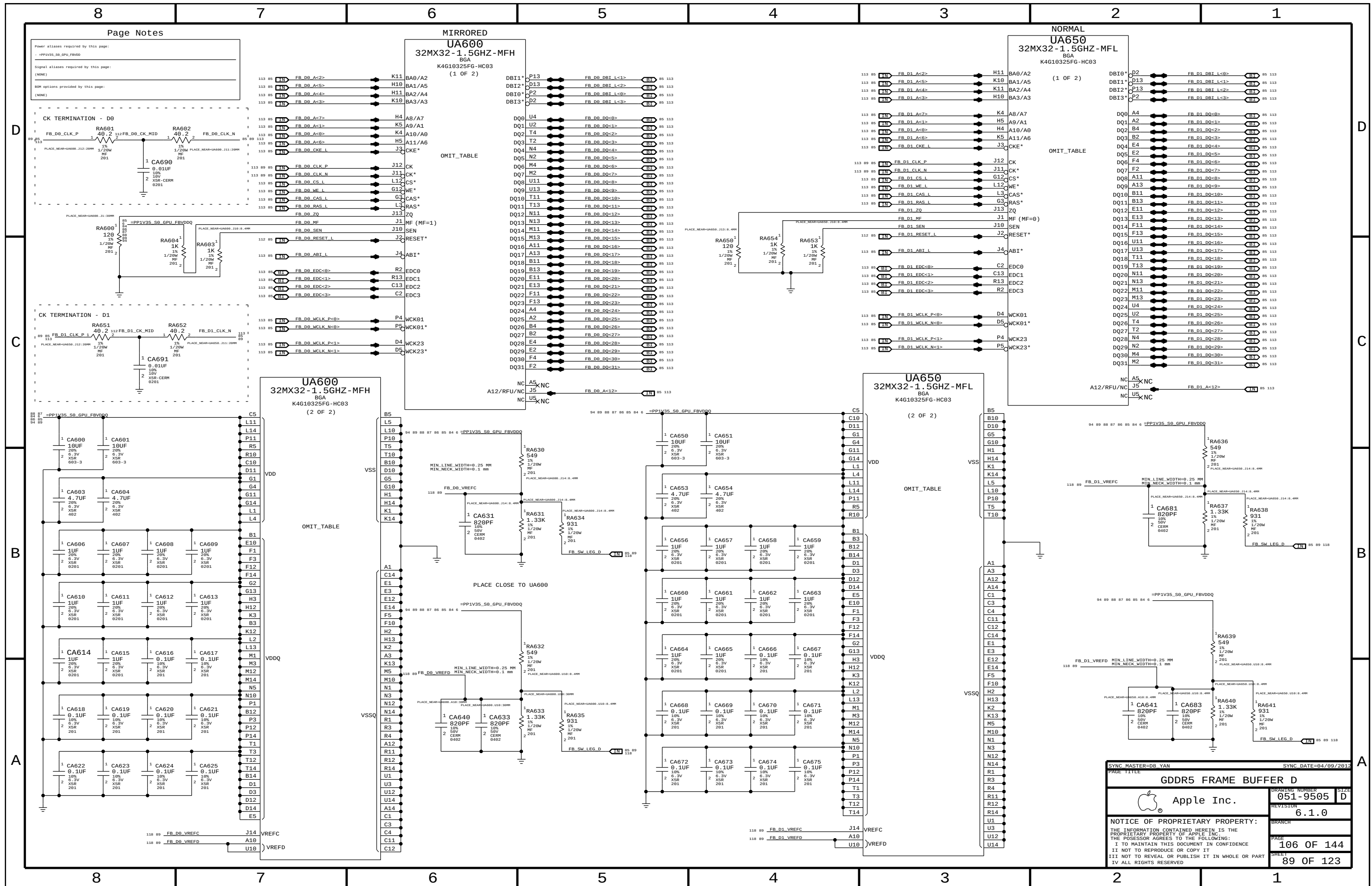
BRANCH

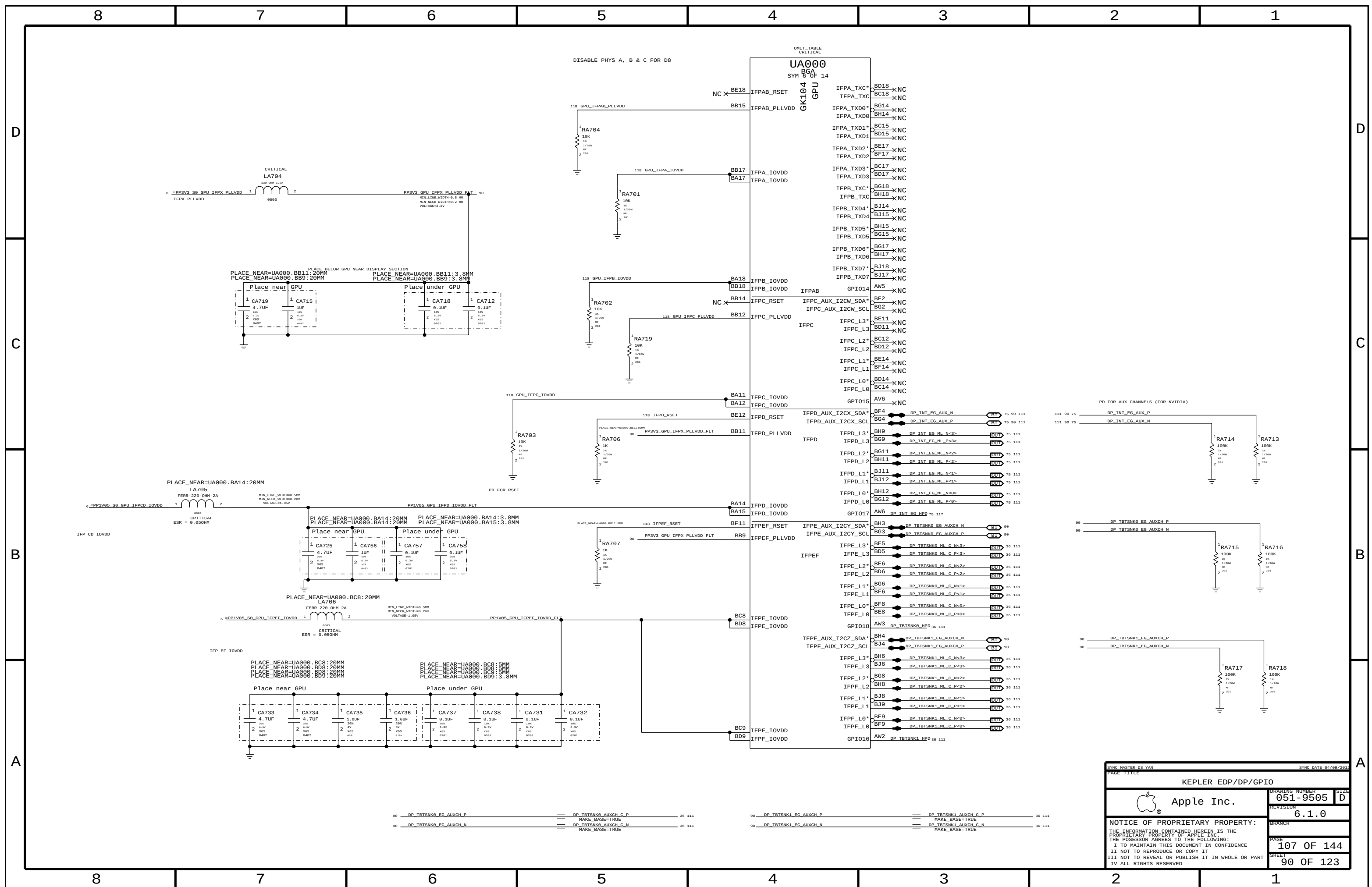
PAGE 103 OF 144

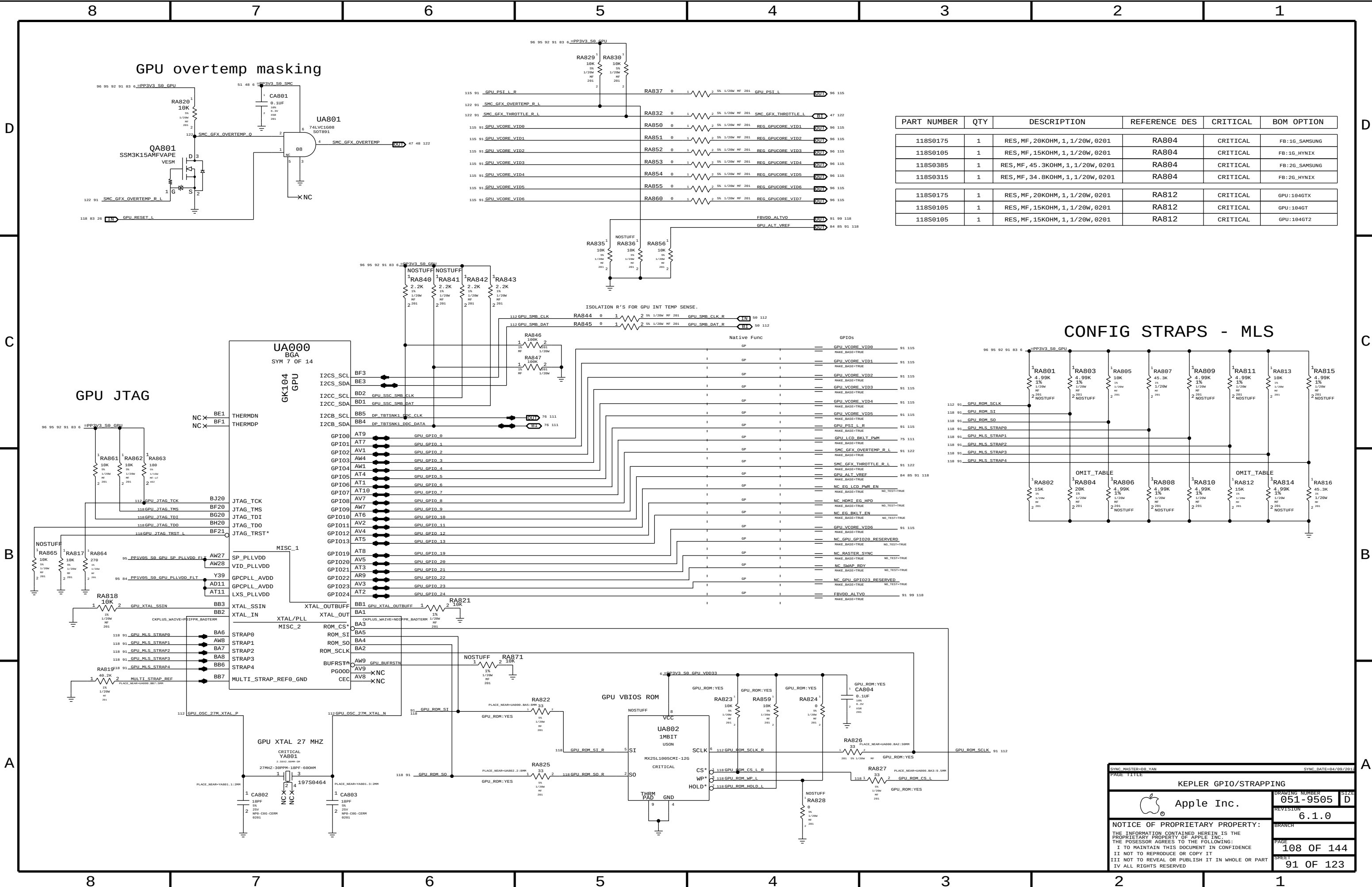
SHEET 86 OF 123



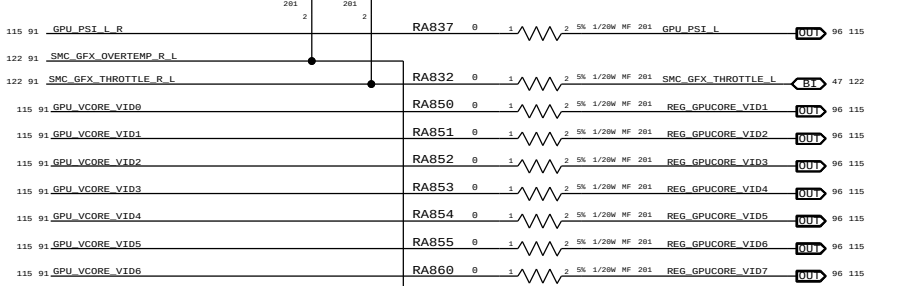
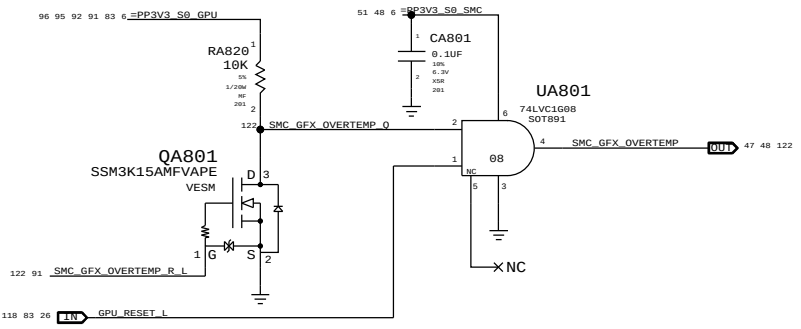






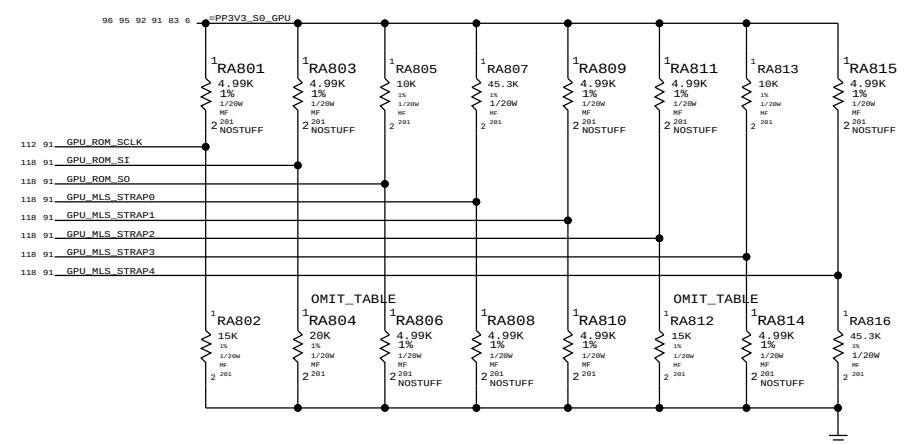


GPU overtemp masking

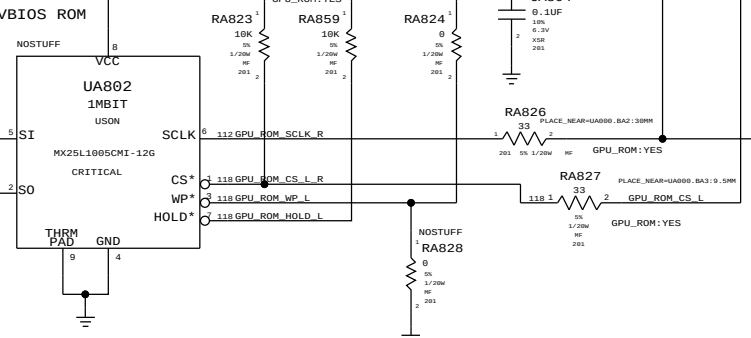
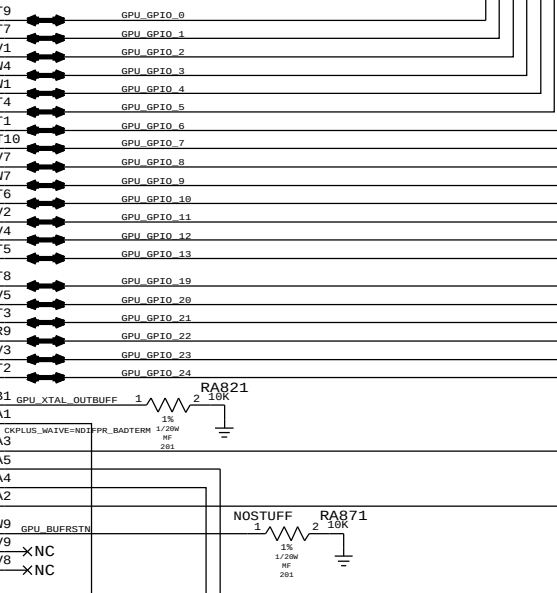
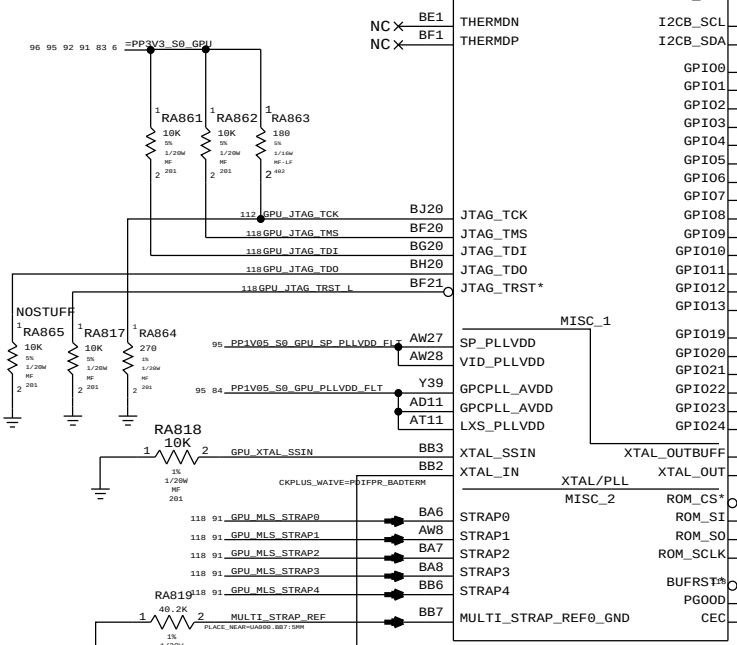


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0175	1	RES,MF,20KOHM,1,1/20W,0201	RA804	CRITICAL	FB:1G_SAMSUNG
118S0105	1	RES,MF,15KOHM,1,1/20W,0201	RA804	CRITICAL	FB:1G_HYNIX
118S0385	1	RES,MF,45.3KOHM,1,1/20W,0201	RA804	CRITICAL	FB:2G_SAMSUNG
118S0315	1	RES,MF,34.8KOHM,1,1/20W,0201	RA804	CRITICAL	FB:2G_HYNIX
118S0175	1	RES,MF,20KOHM,1,1/20W,0201	RA812	CRITICAL	GPU:104GT
118S0105	1	RES,MF,15KOHM,1,1/20W,0201	RA812	CRITICAL	GPU:104GT
118S0105	1	RES,MF,15KOHM,1,1/20W,0201	RA812	CRITICAL	GPU:104GT2

CONFIG STRAPS - MLS



GPU JTAG



SYNC MASTER=DR YAN

PAGE TITLE

KEPLER GPIO/STRAPPING

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER
051-9505

REVISION
6.1.0

PAGE
108 OF 144

SHEET
91 OF 123

SIZE
D

BRANCH

D

D

C

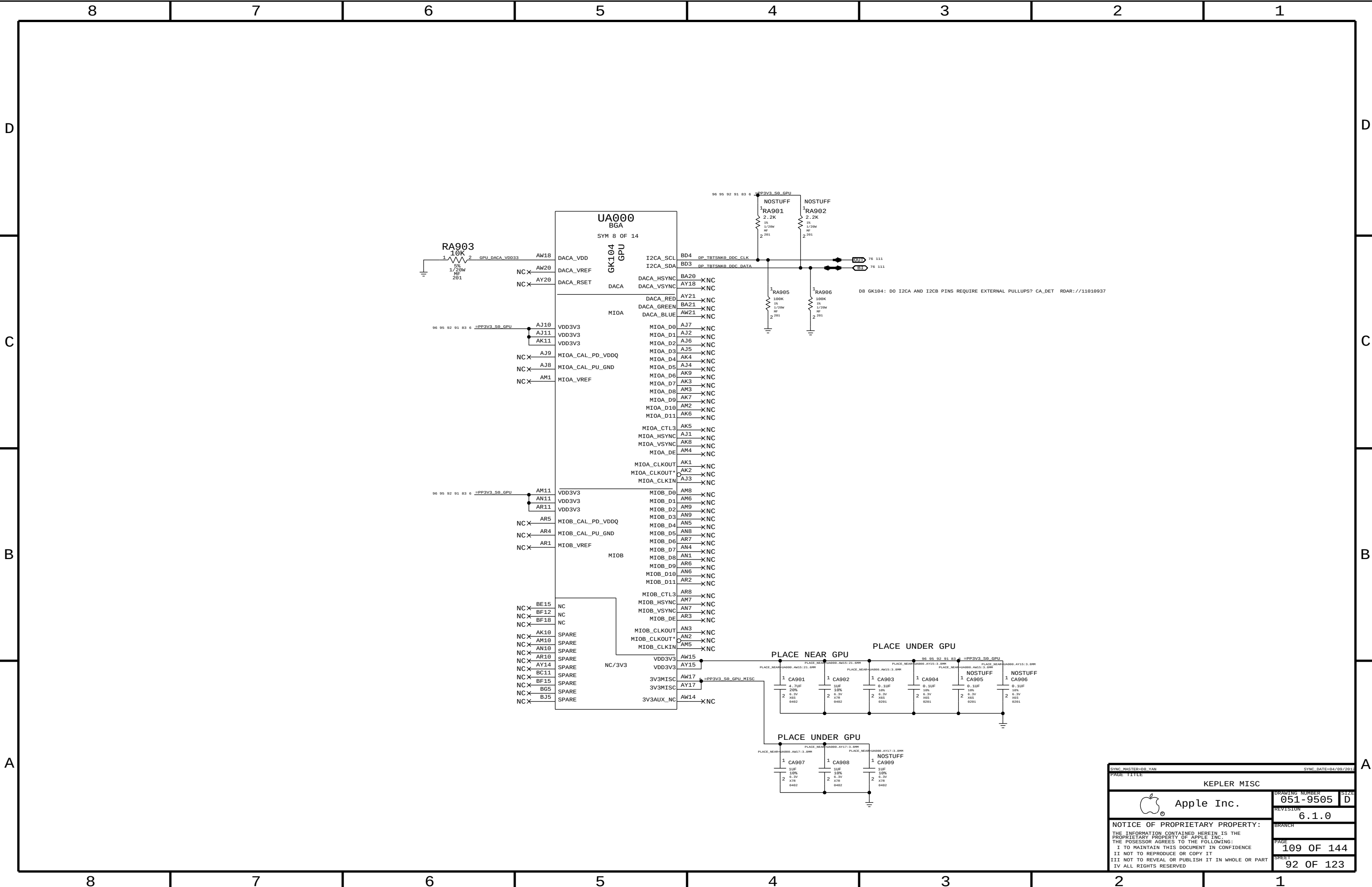
C

B

B

A

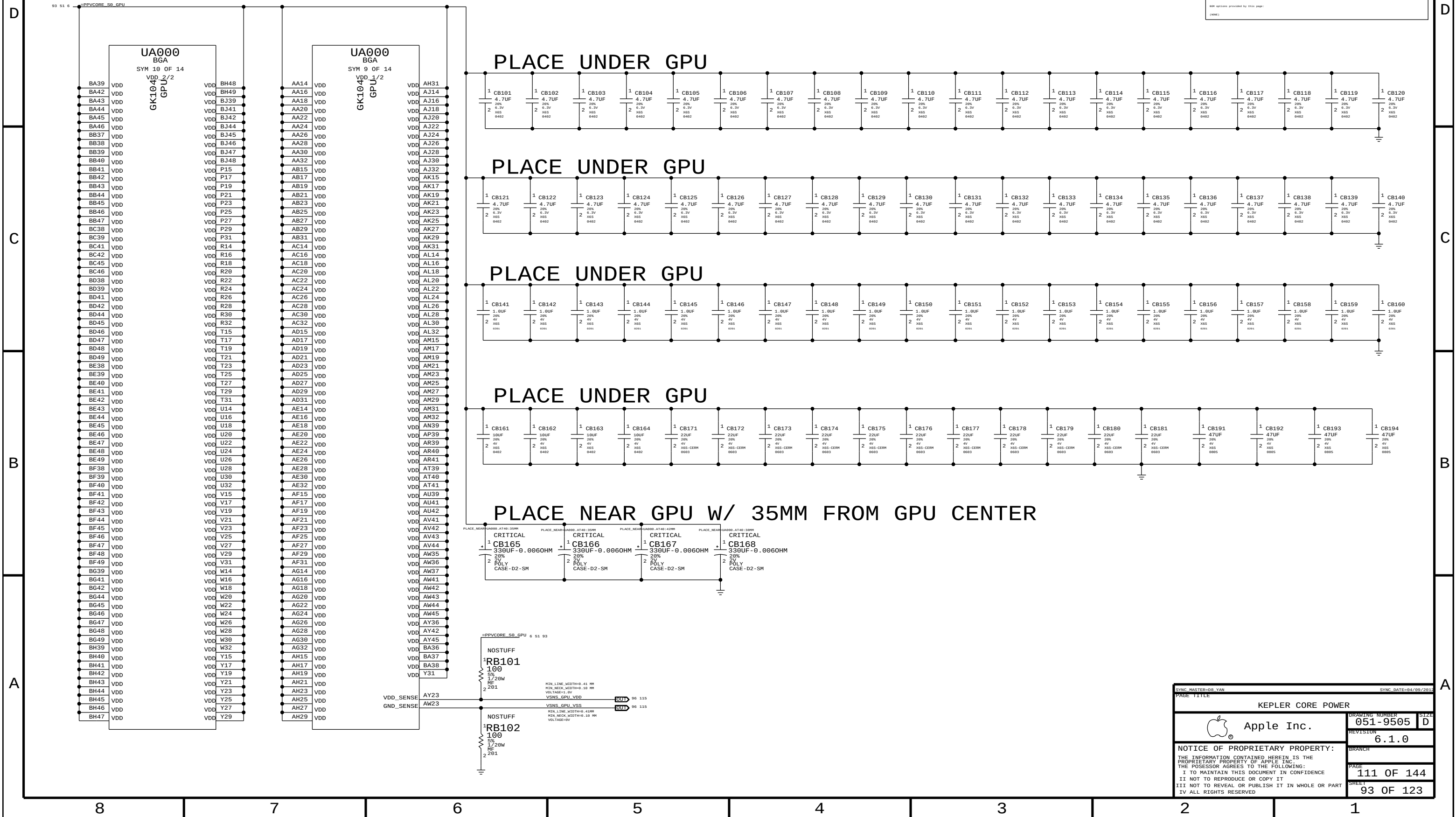
A



PAGE TITLE		KEPLER MISC	
DRAWING NUMBER		051-9505	SIZE D
REVISION		6.1.0	BRANCH
PAGE		109 OF 144	SHEET
SHEET		92 OF 123	

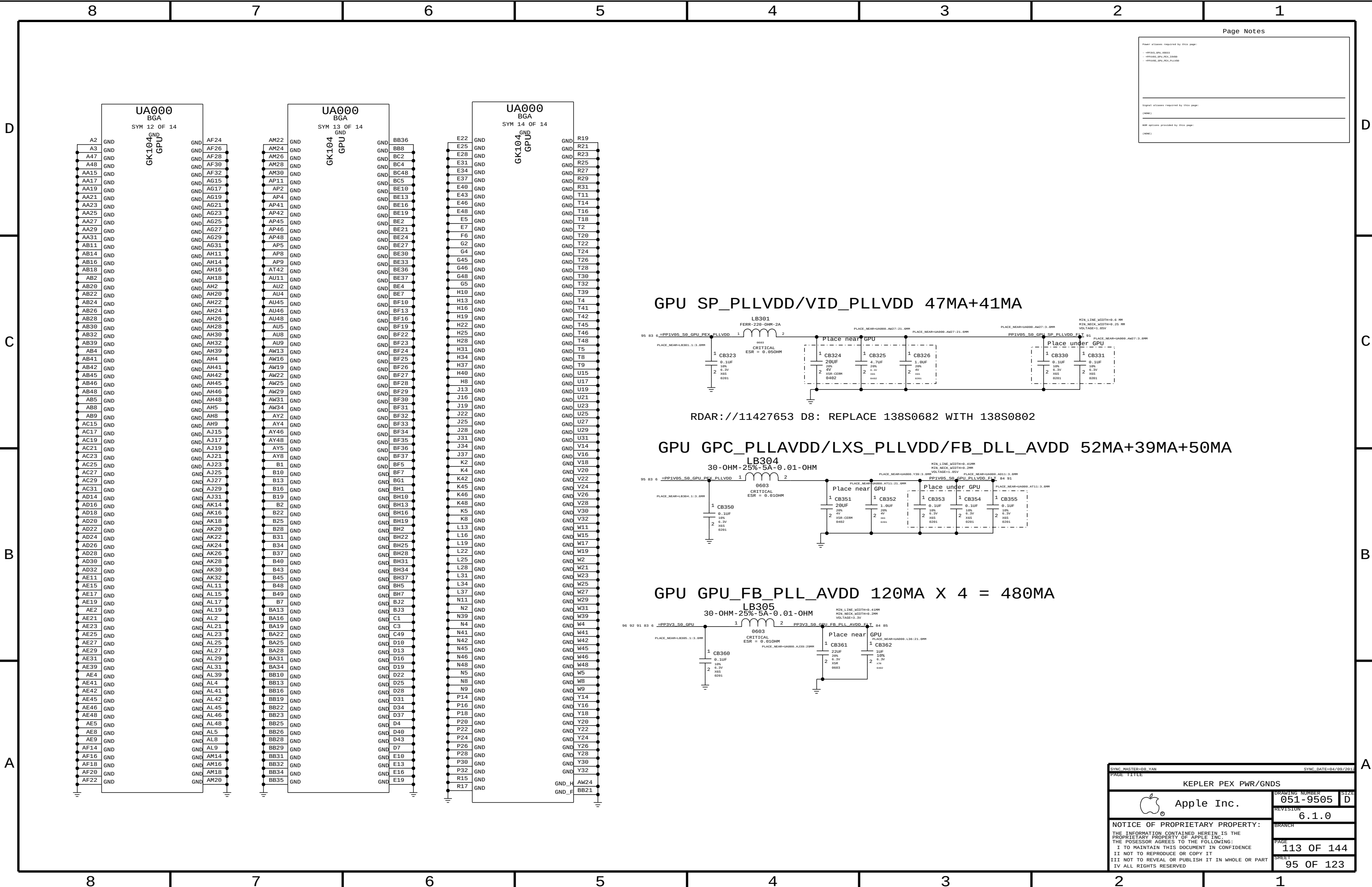
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

GPU NVVDD DECOUPLING I(EDP)=95A



D





Page Notes

Power aliases required by this page:

- NP000_GPU_VDD01

- NP000_GPU_PEX_1000

- NP000_GPU_PEX_PL1VDD

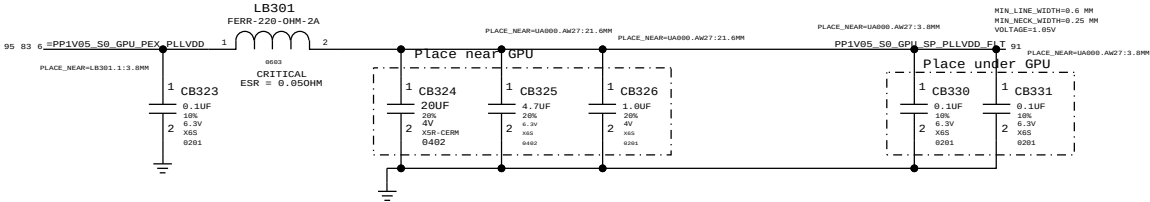
Signal aliases required by this page:

(NONE)

ADP options provided by this page:

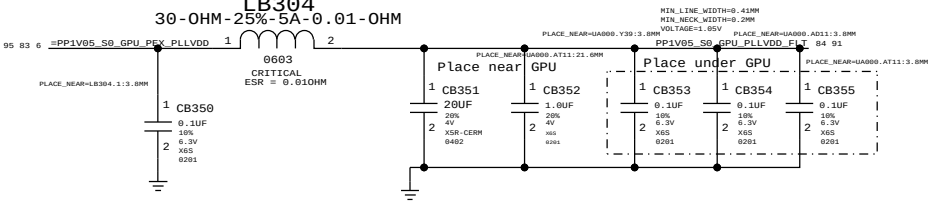
(NONE)

GPU SP_PLLVDD/VID_PLLVDD 47MA+41MA

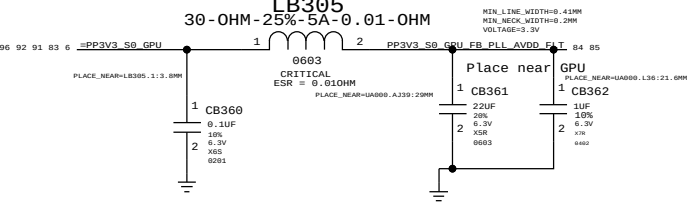


RDAR:///11427653 D8: REPLACE 138S0682 WITH 138S0802

GPU GPC_PLLAVDD/LXS_PLLVDD/FB_DLL_AVDD 52MA+39MA+50MA



GPU GPU_FB_PLL_AVDD 120MA X 4 = 480MA



SYNC MASTER=DR YAN

SYNC DATE=04/09/2015

PAGE TITLE

KEPLER PEX PWR/GNDS

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9505

SIZE

D

REVISION

6.1.0

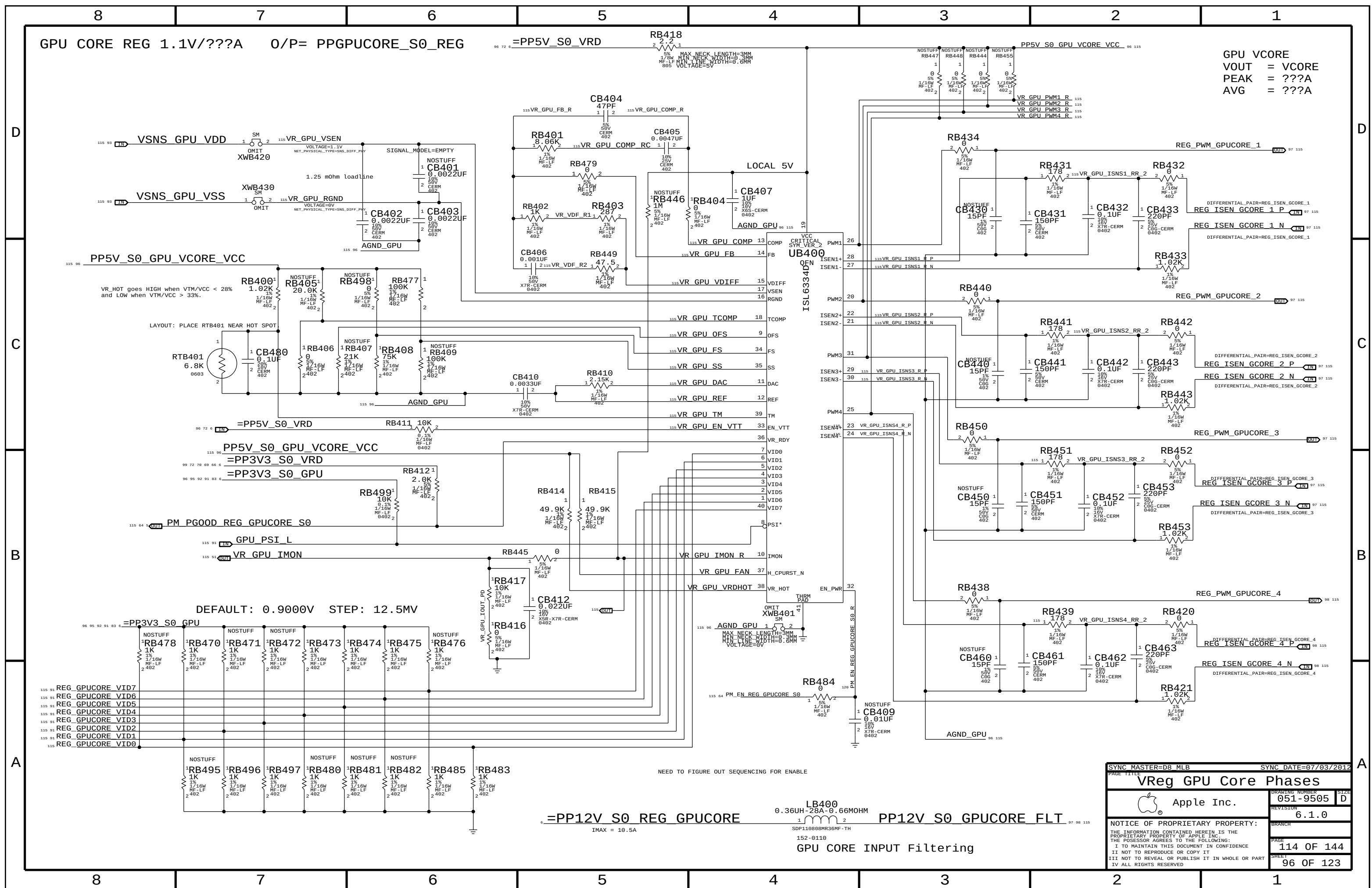
BRANCH

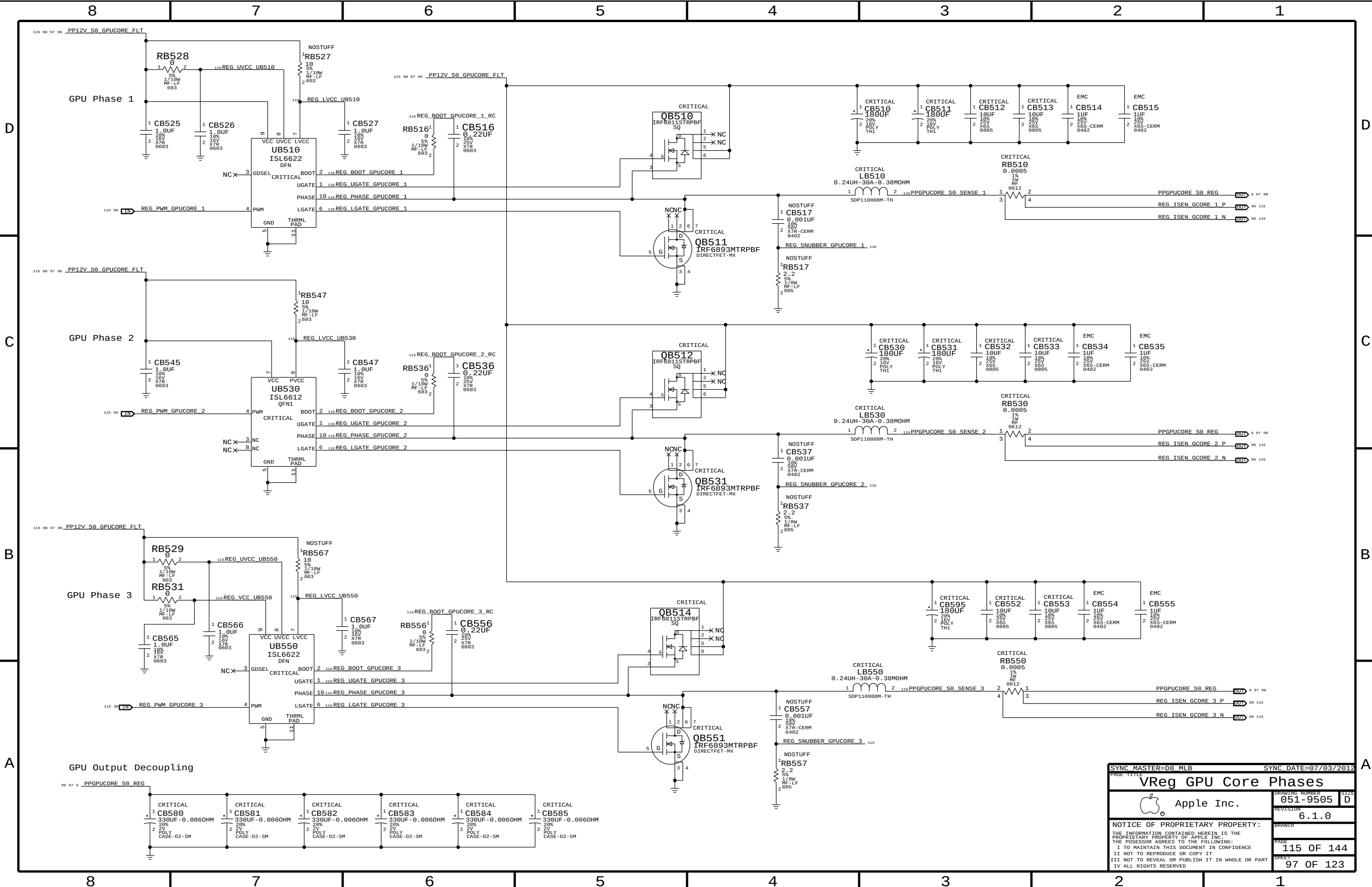
PAGE

113 OF 144

SHEET

95 OF 123





SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

VReg GPU Core Phases

 Apple Inc.

DRAWING NUMBER
051-9505

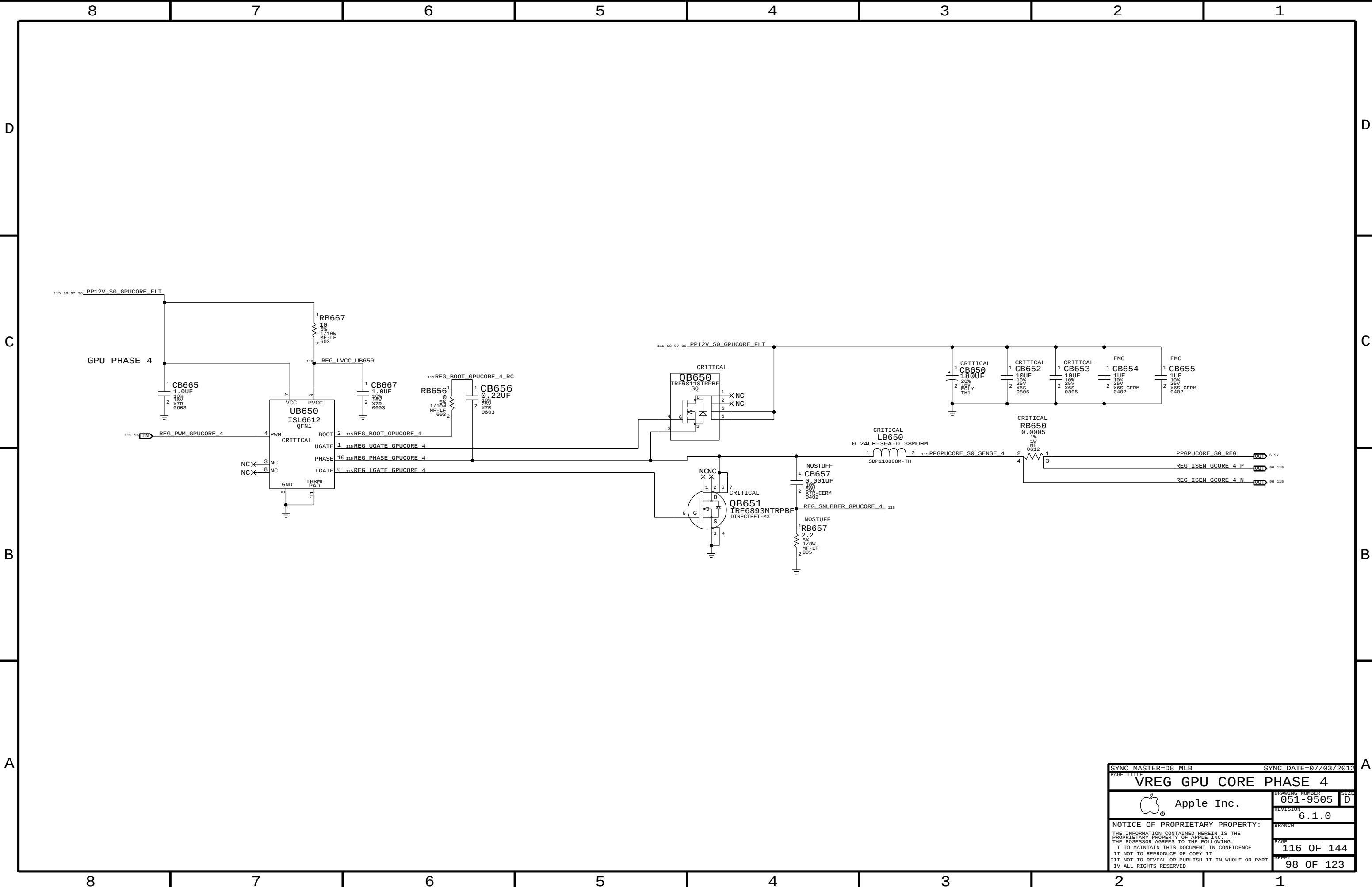
REVISION
6.1.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE
115 OF 144

SHEET
97 OF 123



SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

VREG GPU CORE PHASE 4



Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9505

SIZE

D

REVISION

6.1.0

BRANCH

PAGE

116 OF 144

SHEET

98 OF 123

87654321

DDR3 Constraints

Apple Inc.

051-9505

6.1.0

121 OF 144

101 OF 123

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

PAGE TITLE

DRAWING NUMBER

SIZE

REVISION

BRANCH

PAGE

SHEET

D

C

B

A

B

A

B

A

B

A

D

C

B

A

B

A

A

CPU Misc.

FDI

XDP

Y18

Y19

D

CB

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_90D
COMP_SATA_PHY	*	SATA_50S

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=7.2X_DIELECTRIC	?
COMP_SATA_ISO	*	=5.4X_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA

Electrical Constraint Set	Physical	Spacing	
PCH SATA Port 0 (HDD)			
E601 SATA_R2D	SATA_PHY	SATA	SATA HDD R2D P 44
E602 SATA_R2D	SATA_PHY	SATA	SATA HDD R2D N 44
E603	SATA_PHY	SATA	SATA HDD R2D C P 19 44
E604	SATA_PHY	SATA	SATA HDD R2D C N 19 44
E605 SATA_D2R	SATA_PHY	SATA	SATA HDD D2R P 19 44
E606 SATA_D2R	SATA_PHY	SATA	SATA HDD D2R N 19 44
E607	SATA_PHY	SATA	SATA HDD D2R C P 44
E608	SATA_PHY	SATA	SATA HDD D2R C N 44
PCH SATA Port 1 (SSD)			
E609 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA SSD R2D P 19 44
E610 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA SSD R2D N 19 44
E611			
E612 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA SSD D2R P 19 44
E613 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA SSD D2R N 19 44
PCH SATA Compensation			
E614	COMP_SATA_PHY	COMP_SATA	PCH SATA1COMP 19
E615	COMP_SATA_PHY	COMP_SATA	PCH SATA3COMP 19
E616	COMP_SATA_PHY	COMP_SATA	PCH SATA3RBIAS 19

Unused

Electrical Constraint Set	Physical	Spacing
---------------------------	----------	---------

USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
USB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D
USB_HUB_PHY	*	USB_50S

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Comments
12.2.1	90	90	12/2.8 mils = 4.29:1	USB 2.0
13.3.1	85	85	20/2.8 mils = 7.14:1	USB 3.0
			50/2.8 mils = 17.9:1	USB 2.0/3.0

SMSC Hub Application Note 15.17

Single-ended impedance range from 45-80 ohm is acceptable

USB 2.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=4.4X_DIELECTRIC	?
USB2_ISO	TOP,BOTTOM	=4.4X_DIELECTRIC	?
USB2_CLK_ISO	*	=18X_DIELECTRIC	?
USB2_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB_HUB_ISO	*	=2:1_SPACING	?

USB 2.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB2	*CLK*	*	USB2_CLK_ISO
USB2	DISPLAYPORT	*	USB2_CLK_ISO
USB2	*TBT*	*	USB2_CLK_ISO
USB2	*ENET*	*	USB2_CLK_ISO
USB2	*SD*	*	USB2_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB_HUB	*	*	USB_HUB_ISO

USB 3.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_CLK_ISO	*	=18X_DIELECTRIC	?
USB3_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB3_USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	ISL10	=5X_DIELECTRIC	?

USB 3.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3	*	*	USB3_ISO
USB3	*CLK*	*	USB3_CLK_ISO
USB3	DISPLAYPORT	*	USB3_CLK_ISO
USB3	*TBT*	*	USB3_CLK_ISO
USB3	*ENET*	*	USB3_CLK_ISO
USB3	*SD*	*	USB3_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB3	USB3	*	USB3_USB3_ISO

CAMERA CONTROLLER

Camera Controller's SMIA Interface & MISC. Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
CAM_SE	*	Y	0.2 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D
CAM_PHY	*	CAM_SE

Camera Controller's SMIA Interface & MISC. Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?
CAM_ISO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF
CAM	*	*	CAM_ISO

USB 3.0 and USB 2.0 Trixies Muxing

Electrical Constraint Set		Physical	Spacing	
External Port A (J4600)				
	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_P 45
	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_N 45
		USB3_PHY	USB3	USB3_EXT_A_RX_F_P 20 45
		USB3_PHY	USB3	USB3_EXT_A_RX_F_N 20 45
	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_P 20 45
	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_N 20 45
		USB3_PHY	USB3	USB3_EXT_A_TX_F_P 45
		USB3_PHY	USB3	USB3_EXT_A_TX_F_N 45
		USB3_PHY	USB3	USB3_EXT_A_TX_C_P 45
		USB3_PHY	USB3	USB3_EXT_A_TX_C_N 45
	USB2_MUXED_M010_CONN	USB2_PHY	USB2	USB_PCH_0_P 20 45
	USB2_MUXED_M010_CONN	USB2_PHY	USB2	USB_PCH_0_N 20 45
		USB2_PHY	USB2	USB2_EXT_A_MUXED_P 45
		USB2_PHY	USB2	USB2_EXT_A_MUXED_N 45
		USB2_PHY	USB2	USB2_EXT_A_MUXED_F_P 45
		USB2_PHY	USB2	USB2_EXT_A_MUXED_F_N 45
External Port B (J4610)				
	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_P 45
	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_N 45
		USB3_PHY	USB3	USB3_EXT_B_RX_F_P 20 45
		USB3_PHY	USB3	USB3_EXT_B_RX_F_N 20 45
	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_P 20 45
	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_N 20 45
		USB3_PHY	USB3	USB3_EXT_B_TX_F_P 45
		USB3_PHY	USB3	USB3_EXT_B_TX_F_N 45
		USB3_PHY	USB3	USB3_EXT_B_TX_C_P 45
		USB3_PHY	USB3	USB3_EXT_B_TX_C_N 45
	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_P 20 45
	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_N 20 45
		USB2_PHY	USB2	USB_PCH_9_P 20 45
		USB2_PHY	USB2	USB_PCH_9_N 20 45
		USB2_PHY	USB2	USB2_EXTB_MUXED_P 45
		USB2_PHY	USB2	USB2_EXTB_MUXED_N 45
		USB2_PHY	USB2	USB2_EXTB_MUXED_F_P 45
		USB2_PHY	USB2	USB2_EXTB_MUXED_F_N 45
External Port C (J4700)				
	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_P 46
	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_N 46
		USB3_PHY	USB3	USB3_EXT_C_RX_F_P 20 46
		USB3_PHY	USB3	USB3_EXT_C_RX_F_N 20 46
	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_P 20 46
	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_N 20 46
		USB3_PHY	USB3	USB3_EXT_C_TX_F_P 46
		USB3_PHY	USB3	USB3_EXT_C_TX_F_N 46
		USB3_PHY	USB3	USB3_EXT_C_TX_C_P 46
		USB3_PHY	USB3	USB3_EXT_C_TX_C_N 46
	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_P 20 46
	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_N 20 46
		USB2_PHY	USB2	USB2_EXT_C_F_P 46
		USB2_PHY	USB2	USB2_EXT_C_F_N 46
External Port D (J4710)				
	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_P 46
	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_N 46
		USB3_PHY	USB3	USB3_EXTD_RX_F_P 20 46
		USB3_PHY	USB3	USB3_EXTD_RX_F_N 20 46
	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_P 20 46
	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_N 20 46
		USB3_PHY	USB3	USB3_EXTD_TX_F_P 46
		USB3_PHY	USB3	USB3_EXTD_TX_F_N 46
		USB3_PHY	USB3	USB3_EXTD_TX_C_P 46
		USB3_PHY	USB3	USB3_EXTD_TX_C_N 46
	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_P 20 46
	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_N 20 46
		USB2_PHY	USB2	USB_PCH_10_P 20 46
		USB2_PHY	USB2	USB_PCH_10_N 20 46
		USB2_PHY	USB2	USB2_EXTD_MUXED_P 46
		USB2_PHY	USB2	USB2_EXTD_MUXED_N 46
		USB2_PHY	USB2	USB2_EXTD_MUXED_F_P 46
		USB2_PHY	USB2	USB2_EXTD_MUXED_F_N 46
Camera (U4200)				
	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_P 42
	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_N 42
PCH USB Compensation				
		PCH_50S	COMP_PCH	PCH_USB_RBIA5 20

Electrical Constraint Set	Physical	Spacing	Voltage	
Camera Controller Local Ground				
	GND_PHY	GND	0V	CAM_AGND 42
	GND_PHY	GND	0V	CAM_PLLGND 42

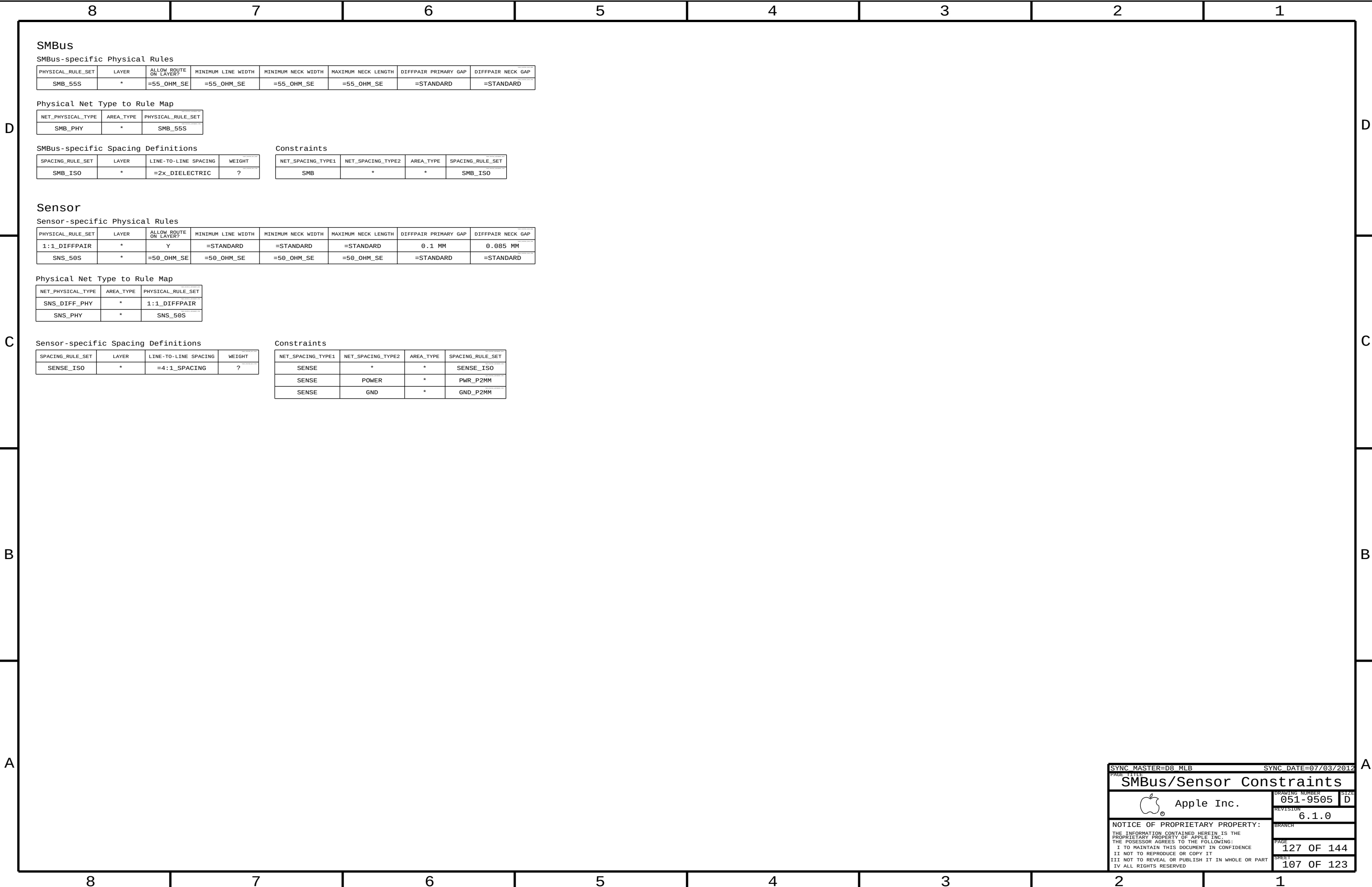
USB Hub

Electrical Constraint Set	Physical	Spacing	
USB 2.0 Hub			
 USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_P 26 27
 USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_N 26 27
 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_P 27 35
 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_N 27 35
 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_P 35
 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_N 35
	USB_HUB_PHY	USB_HUB	USB_HUB_2P 27
	USB_HUB_PHY	USB_HUB	USB_HUB_2N 27
USB 2.0 Hub Misc.			
	USB_HUB_PHY	USB_HUB	USB_HUB_RESET_L 27
	USB_HUB_PHY	USB_HUB	USB_HUB_VBUS_DET 27
	USB_HUB_PHY	USB_HUB	USB_HUB_NON_REMO 27
	USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM1 27
	USB_HUB_PHY	USB_HUB	USB_HUB_HS_IND 27
USB 2.0 Hub Compensation			
	PCH_50S	COMP_PCH	USB_HUB_RB1AS 27
USB 2.0 Hub Crystal			
	CLK_XTAL	XTAL	USB_HUB_XTAL1 27
	CLK_XTAL	XTAL	USB_HUB_XTAL2 27
	CLK_XTAL	XTAL	USB_HUB_XTAL2_R 27

Camera Controller

Electrical Constraint Set		Physical	Spacing	
SMIA				
HS08	SMIA_DATA	SMIA_DFF_PHY	SMIA_DFF	SMIA_DATA_P 42
HS08	SMIA_DATA	SMIA_DFF_PHY	SMIA_DFF	SMIA_DATA_N 42
HS08	SMIA_CLK	SMIA_DFF_PHY	SMIA_DFF	SMIA_CLK_P 42
HS08	SMIA_CLK	SMIA_DFF_PHY	SMIA_DFF	SMIA_CLK_N 42
MISC				
HS08	SPT_5GS	SPT	CAM_SF_CLK	42
HS08	SPT_5GS	SPT	CAM_SF_CLK_R	42
HS08	SPT_5GS	SPT	CAM_SF_DIN	42
HS08	SPT_5GS	SPT	CAM_SF_DIN_R	42
HS08	SPT_5GS	SPT	CAM_SF_CS_L	42
HS08	SPT_5GS	SPT	CAM_SF_WP_L	42
HS08	SPT_5GS	SPT	CAM_SF_DOUT	42
HS08	SPT_5GS	SPT	CAM_SF_DOUT_R	42
HS08	SPT_5GS	SPT	CAM_SF_HOLD_L	42
HS08	CAM_PHY	CAM	CAM_USB_VRES	42
HS08	CAM_PHY	CAM	MIPI_RESISTOR	42
HS08		PM	CAM_EXT_BOOT_L	43
HS08		PM	PCH_CAM_EXT_BOOT_R_L	21 43
HS08		PM	CAM_P1V2_RST_HOLDOFF	43
HS08		PM	CAM_P1V2_RST_HOLDOFF_L	43
I2C				
HS08	SMB_PHY	SMB	I2C_CAMSENSOR_SDA	42
HS08	SMB_PHY	SMB	I2C_CAMSENSOR_SCL	42
HS08	SMB_PHY	SMB	SMB_ALS_F_SDA	42
HS08	SMB_PHY	SMB	SMB_ALS_F_SCL	42
Camera Controller Crystal				
HS08	CLK_XTAL	XTAL	CAM_XTAL_IN	42
HS08	CLK_XTAL	XTAL	CAM_XTAL_OUT	42
HS08	CLK_XTAL	XTAL	CAM_XTAL_OUT_R	42

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
USB/Camera Constraints			
 Apple Inc.		DRAWING NUMBER	D
		051-9505	
		REVISION	
		6.1.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
PAGE		126 OF 144	
SHEET		106 OF 123	



D

D

C

C

B

B

A

A

SYNC_MASTER=D8_MLB

SYNC_DATE=07/03/2012

PAGE_TITLE

SMBus/Sensor Constraints

 Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9505

REVISION

6.1.0

BRANCH

PAGE

127 OF 144

SHEET

107 OF 123

SIZE

D

DC - DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_CTL_PHY	*	POWER_P5MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
POWER_PHY	*	POWER_P6MM
GND_PHY	*	GND_P5MM

Constraints

Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

Power-specific Spacing Definitions

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_TSO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?
SWNODE_LG2SW	*	=3:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND
VR_LGATE	*	*	SWNODE_ISO
VR_LGATE	VR_SWITCH	*	SWNODE_LG2SW

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

CPU VccSA

Electrical Constraint Set	Physical	Spacing	Voltage	IDT	NO_TEST	
Input Bus						
I22	POWER_PHY	POWER	5V			REG_VCC_U7500 70
I25	POWER_PHY	POWER	5V			REG_PVCC_U7500 70
Local Ground						
I26	GND_PHY	GND	0V			AGND_VCCSA0 70
VCCIO						
I27	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VCCSA0 70
I28	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VCCSA0 70
I29	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VCCSA0_RC 70
I30	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VCCSA0 70
I31	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VCCSA0 70
I33	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_VCCSA0 70
I39	VR_CTL_PHY	VR_CTL				REG_VCCSA0_OCSET 70
I35	VR_CTL_PHY	VR_CTL				REG_VCCSA0_VO 70
I34		SENSE				SNS_CPU_VCCSA 13 70
I37 VSNS_CPU_VCCSA	SNS_DTFE_PHY	SENSE				SNS_VCCSA0_XW_P 70
I36 VSNS_CPU_VCCSA	SNS_DTFE_PHY	SENSE				SNS_VCCSA0_XW_N 70
I38		SENSE				REG_VCCSA0_FB 70
I39		SENSE				REG_VCCSA0_RTN 70
I40	VR_CTL_PHY	VR_CTL				REG_VCCSA0_SREF 70
I41	VR_CTL_PHY	VR_CTL				REG_VCCSA0_FSEL 70
Output Bus						
I42	POWER_PHY	POWER	0_925V			PPVCCSA_S0 6

```
CPU VccIO PCH 1.05V S0
```

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
	POWER_PHY	POWER	5V			REG_VCC_U7400 69
	POWER_PHY	POWER	5V			REG_PVCC_U7400 69
Local Ground						
	GND_PHY	GND	0V			AGND_CPU_P1V05S0 69
1.05V S0						
	POWER_PHY	VB_SWITCH	12V	TRUE		REG_CPU_PHASE_P1V05S0 69
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_BOOT_P1V05S0 69
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_BOOT_P1V05S0_RC 69
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0 69
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0_R 69
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_CPU_LGATE_P1V05S0 69
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_SNUBBER_P1V05S0 69
	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_OCSET 69
	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_VD 69
	VSNS_CPU_VCCIO	SENSE				SNS_CPU_VCCIO_P 13 69
	VSNS_CPU_VCCIO	SENSE				SNS_CPU_VCCIO_N 13 69
	SNS_DIEF_PHY	SENSE				SNS_CPU_P1V05S0_XW_P 69
	SNS_DIEF_PHY	SENSE				SNS_CPU_P1V05S0_XW_N 69
		SENSE				REG_CPU_P1V05S0_FB 69
		SENSE				REG_CPU_P1V05S0_RTN 69
	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_SREF 69
	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_FSEL 69
Output Bus						
	POWER_PHY	POWER	1.05V			PP1V05_S0_CPU 6

PCH/GPU/TBT 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
REG	POWER_PHY	POWER	5V			REG VCC UB700 99
Local Ground						
LGND	GND_PHY	GND	0V			P1V05 AGND 99
1.05V S0						
PHV	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE P1V05S0 99
FEH	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE P1V05S0_L 99
HEH	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG BOOT P1V05S0 99
HEH	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG BOOT P1V05S0_RC 99
HEH	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG UGATE P1V05S0 99
HEH	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG UGATE P1V05S0_R 99
HEH	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG LGATE P1V05S0 99
HEH	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P1V05S0 99
HEH	VR_CTL_PHY	VR_CTL				REG P1V05S0_OCSET 99
HEH	VR_CTL_PHY	VR_CTL				REG P1V05S0_VO 99
FEH	POWER_PHY	VR_SWITCH	12V			REG P1V05S0_VO_R 99
FEH	POWER_PHY	VR_SWITCH	12V			P1V05_OCSET_R 99
HEH		SENSE				REG P1V05S0_FB 99
HEH		SENSE				REG P1V05S0_RTN 99
HEH	VR_CTL_PHY	VR_CTL				REG P1V05S0_SREF 99
HEH	VR_CTL_PHY	VR_CTL				REG P1V05S0_FSEL 99
Output Bus						
HEH	POWER_PHY	POWER	1.05V			PP1V05_S0_PCH 6
HEH	POWER_PHY	POWER	1.05V			PP1V05_S0 6
FET Switched						
HEH	POWER_PHY	POWER	1.05V			PP1V05_TBTL 6 3H
HEH	POWER_PHY	POWER	1.05V			PP1V05_TBTCIO 6

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
VReg Constraints			
 Apple Inc.		DRAWING NUMBER	051-9505
		SIZE	D
		REVISION	6.1.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE TWO. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
		PAGE	128 OF 144
		SHEET	108 OF 123

8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
CPU Core Phases																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
<table><thead><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th><th></th></tr></thead><tbody><tr><td colspan="8">Input Bus</td></tr><tr><td>REG1</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>PP12V_S0_CPUCORE_FLT</td><td>66 67 68</td></tr><tr><td>REG1</td><td>POWER_PHY</td><td>POWER</td><td>5V</td><td></td><td></td><td>REG_VCC_U7100</td><td>66</td></tr><tr><td colspan="8">Local Ground</td></tr><tr><td>REG2</td><td>GND_PHY</td><td>GND</td><td>0V</td><td></td><td></td><td>AGND_CPU</td><td>66 67 68</td></tr><tr><td colspan="8">Phase 1</td></tr><tr><td>REG3</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7210</td><td>67</td></tr><tr><td>REG4</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_1</td><td>66 67</td></tr><tr><td>REG5</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_1_R</td><td>66</td></tr><tr><td>REG6</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_1</td><td>67</td></tr><tr><td>REG7</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_1</td><td>67</td></tr><tr><td>REG8</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_1_RC</td><td>67</td></tr><tr><td>REG9</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_1</td><td>67</td></tr><tr><td>REG10</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_1</td><td>67</td></tr><tr><td>REG11</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_1</td><td>67</td></tr><tr><td>REG12</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_1</td><td>67</td></tr><tr><td>REG13</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_1_P</td><td>66 67</td></tr><tr><td>REG14</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_1_N</td><td>67</td></tr><tr><td>REG15</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENCORE_1_NR</td><td>66 67</td></tr><tr><td colspan="8">Phase 2</td></tr><tr><td>REG16</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7230</td><td>67</td></tr><tr><td>REG17</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_2</td><td>66 67</td></tr><tr><td>REG18</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_2_R</td><td>66</td></tr><tr><td>REG19</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_2</td><td>67</td></tr><tr><td>REG20</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_2</td><td>67</td></tr><tr><td>REG21</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_2_RC</td><td>67</td></tr><tr><td>REG22</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_2</td><td>67</td></tr><tr><td>REG23</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_2</td><td>67</td></tr><tr><td>REG24</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_2</td><td>67</td></tr><tr><td>REG25</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_2</td><td>67</td></tr><tr><td>REG26</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_2_P</td><td>66 67</td></tr><tr><td>REG27</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_2_N</td><td>67</td></tr><tr><td>REG28</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENCORE_2_NR</td><td>66 67</td></tr><tr><td colspan="8">Phase 3</td></tr><tr><td>REG29</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7250</td><td>67</td></tr><tr><td>REG30</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_3</td><td>66 67</td></tr><tr><td>REG31</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_3_R</td><td>66</td></tr><tr><td>REG32</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_3</td><td>67</td></tr><tr><td>REG33</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_3</td><td>67</td></tr><tr><td>REG34</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_3_RC</td><td>67</td></tr><tr><td>REG35</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_3</td><td>67</td></tr><tr><td>REG36</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_3</td><td>67</td></tr><tr><td>REG37</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_3</td><td>67</td></tr><tr><td>REG38</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_3</td><td>67</td></tr><tr><td>REG39</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_3_P</td><td>66 67</td></tr><tr><td>REG40</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_3_N</td><td>67</td></tr><tr><td>REG41</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENCORE_3_NR</td><td>66 67</td></tr><tr><td colspan="8">Phase 4</td></tr><tr><td>REG42</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7310</td><td>68</td></tr><tr><td>REG43</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_4</td><td>66 68</td></tr><tr><td>REG44</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_4_R</td><td>66</td></tr><tr><td>REG45</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_4</td><td>68</td></tr><tr><td>REG46</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_4</td><td>68</td></tr><tr><td>REG47</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_4_RC</td><td>68</td></tr><tr><td>REG48</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_4</td><td>68</td></tr><tr><td>REG49</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_4</td><td>68</td></tr><tr><td>REG50</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_4</td><td>68</td></tr><tr><td>REG51</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_4</td><td>68</td></tr><tr><td>REG52</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_4_P</td><td>66 68</td></tr><tr><td>REG53</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_4_N</td><td>68</td></tr><tr><td>REG54</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENCORE_4_NR</td><td>66 68</td></tr><tr><td>REG55</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CORE_XW_P</td><td>66</td></tr><tr><td>REG56</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CORE_XW_N</td><td>66</td></tr><tr><td>REG57</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CORE_R_P</td><td>66</td></tr><tr><td>REG58</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CORE_R_N</td><td>66</td></tr><tr><td>REG59</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CPU_VAXG_P</td><td>13 66</td></tr><tr><td>REG60</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CPU_VAXG_N</td><td>13 66</td></tr><tr><td>REG61</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CPU_VCORE_P</td><td>13 66</td></tr><tr><td>REG62</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CPU_VCORE_N</td><td>13 66</td></tr><tr><td>REG63</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_P1V05_IOVDD_XW_P</td><td>99</td></tr><tr><td>REG64</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_P1V05_IOVDD_XW_N</td><td>99</td></tr></tbody></table>								Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST			Input Bus								REG1	POWER_PHY	POWER	12V			PP12V_S0_CPUCORE_FLT	66 67 68	REG1	POWER_PHY	POWER	5V			REG_VCC_U7100	66	Local Ground								REG2	GND_PHY	GND	0V			AGND_CPU	66 67 68	Phase 1								REG3	POWER_PHY	POWER	12V			REG_LVCC_U7210	67	REG4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1	66 67	REG5	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1_R	66	REG6	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_1	67	REG7	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_1	67	REG8	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_1_RC	67	REG9	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_1	67	REG10	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_1	67	REG11	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_1	67	REG12	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_1	67	REG13	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_1_P	66 67	REG14	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_1_N	67	REG15		SENSE				REG_ISENCORE_1_NR	66 67	Phase 2								REG16	POWER_PHY	POWER	12V			REG_LVCC_U7230	67	REG17	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2	66 67	REG18	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2_R	66	REG19	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_2	67	REG20	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_2	67	REG21	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_2_RC	67	REG22	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_2	67	REG23	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_2	67	REG24	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_2	67	REG25	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_2	67	REG26	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_2_P	66 67	REG27	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_2_N	67	REG28		SENSE				REG_ISENCORE_2_NR	66 67	Phase 3								REG29	POWER_PHY	POWER	12V			REG_LVCC_U7250	67	REG30	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3	66 67	REG31	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3_R	66	REG32	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_3	67	REG33	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_3	67	REG34	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_3_RC	67	REG35	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_3	67	REG36	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_3	67	REG37	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_3	67	REG38	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_3	67	REG39	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_3_P	66 67	REG40	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_3_N	67	REG41		SENSE				REG_ISENCORE_3_NR	66 67	Phase 4								REG42	POWER_PHY	POWER	12V			REG_LVCC_U7310	68	REG43	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4	66 68	REG44	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4_R	66	REG45	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_4	68	REG46	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_4	68	REG47	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_4_RC	68	REG48	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_4	68	REG49	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_4	68	REG50	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_4	68	REG51	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_4	68	REG52	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_4_P	66 68	REG53	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_4_N	68	REG54		SENSE				REG_ISENCORE_4_NR	66 68	REG55	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_XW_P	66	REG56	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_XW_N	66	REG57	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_R_P	66	REG58	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_R_N	66	REG59	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VAXG_P	13 66	REG60	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VAXG_N	13 66	REG61	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VCORE_P	13 66	REG62	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VCORE_N	13 66	REG63	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_P	99	REG64	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_N	99
Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
Input Bus																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG1	POWER_PHY	POWER	12V			PP12V_S0_CPUCORE_FLT	66 67 68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG1	POWER_PHY	POWER	5V			REG_VCC_U7100	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Local Ground																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG2	GND_PHY	GND	0V			AGND_CPU	66 67 68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Phase 1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG3	POWER_PHY	POWER	12V			REG_LVCC_U7210	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG5	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1_R	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG6	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG7	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG8	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_1_RC	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG9	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG10	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG11	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG12	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG13	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_1_P	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG14	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_1_N	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG15		SENSE				REG_ISENCORE_1_NR	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Phase 2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG16	POWER_PHY	POWER	12V			REG_LVCC_U7230	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG17	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG18	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2_R	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG19	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_2	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG20	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_2	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG21	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_2_RC	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG22	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_2	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG23	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_2	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG24	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_2	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG25	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_2	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG26	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_2_P	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG27	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_2_N	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG28		SENSE				REG_ISENCORE_2_NR	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Phase 3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG29	POWER_PHY	POWER	12V			REG_LVCC_U7250	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG30	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG31	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3_R	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG32	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_3	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG33	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_3	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG34	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_3_RC	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG35	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_3	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG36	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_3	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG37	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_3	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG38	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_3	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG39	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_3_P	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG40	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_3_N	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG41		SENSE				REG_ISENCORE_3_NR	66 67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Phase 4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG42	POWER_PHY	POWER	12V			REG_LVCC_U7310	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG43	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4	66 68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG44	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4_R	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG45	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_4	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG46	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_4	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG47	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_4_RC	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG48	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_4	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG49	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_4	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG50	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_4	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG51	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_4	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG52	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_4_P	66 68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG53	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_4_N	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG54		SENSE				REG_ISENCORE_4_NR	66 68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG55	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_XW_P	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG56	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_XW_N	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG57	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_R_P	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG58	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_R_N	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG59	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VAXG_P	13 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG60	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VAXG_N	13 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG61	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VCORE_P	13 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG62	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VCORE_N	13 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG63	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_P	99																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG64	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_N	99																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
CPU AXG Phase and Core Controller																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
<table><thead><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th><th></th></tr></thead><tbody><tr><td colspan="8">AXG</td></tr><tr><td>REG65</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7330</td><td>68</td></tr><tr><td>REG66</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUAXG</td><td>66 68</td></tr><tr><td>REG67</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUAXG_R</td><td>66</td></tr><tr><td>REG68</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUAXG</td><td>68</td></tr><tr><td>REG69</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUAXG</td><td>68</td></tr><tr><td>REG70</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUAXG_RC</td><td>68</td></tr><tr><td>REG71</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUAXG</td><td>68</td></tr><tr><td>REG72</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUAXG</td><td>68</td></tr><tr><td>REG73</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUAXG</td><td>68</td></tr><tr><td>REG74</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUAXG_S0_SENSE</td><td>68</td></tr><tr><td>REG75</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENAXG_P</td><td>68</td></tr><tr><td>REG76</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENAXG_N</td><td>68</td></tr><tr><td>REG77</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENAXG_PR</td><td>66 68</td></tr><tr><td>REG78</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENAXG_NR</td><td>66 68</td></tr><tr><td>REG79</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_R_P</td><td>66</td></tr><tr><td>REG80</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_R_N</td><td>66</td></tr><tr><td>REG81</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_XW_P</td><td>66</td></tr><tr><td>REG82</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_XW_N</td><td>66</td></tr><tr><td colspan="8">ISL6364</td></tr><tr><td>REG83</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_COMP</td><td>66</td></tr><tr><td>REG84</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_COMP_RC</td><td>66</td></tr><tr><td>REG85</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_FB</td><td>66</td></tr><tr><td>REG86</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_RC</td><td>66</td></tr><tr><td>REG87</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_R_1</td><td>66</td></tr><tr><td>REG88</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_R_2</td><td>66</td></tr><tr><td>REG89</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_PSICOMP_RC</td><td>66</td></tr><tr><td>REG90</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_PSICOMP</td><td>66</td></tr><tr><td>REG91</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_HFCOMP</td><td>66</td></tr><tr><td>REG92</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUCORE_VSEN</td><td>66</td></tr><tr><td>REG93</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUCORE_RGND</td><td>66</td></tr><tr><td>REG94</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_IMON</td><td>51 66</td></tr><tr><td>REG95</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_IMON_R</td><td>66</td></tr><tr><td>REG96</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_TM</td><td>66</td></tr><tr><td>REG97</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_SUTH</td><td>66</td></tr><tr><td>REG98</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_NPSI</td><td>66</td></tr><tr><td>REG99</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_FVID</td><td>66</td></tr><tr><td>REG100</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_IAUTO</td><td>66</td></tr><tr><td>REG101</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_SW_FREQ</td><td>66</td></tr><tr><td>REG102</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_RAMPADJ</td><td>66</td></tr><tr><td>REG103</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_EN_PWR</td><td>66</td></tr><tr><td>REG104</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_EN_PWR_R</td><td>66</td></tr><tr><td>REG105</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_RSET</td><td>66</td></tr><tr><td>REG106</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_COMP</td><td>66</td></tr><tr><td>REG107</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_COMP_RC</td><td>66</td></tr><tr><td>REG108</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_FB</td><td>66</td></tr><tr><td>REG109</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_RC</td><td>66</td></tr><tr><td>REG110</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_R_1</td><td>66</td></tr><tr><td>REG111</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_R_2</td><td>66</td></tr><tr><td>REG112</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_HFCOMP</td><td>66</td></tr><tr><td>REG113</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUAXG_VSEN</td><td>66</td></tr><tr><td>REG114</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUAXG_RGND</td><td>66</td></tr><tr><td>REG115</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_IMON</td><td>51 66</td></tr><tr><td>REG116</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_IMON_R</td><td>66</td></tr><tr><td>REG117</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_TM</td><td>66</td></tr><tr><td>REG118</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_TCOMP</td><td>66</td></tr><tr><td>REG119</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_SW_FREQ</td><td>66</td></tr><tr><td>REG120</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDSCLK</td><td>13 66</td></tr><tr><td>REG121</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDSCLK_R</td><td>13</td></tr><tr><td>REG122</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDALERT_L</td><td>13 66</td></tr><tr><td>REG123</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDALERT_R_L</td><td>13</td></tr><tr><td>REG124</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDSOUT</td><td>13 66</td></tr><tr><td>REG125</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDSOUT_R</td><td>13</td></tr><tr><td colspan="8">Output Bus</td></tr><tr><td>REG126</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPVCORE_S0_CPU</td><td>6</td></tr><tr><td>REG127</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPVAXG_S0</td><td>6</td></tr></tbody></table>								Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST			AXG								REG65	POWER_PHY	POWER	12V			REG_LVCC_U7330	68	REG66	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG	66 68	REG67	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R	66	REG68	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUAXG	68	REG69	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUAXG	68	REG70	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUAXG_RC	68	REG71	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUAXG	68	REG72	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUAXG	68	REG73	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUAXG	68	REG74	POWER_PHY	POWER	1.1V			PPCPUAXG_S0_SENSE	68	REG75	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			REG_ISENAXG_P	68	REG76	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			REG_ISENAXG_N	68	REG77		SENSE				REG_ISENAXG_PR	66 68	REG78		SENSE				REG_ISENAXG_NR	66 68	REG79	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_P	66	REG80	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_N	66	REG81	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_P	66	REG82	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_N	66	ISL6364								REG83	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP	66	REG84	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC	66	REG85	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB	66	REG86	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC	66	REG87	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1	66	REG88	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2	66	REG89	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC	66	REG90	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP	66	REG91	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP	66	REG92		SENSE				REG_CPUCORE_VSEN	66	REG93		SENSE				REG_CPUCORE_RGND	66	REG94	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON	51 66	REG95	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R	66	REG96	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM	66	REG97	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH	66	REG98	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI	66	REG99	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FVID	66	REG100	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO	66	REG101	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ	66	REG102	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ	66	REG103	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR	66	REG104	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R	66	REG105	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET	66	REG106	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP	66	REG107	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC	66	REG108	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB	66	REG109	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC	66	REG110	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1	66	REG111	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2	66	REG112	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP	66	REG113		SENSE				REG_CPUAXG_VSEN	66	REG114		SENSE				REG_CPUAXG_RGND	66	REG115	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON	51 66	REG116	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R	66	REG117	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM	66	REG118	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP	66	REG119	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ	66	REG120	VR_VTD_PHY	VR_VTD				CPU_VIDSCLK	13 66	REG121	VR_VTD_PHY	VR_VTD				CPU_VIDSCLK_R	13	REG122	VR_VTD_PHY	VR_VTD				CPU_VIDALERT_L	13 66	REG123	VR_VTD_PHY	VR_VTD				CPU_VIDALERT_R_L	13	REG124	VR_VTD_PHY	VR_VTD				CPU_VIDSOUT	13 66	REG125	VR_VTD_PHY	VR_VTD				CPU_VIDSOUT_R	13	Output Bus								REG126	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU	6	REG127	POWER_PHY	POWER	1.1V			PPVAXG_S0	6																																								
Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
AXG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG65	POWER_PHY	POWER	12V			REG_LVCC_U7330	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG66	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG	66 68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG67	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG68	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUAXG	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG69	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUAXG	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG70	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUAXG_RC	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG71	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUAXG	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG72	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUAXG	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG73	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUAXG	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG74	POWER_PHY	POWER	1.1V			PPCPUAXG_S0_SENSE	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG75	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			REG_ISENAXG_P	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG76	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			REG_ISENAXG_N	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG77		SENSE				REG_ISENAXG_PR	66 68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG78		SENSE				REG_ISENAXG_NR	66 68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG79	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_P	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG80	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_N	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG81	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_P	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG82	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_N	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
ISL6364																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG83	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG84	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG85	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG86	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG87	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG88	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG89	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG90	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG91	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG92		SENSE				REG_CPUCORE_VSEN	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG93		SENSE				REG_CPUCORE_RGND	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG94	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON	51 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG95	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG96	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG97	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG98	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG99	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FVID	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG100	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG101	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG102	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG103	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG104	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG105	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG106	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG107	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG108	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG109	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG110	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG111	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG112	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG113		SENSE				REG_CPUAXG_VSEN	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG114		SENSE				REG_CPUAXG_RGND	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG115	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON	51 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG116	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG117	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG118	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG119	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ	66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG120	VR_VTD_PHY	VR_VTD				CPU_VIDSCLK	13 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG121	VR_VTD_PHY	VR_VTD				CPU_VIDSCLK_R	13																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG122	VR_VTD_PHY	VR_VTD				CPU_VIDALERT_L	13 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG123	VR_VTD_PHY	VR_VTD				CPU_VIDALERT_R_L	13																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG124	VR_VTD_PHY	VR_VTD				CPU_VIDSOUT	13 66																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG125	VR_VTD_PHY	VR_VTD				CPU_VIDSOUT_R	13																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Output Bus																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
REG126	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU	6																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
REG127	POWER_PHY	POWER	1.1V			PPVAXG_S0	6																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																

SYNC_MASTER=D8_MLB

SYNC_DATE=07/03/2012

CPU VReg Constraints

Apple Inc.

051-9505

6.1.0

129 OF 144

109 OF 123

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

3.3V S5/5V S4

	Physical	Spacing	Voltage	D1D2	NO_TEST	
Input Bus						
H065	POWER_PHY	POWER	12V			REG_VIN_U7600
H125	POWER_PHY	POWER	5V			REG_VCC1_U7600
H135	POWER_PHY	POWER	5V			REG_VCC2_U7600
3.3V S5						
H045	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P3V3S5
H075	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_P3V3S5
H085	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_P3V3S5_RC
H045	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P3V3S5
H055	VR_CTL_PHY	VR_IGATE	12V	TRUE		REG_LGATE_P3V3S5
H055	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P3V3S5
H107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_ISEN
H107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_OCSET
H095	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FSET
H095	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT
H107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT_R
H107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FB
5V S3						
H085	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P5VS4
H135	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_P5VS4
H135	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_P5VS4_RC
H135	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P5VS4
H135	VR_CTL_PHY	VR_IGATE	12V	TRUE		REG_LGATE_P5VS4
H135	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P5VS4
H135	VR_CTL_PHY	VR_CTL				REG_P5VS4_ISEN
H135	VR_CTL_PHY	VR_CTL				REG_P5VS4_OCSET
H135	VR_CTL_PHY	VR_CTL				REG_P5VS4_FSET
H095	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT
H095	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT_R
H135	VR_CTL_PHY	VR_CTL				REG_P5VS4_FB
Output Bus						
H015	POWER_PHY	POWER	5V			PP5V_S5
H015	POWER_PHY	POWER	5V			PP5V_S4
H015	POWER_PHY	POWER	3.3V			PP3V3_S5
FET Switched						
H125	POWER_PHY	POWER	5V			PP5V_S0
H125	POWER_PHY	POWER	3.3V			PP3V3_S4
H125	POWER_PHY	POWER	3.3V			PP3V3_S0
H125	POWER_PHY	POWER	3.3V			PP3V3_S4_ENET
H135	POWER_PHY	POWER	3.3V			PP3V3_TBTLIC
H055	POWER_PHY	POWER	3.3V			PP3V3_S4_AP
H135	POWER_PHY	POWER	3.3V			PP3V3_S0_SSD

VDDQ S3 (1.5V)/VTT S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
H139 POWER_PHY	POWER	5V			REG_V5IN_U7700 72
Local Ground					
H139 GND_PHY	GND	0V			AGND_VDDQS3 73
VDDQ S3					
H139 POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3 72
H139 POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_VDDQS3 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_VDDQS3_RC 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R 72
H139 VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VDDQS3 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3 72
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3_SENSE 72
H134 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP 72
H139 VR_CTL_PHY	VR_CTL				LDO_DDRVITS0_SNS 72
Output Bus					
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3 6
H139 POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVTI_S0 6
FET Switched					
H139 POWER_PHY	POWER	1.5V			PP1V5_S0 6
Sensed					
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3_DDR 6
H139 POWER_PHY	POWER	1.5V			PP1V5_S0_CPU_MEM 6
H139 POWER_PHY	POWER	1.5V			PPFBVDDQ_S0_GPU 6

DDR3 Vref

	Physical	Spacing	Voltage	DIDT	NO_TEST	
Memory Vref						
PP3V3	POWER_PHY	POWER	3.3V		PP3V3_S4_VREFMRGN_DAC	34
PP3V3	POWER_PHY	POWER	3.3V		PP3V3_S4_VREFMRGN_CTRL	34
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V		PPDDR_VREF_DQ_MEM_A_S3	6
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V		PPDDR_VREF_DQ_MEM_B_S3	6
CPU	POWER_DDR_PHY	POWER_DDR	0.75V		CPU_DIMM_VREF_DAC_A	11
CPU	POWER_DDR_PHY	POWER_DDR	0.75V		CPU_DIMM_VREF_DAC_B	11
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V		PPDDR_VREF_CA_MEM_A_S3	6
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V		PPDDR_VREF_CA_MEM_B_S3	6
CPU	POWER_DDR_PHY	POWER_DDR	0.75V		CPU_DDR_VREF	11
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V		PPDDRVTT_S3	6

1.8V S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
1.8V S0					
 POWER_PHY	VR_SWITCH	5V	TRUE		REG_PHASE_P1V8S0
 VR_CTL_PHY	VR_CTL				REG_P1V8S0_VFB
 VR_CTL_PHY	VR_CTL				REG_P1V8S0_SYNC
Output Bus					
 POWER_PHY	POWER	1.8V			PP1V8_S0

HDD S0

Physical	Spacing	Voltage	IDIT	NO_TEST	
HDD S0 POWER_PHY	POWER	5V			PP5V_S0_HDD 0

12V S5

Physical	Spacing	Voltage	DIOT	NO_TEST
Input Bus				
 POWER_PHY	POWER	12V		PP12V ACDC
FET Switched				
 POWER_PHY	POWER	12V		PP12V S0
Sensed				
 POWER_PHY	POWER	12V		PP12V S5
 POWER_PHY	POWER	12V		PP12V G3H
 POWER_PHY	POWER	12V		PP12V G3H P3V42
 POWER_PHY	POWER	12V		PP12V S0 FBVDDQ
 POWER_PHY	POWER	12V		PP12V S0 CPU P1V05
 POWER_PHY	POWER	12V		PP12V S0 VCCSA
 POWER_PHY	POWER	12V		PP12V S0 P1V05
 POWER_PHY	POWER	12V		PP12V S0 HDD
 POWER_PHY	POWER	12V		PP12V S0 BLC

Ground/Common

Physical	Spacing	Voltage	IDT	NO_TEST	
Common					
1820 GND_PHY	GND	6V			GND

3V42 S0

Physical		Spacing	Voltage	IDT	NO_TEST	
3V42 S0						
TEST	POWER_PHY	VR_SWITCH	5V	TRUE		P3V42G3H_SW 7
TEST	VR_CTL_PHY	VR_CTL				P3V42G3H_FB 7
TEST	VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L 7
TEST	POWER_PHY	POWER	3.3V			PP3V3_G3 6
TEST	POWER_PHY	POWER	3.3V			PP3V42_G3H 6

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

[illegible]

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3X_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=3X_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3X_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5x_DIELECTRIC	?

Constraints ($x \in \{A, B\}$, $y \in \{0, 1\}$)
Memory Address: $MA_{xy}[8:0]$

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

Electrical Constraint Set		Physical	Spacing		
Memory Address					
H642	GDDR A0_MA	GDDR_MA_PHY	GDDR_A_0_MA	FB A0 A<8...0>	NO_TEST=TRUE
H643	GDDR A1_MA	GDDR_MA_PHY	GDDR_A_1_MA	FB A1 A<8...0>	NO_TEST=TRUE
Address Dynamic Bus Inv					
H644	GDDR A0_ADBT	GDDR_ADBT_PHY	GDDR_A_0_ADBT	FB A0 ABT_L	NO_TEST=TRUE
H645	GDDR A1_ADBT	GDDR_ADBT_PHY	GDDR_A_1_ADBT	FB A1 ABT_L	NO_TEST=TRUE
Control					
H646	GDDR A0_CKE	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB A0 CKE_L	NO_TEST=TRUE
H647	GDDR A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB A0 CS_L	NO_TEST=TRUE
H648	GDDR A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB A0 WE_L	NO_TEST=TRUE
H649	GDDR A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB A0 CAS_L	NO_TEST=TRUE
H650	GDDR A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB A0 RAS_L	NO_TEST=TRUE
H651	GDDR A1_CKE	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB A1 CKE_L	NO_TEST=TRUE
H652	GDDR A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB A1 CS_L	NO_TEST=TRUE
H653	GDDR A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB A1 WE_L	NO_TEST=TRUE
H654	GDDR A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB A1 CAS_L	NO_TEST=TRUE
H655	GDDR A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB A1 RAS_L	NO_TEST=TRUE
Clock					
H656	GDDR A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB A0 CLK_P	NO_TEST=TRUE
H657	GDDR A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB A0 CLK_N	NO_TEST=TRUE
H658	GDDR A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB A1 CLK_P	NO_TEST=TRUE
H659	GDDR A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB A1 CLK_N	NO_TEST=TRUE
Data					
H660	GDDR A0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<7...0>	NO_TEST=TRUE
H661	GDDR A0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<15...8>	NO_TEST=TRUE
H662	GDDR A0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<23...16>	NO_TEST=TRUE
H663	GDDR A0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<31...24>	NO_TEST=TRUE
H664	GDDR A1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<7...0>	NO_TEST=TRUE
H665	GDDR A1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<15...8>	NO_TEST=TRUE
H666	GDDR A1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<23...16>	NO_TEST=TRUE
H667	GDDR A1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<31...24>	NO_TEST=TRUE
Error Detection					
H668	GDDR A0_EDC0	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<0>	NO_TEST=TRUE
H669	GDDR A0_EDC1	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<1>	NO_TEST=TRUE
H670	GDDR A0_EDC2	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<2>	NO_TEST=TRUE
H671	GDDR A0_EDC3	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<3>	NO_TEST=TRUE
H672	GDDR A1_EDC0	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<0>	NO_TEST=TRUE
H673	GDDR A1_EDC1	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<1>	NO_TEST=TRUE
H674	GDDR A1_EDC2	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<2>	NO_TEST=TRUE
H675	GDDR A1_EDC3	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<3>	NO_TEST=TRUE
Data Dynamic Bus Inv					
H676	GDDR A0_DBI0	GDDR_DBI_PHY	GDDR_A_0_DBI	FB A0 DBI_L<0>	NO_TEST=TRUE
H677	GDDR A0_DBI1	GDDR_DBI_PHY	GDDR_A_0_DBI	FB A0 DBI_L<1>	NO_TEST=TRUE
H678	GDDR A0_DBI2	GDDR_DBI_PHY	GDDR_A_0_DBI	FB A0 DBI_L<2>	NO_TEST=TRUE
H679	GDDR A0_DBI3	GDDR_DBI_PHY	GDDR_A_0_DBI	FB A0 DBI_L<3>	NO_TEST=TRUE
H680	GDDR A1_DBI0	GDDR_DBI_PHY	GDDR_A_1_DBI	FB A1 DBI_L<0>	NO_TEST=TRUE
H681	GDDR A1_DBI1	GDDR_DBI_PHY	GDDR_A_1_DBI	FB A1 DBI_L<1>	NO_TEST=TRUE
H682	GDDR A1_DBI2	GDDR_DBI_PHY	GDDR_A_1_DBI	FB A1 DBI_L<2>	NO_TEST=TRUE
H683	GDDR A1_DBI3	GDDR_DBI_PHY	GDDR_A_1_DBI	FB A1 DBI_L<3>	NO_TEST=TRUE
Forwarded Clock					
H684	GDDR A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK_P<0>	NO_TEST=TRUE
H685	GDDR A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK_N<0>	NO_TEST=TRUE
H686	GDDR A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK_P<1>	NO_TEST=TRUE
H687	GDDR A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK_N<1>	NO_TEST=TRUE
H688	GDDR A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK_P<0>	NO_TEST=TRUE
H689	GDDR A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK_N<0>	NO_TEST=TRUE
H690	GDDR A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK_P<1>	NO_TEST=TRUE
H691	GDDR A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK_N<1>	NO_TEST=TRUE

GPU

Electrical Constraint Set	Physical	Spacing	
Clocks			
H227	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_PL 83
H221	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_NG 83
H249	CLK_PCTE_PHY	CLK_PCTE	GPU_TESTMODE 83
H255	CLK_PCTE_PHY	CLK_PCTE	GPU_PEX_TERM 83
H245	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID NO_TEST=TRUE 86
H272	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID NO_TEST=TRUE 86
H246	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID NO_TEST=TRUE 87
H270	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID NO_TEST=TRUE 87
H247	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID NO_TEST=TRUE 88
H240	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID NO_TEST=TRUE 88
H259	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID NO_TEST=TRUE 89
H277	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID NO_TEST=TRUE 89
H273	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK 91
H274	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK 91
H275	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R 91
H276	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_P 91
	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_N 91
SMB			
H250	SMB_PHY	SMB	GPU_SMB_CLK 91
H258	SMB_PHY	SMB	GPU_SMB_DAT 91
H243	SMB_PHY	SMB	GPU_SMB_CLK_R 50 91
H252	SMB_PHY	SMB	GPU_SMB_DAT_R 50 91
PCIe Compensation			
H257	PCIE_50S	COMP_PCTE	FB_CAL_PD_VDDQ 94
H256	PCIE_50S	COMP_PCTE	FB_CAL_PU_GND 94
H253	PCIE_50S	COMP_PCTE	FB_CAL_TERM_GND 94

GDDR5 Frame Buffer B

Electrical Constraint Set		Physical	Spacing		
Memory Address					
MEM0	GD0R_B0_MA	GD0R_MA_PHY	GD0R_B_0_MA	FB B0 A<8...0>	NO TEST=TRUE
MEM0	GD0R_B1_MA	GD0R_MA_PHY	GD0R_B_1_MA	FB B1 A<8...0>	NO TEST=TRUE
Address Dynamic Bus Inv					
MEM0	GD0R_B0_ADBT	GD0R_ADBT_PHY	GD0R_B_0_ADBT	FB B0 ABT_L	NO TEST=TRUE
MEM0	GD0R_B1_ADBT	GD0R_ADBT_PHY	GD0R_B_1_ADBT	FB B1 ABT_L	NO TEST=TRUE
Control					
CTRL0	GD0R_B0_CKE	GD0R_CTRL_PHY	GD0R_B_0_CTRL	FB B0 CKE_L	NO TEST=TRUE
CTRL0	GD0R_B0_CTRL1	GD0R_CTRL_PHY	GD0R_B_0_CTRL1	FB B0 CS_L	NO TEST=TRUE
CTRL0	GD0R_B0_CTRL1	GD0R_CTRL_PHY	GD0R_B_0_CTRL1	FB B0 WE_L	NO TEST=TRUE
CTRL0	GD0R_B0_CTRL1	GD0R_CTRL_PHY	GD0R_B_0_CTRL1	FB B0 CAS_L	NO TEST=TRUE
CTRL0	GD0R_B0_CTRL1	GD0R_CTRL_PHY	GD0R_B_0_CTRL1	FB B0 RAS_L	NO TEST=TRUE
CTRL0	GD0R_B1_CKE	GD0R_CTRL_PHY	GD0R_B_1_CTRL	FB B1 CKE_L	NO TEST=TRUE
CTRL0	GD0R_B1_CTRL1	GD0R_CTRL_PHY	GD0R_B_1_CTRL1	FB B1 CS_L	NO TEST=TRUE
CTRL0	GD0R_B1_CTRL1	GD0R_CTRL_PHY	GD0R_B_1_CTRL1	FB B1 WE_L	NO TEST=TRUE
CTRL0	GD0R_B1_CTRL1	GD0R_CTRL_PHY	GD0R_B_1_CTRL1	FB B1 CAS_L	NO TEST=TRUE
CTRL0	GD0R_B1_CTRL1	GD0R_CTRL_PHY	GD0R_B_1_CTRL1	FB B1 RAS_L	NO TEST=TRUE
Clock					
CLK0	GD0R_B0_CLK	GD0R_CLK_PHY	GD0R_B_0_CLK	FB B0 CLK_P	NO TEST=TRUE
CLK0	GD0R_B0_CLK	GD0R_CLK_PHY	GD0R_B_0_CLK	FB B0 CLK_N	NO TEST=TRUE
CLK0	GD0R_B1_CLK	GD0R_CLK_PHY	GD0R_B_1_CLK	FB B1 CLK_P	NO TEST=TRUE
CLK0	GD0R_B1_CLK	GD0R_CLK_PHY	GD0R_B_1_CLK	FB B1 CLK_N	NO TEST=TRUE
Data					
DATA0	GD0R_B0_DQ_RYTE0	GD0R_DQ_PHY	GD0R_B_0_DQ	FB B0 DQ<7...0>	NO TEST=TRUE
DATA0	GD0R_B0_DQ_RYTE1	GD0R_DQ_PHY	GD0R_B_0_DQ	FB B0 DQ<15...8>	NO TEST=TRUE
DATA0	GD0R_B0_DQ_RYTE2	GD0R_DQ_PHY	GD0R_B_0_DQ	FB B0 DQ<23...16>	NO TEST=TRUE
DATA0	GD0R_B0_DQ_RYTE3	GD0R_DQ_PHY	GD0R_B_0_DQ	FB B0 DQ<31...24>	NO TEST=TRUE
DATA0	GD0R_B1_DQ_RYTE0	GD0R_DQ_PHY	GD0R_B_1_DQ	FB B1 DQ<7...0>	NO TEST=TRUE
DATA0	GD0R_B1_DQ_RYTE1	GD0R_DQ_PHY	GD0R_B_1_DQ	FB B1 DQ<15...8>	NO TEST=TRUE
DATA0	GD0R_B1_DQ_RYTE2	GD0R_DQ_PHY	GD0R_B_1_DQ	FB B1 DQ<23...16>	NO TEST=TRUE
DATA0	GD0R_B1_DQ_RYTE3	GD0R_DQ_PHY	GD0R_B_1_DQ	FB B1 DQ<31...24>	NO TEST=TRUE
Error Detection					
ED00	GD0R_B0_EDC0	GD0R_EDC_PHY	GD0R_B_0_EDC	FB B0 EDC<0>	NO TEST=TRUE
ED00	GD0R_B0_EDC1	GD0R_EDC_PHY	GD0R_B_0_EDC	FB B0 EDC<1>	NO TEST=TRUE
ED00	GD0R_B0_EDC2	GD0R_EDC_PHY	GD0R_B_0_EDC	FB B0 EDC<2>	NO TEST=TRUE
ED00	GD0R_B0_EDC3	GD0R_EDC_PHY	GD0R_B_0_EDC	FB B0 EDC<3>	NO TEST=TRUE
ED00	GD0R_B1_EDC0	GD0R_EDC_PHY	GD0R_B_1_EDC	FB B1 EDC<0>	NO TEST=TRUE
ED00	GD0R_B1_EDC1	GD0R_EDC_PHY	GD0R_B_1_EDC	FB B1 EDC<1>	NO TEST=TRUE
ED00	GD0R_B1_EDC2	GD0R_EDC_PHY	GD0R_B_1_EDC	FB B1 EDC<2>	NO TEST=TRUE
ED00	GD0R_B1_EDC3	GD0R_EDC_PHY	GD0R_B_1_EDC	FB B1 EDC<3>	NO TEST=TRUE
Data Dynamic Bus Inv					
DBI0	GD0R_B0_DBI0	GD0R_DBI_PHY	GD0R_B_0_DBI	FB B0 DBI_L<0>	NO TEST=TRUE
DBI0	GD0R_B0_DBI1	GD0R_DBI_PHY	GD0R_B_0_DBI	FB B0 DBI_L<1>	NO TEST=TRUE
DBI0	GD0R_B0_DBI2	GD0R_DBI_PHY	GD0R_B_0_DBI	FB B0 DBI_L<2>	NO TEST=TRUE
DBI0	GD0R_B0_DBI3	GD0R_DBI_PHY	GD0R_B_0_DBI	FB B0 DBI_L<3>	NO TEST=TRUE
DBI0	GD0R_B1_DBI0	GD0R_DBI_PHY	GD0R_B_1_DBI	FB B1 DBI_L<0>	NO TEST=TRUE
DBI0	GD0R_B1_DBI1	GD0R_DBI_PHY	GD0R_B_1_DBI	FB B1 DBI_L<1>	NO TEST=TRUE
DBI0	GD0R_B1_DBI2	GD0R_DBI_PHY	GD0R_B_1_DBI	FB B1 DBI_L<2>	NO TEST=TRUE
DBI0	GD0R_B1_DBI3	GD0R_DBI_PHY	GD0R_B_1_DBI	FB B1 DBI_L<3>	NO TEST=TRUE
Forwarded Clock					
WCK0	GD0R_B0_WCK0	GD0R_WCK_PHY	GD0R_B_0_WCK	FB B0 WCLK_P<0>	NO TEST=TRUE
WCK0	GD0R_B0_WCK0	GD0R_WCK_PHY	GD0R_B_0_WCK	FB B0 WCLK_N<0>	NO TEST=TRUE
WCK0	GD0R_B0_WCK1	GD0R_WCK_PHY	GD0R_B_0_WCK	FB B0 WCLK_P<1>	NO TEST=TRUE
WCK0	GD0R_B0_WCK1	GD0R_WCK_PHY	GD0R_B_0_WCK	FB B0 WCLK_N<1>	NO TEST=TRUE
WCK0	GD0R_B1_WCK0	GD0R_WCK_PHY	GD0R_B_1_WCK	FB B1 WCLK_P<0>	NO TEST=TRUE
WCK0	GD0R_B1_WCK0	GD0R_WCK_PHY	GD0R_B_1_WCK	FB B1 WCLK_N<0>	NO TEST=TRUE
WCK0	GD0R_B1_WCK1	GD0R_WCK_PHY	GD0R_B_1_WCK	FB B1 WCLK_P<1>	NO TEST=TRUE
WCK0	GD0R_B1_WCK1	GD0R_WCK_PHY	GD0R_B_1_WCK	FB B1 WCLK_N<1>	NO TEST=TRUE

Frame Buffer Reset

Electrical Constraint Set		Physical	Spacing	
Reset				
UNCL	GDDR_A0_RESET	GDDR_50S	GDDR_CTRL	FB_A0_RESET_L
UNCL	GDDR_A1_RESET	GDDR_50S	GDDR_CTRL	FB_A1_RESET_L
UNCL	GDDR_B0_RESET	GDDR_50S	GDDR_CTRL	FB_B0_RESET_L
UNCL	GDDR_B1_RESET	GDDR_50S	GDDR_CTRL	FB_B1_RESET_L
UNCL	GDDR_C0_RESET	GDDR_50S	GDDR_CTRL	FB_C0_RESET_L
UNCL	GDDR_C1_RESET	GDDR_50S	GDDR_CTRL	FB_C1_RESET_L
UNCL	GDDR_D0_RESET	GDDR_50S	GDDR_CTRL	FB_D0_RESET_L
UNCL	GDDR_D1_RESET	GDDR_50S	GDDR_CTRL	FB_D1_RESET_L

SYNC MASTER=D8 MLB		SYNC DATE=01/11/2012	
PAGE TITLE			
GDDR5/GPU Constraints			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-9505	D	
	REVISION	6.1.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 132 OF 144 SHEET 112 OF 123	

D

C

B

A

Constraints

BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

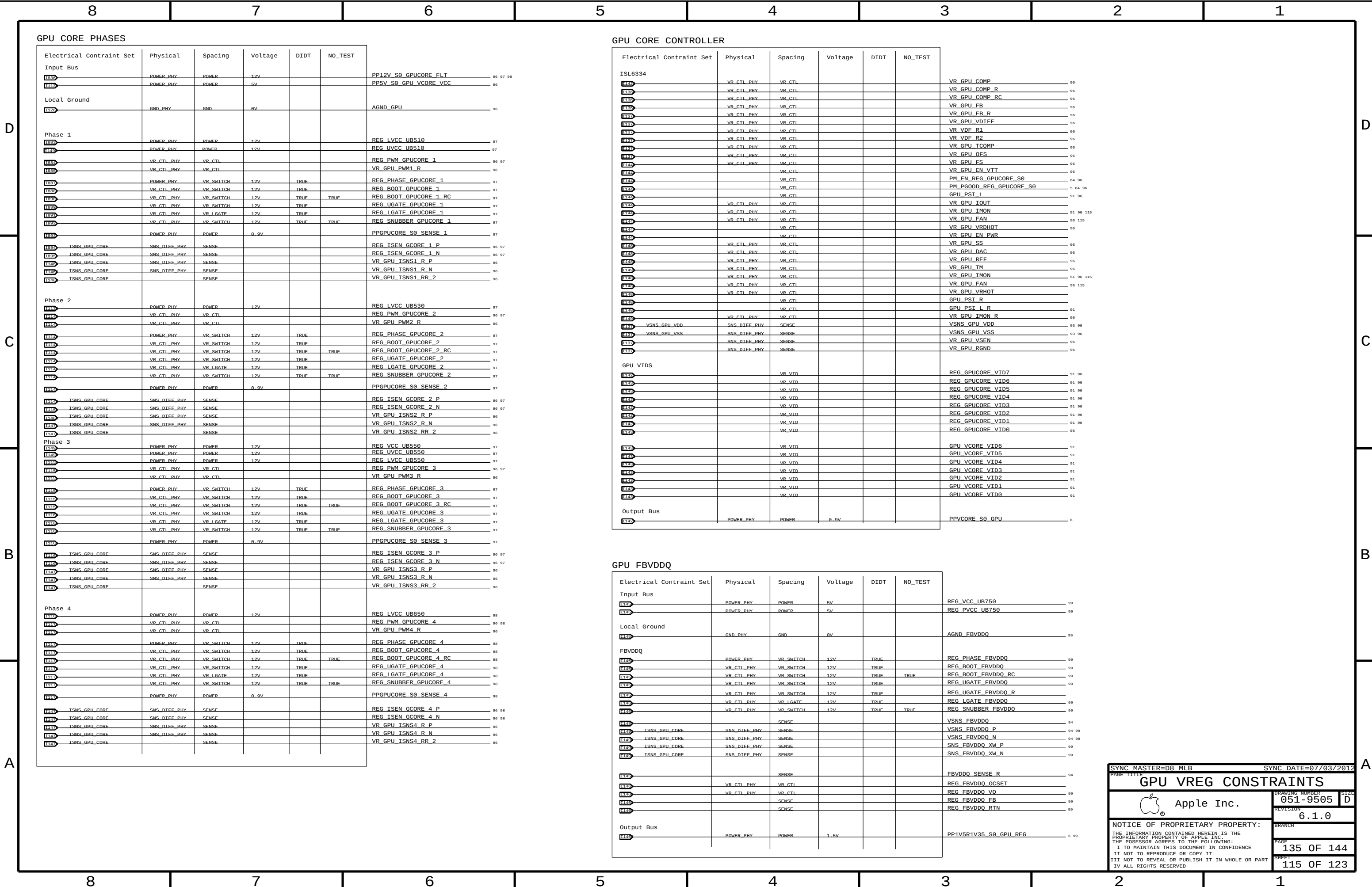
BLC Control

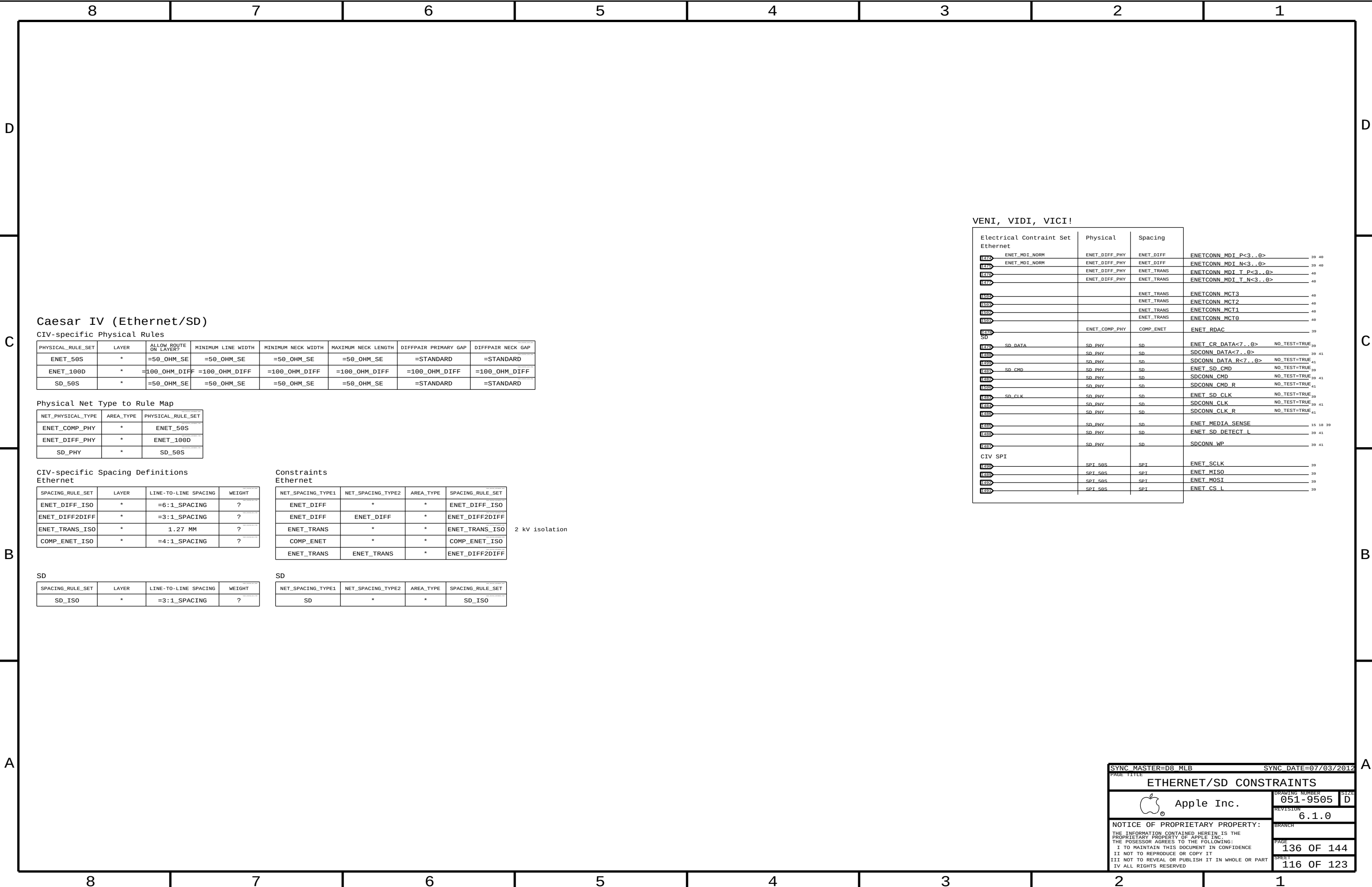
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Electrical Constraint Set	Physical	Spacing	
SPI			
MEM0	SMB_PHY	SMB	SMB_PCH_BLC_SCL 50 00
MEM0	SMB_PHY	SMB	SMB_PCH_BLC_SDA 50 00
MEM0	SMB_PHY	SMB	SMB_TCON_BLC_SCL 50 00
MEM0	SMB_PHY	SMB	SMB_TCON_BLC_SDA 50 00
12M REFERENCE CRYSTAL			
MEM0	CLK_XTAL	XTAL	BLC_MCU_XTAL_IN 00
MEM0	CLK_XTAL	XTAL	BLC_MCU_XTAL_OUT 00
MEM0	CLK_XTAL	XTAL	BLC_MCU_XTAL_OUT_R 00
250K REFERENCE CLOCKS			
MEM0	CLK_PCH_50S	CLK_PCH	STRCLK_R1 00 01
MEM0	CLK_PCH_50S	CLK_PCH	LED_DVR CLK 01

DCB

Δ





Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

VENI, VIDI, VICI!

Electrical Constraint Set	Physical	Spacing	
ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN MDI P<3..0>
ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN MDI N<3..0>
	ENET_DIFF_PHY	ENET_TRANS	ENETCONN MDI T P<3..0>
	ENET_DIFF_PHY	ENET_TRANS	ENETCONN MDI T N<3..0>
		ENET_TRANS	ENETCONN MCT3
		ENET_TRANS	ENETCONN MCT2
		ENET_TRANS	ENETCONN MCT1
		ENET_TRANS	ENETCONN MCT0
ENET_COMP_PHY	ENET_COMP_PHY	COMP_ENET	ENET_RDAC
SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0> NO_TEST=TRUE
	SD_PHY	SD	SDCONN_DATA<7..0>
	SD_PHY	SD	SDCONN_DATA R<7..0> NO_TEST=TRUE
SD_CMD	SD_PHY	SD	ENET_SD_CMD NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CMD NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CMD R NO_TEST=TRUE
SD_CLK	SD_PHY	SD	ENET_SD_CLK NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CLK NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CLK R NO_TEST=TRUE
	SD_PHY	SD	ENET_MEDIA_SENSE
	SD_PHY	SD	ENET_SD_DETECT L
	SD_PHY	SD	SDCONN_WP
CIV SPI	SPI_50S	SPI	ENET_SCLK
	SPI_50S	SPI	ENET_MISO
	SPI_50S	SPI	ENET_MOSI
	SPI_50S	SPI	ENET_CS_L

SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

PAGE TITLE

ETHERNET/SD CONSTRAINTS



Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9505

REVISION

6.1.0

BRANCH

PAGE

136 OF 144

SHEET

116 OF 123

SIZE

D

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

DCBABA

BOOST_*		
	Spacing	Netname
1600	GENERIC ISO	BOOST_BP
1600	GENERIC ISO	BOOST_BYPASS
1600	GENERIC ISO	BOOST_BYPASS_GATE
1600	PM	BOOST_EN_GATE
1600	PM	BOOST_EN_L
1600	GENERIC ISO	BOOST_RC
1600	GENERIC ISO	BOOST_SS

DP_*			
	Physical	Spacing	Netname
FE80		PM	DP_AUXIO_EN
FE80		PM	DP_GPU_MUX_EN
FE80		GENERIC_ISO	DP_INTPNL_HPD
FE80		GENERIC_ISO	DP_INT_EG_HPD
FE80	TBT_GEN_55S	TBT_GEN	DP_TBTPA_ODC_CLK
FE80	TBT_GEN_55S	TBT_GEN	DP_TBTPA_ODC_DATA
FE80		GENERIC_ISO	DP_TBTPA_HPD
FE80	TBT_GEN_55S	TBT_GEN	DP_TBTPB_ODC_CLK
FE80	TBT_GEN_55S	TBT_GEN	DP_TBTPB_ODC_DATA
FE80		GENERIC_ISO	DP_TBTPB_HPD

AUTO-CONSTRAINTS PG 2

ENETCONN_*

Spacing	Netname
1E4	GENERIC ISO ENETCONN_TCT 40

ENET_*

Physical	Spacing	Netname
1E2	GENERIC ISO	ENET_ACT 40
1E3	GENERIC ISO	ENET_ASF_GPIO 47 48
1E4	GENERIC ISO	ENET_CLKREQ_L 15 18 40
1E5	PM	ENET_CLKREQ_L_Q 39 40
1E6	PM	ENET_CR_1V8_EN 39 40
1E7	PM	ENET_CR_1V8_EN_R 39 40
1E8	PM	ENET_CR_3V3_EN_L 39 40
1E9	PM	ENET_CR_3V3_EN_L_R 39 40
1E10	PM	ENET_CR_PWREN 39 41
1E11	XDP_PHY	ENET_LOW_PWR 26 39 41
1E12	PM	ENET_PWR_EN_L 40
1E13	PM	ENET_PWR_EN_L_R 40
1E14	PM	ENET_RESET_L 39 41
1E15	PM	ENET_SD_RESET_L 26 41
1E16	GENERIC ISO	ENET_SR_DISABLE 39
1E17	GENERIC ISO	ENET_SR_LX 39 40
1E18	PM	ENET_WAKE_L 40

FAN_*

Spacing	Netname
1E19	GENERIC ISO FAN_0_PWM_FET 54
1E20	GENERIC ISO FAN_0_PWM_FILT 54
1E21	GENERIC ISO FAN_0_TACH_FET 54
1E22	GENERIC ISO FAN_0_TACH_FILT 54

FBVDD_*

Spacing	Netname
1E23	GENERIC ISO FBVDD_ALTVO 91 99

FB_*

Spacing	Netname
1E24	GENERIC ISO FB_A0_VREFC 86
1E25	GENERIC ISO FB_A0_VREFD 86
1E26	GENERIC ISO FB_A1_VREFC 86
1E27	GENERIC ISO FB_A1_VREFD 86
1E28	GENERIC ISO FB_B0_VREFC 87
1E29	GENERIC ISO FB_B0_VREFD 87
1E30	GENERIC ISO FB_B1_VREFC 87
1E31	GENERIC ISO FB_B1_VREFD 87
1E32	GENERIC ISO FB_C0_VREFC 88
1E33	GENERIC ISO FB_C0_VREFD 88
1E34	GENERIC ISO FB_C1_VREFC 88
1E35	GENERIC ISO FB_C1_VREFD 88
1E36	GENERIC ISO FB_D0_VREFC 89
1E37	GENERIC ISO FB_D0_VREFD 89
1E38	GENERIC ISO FB_D1_VREFC 89
1E39	GENERIC ISO FB_D1_VREFD 89
1E40	GENERIC ISO FB_SW_LEG_A 84 86

FB_*

Spacing	Netname
1E41	GENERIC ISO FB_SW_LEG_B 84 87
1E42	GENERIC ISO FB_SW_LEG_C 85 88
1E43	GENERIC ISO FB_SW_LEG_D 85 89
1E44	GENERIC ISO FB_VREF_GPU 94

FET_*

Physical	Spacing	Voltage	Netname
1E45	GENERIC ISO		FET_EN_P12V_S0 74
1E46	GENERIC ISO		FET_EN_P12V_S0_BLC 74
1E47	GENERIC ISO		FET_EN_P12V_S0_BLC_R 74
1E48	GENERIC ISO		FET_EN_P12V_S0_R 74
1E49	GENERIC ISO		FET_EN_P12V_S5 74
1E50	GENERIC ISO		FET_EN_P12V_S5_R 74
1E51	GENERIC ISO		FET_EN_VDDQ_S0 74
1E52	GENERIC ISO		FET_HDD_SLGSW 52
1E53	POWER_PHY	POWER 12V	FET_VCC_U7950 74
1E54	POWER_PHY	POWER 12V	FET_VCC_U7970 74
1E55	POWER_PHY	POWER 12V	FET_VCC_U7980 74

FLAG_*

Spacing	Netname
1E56	GENERIC ISO FLAG_V 89 91 92

G3_*

Spacing	Netname
1E57	GENERIC ISO G3_POWERON_L 47 48

GND_*

Spacing	Netname
1E58	SENSE GND_SMC_AVSS 47 48 51 55

GPU_*

Physical	Spacing	Netname
1E59	GENERIC ISO	GPU_ALT_VREF 84 85 91
1E60	GENERIC ISO	GPU_BUFRSTN 91
1E61	GENERIC ISO	GPU_IFPAB_PLLVDD 90
1E62	GENERIC ISO	GPU_IFPA_IOVDD 90
1E63	GENERIC ISO	GPU_IFPB_IOVDD 90
1E64	GENERIC ISO	GPU_IFPC_IOVDD 90
1E65	GENERIC ISO	GPU_IFPC_PLLVDD 90
1E66	XDP_PHY	GPU_JTAG_TDI 91
1E67	XDP_PHY	GPU_JTAG_TDO 91
1E68	XDP_PHY	GPU_JTAG_TMS 91
1E69	XDP_PHY	GPU_JTAG_TRST_L 91
1E70	GENERIC ISO	GPU_MLS_STRAP0 91
1E71	GENERIC ISO	GPU_MLS_STRAP1 91

GPU_*

Physical	Spacing	Netname
1E72	GENERIC ISO	GPU_MLS_STRAP2 91
1E73	GENERIC ISO	GPU_MLS_STRAP3 91
1E74	GENERIC ISO	GPU_MLS_STRAP4 91
1E75	GENERIC ISO	GPU_RESET_L 26 83 91
1E76	SPI_50S	SPI GPU_ROM_CS_L 91
1E77	SPI_50S	SPI GPU_ROM_CS_L_R 91
1E78	GENERIC ISO	GPU_ROM_HOLD_L 91
1E79	SPI_50S	SPI GPU_ROM_SI 91
1E80	SPI_50S	SPI GPU_ROM_SI_R 91
1E81	SPI_50S	SPI GPU_ROM_S0 91
1E82	SPI_50S	SPI GPU_ROM_S0_R 91
1E83	SPI_50S	SPI GPU_ROM_WP_L 91

HDD_*

Spacing	Netname
1E84	GENERIC ISO HDD_12V_S0_GATE 52
1E85	GENERIC ISO HDD_00B_1V00_REF 52
1E86	PM HDD_PWR_EN 21 52
1E87	GENERIC ISO HDD_PWR_EN_L 52
1E88	GENERIC ISO HDD_PWR_EN_R 52

I2C_*

Physical	Spacing	Netname
1E89	SMB_PHY	SMB I2C_TCON_MAS_SCL 50 78
1E90	SMB_PHY	SMB I2C_TCON_MAS_SDA 50 78

IFPD_*

Spacing	Netname
1E91	GENERIC ISO IFPD_RSET 99

IFPEF_*

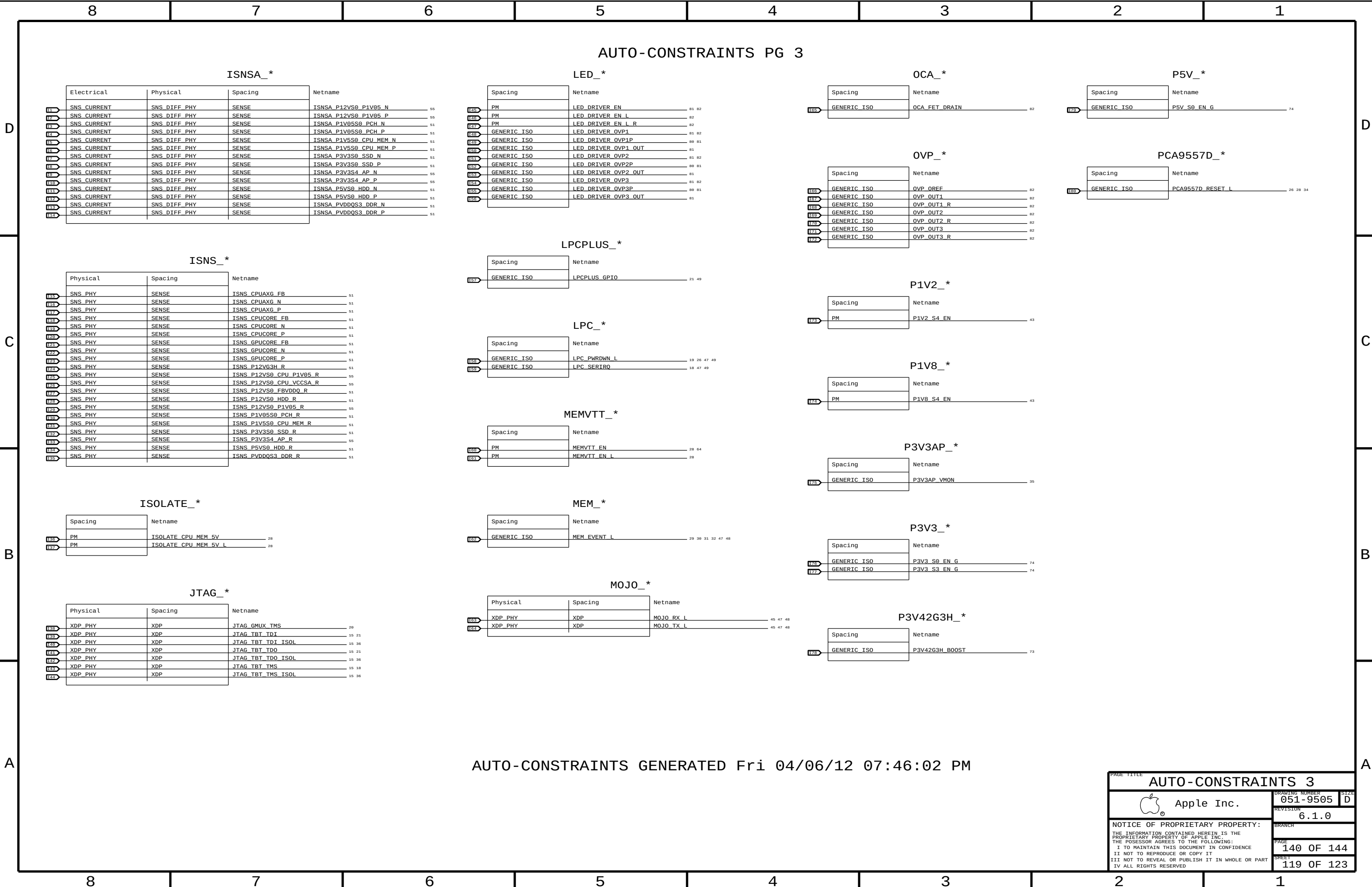
Spacing	Netname
1E92	GENERIC ISO IFPEF_RSET 99

ISNSA_*

Electrical	Physical	Spacing	Netname
1E93	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VG3H_N 91
1E94	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VG3H_P 91
1E95	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_CPU_P1V05_N 95
1E96	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_CPU_P1V05_P 95
1E97	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_CPU_VCCSA_N 95
1E98	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_CPU_VCCSA_P 95
1E99	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_FBVDDQ_N 91
1E100	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_FBVDDQ_P 91
1E101	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_HDD_N 91
1E102	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_HDD_P 91

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

PAGE TITLE			
AUTO-CONSTRAINTS 2			
DRAWING NUMBER		SIZE	
051-9505		D	
REVISION		6.1.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
PAGE		139 OF 144	
SHEET		118 OF 123	



AUTO-CONSTRAINTS PG 4

PCH_*

Physical	Spacing	Netname	
	PM	PCH_BLC_EXT_BOOT	21 80
	PM	PCH_BLC_EXT_BOOT_R	15 21
	PM	PCH_BLC_MCU_RESET	21 80
	PM	PCH_BLC_MCU_RESET_R	15 21
	PM	PCH_CAM_RESET	15 21
	PM	PCH_CAM_RESET_R	21 43
	GENERIC_ISO	PCH_DSMVRMEN	19
CPU_PHY	CPU	PCH_PECT	21
CPU_PHY	CPU	PCH_PROCPWRGD	21
	GENERIC_ISO	PCH_RCIN_L	21
	GENERIC_ISO	PCH_RI_L	19
	GENERIC_ISO	PCH_SMBALERT_L	19 18
	GENERIC_ISO	PCH_SRTCST_L	18
	GENERIC_ISO	PCH_STRP_TOPBLK_SWP_L	20
	GENERIC_ISO	PCH_SUSWARN_L	15 19

PCIE_*

Spacing	Netname	
GENERIC_ISO	PCIE_CLKREQ05_GPIO44_L	18
PM	PCIE_WAKE_L	19 35 48

PEG_*

Spacing	Netname	
GENERIC_ISO	PEG_CLKREQ_L	15 18 83

PGOOD_*

Spacing	Netname	
PM	PGOOD_P1V5_S0_DLY	28

PLT_*

Spacing	Netname	
PCH	PLT_RESET_L	20 26
PCH	PLT_RST_BUF_L	26

PM_*

Spacing	Netname	
PM	PM_CLKRUN_L	47 48 49
PM	PM_DSW_PWRGD	47 48 65
PM	PM_EN_FET_P12V_S0	64 74
PM	PM_EN_FET_P12V_S0_R	74
PM	PM_EN_FET_P3V3_S0	64 74
PM	PM_EN_FET_P3V3_S4	64 74
PM	PM_EN_FET_P5V_S0	64 74
PM	PM_EN_FET_VDDQ_S0	64 74
PM	PM_EN_LDO_DDRVTT_S0	64 72
PM	PM_EN_REG_CPUCORE_S0	64 66

PM_*

Spacing	Netname	
PM	PM_EN_REG_CPU_P1V05_S0	64 69
PM	PM_EN_REG_FBVDDQ_S0	64 99
PM	PM_EN_REG_GPUCORE_S0_R	96
PM	PM_EN_REG_P1V05_S0	64 99
PM	PM_EN_REG_P1V8_S0	64 72
PM	PM_EN_REG_P3V3_S5	64 71 74
PM	PM_EN_REG_P5V_S4	64 71
PM	PM_EN_REG_VCCSA_S0	64 70
PM	PM_EN_REG_VDDQ_S3	64 72
PM	PM_EN_S0	64
PM	PM_EN_S4	64
PM	PM_EN_USB_PWR	45 46 64
PM	PM_LED_A_ALL_SYS_PWRGD	5
PM	PM_LED_A_BLC_G00D	5
PM	PM_LED_A_CPUAXG_PG00D	5
PM	PM_LED_A_GPU_G00D	5
PM	PM_LED_A_PG00D_CPUCORE_S0	5
PM	PM_LED_A_PG00D_CPU_P1V05_S0	5
PM	PM_LED_A_PG00D_REG_FBVDDQ_S0	5
PM	PM_LED_A_PG00D_REG_GPUCORE_S0	5
PM	PM_LED_A_PG00D_REG_P1V05	5
PM	PM_LED_A_PG00D_REG_VDDQ_S3	5
PM	PM_LED_A_S4	5
PM	PM_LED_A_S5	5
PM	PM_LED_A_SLP_S3	5
PM	PM_LED_A_VIDEO_ON	5
PM	PM_LED_K_ALL_SYS_PWRGD	5
PM	PM_LED_K_BLC_G00D	5
PM	PM_LED_K_CPUAXG_PG00D	5
PM	PM_LED_K_GPU_G00D	5
PM	PM_LED_K_PG00D_CPUCORE_S0	5
PM	PM_LED_K_PG00D_CPU_P1V05_S0	5
PM	PM_LED_K_PG00D_REG_FBVDDQ_S0	5
PM	PM_LED_K_PG00D_REG_GPUCORE_S0	5
PM	PM_LED_K_PG00D_REG_P1V05	5
PM	PM_LED_K_PG00D_REG_VDDQ_S3	5
PM	PM_LED_K_SLP_S3	5
PM	PM_MEM_PWRGD_L	28
PM	PM_PCH_APWROK	19 65
PM	PM_PCH_PWROK	15 19 26 35 43 65 88
PM	PM_PCH_PWROK_APWROK	65
PM	PM_PCH_SYS_PWROK	19 48 65
PM	PM_PG00D_FBVDDQ_VDDQ_S0	64
PM	PM_PG00D_FET_P12V_S0	74
PM	PM_PG00D_FET_P12V_S0_BLC	74 82
PM	PM_PG00D_FET_P12V_S5	74
PM	PM_PG00D_FET_P3V3_S0	64 74
PM	PM_PG00D_FET_P5V_S0	64 74
PM	PM_PG00D_FET_VDDQ_S0	28 64 74
PM	PM_PG00D_P3V3_S4_FET	27 35 64 74
PM	PM_PG00D_REG_ALL_P1V05_S0	64 65
PM	PM_PG00D_REG_ALL_P1V05_S0_R	28 64
PM	PM_PG00D_REG_CPUCORE_S0	5 25 65 66
PM	PM_PG00D_REG_CPU_P1V05_S0	5 64 69
PM	PM_PG00D_REG_FBVDDQ_S0	5 64 99
PM	PM_PG00D_REG_P1V05_S0	5 64 99
PM	PM_PG00D_REG_P1V8_S0	64 72
PM	PM_PG00D_REG_P3V3_S5	65 71
PM	PM_PG00D_REG_P5V_S4	64 71
PM	PM_PG00D_REG_VCCSA_S0	64 65 70
PM	PM_PG00D_REG_VDDQ_S3	5 64 72
PM	PM_PWRBTN_L	15 19 25 47
PM	PM_RSMRST_PCH_L	19 65
PM	PM_RSMRST_PCH_L_R	65
PM	PM_SLP_S3_L	5 15 19 28 48 47 48 64
PM	PM_SLP_S4_L	15 19 47 64
PM	PM_SLP_S5_L	15 19 47 64
PCH	PM_SYSRST_L	19 25 26 47
PCH	PM_THRMTRIP_L	21 48
PM	PG00D_P12V_S0_R	65
PM	PG00D_P12V_S0	64 65 74

PP12V_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	12V	PP12V_LCD	78
POWER_PHY	POWER	12V	PP12V_LCD_EXT	78
POWER_PHY	POWER	12V	PP12V_S0_FAN_0_FILT	54
POWER_PHY	POWER	12V	PP12V_S0_HDD_FET	44 52

PP1V05_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA_F	17 22
POWER_PHY	POWER	1.05V	PP1V05_S0_PCH_VCCADPLL_B_F	17 22

PP1V2_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	1.2V	PP1V2_ENET_INTREG	49
POWER_PHY	POWER	1.2V	PP1V2_G3H_SMC_VDDC	47
POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_AVDDL	39
POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_GPHYPLL	39
POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_PCIEPLL	39
POWER_PHY	POWER	1.2V	PP1V2_USB_HUB_CRFLT	27
POWER_PHY	POWER	1.2V	PP1V2_USB_HUB_PLIFILT	27

PP1V5_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	1.5V	PP1V5_S0_DP_BIAS	75

PP1V8_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	1.8V	PP1V8_S0_PCH_VCCVRM_F	22 24

PP3V3R1V8_*

Physical	Spacing	Netname	
POWER_PHY	POWER	PP3V3R1V8_ENET_LR_OUT_REG	39

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

PAGE TITLE AUTO-CONSTRAINTS 4			
 Apple Inc.	DRAWING NUMBER	051-9505	SIZE D
	REVISION	6.1.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
PAGE 141 OF 144		SHEET 120 OF 123	

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

D

CBA

6

6

6

6

666

6

66666666

6

6666

6

6

6

D

C

B

A

